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May 2021

LETTER OF PROMULGATION

1. NWP 5-01 (MAY 2021), Navy Planning, is hereby promulgated as part of the Navy Warfare Library as a doctrinal publication.

2. NWP 5-01 (MAY 2021) is the foundation for Navy planning. It establishes Navy doctrine and governs the application of Navy planning methodologies to identify problems, develop solutions and communicate conclusions. It outlines a structured process to understand a problem and develop the information required to support decisions across the range of Navy, Joint and multinational operations.

3. NWP 5-01 (MAY 2021) was developed in accordance with NTRP 1-01 (JUN 2019), The Navy Warfare Library User Manual, and supersedes NWP 5-01 (DEC 2013), Navy Planning. Destroy superseded material in accordance with DoDM 5200.01.

4. NWP 5-01 (MAY 2021) is unclassified and approved for public release. Distribution is unlimited.

D. L. AAMODT

April 2025

ERRATUM NOTIFICATION

1. PURPOSE. Erratum 1 to NWP 5-01 (MAY 2021), Navy Planning, is developed in accordance with NTRP 1-01 (MAR 2024), The Navy Warfare Library User Manual.

2. SUPERSEDURE. Erratum 1 supersedes NWP 5-01 (MAY 2021), Navy Planning.

3. CONTENTS OF CHANGE. This change updates terminology to align with ALNAV 017/25.

a. Changes ‘gender’ to ‘sex’ on page H-8.

b. Updates the front cover and List of Effective Pages (LEP).

4. ENTRY OF CHANGE. This change is developed with the intent that the entire publication is to be reprinted and distributed in its entirety. Affected pages are identified in the LEP. Changes throughout are identified by change bars and/or updated footers.

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FOREWORD

This revised edition of Navy Planning, NWP 5-01, reflects the current guidance for conducting maritime activities and has been refined to ensure accuracy of content and consistency with joint and other Service doctrine. This publication serves as the Navy's capstone planning document for commanders, their staffs, and planners at all echelons. Further, this document forms the core of Navy planning doctrine and establishes the framework for our forces' ability to plan, prepare, and execute maritime operations.

This publication reflects recent Navy planning experiences and incorporates the following chapter changes:

- Streamlined the Navy Planning Overview chapter.
- Revised and expanded examples in the Mission Analysis step.
- Revised the Course of Action (COA) development process to reflect a more intuitive workflow.
- Streamlined the COA analysis step and generated an appendix to guide the conduct of war gaming.
- Elaborated on preparing the concept of operations in the COA Comparison and Decision step.
- The Plan/Order Development step title has been changed to "Directive Development" and has been revised to describe current practice.
- The Transition step has been revised to expand outputs and better explain transitions across the planning horizons.

Appendices have been:

- Restructured to move crucial content into applicable chapters.
- Deleted from the previous version of NWP 5-01 due to redundancies with other doctrinal publications and inclusion in the revised publication's chapters.
- Revised to be consistent with associated chapter information and updated to be consistent with current formats and the scenario example used throughout the document.
- Revised to describe best practices and to elaborate on methodologies.
- Edited for out-of-date information.

NWP 5-01 is focused at the operational level of war, but the methodologies upon which this publication are built can and should be used to perform Navy planning at all levels. I offer that it is a valuable tool for the planning practitioner, enabling the execution of operations throughout the full range of military operations, and should be studied closely.



SHOSHANA S. CHATFIELD
Rear Admiral, U.S. Navy
President, Naval War College

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PREFACE

SCOPE

This publication is the foundation for Navy planning. It establishes doctrine and puts forward the fundamental principles to guide planning within the framework of Service, joint, interagency, or multinational operations. It supersedes NWP 5-01 (DEC 2013).

PURPOSE

This publication sets forth Navy doctrine to govern the application of Navy planning methodologies to identify problems, develop solutions, and communicate conclusions. It outlines a structured process to understand a problem and develop the information required to support decisions across the range of Navy operations. This publication is in essence about the concepts of decision making, and it provides the basic process for planning Navy operations. The principles contained within can be applied by any echelon—on a Fleet staff, in a strike group, a task force, or a ship. The process can be applied by a commander facing a conventional threat such as another nation's navy, or in a conflict dominated by asymmetric tactics and an unconventional adversary. Moreover, the principles contained within are applicable to a variety of other evolutions and activities, some not necessarily operational in nature.

Though this publication likely will find its greatest utility with a Navy component commander, Navy numbered fleet commander, or a joint/combined force maritime component commander, it refers to Navy organizations in general, regardless of level of command.

The Navy planning process (NPP) is compatible with joint planning guidelines and procedures of JP 5-0, Joint Planning, as well as the United States Marine Corps' planning process, as reflected in MCWP 5-1, Marine Corps Planning Process. This publication alerts the reader when the NPP deviates from either of these two planning processes. Thus, Navy commands following the process can effectively integrate into and operate as part of a joint force.

APPLICATION

Doctrine and procedures established in this publication apply to all Navy commands and also may apply to commands predominantly composed of Navy forces (NAVFOR) but headed by another Service.

Guidance in this publication is authoritative. It will be followed unless, in the judgment of the commander, circumstances do not permit its use, mandate substantial changes, or require that other planning doctrine be used. As stated previously, this publication mirrors the guidance in joint and other Service documents; however, conflicts between the contents of this publication and that contained in other planning documents may arise given the dynamic and unpredictable nature of combat, potentially unforeseen changes in joint and other Service doctrine, as well as intricacies of joint and multinational operations. Navy commanders should take every measure to resolve significant disagreements between this publication and other planning doctrine that can adversely affect the planning and conduct of effective operations.

INSTRUCTIONS FOR USE

Printed copies may be ordered by following the directions included in Appendix C of NTRP 1-01.

Report urgent changes, routine changes, and administrative discrepancies by letter, general administrative message, the Navy Warfare Library (NWL) portal, or email to COMMANDER, NAVY WARFARE DEVELOPMENT COMMAND, ATTN: DOCTRINE, 1528 PIERSEY STREET, BLDG O-27, NORFOLK, VA 23511-2723.

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CHANGE BARS

Revised text is indicated by a black vertical line in the outside margin of the page, like the one printed next to this paragraph. The change bar indicates added or restated information. A change bar in the margin adjacent to the chapter number and title indicates a new or completely revised chapter.

WARNINGS, CAUTIONS, AND NOTES

The following definitions apply to warnings, cautions, and notes used in this publication:



WARNING

An operating procedure, practice, or condition that may result in injury or death if not carefully observed or followed.



CAUTION

An operating procedure, practice, or condition that may result in damage to equipment if not carefully observed or followed.

Note

An operating procedure, practice, or condition that requires emphasis.

WORDING

Word usage and intended meaning throughout this publication are as follows:

“Shall” and “must” indicate the application of a procedure is mandatory.

“Should” indicates the application of a procedure is recommended.

“May” and “need not” indicate the application of a procedure is optional.

“Will” indicates future time. It never indicates any degree of requirement for application of a procedure.

CHAPTER 1
Navy Planning Overview

In many ways the solution of a military problem is similar to the problems of everyday life. When one is faced with a problem, he [or she] studies the situation from all angles and decides what must be done. He then determines **how** it should be done—and proceeds to act, governing his actions according to the situation. Hundreds of such problems are solved by everyone. The great majority of them are not committed to writing, nor are they consciously solved in phases. They are solved by the natural, logical thought process normally pursued by the human mind.

In military problems the situation is much more complex, and the stakes are much higher. In order to insure a logical thought process, to guard against the oversight of important details, and to form a readily available record, military problems are usually solved in written form.

It is the purpose of this Manual to aid in the development of sound reasoning. There is no substitute for good sound reasoning in military planning. No check-off list nor diagram can be the key to a sound solution. With the exception of the Operation Plan [and other formal directives], this Manual does not attempt to prescribe a standard form for solving military problems. It merely outlines a logical procedure for arriving at a sound decision that can be translated into successful action. The emphasis is placed on sound reasoning–on the logical approach to military problems.

Throughout the entire planning phase and during [execution], a military situation remains fluid–never static. Open-mindedness on the part of military planners is therefore indispensable. Changing conditions will often require reversal or revision of previous decisions. Instead of striving to justify questionable decisions, the successful commander must be quick to recognize and correct their weaknesses.

Naval Manual of Operational Planning (1948)

1.1 INTRODUCTION

Military planning and, by extension, Navy planning, is a process of determining how (the ways) to use military capabilities (the means) in time and space to achieve objectives (OBJs) (the ends) while considering the associated risks. The conduct of planning includes the application of the art and science of understanding a situation, envisioning a desired future, and laying out effective ways of bringing that future about. Military planning is critical at every level of warfare—strategic, operational, and tactical—across the range of military operations (OPS) and continuum of conflict. Navy planning can be applied whether conditions permit a lengthy, deliberate process or if the situation forces a compressed timeline.

Planning provides commanders, staff, and planners a method of figuring out how to move from the current state to a more desirable future state. By its nature, planning is a learning process, as well as a creative process. It improves the collective understanding of a given situation and provides a mechanism to ensure a common understanding by all involved. Specifically, planning helps commanders, their staffs, and planners develop, direct, coordinate, and synchronize actions of a force. Planning involves projecting actions forward in time and space to influence events before they occur rather than merely responding to events as they occur. Ultimately, the product of planning is a plan or order—a directive issued to subordinates for future action. Commanders issue plans and orders to subordinates to communicate their understanding of the situation and their visualization of an operation.

1.2 PLANNING FOR A CAMPAIGN, CONTINGENCY, OR CRISIS

Planning takes place in a collaborative environment, which facilitates dialogue among leaders and planners across multiple planning levels. Concurrent and parallel planning facilitate a shared understanding of the situation, the

problem, and OBJs. Interactions with other components and higher headquarters (HHQ) enable the Navy planner to translate strategic guidance and direction into the maritime portion of campaign plans, contingency plans, or operation orders (OPORDs). HHQ defines tasks in guidance and directives that form a basis for planning. Alternatively, an unforeseen event, emergency, or time-sensitive crisis may trigger the need for Navy planning.

Campaign, contingency, and crisis planning use the same planning process, although the time available and specific products may differ. Campaign and contingency planning encompasses the preparation of plans during noncrisis situations with a timeline generally not driven by external events. Commanders, staff, and planners prepare supporting plans in concept plan (CONPLAN) or operation plan (OPLAN) format that describe how they intend to achieve their assigned OBJs and tasks.

Planning initiated in response to an emergent event or crisis uses the same process as all other planning. The specified steps, however, may be compressed or conducted in parallel to enable the time-sensitive development of OPLANs or OPORDs. Although planning is process-focused, the products of planning are critical to the successful execution of OPS. Whether it is for a campaign, contingency, or crisis, the ultimate result of Navy planning is a directive.

1.3 PLANNING AND SEA CONTROL

Navy planning incorporates naval seapower, and naval seapower encompasses the concepts of combat OPS and stability OPS. Policy, strategy, and doctrine (based on theory and practice) promulgate concepts and lexicon about naval seapower to inform how the Navy thinks, plans, communicates, and operates. However, this subject is fragmented across Service, joint, and combined publications. Consequently, planners must research, interpret, and translate doctrine for their particular situation.

Chief among doctrinal Navy seapower concepts is sea control. Invariably, Navy planning focuses on establishing and maintaining sea control, and exploiting sea control to influence events ashore. Historically, sea control has had several aliases. During the 19th and early 20th centuries, sea control was referred to as command of the sea. Joint Publication (JP) 3-32, Command and Control of Joint Maritime Operations, uses the terms maritime superiority and maritime supremacy to define degrees of sea control, and the term sea control OPS to describe the tasks required to establish maritime superiority or supremacy. Within this publication, sea control refers to the ability to prevail in open hostilities in an area of real or potential conflict and use that sea-space and the airspace above to accomplish mission OBJs. Sea control is not an aspect of peacetime power projection. Naval forces obtain sea control through combat or the immediate threat of combat. Hence, the United States Navy's forward presence in itself is insufficient to achieve sea control. However, forward deployment of Navy forces creates favorable conditions to obtain and then maintain sea control quickly after the start of hostilities. In practice, sea control planning requires precise articulation of the specific types of OPS, their desired outcomes, and their purpose.

Sea denial pertains to one's ability to deny partially or completely the enemy's use of the sea and associated airspace for military and commercial purposes. It is possible that the Navy will be forced to have sea denial as its principal strategic or operational OBJ depending the OBJs of the joint force, the maritime resources available, and the desired end state of an operation. It is an oversimplification to say that denying the use of the sea to an opponent is the opposite of sea control. If a weaker side denies control of the sea to a stronger opponent, this does not mean that it necessarily obtains control itself. Sea control and sea denial are often complementary OBJs.

Planners must balance the operational factors of space, time, and force to achieve the degree of sea control required to accomplish ultimate and intermediate OBJs. In some situations, mission OBJs can best be accomplished by denying the enemy use of specific sea areas vice exercising complete control of those areas. Therefore, maritime OPS will often consist of a mix of sea control and sea denial tasks.

1.4 OPERATIONAL ART AND DESIGN METHODOLOGY

The commander, staff, and planners develop plans and orders through the application of operational art and operational design in conjunction with (ICW) the Navy planning process (NPP). Leveraging operational art (Figure 1-1), commanders can conceptualize how the end states and desired OBJs set forth in HHQ guidance are

achieved. Operational art is the application of intuition and creative imagination by commanders and staffs. Supported by their skill, knowledge, experience, creativity, and judgment, commanders seek to understand the operational environment (OE), visualize and describe the desired end state, and employ assigned resources to achieve OBJs. Moreover, to appreciate operational art, the commander can think of Navy planning as having aspects of both science and art. The science involves such tangible aspects as disposition and number of ships, aircraft, weapons, supplies, and consumption rates as well as the interplay of operational factors, such as time and space, which affect employment of the naval force. The art lies in the creative application of operational art's theoretical underpinnings to build the operational approach.[1]

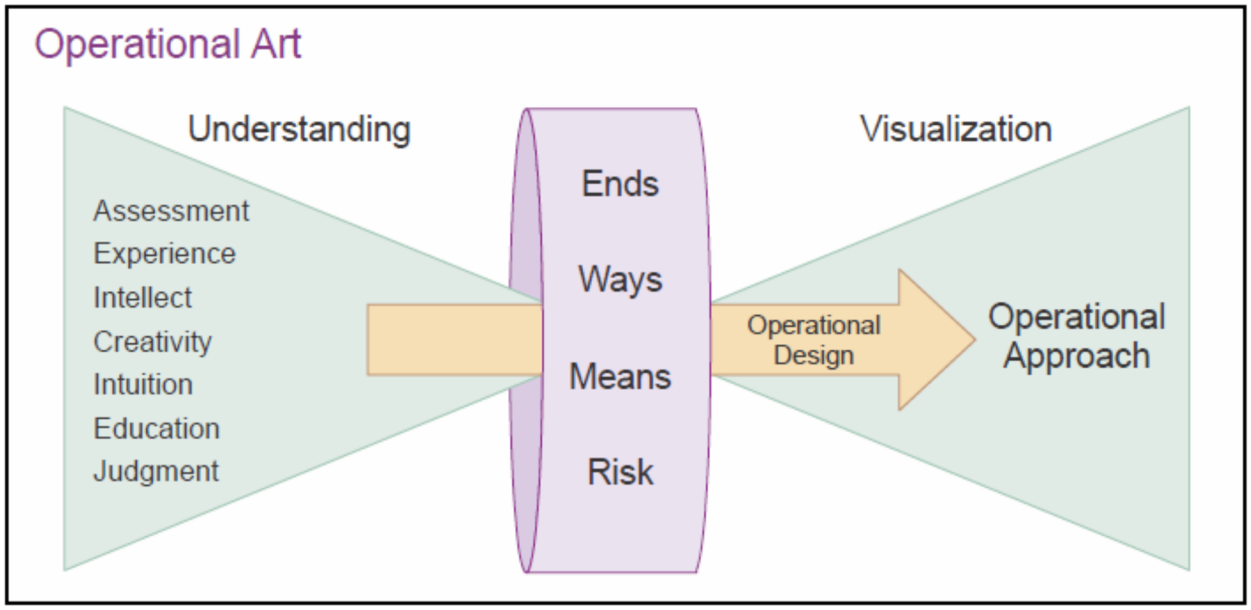


Figure 1-1. Operational Art and Operational Design[2]

Design methodology is a process that applies operational art to conceptualize and articulate an operational approach. This methodology may be used for any situation but is particularly useful when planning for an unfamiliar or complex and ill-structured situation. Planning has two closely related components: conceptual and detailed. Design methodology directly supports the conceptual aspect of planning by assisting the commander in visualizing the OE and the nature of the problem. This conceptualization sets the framework for the detailed approach to planning (NPP) that follows by deriving maritime OBJs, gaining a better appreciation of the environment, establishing initial guidance and intent, and articulating a broad operational approach to address the problem(s). Design methodology process has four components, each of which is reframed as circumstances change: (1) understanding the operational direction, (2) understanding the OE, (3) defining the problem, and (4) determining the operational approach. Appendix B provides a detailed explanation of these aspects and the process of design methodology.

1.5 IDENTIFYING THE OBJECTIVE(S) AND END STATE

The desired end state and OBJ(s) are critical to all aspects of planning and must be provided in the commander's initial planning guidance. Examining the HHQ's end state, OBJs, assumptions, and intent determines how the commander's portion of the operational problem relates to the HHQ's direction. Design methodology (described in Appendix B) facilitates initial development of OBJs for use and validation during the NPP. As planning progresses and deeper understanding occurs, the commander and staff frequently revisit the HHQ's stated end state and OBJs to ensure that the ends are achievable and properly framed.

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[1] **operational approach**—a broad description of the mission, operational concepts, tasks, and actions required to accomplish the mission. (DOD Dictionary. Source: JP 5-0.).

[2] JP 5-0, Joint Planning, 16 June 2017, fig. IV-2, p. IV-5.

The commander must first determine the strategic end state and then the operational and intermediate OBJs, ensuring that they all nest. An OBJs is clearly defined, decisive, and attainable. OBJs and their supporting effects provide the basis for identifying tasks. Military OBJs specify what must be accomplished and provide the basis for describing desired effects. OBJs describe what must be achieved to reach or attain the end state. Achieving operational OBJs ties execution of tactical tasks to reaching the military end state.

There are four primary considerations for an OBJ.

1. An OBJ establishes a single desired result (a goal).

2. An OBJ should link directly or indirectly to higher-level OBJs or to the end state.

3. An OBJ is specific and unambiguous.

4. An OBJ does not infer ways and/or means—it is not written as a task.

1.6 NAVY PLANNING PROCESS

The NPP (Figure 1-2) is the specific process for planning naval OPS. The NPP ensures that the employment of forces is linked to OBJs and integrates naval OPS with the actions of the joint force. Accordingly, the terminology, products, and concepts in the NPP are consistent with joint planning and joint doctrine and they are compatible with other Services' doctrine.

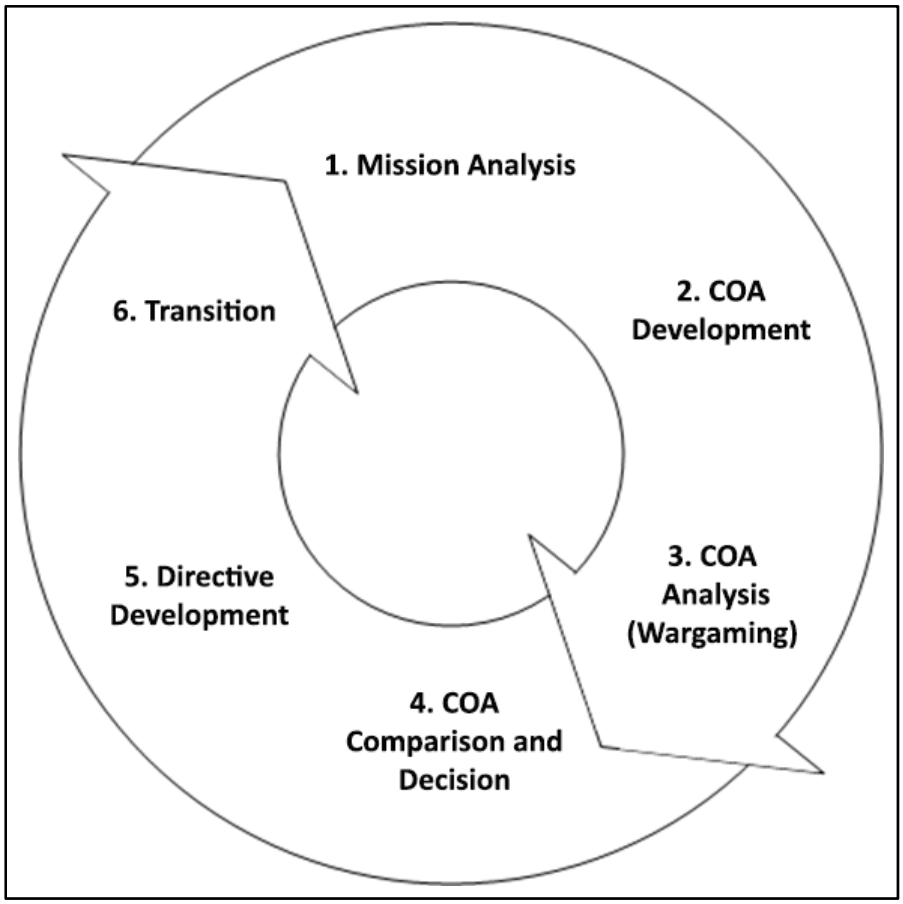


Figure 1-2. The Navy Planning Process

The NPP assists commanders, their staff, and planners in analyzing the OE. It helps distill a multitude of planning information to provide the commander with a coherent framework for determining the what and why (ends) as well as developing the method for execution (ways), given the forces and resources available (means) and the level of risk to the mission and forces. The NPP is an iterative process designed to gain decisions from the commander on how to proceed toward a solution. The process is thorough and repeatable, and it helps apply clarity, sound judgment, logic, and professional expertise to identifying problems, developing solutions, and communicating directions. The NPP can be time-consuming, but through consistent use commanders, their staff, and planners will become more proficient. Therefore, in the event experienced planners face a short timeline, they can easily flex the NPP to support planning in crises. Appendix A provides further detail on this concept of time-compressed planning.

The NPP establishes procedures to progressively analyze HHQ tasking; craft a mission statement; develop and analyze courses of action (COAs) against projected enemy actions (in some cases enemies could be forces of nature or other emerging nonmilitary threats); compare friendly COAs against the commander's criteria and each other; recommend a COA for decision; refine the concept of operation; prepare a directive; and transition the plan or order to subordinates tasked with its execution. The NPP organizes these procedures into six steps, shown in Figure 1-2, that provide commanders, their staff, and planners a means to organize planning activities, transmit plans to subordinates, and share a critical common understanding of the mission. Interactions among stakeholders during the various planning steps ensure a complete, concurrent, and coordinated effort that ensures flexibility, makes efficient use of available time, and facilitates continuous information sharing.

The result of the NPP is a military decision that is translated into a directive such as an OPLAN or OPORD. Frequently, products created during the NPP can and should be used during subsequent planning efforts when planning time is limited. It must be emphasized that while the time available to plan may change, the process does not.

1.6.1 Mission Analysis

Mission analysis forms the foundation for planning. As the first step of the process, its main purpose is to gain an understanding of the situation and produce a mission statement, commander's intent, and commander's planning guidance. The planning team and staff review and analyze orders, products from design methodology, guidance, intelligence, and other information in order to (IOT) gain knowledge and situational understanding to support the commander's decision making and subsequent planning steps.

1.6.2 Course of Action Development

Planners use the mission statement, commander's intent, and planning guidance with the commander's governing factors to generate and develop options for accomplishing the mission. Then they examine each prospective COA for validity by ensuring suitability, feasibility, acceptability, distinguishability, and completeness with respect to the current and anticipated situation, the mission, and the commander's guidance and intent.

1.6.3 Course of Action Analysis (Wargaming)

COA analysis involves a detailed assessment of each COA as it pertains to the enemy and the OE. Each friendly COA is war-gamed against selected enemy COAs. This step assists planners in identifying strengths, weaknesses, and associated risks, and in assessing shortfalls for each prospective friendly COA. Wargaming can identify potential branches and sequels that may require additional planning. Short of execution, COA wargaming provides the most reliable basis for understanding and improving each COA. This step also allows the staff to update its initial estimates based on a more refined understanding of the COA that is gained through the war game.

1.6.4 Course of Action Comparison and Decision

COA comparison is an OBJ process to evaluate COAs independently of each other and against set evaluation criteria approved by the commander. The goal is to identify the strengths and weaknesses of COAs that enable selecting a COA with the highest probability of success. All retained friendly COAs are evaluated ultimately leading to a decision by the commander.

1.6.5 Directive Development

Military directives, including plans and orders, are the principal output that commanders use to communicate the decisions reached through the planning process. These military directives may be formal, informal, written, or oral, depending on the time available and the complexity of the situation. They are the linchpins connecting concepts developed in the planning process with achieving the stated OBJs. A complete plan provides for common situational awareness and a point of common understanding that allows the force to influence the situation and use initiative to better react to changes that may occur during the course of OPS in execution.

1.6.6 Transition

Transition is the orderly handover of a directive to those tasked with execution of the operation. It provides staffs with the situational understanding and rationale for key decisions necessary to ensure that there is a coherent transition from planning to execution. Internal and external transition may be in the in the form of briefs or rehearsal of concept (ROC) drills. Internally, transition occurs between future plans (FUPLANS) and future/current operations (COPS) cells. Externally, transition occurs between the commander and the subordinate and supporting commands. The process, however, does not end here. As depicted in Figure 1-2 the process is continuous. Staffs maintain running estimates of the enemy, environment and risks that allow for plans and orders refinement. The planning team and staff continue to examine risks and assumptions and develop branches and sequels to the plans and orders. Key to this continuous process is the ongoing assessment of the operation's progress.

1.7 MISSION COMMAND

The Navy's heritage has inculcated an expectation of commanders to operate independently while following their superior commander's intent; to act when an opportunity presents itself, and to feel comfortable in conditions of ambiguity. These are attributes honed by mutual trust, confidence, and years of experience at sea. This description of disciplined initiative is also known as mission command in joint doctrine, which defines it as "the conduct of military operations through decentralized execution based upon mission type orders."[3] To ensure that planning does not stifle mission command, the superior Navy commander and staff focus more on the purpose of OPS rather than the details of how subordinates will execute the tasks, and avoid overly restrictive command and control (C2) concepts. The commander's intent cannot be a staff product; rather it must be a true embodiment of the commander's vision and the centerpiece of the commander's discussions with their staff and subordinate commanders.

1.8 NESTING OF NAVY PLANNING WITHIN OTHER PLANNING PROCESSES

Navy forces (NAVFOR) seldom operate independently without integration and coordination with other Services or agencies. Navy staffs should be well-versed in joint doctrine, particularly JP 3-0, Joint Operations; JP 5-0, Joint Planning; JP 3-32; Chairman of the Joint Chiefs of Staff manual (CJCSM) 3130 (series), Adaptive Planning and Execution (APEX) System; and approved joint terminology. Additionally, Navy planners shall integrate with the National Incident Management System (NIMS) during responses to domestic incidents.

Navy planning links commanders, as well as their staffs, with HHQ, laterally to other Service and functional component commands, and to subordinate commanders and their staffs. In the case of a Navy component commander (NCC) or joint force maritime component commander (JFMCC), for instance, a similar concurrent, collaborative planning technique also should be used with subordinate forces such as carrier strike groups (CSGs), expeditionary strike groups (ESGs) or amphibious ready groups (ARGs), surface action groups (SAGs), as well as task forces (TFs) and task groups (TGs) organized for specific missions. Likewise, in the case of a CSG or ESG (or ARG), planning among the warfare commanders, ships, squadron, and other attached forces must be linked.

Liaison provides a critical collaboration link to other commands associated with the operation. IOT ensure that planning efforts remain in concert with those of senior, subordinate, and adjacent commands, continuous and reliable communications are indispensable. Often, due to time and space considerations, it is impossible to be in

[3] JP 3-32.

the same location as other commands. Liaison can take many forms. It can be, and often is, established informally among corresponding functional staff elements. However, to ensure true unity of purpose and to avoid contradictory planning efforts, a formal liaison relationship is best. A HHQ should set forth guidance for liaison among subordinates, but a commander should arrange to have a liaison from other commands (and agencies if applicable), particularly subordinates, as part of the planning team. Liaison is a critical function and only the best staff officers should serve in liaison positions.

While Navy components of a joint force assist joint force commanders (JFCs) in developing a campaign plan, NAVFOR do not develop independent campaign plans. A campaign plan is a joint OPLAN for a series of related major OPS aimed at achieving strategic or operational OBJs within a given time and space (JP 3-0). NAVFOR develop supporting plans (OPLANs and orders) that nest with the JFC's campaign plan. Navy and maritime operational planning usually focuses on major or minor maritime OPS within specific portions (phases or multiple phases) of a campaign. As noted in Navy Warfare Publication (NWP) 3-56, Composite Warfare: Maritime Operations at the Tactical Level of War, tactical-level commanders and staffs utilize the NPP to make informed decisions, solve complex problems, and ultimately accomplish assigned missions. NWP 3-56 describes the details of when and why subordinate commanders nest into HHQ planning efforts, as well as what products (e.g., operational tasking (messages) (OPTASKs)) should be reviewed and modified.

Military planning occurs at all levels—from the strategic level and the Chairman of the Joint Chiefs of Staff (CJCS), through the combatant commanders (CCDRs) at the theater strategic level, across to other Service or functional components, and down to the individual ship, submarine (sub), or aircraft. The levels of planning differ in their complexity, scope, and purpose; however, they are all linked. In particular, the operational level, where the NPP ICW the joint planning process can be most effectively implemented, is where the crucial link from strategic military OBJs to tactical warfighting is established.

1.9 THE ROLE OF THE COMMANDER, STAFF, AND PLANNERS IN PLANNING

1.9.1 General

Commanders are making decisions constantly. Every day, commanders, their staffs, and planners resolve simple, routine, and complex problems. To help consider the best option(s) when faced with a force employment decision they apply knowledge, experience, and judgment coupled with the planning process. Navy staffs use this process to provide maritime planners with the procedures required for solving problems across the range of military OPS. Navy planning is a dynamic process that requires close cooperation and involvement between the commander, staff, and the planning team to ensure that time is used efficiently and that the most effective plan to meet the commander's guidance and intent is developed. The synergy among the commander, staff, and planning team is critical; however, with the collaborative nature of planning, the relationships with the JFC, Service component commands, functional component commands, adjacent supporting commands, and subordinate commands are equally important during planning.

Time for planning will often be insufficient. Effective interaction and a flexible approach to the process are required when time becomes critical. Commanders must understand how they can optimize the process through more specific guidance and increased participation. Generally, the less time to plan, the more involved the commander must become.

1.9.2 The Commander's Role in Planning

The commander plays a central role in planning. The responsibility for planning is inherent in command, and planning is a principal tool for exercising C2. Commanders are planners—the most important participants in effective planning. They are ultimately responsible for execution and often have the most experience. The commander provides the necessary focus and guidance to the planning team and staff. Additionally, the commander's relationships with other commanders and the ability to gain insights and expectations from adjacent and HHQ commanders contribute to the effectiveness of the process.

At the operational level, problems are often more complex and ill-defined. Solutions will incur positive and negative effects and there will be a need to synchronize diverse military and nonmilitary organizations. Since the operational-level commander's shaping of future tactical actions occurs while still planning, execution decisions and the direction of forces and functions occur throughout planning. The long geographical distances involved make timely direction during planning critical to future action. Delayed decisions can have adverse effects days and weeks into the future.

The commander's participation in planning lends credibility and value to the process. The commander's role becomes more critical when requirements compete with one another or when there is less time available for planning. Generally, the more involved commanders are in planning, the faster the commander, staff, and planners can plan. Through personal involvement, commanders learn from the staff about the situation while ensuring that the plan reflects the commander's intent. The commander can help in gaining efficiencies in the process by being more available to staff and planners and by increasing frequency of guidance, direction, and decision making. This is achieved through sound, precise guidance and up-front decisions on operational approach, COAs, wargaming, and with decreased formality of the process. Refer to Figure 1-3 for an illustration of the planning responsibilities for a JFMCC or NCC and the planning staff.

1.9.3 The Planning Team and Staff's Role in Planning

The staff must be properly organized to support planning. However, staffs are functional organizations and often support a command's efforts across multiple time horizons: far, mid, and near and through their respective staff directorates; FUPLANS, future operations (FOPS), and COPS. Therefore, the planning team(s) should draw personnel from the various staff codes and directorates. This cross-functional approach provides a holistic and dedicated effort to planning. Planning initiation can come from various sources depending on the level of command. Generally, an HHQ will start the process, but the commander, deputy, chief of plans, or other key leaders within a command could initiate a planning effort. Based upon their specific organization and staff resources, smaller units may direct another staff to guide planning.[4]

Additionally, each functional staff code should concurrently develop staff estimates in support of (ISO) the planning effort to ensure that proposed COAs are valid from each functional perspective (see Appendix I for selected examples of staff estimates). Staff estimates are essential throughout planning. They form the basis for supporting annexes and appendixes of an OPLAN or OPORD and are continuously updated as the situation or conditions change within the OE.

One of the staff's more important functions is to support and advise the commander throughout the planning process. The staff collects, analyzes, and presents relevant information to the commander to assist in achieving situational understanding and to aid in decision making. The planning team and staff representatives must continually communicate and synchronize their efforts to ensure that planning is coordinated and all staff sections are providing the commander with the most relevant information.

As the plan is briefed and discussed, it establishes a common purpose and OBJs within the organization and chain of command. The more effectively that staffs plan and exercise the plan, the more efficiently they can react to changing circumstances. Ultimately, planning enhances operational success by enabling the command to react faster and more effectively than the enemy. This facilitates planning FOPS and anticipating changes as they arise in a dynamic environment. The planning team and staff have several critical functions in relation to the commander and to their staff function. They all must fully understand the importance of the commander's planning guidance in their contributions from both perspectives.

[4] See Navy Tactics, Techniques, and Procedures (NTTP) 3-32.1, Maritime Operations Center, for a more detailed discussion of the operation of a maritime operations center.

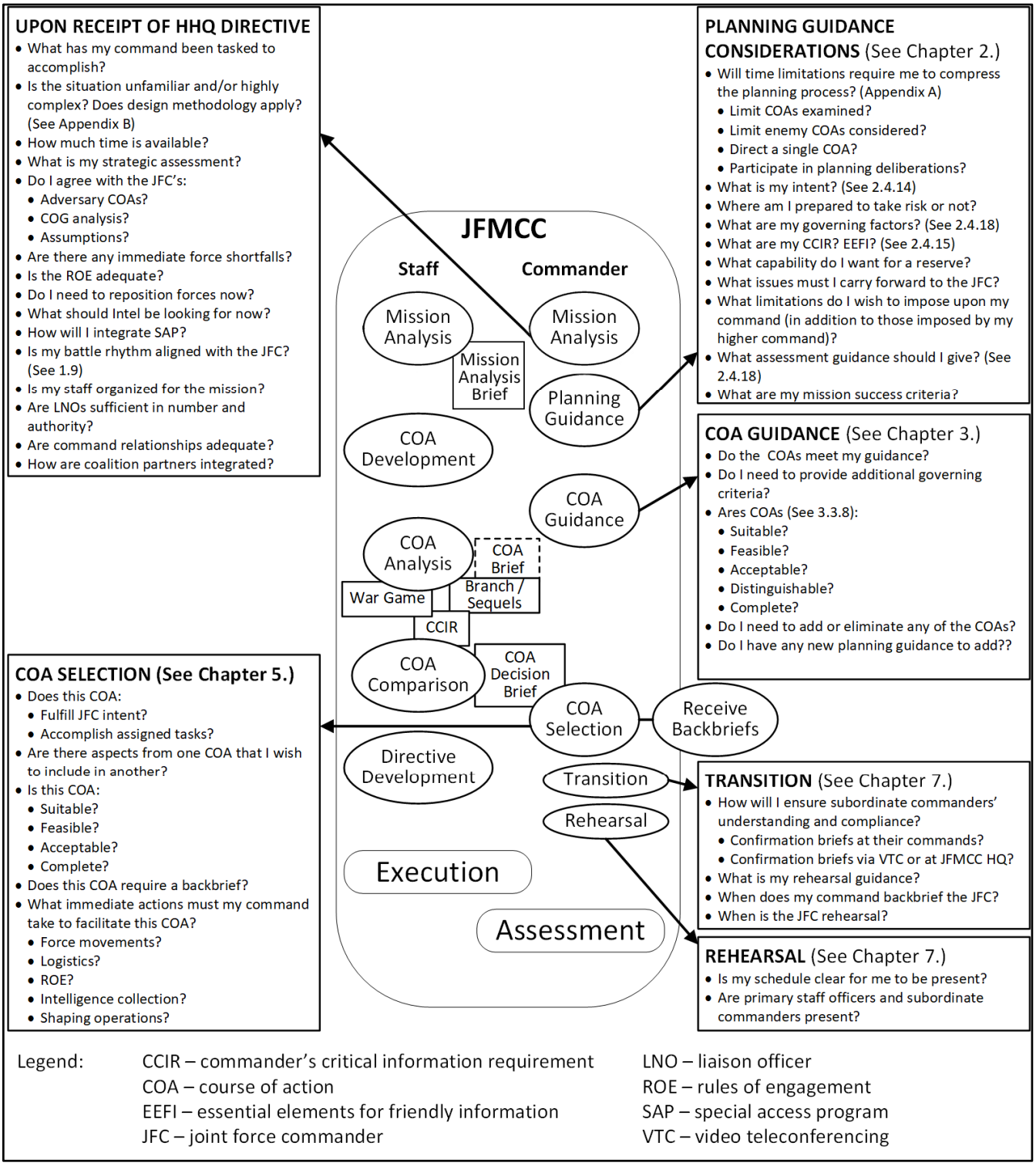


Figure 1-3. Commander's Role in the Navy Planning Process

Planning team and staff personnel should solicit, expect, and incorporate the commander's guidance. They should not move forward in the process without a clear understanding of the commander's desires. As time and requirements begin to compete with each other, making decisions and assumptions at the planning team level becomes imperative. However, if the team does not have a clear understanding of the commander's intent and guidance, those decisions and assumptions may move them farther away from the final product and may cause the planning to diverge from the naval force's OBJs.

The planning team and staff must know how to obtain the correct guidance from the commander, who is frequently involved in other urgent issues and may have competing priorities impacting time to adequately contemplate guidance for planning. Planners should be prepared to ask questions and make recommendations to the commander. Going to the commander with recommendations moves planning along while avoiding the pitfall of making decisions for the commander. Planners need frequent guidance from the commander, even after the initial order is published. This is especially true for planning in a time-constrained environment.

It is understood that echelons below an NCC or JFMCC do not always have the same robust staff numbers of higher echelons. However, that should not preclude them from employing Navy planning. Often, lower level commands find that the bulk of the planning effort resides with the commander and a select number of staff or command personnel. Even smaller naval organizations will find the planning process a useful framework for the analysis of an operational task and the development of COAs to address force employment.

CHAPTER 2
Mission Analysis

Before undertaking a task, the commander makes an estimate of the situation and formulates a plan of action . . . Even when time is so short as to permit only a mental estimate, the same logical process is used.

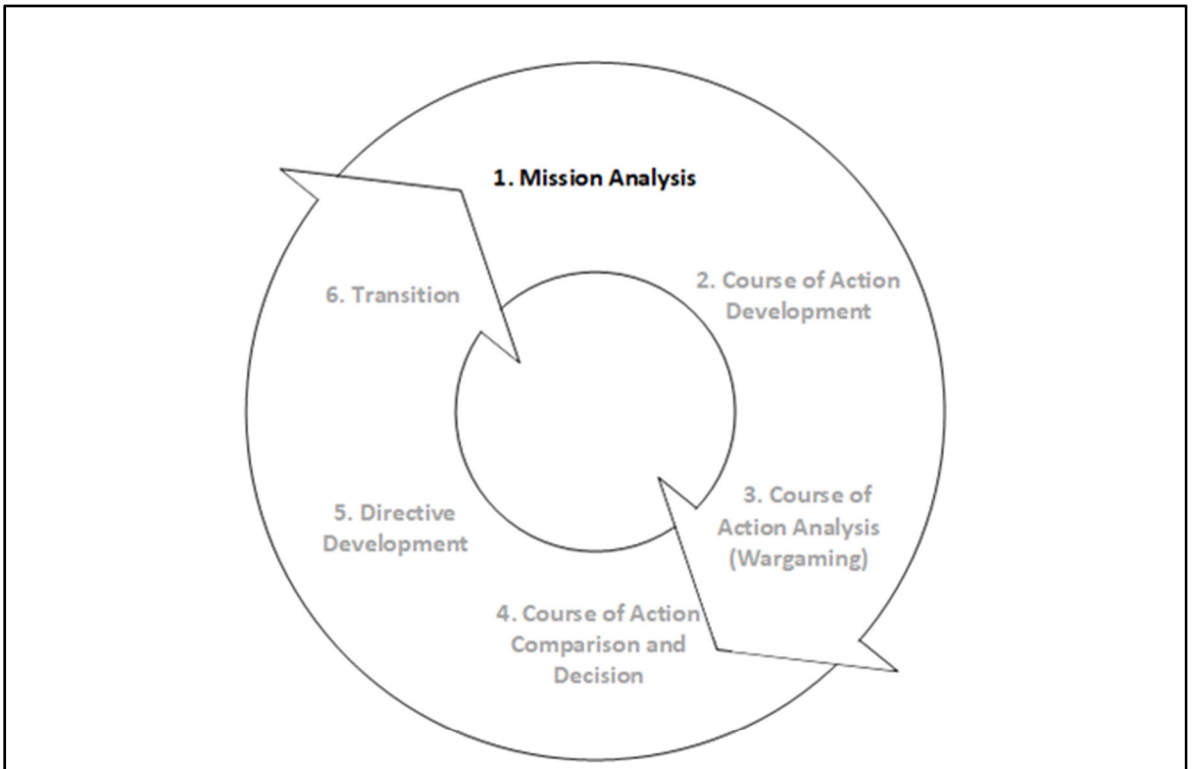
War Instructions, U.S. Navy, 1944

2.1 INTRODUCTION

As the first step of the NPP (Figure 2-1), mission analysis builds the foundation for the entire planning process. Its purpose is to give the commander, staff, and planning team an overall understanding and appreciation of the tasking from HHQ, as well as identifying the ends, ways, and means to accomplish that HHQ tasking. Mission analysis, when completed correctly, provides the who, what, when, where, and why for the commanders, staffs and planners and makes the development of the how possible. As mission analysis occurs, commanders, staffs, and planners must ensure the cross-functional integration of all staff sections and functional capabilities.

The commander directs the formation of a planning team and guides commencement of mission analysis. Mission analysis begins with a review of orders, plans, intelligence products, and guidance provided by HHQ to determine what tasks are assigned to the Navy commander. A detailed review of the component and subordinate capabilities, strengths, and weaknesses follows, providing the commander and staff an understanding of what can be accomplished and the risks associated with possible actions. This results in the production of a mission statement for the component and identifies tasks necessary to accomplish that mission. Mission analysis leads to the delivery of a briefing to the commander. The mission analysis briefing ensures that the commander and staff have a common understanding of mission, intent, and planning guidance, which allows for the formation of any additional detailed commander's planning guidance and for COA development (the how) to begin. Further, the warning order (WARNORD) produced at the conclusion of mission analysis becomes the impetus and foundation for subordinate planning. Although this list is not all-inclusive, effective mission analysis should address:

1. Purpose of the mission and tasks assigned
2. Understanding the OE
3. Tasks that the command must complete to accomplish the mission
4. Limitations that have been placed on our own forces' actions
5. Knowledge gaps that must be resolved for planning to continue
6. Forces/assets available to conduct the operation
7. Capability shortfalls to be filled to conduct the operation
8. Risks that must be mitigated.



Inputs

Higher Headquarters
- Plans, orders, and guidance
- Intelligence products
- Staff estimates

Commander
- Initial planning guidance
- Initial commander's intent
- Design products (if used)

Staff
- Initial staff estimates

Process

- Identify source(s) of mission.
- Review the commander's initial planning guidance.
- Receive intelligence preparation of the operational environment briefing.
- Identify command relationships.
- Analyze higher command's mission and intent.
- Determine specified, implied, and essential tasks.
- State the purpose of the operation.
- Identify externally imposed limitations.
- Identify facts and develop planning assumptions.
- Analyze available forces and assets.
- Determine friendly force center of gravity and decisive points.
- Conduct initial risk assessment.
- Develop proposed mission statement.
- Develop proposed updates to commander's intent.
- Develop initial commander's critical information requirements.
- Conduct mission analysis briefing.
- Develop warning order(s).
- Develop commander's planning guidance.

Outputs

- Approved mission statement
- Approved commander's intent
- Approved commander's critical information requirements
- Approved commander's planning guidance
- Warning order(s)
- Updated staff estimates

Figure 2-1. Mission Analysis

2.2 THE COMMANDER'S ROLE AND COMMENCING MISSION ANALYSIS

The commander's role in guiding the planning team and the planning process cannot be overstated. While the commander's close involvement is imperative throughout planning to ensure the concept unfolds as visualized, the commander's early engagement shapes all subsequent planning actions. The commander is instrumental in framing the nature of the operation and providing the initial planning guidance to the planning team and the staff. If planning

is straightforward—albeit potentially complex—and something the commander and staff have encountered before, the commander's initial planning guidance might be easily crafted. If, however, the situation is ill-defined and unfamiliar, the commander may direct a closer examination of the problem before or ICW mission analysis. One approach used to gain a deeper understanding of the situation is encompassed in the methodologies of operational design. This approach can provide a framework for understanding the environment and the nature of the problem IOT provide greater clarity when creating an operational approach (see Appendix B). While tactical maritime OPS will seldom necessitate this deeper exploration of the nature of the problem, the requirements faced by component commanders at the operational level of warfare demand consideration of methodologies to develop clear and adequate initial planning guidance for a planning team to commence the NPP. No matter what methodologies are used, the commander's initial planning guidance is an important orientation for the planning staff as it initiates any planning activity. Regardless of whether the situation has evolved over time or instantaneously, the commander has a responsibility to guide the planning team.

2.3 INPUTS

Before the planning team can effectively commence mission analysis, several inputs should be on hand to ensure planning team efforts are not wasted. As reflected in Figure 2-1, the planning team must gather information from several levels as a prelude to the actual initiation of the mission analysis process. As the planning process continues, the planning team and staff will need to continue to update the inputs and other sources as the current situation changes.

2.3.1 Higher Headquarters Plans, Orders, and Guidance

A verbal or written directive from HHQ provides necessary initial information for the initiation of mission analysis. This information is normally contained in OPLANs, CONPLANs, WARNORDs, OPORDs, etc. If HHQ directives or guidance are unclear, the commander, staff, or planning team should immediately seek clarification. Liaison officers (LNOs) can provide valuable information and should actively participate in planning.

2.3.2 Higher Headquarters Intelligence Products

HHQ intelligence products form the basis of the commander's own intelligence support. Much of this information is derived from the intelligence estimate, intelligence summaries, and Annex B of the HHQ OPORD. At a minimum, HHQ intelligence preparation of the operational environment (IPOE) products should provide identification and analysis of the enemy's OBJs, critical strengths and critical weaknesses, centers of gravity (COGs), critical capabilities, critical requirements, and critical vulnerabilities. The IPOE products, based on the current situation, should provide an analysis of the enemy's most likely and most dangerous COAs. The IPOE products from HHQs and those from the commander's intelligence staff may include the maritime modified combined obstacle overlay (MCOO) and enemy situation templates. See NWP 2-0, Naval Intelligence, for additional information about intelligence products.[5]

2.3.3 Higher Headquarters Staff Estimates

Based upon missions, OE situations, and enemy and friendly forces, staff estimates are documents that provide a current status and an assessment of the ability of the command to meet the requirements of the mission assigned while identifying shortfalls and potential issues. Every functional staff area has a responsibility to create and maintain a staff estimate. Depending on the level of command and the time available, the staff estimate may be a formal detailed written document, an informal oral presentation, or even a simple discussion with the commander and planning team. Whatever the form, the commander and planning team should have access to HHQ's staff estimates. Staff estimates provide context for a commander's design; identify critical information for mission analysis; form the basis for a staff section's recommendation of how a mission should be accomplished; and, most importantly, assist subordinate commands in developing their own staff estimates.

[5] JP 2-01.3, Joint Intelligence Preparation of the Operational Environment, provides a detailed examination of JIPOE products.

2.3.4 Commander's Initial Planning Guidance

The commander not only receives guidance and products from higher authority, but is also responsible for developing initial planning guidance to focus mission analysis. The commander must also consider the capabilities of the forces assigned/attached—combat readiness, material support, unit morale, and other factors—to accomplish the mission. This mental exercise can consist simply of the commander's thoughts or can be a more rigorous effort that uses design methodology (Appendix B). Regardless, the commander develops a vision of the OE that then helps to produce guidance to the planning team.

Based on this initial assessment of available information (the commander may continue to refine these factors as planning continues), the commander provides the initial planning guidance. Ideally, the commander personally issues this guidance to the staff, planning team, and subordinate commanders and seeks immediate feedback to ensure understanding of that vision of the OE and guidance. Depending on the time available, the commander may provide either general or specific guidance for the planning team, staff, and subordinate commanders to consider (e.g., an enemy COG, a certain C2 relationship, etc.). The commander's initial planning guidance may also include a timeline to which the planning team must adhere, including time allocated for subordinate planning, such as the 1/3–2/3 rule.[6] Generally, the commander's initial planning guidance should include general assessment of the problem, any initial ideas regarding commander's critical information requirements (CCIRs),[7] any specific ideas regarding operational functions (C2, intelligence, fires, movement and maneuver, sustainment, protection),[8] specific force employment considerations, and planning limitations.

2.3.5 Commander's Initial Intent

A planning team should also receive an initial idea regarding commander's intent. While planning teams will review and recommend revisions through the mission analysis process, any thoughts the commander may have regarding the purpose, method and desired end state of the mission will also help focus the planning efforts. Ideally, a commander is able to provide initial guidance regarding risk(s) the commander is willing to accept in the execution of the tasks from HHQ. Commander's intent is addressed in greater detail in 2.4.14.

2.3.6 Staff Estimates

The value of staff estimates to the planning process cannot be overemphasized. In addition to HHQ staff estimates, a planning team should also have access to its own staff section estimates. Each section of a headquarters (HQ) staff should prepare a staff estimate at the outset of planning. If the short-fused nature of the mission prevents a staff estimate prior to the initiation of mission analysis, staff members must take advantage of the information determined in mission analysis to produce one before COA development begins, and then continue refinement of the staff estimate at the conclusion of each step in the planning process. In the absence of an internal staff estimate, the HHQ staff estimate may become the starting point. Appendix I contains formats and content for selected staff section estimates and can be used as a foundation for any staff section staff estimate.

2.4 PROCESS

Mission analysis provides an overall understanding and appreciation of the tasking from HHQ, identifies tasks needed to accomplish the mission, provides direction needed for planning, and identifies ends, ways and means to accomplish that HHQ tasking. The steps in mission analysis are not necessarily sequential. All of the steps, however, must be completed, regardless of time available. The process is neither rigid nor static; it is continuous, evolving, iterative, and dynamic. Several of the steps lend themselves to being done sequentially. Planning team leaders must be careful, however, to reconcile these simultaneous efforts to ensure the entire team is aware of the results. Other steps, such as determining critical factors, friendly COG, and decisive points (DPs) (2.4.11) cannot be completed successfully without a full understanding of available forces and assets (2.4.10).

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[6] This reflects a goal of the HHQ using no more than one-third of the allocated preparation time, allowing the subordinate commands the remainder of the time to plan for the operation.

[7] CCCIRs are information requirements identified by the commander as being critical to facilitating timely decision making. See 2.4.15.

[8] At time of publication, Navy doctrine recognizes six operational functions and has not addressed the joint function of information described in JP 3-0. Without approved Navy doctrine regarding how operational level of warfare commanders (i.e., numbered fleet commander (NFC)/NCC/MCC) will implement the new joint function of Information, the NPP continues to address the Navy-approved six operational functions.

Planning team leaders must develop their "plan to plan" to determine how efforts of the planning team will be oriented, which steps will be done sequentially and/or simultaneously (if any), and how much time will be spent on each. The planning team develops and gathers information that is continuously refined throughout the planning process. The staff uses this information to prepare functional staff estimates, which provide a logical and orderly examination of all factors that affect mission accomplishment. At the conclusion of mission analysis, the planning team should be able to present a sufficient level of detail to the commander for planning to advance to the next step in the process. The commander, staff, and planning team should fully understand the who, what when, where and why and be prepared to mitigate risks identified through COA development, the how. Additionally, during this step of the process, planners start to build the synchronization matrix and decision support tools that become fully developed later in the NPP. The following 18 subparagraphs review each step of mission analysis in greater detail IOT describe the requirements to complete each step.

2.4.1 Identify Source(s) of the Mission

The source of the mission is normally found in an HHQ directive (e.g., OPLAN, OPORD, or WARNORD). To plan effectively, however, planners must also aggressively identify and analyze all documents relative to the mission and area of operations (AO) such as: on-the-shelf OPLANs, standing and supplemental rules of engagement (ROE), standard operating procedures (SOPs), and existing IPOE documents. Many of these documents may be long-standing, but other shorter-term documents may be equally as important. Planners should, therefore, incorporate standing orders and directives from both their higher and their own HQ into the process. Depending on the scope of the operation, consider also reviewing applicable United Nations Security Council resolutions (UNSCRs), alliance directives, Presidential policy, and other authoritative sources for additional information. Regardless of the time frame available for the planning process and mission analysis, planners must refer to the appropriate documents IOT support their commander.

2.4.2 Review Commander's Initial Planning Guidance

The importance of the commander in the planning process at this point cannot be overstated. The commander's initial planning guidance is critical for the planning team to be able to effectively execute mission analysis. Ideally, the planning team should already have the commander's initial planning guidance. Planners should analyze the guidance, considering identified OBJs,[9] HHQ products, and other available information. If aspects of the guidance are inconsistent with known information or unclear, planners must clarify with the commander. In a time-constrained environment, all members of the planning team may not have been present when the commander's initial planning guidance was given. If so, the planning team lead must review the guidance for the benefit of all members. If, for some reason, the commander has not developed or presented initial planning guidance, the planning team must solicit that guidance. To go forward in mission analysis without commander's initial planning guidance may result in wasted time and effort looking at various aspects of the who, what, when, where, and why that are not aligned with the commander's expectations.

2.4.3 Receive Intelligence Preparation of the Operational Environment Briefing

As with commander's initial planning guidance, all members of the planning team should have awareness of the OE. The inputs required to begin mission analysis include many of the documents and products needed to raise the awareness of the planning team. Ideally, this would come in the form of a formal IPOE briefing to the planning team. The briefing provides the planning team with a review of the OE and the enemy, and offers the planning team a means to focus on many of the factors that will influence the remainder of planning. In some circumstances, a formal briefing may not be possible. Members of the planning team may have to assimilate the information on their own through thorough study of the various input documents. Regardless of the format, planning team leaders must ensure that all members of the team have a detailed understanding of the OE IOT proceed. Members of the planning team must also understand that the OE may change while their planning efforts proceed. The planning team must vigorously communicate with the staff sections to receive timely updates to the OE.

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[9] An objective is clearly defined, decisive, and attainable. Objectives and their supporting effects provide the basis for identifying tasks to be accomplished. Military objectives specify what must be accomplished and provide the basis for describing desired effects. Objectives describe what must be achieved to reach or attain the end state. Appendix C explains how objectives are identified.

2.4.4 Identify Command Relationships

The staff and planning team must be clear in their understanding of command relationships that exist at all levels and phases of a given operation. Depending on the level of command, the commander could operate with a range of command authorities from having forces under operational control (OPCON), tactical control (TACON), or in a support (supported/supporting) relationship.[10] For example, at the Service or functional component level, the JFC should promulgate a directive that outlines the command relationships across the various phases of an operation.

The primary focus of this step of mission analysis is on command relationships, the associated authorities, and how forces may be employed—i.e., the command relationships between the commander's assigned forces, adjacent components/commands, coalition forces, and HHQ. Traditional NAVFOR or platforms (such as subs or ballistic missile defense (BMD)-capable ships) may be allocated TACON to other components or to the JFC. Other nations' forces working as a part of a coalition may only be in a TACON or support relationship. Support relationships often exist at the operational level of warfare between Service and functional component commanders (the JFMCC may be designated by the JFC to be supporting the joint force air component commander (JFACC) for area air defense (AD). Simultaneously, the JFC may designate the JFACC to be supporting the JFMCC during amphibious OPS). There will likely be multiple supported and supporting relationships that exist between Service or functional components for other tasks that are being conducted in each phase. Careful analysis of command relationships and the impact on the commander is critical to mission analysis and will enable the staff to approach the next step in the planning process, COA development, from an informed position.

The planning team should also develop the understanding of what command relationships exist within the JFMCC organization. Familiarity with NWP 3-56, will help provide doctrinal employment options and tactical structure of maritime forces. Awareness of the currently existing tactical-level command relationships must be known and understood so planners can determine if changes are possible and necessary to maximize efficiency in the task organization. Specific command relationship adjustments may become a separate COA. The C2 structure for the Navy/maritime component should be developed to best match the mission, within the bounds of command relationships established by the JFC.

2.4.5 Analyze the Higher Commander's Mission and Intent

The higher commander's mission statement, normally found in the HHQ's directive, must be studied. The commander and planning team must draw broad conclusions as to the character of the forthcoming military action. Neither, however, should make hasty assumptions about issues not addressed in the HHQ directive. If the HHQ directive is unclear, ambiguous, or confusing, the commander must seek clarification. Figure 2-2 is an example of a mission statement from a joint force HHQ.

> On order (O/O), Joint Task Force (JTF) BLUE SWORD defeats the Redland 23rd Guards Division (Div) and destroys terrorist forces and their infrastructure in Redland IOT eliminate the terrorist base of OPS in the area, secure the territorial integrity of Pinkland, and assist in stabilizing the region.

Figure 2-2. Example of a Joint Task Force Mission Statement

The higher commander's intent is normally found after the mission statement of the HHQ directive. Sometimes the higher commander's intent may not be transmitted at all. When this occurs, the subordinate commander and staff should derive an intent statement and confirm it with the HHQ. The intent statement of the higher commander should then be repeated in the situation paragraph of the commander's own directive to ensure that the staff, supporting commanders, and subordinates understand it. Each subordinate commander's intent must be framed and embedded within the context of the higher commander's intent, and nested both vertically and horizontally to achieve a common military end state throughout the command. Figure 2-3 provides an example of a JFC's intent.

[10] See JP 1, Doctrine for the Armed Forces of the United States, and NWP 3-32, Maritime Operations at the Operational Level of War, for a detailed examination of the various command relationships and associated authorities.

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GENTEXT/EXECUTION//

(U) COMMANDER'S INTENT.

(U) PURPOSE: Enable a stable and secure environment in which Redland
terrorist and conventional forces no longer threaten the government
of Pinkland.

(U) METHOD: There will be three primary focus areas: 1) neutralizing
Redland C2, 2) destroying Redland conventional forces, and
3) eliminating terrorist groups and their infrastructure. We will
respect the boundaries of neighboring neutral countries and minimize
damage to civilian infrastructure within Redland. Operating within a
joint organization construct, U.S. forces will cooperate with
multinational partners and synchronize with Pinkland to ensure unity
of effort. Maximize employment of all JTF capabilities, especially
our precision and mobility, to dominate the OE. Speed and timing are
essential to mission success.

(U) END STATE: The 23rd Guards Division is defeated and terrorist
forces and their camps in Redland destroyed. Territorial integrity
of Pinkland is secured, and Pinkland is prepared to assume
responsibility for security OPS.

(U) RISK: I am prepared to accept risk by foregoing completion of
shaping OPS to neutralize conventional Redland military forces
should an early opportunity present itself to quickly eliminate the
terrorists and their bases.
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Figure 2-3. Example of a Joint Force Commander's Intent

2.4.6 Determine Specified, Implied, and Essential Tasks

Every mission consists of two elements: the tasks to be accomplished by one's own forces, and the purpose of those tasks. If an operation has multiple tasks, then the priority of each task should be clearly established. All levels of command develop mission statements. Using information provided by HHQ and the commander's initial planning guidance, the planning team identifies specified and implied tasks, and organizes them by phase. From these two groups of tasks, the planning team will discern the essential tasks. Figure 2-4 provides examples of specified, implied, and essential tasks by phase.

2.4.6.1 Specified Tasks

Specified tasks are specifically assigned to a command by HHQ. They are derived primarily from the execution paragraphs of the directive but may be found elsewhere, such as in the coordinating instructions, annexes, or appendixes. Specified tasks may be identified but not have a corresponding purpose. The planning team must identify the corresponding purpose for any specified task without one and confirm the purpose with the HHQ issuing the task.

Examples of Specified Tasks (ST), Implied Tasks (IT) and Essential Tasks (*)

Phase I ST1: Deploy forces and establish a maritime presence in the joint operations area (JOA) IOT deter Redland aggression.

IT1.1 Be prepared to transition to hostilities and establish sea control in the vicinity of Pinkland IOT demonstrate U.S. resolve.

IT1.2 Conduct maritime security operations ISO Pinkland IOT deter Redland maritime aggression.

IT1.3 Conduct intelligence, surveillance, and reconnaissance (ISR) IOT determine Redland maritime disposition and intentions.

Phase I ST2: Counter air and missile threats in the vicinity of Pinkland seaports of debarkation (SPODs) IOT protect the buildup of combat power at intermediate staging bases (ISBs).

Phase II ST3: Conduct sea-control OPS in the JOA IOT gain maritime superiority to set the conditions for amphibious OPS.

IT3.1* Establish sea control within the amphibious objective area (AOA) and adjacent portions of the JOA IOT provide conditions suitable for amphibious assault.

IT3.2 Conduct air interdiction in the vicinity of central Redland IOT shape the environment for an amphibious assault.

IT3.3 Conduct ISR IOT target Redland maritime forces.

IT3.4 Conduct strike OPS against the Redland navy's shore infrastructure IOT degrade its ability threaten friendly maritime OPS.

Phase III ST4: *Conduct amphibious operation in AOA 2 IOT fix Redland forces in OBJ RAT and seize Port Yankee.

IT4.1 Plan and rehearse an amphibious assault into central Redland IOT support joint force land component commander (JFLCC) ground OBJs (phase I).

IT4.2 Be prepared to conduct mine countermeasures (MCM) OPS in the AOA IOT ensure execution of an amphibious assault (phase II).

IT4.3 Provide fire support to the landing force IOT support landing force OBJs.

Phase IV ST5: Conduct stability OPS IOT provide regional security.

IT5.1 Be prepared to conduct MCM OPS in the vicinity of Port Yankee IOT ensure its rapid opening as an SPOD.

IT5.2 Secure and open Port Yankee in southern Redland as an SPOD IOT allow the introduction of follow-on forces.

IT5.3 Be prepared to conduct port reconstruction OPS in port Yankee IOT ensure the rapid opening of the SPOD.

All phases ST6: Be prepared to conduct a noncombatant evacuation operation from Greyland IOT ensure the safety of U.S. citizens and other designated personnel.

All phases IT7: Provide logistics (LOG) support to maritime forces IOT extend their operational reach.

Figure 2-4. Examples of Specified, Implied, and Essential Tasks

2.4.6.2 Implied Tasks

Implied tasks are not specifically stated in the HHQ order but are additional tasks that must be performed IOT accomplish specified tasks or the mission. Implied tasks emerge from analysis of the order and other relevant sources of guidance and insight, such as multinational planning documents, enemy and friendly COG analysis, IPOE products, relevant doctrinal publications, interviews with subject matter experts, and the commander's guidance and operational approach. Implied tasks do not include routine tasks or SOPs that are inherent in most OPS, such as reporting requirements, sending LNOs, or conducting unit training. Commanders and planners should exercise caution when developing implied tasks to ensure that they include only those tasks necessary to accomplish assigned tasks or obtain mission success. One test to determine if an implied task is valid and necessary is to determine whether that proposed implied task will require the allocation of resources and must be further tasked to a subordinate unit. Any developed implied tasks must also have a corresponding purpose.

Note

Specified and implied tasks may be conditional before the execution is required. These tasks should be qualified by designating them as "be prepared to" (BPT). The BPT tasks may never be executed based upon the conditions required. If the task (whether specified or implied) is fully expected to be executed as a part of the operation, regardless of conditions, it should not be a BPT task.

2.4.6.3 Essential Tasks

Those tasks that most contribute to mission success are deemed essential. They become the central focus for OPS planning. Essential tasks are those that define mission success and apply to the force as a whole. Essential tasks can come from either specified or implied tasks. If a task must be successfully completed for the commander to accomplish an OBJ, it is an essential task. Only essential tasks are included in the proposed mission statement. If justified, a BPT task (specified or implied) can become an essential task when the required conditions exist.

Note

Key references that should be considered when generating tasks for the Navy or a naval force are the Universal Joint Task List (UJTL) and the Chief of Naval Operations Instruction (OPNAVINST) 3500.38B, Universal Naval Task List (UNTL). The UNTL includes the Navy Tactical Task List and the Marine Corps Task List. The UJTL and UNTL provide a wide range of tasks that joint and naval forces, respectively, can perform across a broad spectrum of OPS. Because both lists use joint terminology, there is an understanding at both the joint and Service or functional component command level of exactly the types of tasks that a naval force can accomplish. When communicating with subordinate commands and other Services, either directly through directives such as orders and plans, commanders should use doctrinally based terms and types of tasks that are delineated in the UJTL and UNTL.

2.4.7 State the Purpose of the Operation

The purpose of the operation is the unifying statement for the operation that answers the question "Why are we conducting this operation?" The purpose statement will later be placed in the proposed mission statement. The purpose must nest with the HHQ purpose, found in the HHQ's mission statement. If the superior's directive contains an intent statement, it should also be reviewed to help analyze the purpose of the operation. The purpose should be enduring, and it always dominates the tasks, since tasks can be accomplished or changed due to unforeseen circumstances, but their purpose remains essentially the same if the original mission remains unchanged. Figure 2-5 shows an example of a purpose statement.

Purpose: Establish a secure and stable maritime environment free from Redland military interference or terrorist threats.

Figure 2-5. Example of a Purpose Statement

2.4.8 Identify Externally Imposed Limitations

Limitations are externally imposed and do not include self-imposed limitations. They may be specifically stated in HHQ directives, guidance or instructions, or in the ROE. Planners should review operational general (message) (OPGEN) and OPTASKs from HHQ (if there is an applicable Navy HQ) as they may inadvertently impose limitations on a commander. Limitations are requirements placed on a command from HHQ that restrict freedom of action. They directly impact mission analysis and the planning process and are divided into two categories: restraints and constraints. Restraints prohibit an action. Constraints dictate an action. Self-imposed limitations

cannot be allowed to prevent accomplishment of an HHQ-directed mission. For instance, the requirement to act within a composite warfare commander (CWC) construct must not undermine the HHQ operational mission. Each member of the planning team must review all documents applicable to the mission and AO. Figure 2-6 provides examples of externally imposed limitations.

Example: Limitations and Implications		
	Limitations	**Implications**
Restraints (Must Not Do)	Do not mine Redland territorial waters (TTW). JTF forces must not conduct offensive OPS from neutral country TTW.	Accept risk of Redland naval forces at sea or use other means to close the port. Restrictions on operational fires and maneuver in the JOA.
Constraints (Must Do)	Strictly limit collateral damage during the deployment of lethal fires. Preserve Redland civilian infrastructure. Incorporate Pinkland military into the planning and execution of OPS.	Increases targeting complexity. Increases targeting complexity. Increases requirements for coordination and number of LNOs while challenging C2 and unity of effort.

Figure 2-6. Examples of Restraints and Constraints

2.4.9 Identify Facts and Develop Planning Assumptions

A fact is a statement of information known to be true (such as specific bases in Pinkland that have been made available for the operation). An assumption is a specific supposition of the OE that is treated as fact, in the absence of positive proof, which is essential for the continuation of planning.[11] Assumptions are made for friendly and enemy situations IOT continue planning. They encompass issues over which a commander normally does not have control. Ideally, for an assumption to be accepted, it should have the following qualities: It is logical, it is realistic, it is essential for planning to continue, and it does not assume away an enemy capability.

Assumptions are used in planning at every level. Subordinate commanders shall treat assumptions given by HHQ as facts IOT ensure they are planning with the same assumptions identified by HHQ. However, commanders and staffs at all levels continuously question and monitor whether these assumptions are validated or invalidated throughout planning and execution. While commanders can assume the success of friendly supporting forces, they cannot assume success for their own. Also, a planning team should never assume away an enemy capability or COA. As planning continues, additional assumptions may be needed, and previous assumptions may be discarded. Keep a record of assumptions IOT track and validate them as they are confirmed or disproved. If assumptions cannot be validated before execution, they become part of the inherent risk of the operation. During COA development, the commander may require the planning team to develop branch plans for all assumptions pertaining to enemy COAs. Figure 2-7 provides examples of facts and assumptions.

[11] Defined by DOD Dictionary. Source: JP 5-0.

Example: JTF Assumptions (Treated as Facts by the JFMCC)

- Orangeland will not freely allow terrorist camps to be established within their borders.
- Whiteland will allow support to terrorist organizations to pass over their border into Redland.
- Coalition nations will remain committed throughout the length of the operation.

Example: JFMCC Assumptions

- Greyland will permit overflight for carrier-based tactical aviation and Tomahawk land-attack missiles (TLAMs).
- Pinkland naval forces will cooperate with the JTF but remain under their national control.
- JFLCC Patriot missile sites are fully mission capable (FMC) at the ISBs before combat OPS begin.
- Pinkland will allow establishment of an SPOD and forward operating base (FOB) at Port X-Ray.
- The State Department-designated safe haven for noncombatant evacuees will be Port X-Ray in Pinkland.

Figure 2-7. Examples of Facts and Assumptions

2.4.10 Analyze Available Forces and Assets

Commanders and their staffs shall review the forces that have been provided for planning, identify their locations (if known), consider their capabilities, and initially determine whether there is a need to modify the current task organization and support relationships. This is also the time to determine what and when reserve forces will be available. When accomplishing this, the staff should refer to the specified and implied tasks to identify what broad force structure and capabilities are necessary to accomplish these tasks. This is just the initial analysis in which the staff identifies any potential shortfalls between the tasks/purposes and forces or subject matter/technical expertise available to carry out the tasks. This analysis also starts to gather information needed to conduct relative combat power analysis (RCPA) described in Appendix D. During this initial analysis, planners should evaluate the ability of the command to successfully accomplish the mission, given available forces and resources, including combat and support. If shortfalls are identified, with the approval of the commander, the request for forces (RFF) and/or request for capabilities (RFC) should begin. Figure 2-8 provides an example.

Example: Availability of Forces

Task: Be prepared to conduct port reconstruction OPS in Port Yankee IOT ensure the rapid opening of the SPOD.

Forces allocated: One CSG (staffed) consisting of one aircraft carrier (nuclear) (CVN) and four CRUDES. One ESG (staffed) consisting of one ARG (one amphibious assault ship (general purpose) (LHA), one dock landing ship (LSD), one amphibious transport dock (LPD), and two cruiser-destroyer (CRUDES) and one embarked Marine expeditionary unit (MEU). One mine countermeasures squadron (MCMRON) (staffed) consisting of three MCM. One maritime patrol and reconnaissance aircraft (MPRA) (staffed) consisting of four P8s.

Shortfall: No allocated construction battalion (BN) units.

Figure 2-8. Example of Available Forces

2.4.11 Determine Friendly Center of Gravity and Decisive Points

The next step in the mission analysis process requires a progressive analysis of determining the friendly center of gravity and DPs. Each of these elements applies to each level of war—strategic, operational, and tactical. The commander must be cognizant of critical factors and COGs at other levels of warfare, while focusing on those at the level of warfare for which the command is planning.

One must know and understand the OBJ(s) before determining the COG(s). OBJs describe what must be achieved to reach or attain the end state. Achieving operational OBJs ties execution of tactical tasks to reaching the military end state.

Determination of critical strengths and weaknesses leads to finding the COGs, which in turn leads to assessing DPs. This analysis should have already been completed for enemy forces as a part of the IPOE process. The commander and planning team must complete the same process for friendly forces. A detailed discussion of the steps to complete a center of gravity analysis is contained in appendix C. The intelligence staff identifies enemy critical strengths and weaknesses and COGs as part of the IPOE process. A complete intelligence staff estimate should contain critical strengths and weaknesses, COGs and decision points for the enemy. In many cases the enemy COG analysis will be complete before the mission analysis process begins. If not, it must be completed before the same process is done for friendly forces. Although the intelligence section completes the enemy COG analysis process, it should be the operations staff members, operational planning team (OPT) members, and subordinate commanders who assist the commander in identifying friendly critical factors and COGs. If the commander and staff employed design methodologies earlier in the planning process, some aspects of the friendly COG analysis may have already been completed.

As detailed in Appendix C, once the initial friendly COG assessment is completed, planners should deconstruct the COG and determine critical capabilities and critical requirements. By performing analysis at this level of detail, planners can then start to determine which critical requirements are deficient or vulnerable to attack. These vulnerable critical requirements become critical vulnerabilities. To be a critical vulnerability, it must allow for decisive or significant results against the COG. It is important to note friendly vulnerabilities IOT properly protect the friendly COG(s). Therefore, enemy critical vulnerabilities begin to define what needs to be attacked, influencing the COA development. Friendly critical vulnerabilities identify what needs to be protected, again, influencing the COA development. A detailed discussion of the steps to complete a center of gravity analysis is contained in Appendix C.

2.4.12 Conduct Initial Risk Assessment

During mission analysis, the commander and staff conduct an initial risk assessment. Risk is inherent in any use of military force or routine military activity. In mission analysis, this consists of identifying and assessing the threats to both the mission and the force. Appendix E has an expanded discussion of risk assessment and mitigation. The outcome of this initial risk assessment may influence considerations for commander's initial intent and planning guidance. At no time during mission analysis should the planning team address how to mitigate the risk identified. Risk mitigation is inherent in the COA development step of the NPP.

2.4.13 Develop Proposed Mission Statement

Based on mission analysis, the planning team drafts a mission statement for the commander to review, edit, and approve in concert with (or following) the mission analysis briefing. The mission statement should be a clear and concise statement of the essential tasks along with the purpose of those tasks. If the mission contains multiple tasks, they should be listed in the sequence that they are to be accomplished. A proper mission statement should be written in the present and imperative tense and contain the following items: who (which forces) will execute the mission, what type of action (e.g., defend) is contemplated (include essential tasks only), when the action will begin, where the action will occur, and why (purpose) each force conducts its part of the operation (including OBJs). Figure 2-9 shows an example of a mission statement.

Mission statement: O/O, JTF BLUE SWORD maritime forces establish sea control in vicinity of central Redland and the amphibious objective area (AOA) IOT establish a secure and stable maritime environment free from Redland military interference or terrorist threats.

Figure 2-9. Example of a Joint Force Maritime Component Commander Mission Statement

2.4.14 Develop Proposed Updates to Commander's Intent

A commander's intent is broader than the mission statement; it is a concise, free-form expression of the purpose of the force's activities, the desired results, and how actions should progress toward that end. It establishes a clear and succinct vision of how to conduct the action. It provides essential focus for planning as it enables the commander to indirectly control events during the execution of the operation. In short, the commander's intent links the mission and the concept of operations (CONOPS).[12] The intent expresses the broader purpose of the action that looks beyond the why of the immediate operation to the broader context of that mission, and it may include how the posture of the force at the end state of the action will transition to follow-on OPS (sequels). The commander's intent often evolves as planning progresses.

Each commander may view the content and timing of the commander's intent differently. While an initial commander's intent should be an input before mission analysis begins, that may not always be the case. Additionally, some commanders may insist that they write their own commander's intent while other commanders may ask their chief of staff (COS), maritime operations center (MOC) director, or even the planning team to craft a commander's intent for approval. Regardless of the commander's approach, a planning team will be hampered going into COA development without a commander's intent. This step in mission analysis, regardless of the form in which commander's intent resides, must be updated and some level of concurrence must be obtained from the commander. Should an initial commander's intent already exist, the results of mission analysis must be integrated into that intent and refined to accurately reflect the OE. If one does not exist, the planning team may draft one or present the current results of mission analysis to the commander for concurrence and then ask the commander to draft an intent. This will allow planning to continue forward. Communication and knowing the desires of the commander, at this point, is essential to the planning team.

Commander's intent is not a summary of the CONOPS. It should not tell specifically how the operation will be conducted but should be crafted to allow subordinate commanders sufficient flexibility and freedom to act in accomplishing their assigned mission(s)—even in the fog of war. While there is no specified format for commander's intent, one generally accepted construct includes the purpose, method, and end state. Commanders should also highlight operational risk.

1. Purpose: This describes the reason for the military action with respect to the mission of the next higher echelon. The purpose explains why the military action is being conducted. The purpose helps the force execute the mission without further orders, even when actions do not unfold as planned. Thus, if an unanticipated situation arises, participating commanders understand the purpose of the forthcoming action well enough to act decisively and within the bounds of the higher commander's intent. The purpose sections of the mission statement and commander's intent should complement each other. Often the commander's intent purpose is an expansion and explanation of the more concise mission statement purpose.

2. Method: The how, in doctrinally concise terminology, explains the offensive form of maneuver, the alternative defense, or other action to be used by the force as a whole. Details of specific subordinate missions are not discussed.

3. End state: Describes what the commander wants to see in military terms of enemy forces, friendly forces, and the operating environment after the completion of the mission.

4. Risk: The commander may clearly indicate where he/she is prepared to accept (or not accept) the operational risk and make risk tradeoffs.

The commander is responsible for formulating the single unifying concept for a mission. If the commander has not already provided an initial intent statement in the initial guidance, an intent statement is completed based on the mission analysis, the intent of the higher commander, and own vision to ensure that subordinate commanders are focused on a common goal. While brevity and clarity are imperative, the intent should be crafted so that it is understood two echelons below the issuing commander, allowing subordinates to have the flexibility to accomplish

[12] **concept of operations**—A verbal or graphic statement that clearly and concisely expresses what the commander intends to accomplish and how it will be done using available resources. (DOD Dictionary. Source: JP 5-0).

their mission in lieu of further guidance. For a JFMCC, this means that a level down to a TG should be considered when writing the commander's intent. For a TF, the commander needs to consider down to the task unit level. When possible, the commander delivers his/her intent, along with the order (or plan), personally, or through video teleconferencing (VTC). Face-to-face delivery ensures the greatest mutual understanding of what the issuing commander wants by allowing immediate clarification of specific points. While intent is more enduring than the CONOPS, the commander can, and should, revise intent when circumstances dictate. Figure 2-10 offers an example of a JFMCC commander's intent.

Example: JFMCC Commander's Intent

GENTEXT/EXECUTION//

(U) PURPOSE: JFMCC supports the JTF and government of Pinkland by controlling all aspects of the maritime environment to provide a stable and secure environment free from any conventional military or terrorist threat emanating from Redland.

(U) METHOD: Commencing combat OPS on D-day, JFMCC conducts sea control OPS to create a condition of maritime advantage in the maritime AO. We will do this by destroying enemy forces, suppressing enemy sea commerce, and protecting vital sea lanes. If required, we will conduct OPS focusing our efforts in the southern portion of the maritime AO to create a degree of sea control that allows for amphibious OPS to commence. Follow-on tasks include developing Port Yankee as an SPOD for follow-on forces and sustainment. Concurrent to these efforts, we will use our strike capabilities to support the land component commander's efforts in defeating the 23rd Guards Div and destroy the terrorists and their infrastructure in the JOA.

(U) END STATE: Redland maritime forces and maritime-based terrorists are neutralized and cannot conduct organized OPS. We retain the ability to strike in and from the maritime environment as well as place amphibious forces ashore into Redland. Port Yankee is a fully functioning SPOD.

(U) RISK: I see a significant threat from Redland forces for our maritime forces at the outset. We must be fully prepared to transition to kinetic OPS quickly to attrite their force and reduce the risk, while holding attrition to civilian population and infrastructure to an absolute minimum.

Figure 2-10. Example of Joint Force Maritime Component Commander's Intent

2.4.15 Develop Initial Commander's Critical Information Requirements

The planning team should then generate a list of proposed initial CCIRs for the commander's approval. The list should not be extensive. It should focus solely on the absolutely critical pieces of information that the commander needs to know IOT make a decision. Initial CCIRs are linked to potential decision points that may fundamentally alter the planning effort going forward, i.e., validating assumptions. These initial CCIRs will be refined based on decision points identified later during COA development and subsequent steps. For example, the initial priority intelligence requirements (PIRs) are used to develop the initial intelligence estimate, i.e., what do we need to start planning to collect, produce and monitor.

They can have a significant impact on the commander's decisions and actions and therefore can influence the course of events for an operation. The key question to answer when thinking of a CCIR is: "What does the commander need to know IOT make a timely decision?" CCIRs focus the staff, information collection, processing, and flow of execution. While the planning team can recommend CCIRs, only the commander can approve them. They are subdivided into two categories: PIRs and friendly force information requirements (FFIRs). A PIR is an intelligence requirement (IR), stated as a priority for intelligence support that the commander and staff need to understand the enemy or other aspects of the OE. A FFIR is information the commander and staff need to understand the status of friendly forces and supporting capabilities. PIRs are usually listed as questions, and FFIRs can be constructed as either questions or statements. No matter what format is used, both are associated with decision points for the commander during execution.

Figure 2-11 shows possible CCIRs for a JFMCC.

- PIRs:

  Have sea mines been placed in major waterways?

  (Note: This PIR is tied to the significance of sea mines preventing maritime maneuver and possibly denying access to key terrain.)

  Are Redland maritime assets maneuvering to or in position to threaten JFMCC security and sea-control OPS in the JOA?

  (Note: This PIR is directly related to phase I implied task 1.2 and phase II specified task 3.)

  Are there indications of air and missile threats in the vicinity of its Pinkland seaports of debarkation (SPODs)?

  (Note: This PIR is directly related to phase I specified task 2.)

  What is the potential air combat power Redland can generate in its central region? (Note: This PIR is directly related to phase II implied task 3.2.)

- FFIRs:

  Have any coalition members withdrawn maritime forces from the coalition?

  (Note: This FFIR is tied to the status of coalition forces.)

  Will any coalition asset ready for tasking times (arrivals) be delayed more than 24 hours?

  (Note: This FFIR is tied to force availability.)

  Has carrier air wing (CVW) readiness dropped below 50 percent for more than 24 hours?

  (Note: This FFIR is tied to force readiness.)

Figure 2-11. Example of Possible Joint Force Maritime Component Commander's Critical Information Requirements

Note

If the answer to any of these CCIRs is yes, then the planners need to revisit their COA to determine if it is still valid under the given conditions and make changes as necessary.

While not a component of CCIRs, a planning team may develop a list of essential elements of friendly information (EEFI). These are critical aspects of a friendly operation that, if known by the enemy, could subsequently compromise, lead to failure, or limit success of the operation, and therefore must be protected from enemy detection. EEFI help commanders understand what enemy commanders want to know about friendly forces and why. They tell commanders what cannot be compromised. Just as CCIRs are the basis for allocating collection assets to answer questions, EEFI are the basis for the commander's operations security (OPSEC) plan. When assets available for OPSEC are limited, the first priority goes to protecting EEFI.

Also during this time, the planning team and staff should be looking for intelligence gaps that, if answered, can help satisfy a PIR. An identified gap in intelligence may require that a request for information (RFI) or collection requirement be submitted to the intelligence section of the command or higher echelon intelligence. Procedures for submitting RFIs or collection requirements are delineated in JP 2-01, Joint and National Intelligence Support to Military Operations, and should be included in Annex B of an OPLAN or OPORD.

Later in the planning process, CCIRs contribute to the development of the decision support matrix (DSM) and decision support template (DST). An effective mission analysis process should produce insight into nearly every sequel or branch plan that may be faced during the operation for which the planning team is planning. CCIRs, along with the DSM and DST, should assist the commander in making good decisions in a timely manner and gain advantage over the enemy.

2.4.16 Conduct Mission Analysis Briefing

The planning team presents a mission analysis briefing to the commander and staff to obtain approval of the mission statement, intent, and follow-on planning guidance. The briefing reviews the specific products developed and refined during mission analysis before proceeding to COA development. Additionally, the briefing can include an OE update, intelligence estimate and IPOE products (including enemy COGs, COAs, and DPs), HHQ mission and commander's intent, and HHQ commander's guidance. The mission analysis briefing ensures a common and thorough understanding of the proposed mission and tasks/purposes, along with the underlying mission analysis. It focuses on relevant conclusions reached through the analysis process and creates a common understanding and direction for the follow-on planning. The briefing format in Figure 2-12 reflects some of the typical information often found in a mission analysis briefing. Exact content varies based upon the level of command, type of operation, SOP, and the commander's needs.

Briefer	Subject
COS or N-5/N-3	Purpose and agenda AO.
N-2	Initial intelligence estimate brief Terrain analysis. Meteorological and oceanographic analysis. Threat integration with situation templates, enemy COGs, and enemy COAs.
N-5/N-3	HHQ's mission and intent Facts: source(s) of the mission, and command relationships. Assumptions. Limitations: restraints and constraints. Specified tasks/implied tasks/essential tasks. Available forces and assets and noted shortfalls (U.S. and coalition). Friendly COGs/DPs. Initial force movement control center and force structure analysis. Initial risk assessment and vulnerability assessment. End state. Proposed mission statement. Proposed initial CCIRs. Time analysis including projected planning milestones. Conclusions: shortfalls and war-stoppers, recommendations.

Figure 2-12. Sample Mission Analysis Briefing (Sheet 1 of 2)

Briefer	Subject
N-1	Current manning Facts: personnel strengths and morale, replacements and medical returned to duty, critical shortages. Assumptions: replacements, coalition support, other. Conclusions: projected strengths on D-day, projected critical Navy enlisted classification status on D-day, shortfalls and recommendations.
N-4	Sustainment Facts: Class I, II, III(p), IV, VI, VII, X status, status of supply services, critical shortages. Assumptions: resupply rates, host-nation support (HNS), other. Conclusions: projected supply level status on D-day, shortfalls, recommendations, ordnance/weapons. Facts: Class V status, distribution system, restrictions, critical shortages. Assumptions: resupply rates, HNS, other. Conclusions: projected supply status on D-day, projected distribution system, shortfalls and war-stoppers, recommendations. Fueling Facts: Class III(b) status, distribution system, restrictions, critical shortages. Assumptions: resupply rates, HNS, other. Conclusions: projected supply status on D-day, projected distribution system, shortfalls and war-stoppers, recommendations. Fixing Facts: maintenance status (equipment readiness); class IX status; repair times, evacuation policy, and assets; critical shortages. Assumptions: coalition support, other. Conclusions: projected maintenance status on D-day.
N-6	Communications architecture and status Facts: operational status of communications circuits and command, control, communications, computers, and intelligence systems (C4I); bandwidth allocation; communications paths for various C2 functions; planned outages and degradations. Assumptions: bandwidth stability, C4I system reliability. Conclusions: projected C4I systems and communications status during OPS, impact of loss or degradation of C4I systems or communications.
Medical	Medical planning considerations Facts: medical evacuation procedures, lay-down of medical treatment capabilities and resources, critical environmental health concerns (prevalent diseases, hazardous animals, pollutants, potability of local water sources). Assumptions: aircraft to move injured or sick, shore-based medical facilities within flying distance. Conclusions: critical shortages in supplies or personnel, number of wounded and sick that organic medical services can handle.
Others COS or N-3	Others as appropriate to the mission (e.g., assessment guidance, legal). Proposed restated mission. Proposed refinements to commander's planning guidance and commander's intent. Commander's guidance requested.

Figure 2-12. Sample Mission Analysis Briefing (Sheet 2 of 2)

2.4.17 Develop Warning Order(s)

Once the commander approves or modifies the results of mission analysis, the planning team may draft and issue a WARNORD to subordinate units. The WARNORD should include the approved mission statement, the commander's intent, the commander's planning guidance, and any other information that will assist subordinate units with their planning (e.g., changes in task organization, earliest time of movement, etc.). The WARNORD should be written in the standard situation, mission, execution, administration and logistics, and command and control (SMEAC) five-paragraph format for a military directive. It serves notice to subordinate commands of forthcoming military OPS. The commander may transmit additional WARNORDs as planning matures and more information becomes available. This process allows parallel planning at multiple levels of command for pending OPS. A sample WARNORD is included in Appendix J.

2.4.18 Develop Commander's Planning Guidance

Mission analysis began with the commander's initial planning guidance. Now the commander, with planning team assistance, should refine the guidance IOT focus the planning team during COA development. It should be sufficiently specific to capture the commander's expectations but not so restrictive that it inhibits creative COA development. The content of the planning guidance is dependent on the commander's leadership style and time available. It can be very detailed and specific or general in nature. This guidance may also be expressed in terms given in detail in NWP 3-56, such as tactical warfighting functions, including air and missile defense (AMD), surface warfare (SUW), or mine warfare (MIW), or by using the JFMCC functions as detailed in JP 3-32.

Planning guidance should include the commander's vision of decisive and shaping actions in the OE. This assists the planning team in determining the main effort (ME), phases of the operation, location and timing of critical events, risk tolerance, and other aspects of the operation the commander deems pertinent to COA development. The commander should identify mission success criteria, so that the staff and subordinate units better understand what constitutes mission success. Mission success criteria can apply to any operation or phase of an operation. These criteria help the commander determine if and when to move to the next operation or phase. The initial set of criteria could be included in the commander's initial planning guidance and then further developed by the staff during mission analysis. These criteria become the basis for assessment (see Appendix B and Appendix H).

Additionally, the planning team may develop draft governing factors. Governing factors are those aspects of the situation (or externally imposed factors) that the commander deems critical to the accomplishment of the mission. The planning team uses the commander's governing factors to guide COA development and form the basis for evaluation criteria (which also include staff-developed criteria from their staff estimates) used to compare COAs. Although the planning team typically drafts the governing factors based upon earlier interactions with the commander, they belong to the commander. The commander may modify the factors at any time and is the ultimate approval authority. The planning team should emphasize the importance of the commander's involvement in shaping the commander's governing factors. Poorly developed governing factors may undermine effective COA development and later analysis. An example of how commander's guidance can become a governing factor is when a staff can take an element of commander's guidance (for example, ensure the plan offers flexibility) and employ it as a governing factor for COA development and later COA comparison (which COA offers the greatest flexibility?). Generic governing factors, expressed in this example to support the COA comparison provided later, may include:

1. Is the least complicated regarding the ROE
2. Allows the greatest flexibility in selecting the time and place of the action
3. Easiest to support from the perspective of command, control, and communication
4. Offers best logistic sustainability
5. Most dependent on weather and oceanographic conditions
6. Offers the best use of our transportation links

7. Has the greatest effect on the enemy's COGs

8. Will most strengthen the cohesion of our multinational operation

9. Offers the most favorable ratio of relative combat power.

Figure 2-13 gives examples of specific governing factors.

- Flexibility: The disposition and capability of the maritime forces to seize opportunities and counter unanticipated actions of threat forces in the maritime domain.

- Restraint: Limit the use of force against civilians IOT prevent neutral actors from supporting terrorist organizations in the area.

- Simplicity: Which COA is most clear and concise IOT minimize misunderstanding and confusion during the execution of maritime OPS? Ensure that the contribution of partner nations is included.

- Sourcing risk: Are any of the COAs demanding any resources beyond what is currently allocated?

- Joint approach: Leverage the capabilities of the other component commands within the JTF to attain OBJs in the maritime environment.

Figure 2-13. Example of Specific Governing Factors

2.5 OUTPUTS

1. Approved mission statement

2. Approved commander's intent

3. Approved CCIRs

4. Approved commander's guidance

5. WARNORD(s)

6. Updated staff estimates.

2.6 KEY POINTS

1. Commanders may or may not use design methodologies IOT develop those products that precede the initiation of mission analysis.

2. A planning team must aggressively pursue the involvement of the commander at the outset and through the mission analysis process.

3. Regardless of time available, all steps must be addressed.

4. Some steps in the mission analysis process lend themselves to simultaneous planning. Some do not. Planning team leads must determine the approach of the team.

5. Mission analysis sets the stage for all follow-on planning.

6. The better prepared the planning team is for the mission analysis briefing, the better the outputs of mission analysis will be for COA development, analysis, and selection.

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CHAPTER 3
Course of Action Development

Never tell people how to do things. Tell them what to do and they will surprise you with their ingenuity.

General George S. Patton, Jr.

3.1 INTRODUCTION

A COA is a broadly stated potential solution that results in the accomplishment of the mission. During this step of the process, the planning team generates options (COAs) for follow-on analysis and comparison. These must satisfy the commander's intent and planning guidance to accomplish identified tasks. COA development should consider all maritime force capabilities and those joint capabilities necessary to achieve OBJs. For each COA, the commander and planning team should visualize the employment of forces as a whole—normally two echelons down—considering externally imposed limitations, the situation, the OE, the enemy, and previously established conclusions (see Figure 3-1). This ensures that each COA is valid and subordinates are properly task-organized and resourced.

Each COA should also address the requirements for support to and support from adjacent commanders (e.g., other components, national, and coalition capabilities) deemed necessary to accomplish the assigned OBJ. Note: Types of support include close, direct, general, and mutual. The output of COA development is valid COAs that describe, in clear terms, what is to be done (task), why (purpose), by whom (which forces), when (timing instructions), and where (place) the actions will occur. A COA statement should provide maximum latitude for initiative by subordinates. It is the responsibility of subordinate commands to ascertain how they will accomplish their assigned tasks.

The commander involves the entire staff in COA development. The commander's guidance and intent focus the planning team's creativity to produce comprehensive, flexible COAs within the time available for planning. When possible, the commander's direct participation in planning helps the staff gain quick, accurate answers to questions that arise during the process. COA development attempts to generate valid COAs for the commander, each of which will accomplish the mission. A good COA provides flexibility to meet unforeseen events during execution and postures the force for FOPS. Over time, products prepared during COA development become the basis of the CONOPS articulated in the implementing directive.

3.2 INPUTS

Prior to COA development, the commander, staff, and planning team should have an approved mission statement, commander's intent and planning guidance, and an increased appreciation for the overall situation and enemy. The commander's planning guidance and HHQ CONOPS provide the baseline for COA development. COA development uses the most recent information and assessments available: updated IPOE products, specified tasks, implied tasks, essential tasks, WARNORDs, commander's governing factors, limitations (restraints and constraints), assumptions, resource shortfalls, COG analysis (friendly and enemy), CCIRs, initial staff estimates, and initial risk assessment.

Staff estimates—discussed further in Appendix I—are rough at the beginning of COA development. However, the functional, cross-functional, and special staff representatives have access to existing information for use in developing initial staff estimates. The available information is adjusted and applied to the specific operation. Each staff estimate is developed throughout planning to support accomplishing the mission. Staff estimates provide critical functional information to the staff during planning.

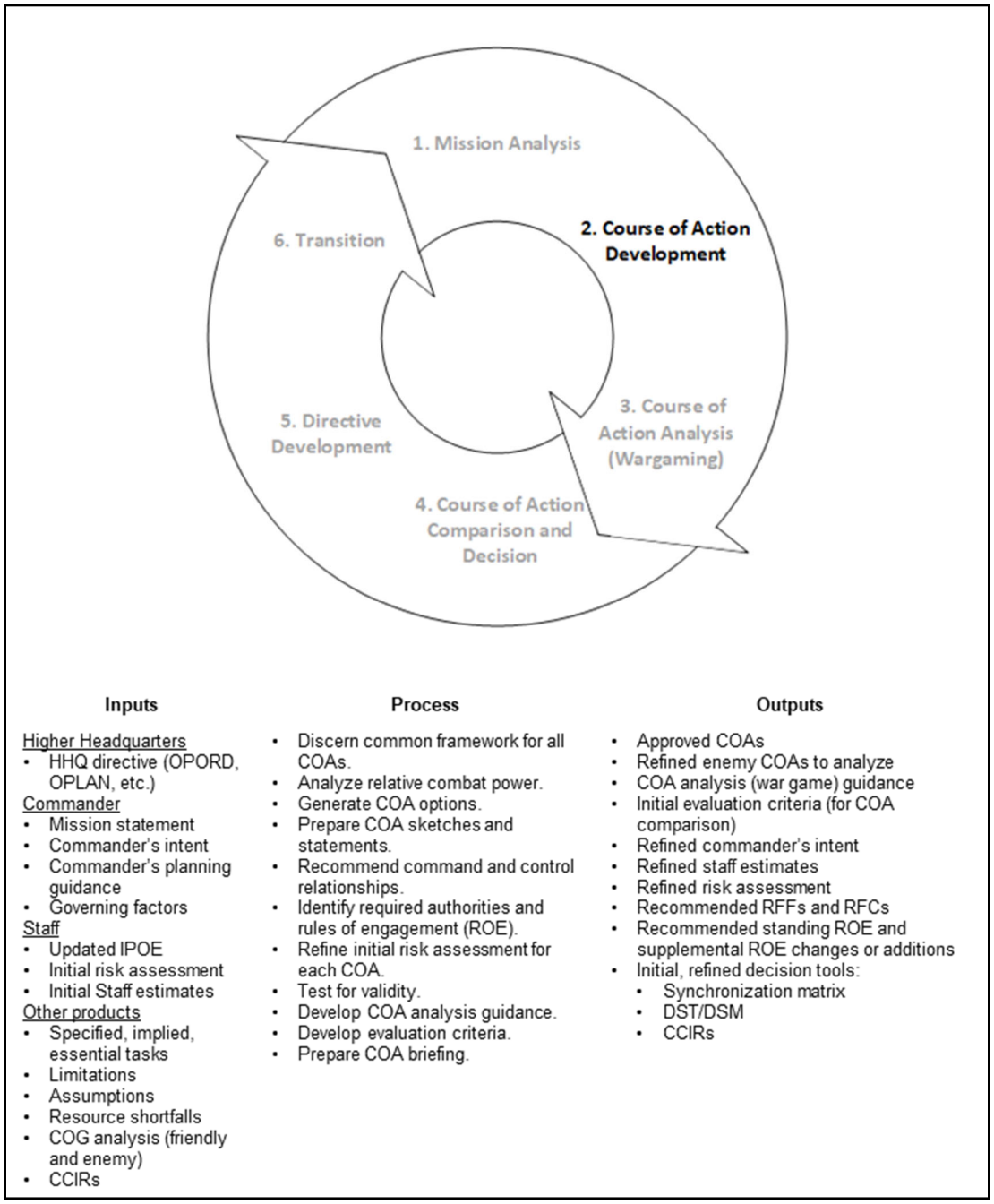


Figure 3-1. Course of Action Development

3.3 PROCESS

The planning team prepares for COA development by reviewing the mission statement, commander's intent and planning guidance, HHQ CONOPS, and other information and direction to discern the common framework within which COAs must be developed. Using these inputs and the most recent information about the OE, planners generate COAs that:

1. Nest within, and are aligned to, the HHQ's CONOPS and phasing construct.

2. Are consistent with the commander's intent and planning guidance.

Whether planning for campaigns or major OPS, commanders and their staffs focus on developing options to achieve the OBJ. Each COA describes a viable way to accomplish the OBJ. The HHQ CONOPS evolved from the COAs presented to the HHQ commander. Ultimately the HHQ commander selected the COA that best achieved the OBJ. The HHQ CONOPS drives the development of subordinates' CONOPS. Subordinates develop their COAs within the framework outlined in the HHQ CONOPS. For example, if the HHQ CONOPS is phased, subordinates develop their COA using the same phasing construct. Ultimately, the COA that best achieves the subordinates' OBJ will become the subordinates' CONOPS.

The OPT leader ensures that each COA developed is valid and distinguishable from the others. While each COA must be distinct, they all share the same common purpose and specified/essential tasks identified during mission analysis. Each COA uses a COA sketch and statement to describe OBJs and the major actions (tasks) to be accomplished in terms of MEs, supporting efforts (SEs), shaping efforts (SHs), and sustaining efforts (SUSs), as well as timing factors for commencing and ending the operation and transitioning from one phase to another. Each COA also includes a description of C2 relationships (task organization), and is accompanied by a synchronization matrix that captures the major actions (critical events), timing factors, and decision support tools that include known decision points. During COA analysis, planners populate these products with details.

Once potential COAs are generated, planners test each COA for validity. The COA must pass the validity test to move on in the process. Any refinements needed to ensure that COAs are valid are made at this time. Upon completion of the validity test, the staff prepares a COA development brief, refines the initial staff estimates, and prepares to move on to COA analysis.

3.3.1 Discern the Common Framework

Discerning the framework is a technique of examining the OE to prioritize major actions (decisive, supporting, shaping, and sustaining). The framework focuses the planners on the most important aspects of the operation. The main consideration when establishing the framework is identifying the operational OBJ. If the operation is phased, OBJs by phase are identified, and a list of prioritized tasks (by phase) is also examined. This technique allows planners to view the OE as an indivisible whole by translating the commander's vision of decisive, supporting, shaping, and sustaining actions into a framework within which to develop initial COA(s).

The commander's planning guidance and HHQ CONOPS establish the common framework for all COAs, i.e., the series of OBJs and actions focused on accomplishing the mission. Planners review the planning guidance and HHQ CONOPS to discern, by phase, the command's role in the HHQ CONOPS (ME, SE, SH, or SUS), the command's OBJ, and the essential task to be accomplished to achieve the command's OBJ. The results of this review describe the framework within which all COAs are to be developed.

Commanders and their staffs use the common framework to help conceptualize and describe each COA's decisive, supporting, shaping, and sustaining actions.

1. The decisive action (identified as an essential task) is the activity that accomplishes the OBJ and becomes the focus of the ME (when the action (task) is assigned to a specific unit (with purpose)).

2. Supporting actions are those that directly support the decisive action, i.e., set conditions that enable accomplishing the OBJ now.

3. Shaping actions set conditions for success of subsequent OPS, i.e., enable accomplishing the OBJ later.

4. Sustaining actions provide resources and support the generation of combat power needed for the decisive, supporting, and shaping actions.

Commanders and their staffs follow this framework to derive COAs. Within the common framework, commanders and staffs nest their CONOPS, synchronize actions and phasing decisions, identify DPs, and determine lines of operations (LOOs) and/or lines of effort (LOEs) when conceptualizing and describing each potential COA. Each COA aims to accomplish specific intermediate and ultimate OBJs identified in the common framework.

3.3.1.1 Example Joint Task Force (Higher Headquarters) Concept of Operations for Operation FREEDOM ASSURANCE

Figure 3-2 depicts the first three phases of an example JTF (HHQ) CONOPS. Each component derives its COAs and ultimately its CONOPS from the JTF's CONOPS. The JFMCC begins COA development by examining the JTF CONOPS to determine what it is directed to do ISO the JTF CONOPS. Within the example JTF CONOPS that follows, the relevant JTF instructions to the JFMCC are italicized to emphasize JFMCC priorities (italics are not used in actual CONOPS). The relevant instructions to JFMCC are extracted from the JTF CONOPS to begin developing the common framework for JFMCC COAs.

Phase I (deter). *Operation FREEDOM ASSURANCE (FA), phase I, commences on C-day with JTF BLUE SWORD deploying forces into the joint operations area (JOA). Hostilities have not been declared. The JTF phase I OBJ is Redland military forces are deterred. JFMCC (ME) deploys forces to establish a maritime presence in the JOA to deter Redland aggression.* JFACC (SE) conducts defensive counterair (DCA) in the JOA IOT protect JTF forces. Joint force land component commander (JFLCC) (SH) deploys forces to the intermediate staging base (ISB) in Pinkland IOT support follow-on JTF offensive OPS in Redland. Joint force special operations component commander (JFSOCC) (SH) conducts special reconnaissance (SR) in Redland IOT set conditions for FOPS. *Phase I ends when Redland agrees to a diplomatic resolution or commences hostilities, and JTF forces have completed deployment to the JOA for FOPS.*

Phase II (seize the initiative). *Operation FA, phase II, begins when hostilities commence and the JTF forces are in position to conduct offensive OPS in the JOA. The JTF phase II OBJ is Redland military forces are degraded. JTF forces conduct offensive OPS in the JOA IOT degrade Redland forces. JFMCC (ME), conducts offensive OPS in the JOA IOT gain sea control to set the conditions for amphibious OPS.* Note: Amphibious OPS include amphibious assaults, amphibious demonstrations, amphibious raids, amphibious support to other OPS, and amphibious withdrawals. JFACC (SE) conducts strike OPS in the JOA IOT degrade Redland capabilities. JFSOCC (SE) conducts direct action (DA) at OBJ RAT IOT degrade Redland C2. JFLCC (SH) stage forces in Pinkland IOT conduct phase III forcible entry OPS. *Phase II ends when Redland forces are degraded in the JOA, sea control is established in the maritime AO, and JTF forces are postured to commence offensive OPS in Redland.*

Phase III (dominate*). Operation FA, phase III, begins with JTF forces conducting simultaneous offensive operation in Redland. The JTF phase III OBJ is Redland military forces are defeated.* JFLCC (ME) conducts airborne forcible entry into Redland IOT destroy Redland forces at OBJs DOG and CAT. JFACC (SE) conduct joint fire support in the JOA IOT degrade Redland forces. *JFMCC (SE) conduct amphibious operation in AOA 2 IOT fix Redland forces in OBJ RAT. JFSOCC conducts DA in the JOA IOT disrupt Redland reinforcements. Phase III ends when Redland forces are defeated, JTF forces have consolidated in Redland, and JTF forces are postured for stability OPS.*

Phase IV (stabilize). *Operation FA, phase IV, begins when sufficient JTF forces are in position to transition from offensive to stability OPS in Redland.* JFLCC (ME) conducts stability OPS throughout Redland IOT reestablish critical infrastructure. *JFMCC (SE) supports stability OPS IOT reestablish critical maritime infrastructure.* JFACC (SE) continues to provide close air support (CAS) to JFLCC and JFSOCC OPS, and O/O redeploys nonessential assets. JFSOCC (SE) continues to kill or capture fugitive terrorists, conducts sensitive site exploitation (SSE), and O/O redeploys nonessential JFSOCC assets. *Phase IV ends when security conditions are adequate to transfer authority to a legitimate Redland government and JTF forces have completed or ready to transition their stabilization tasks to follow-on forces.*

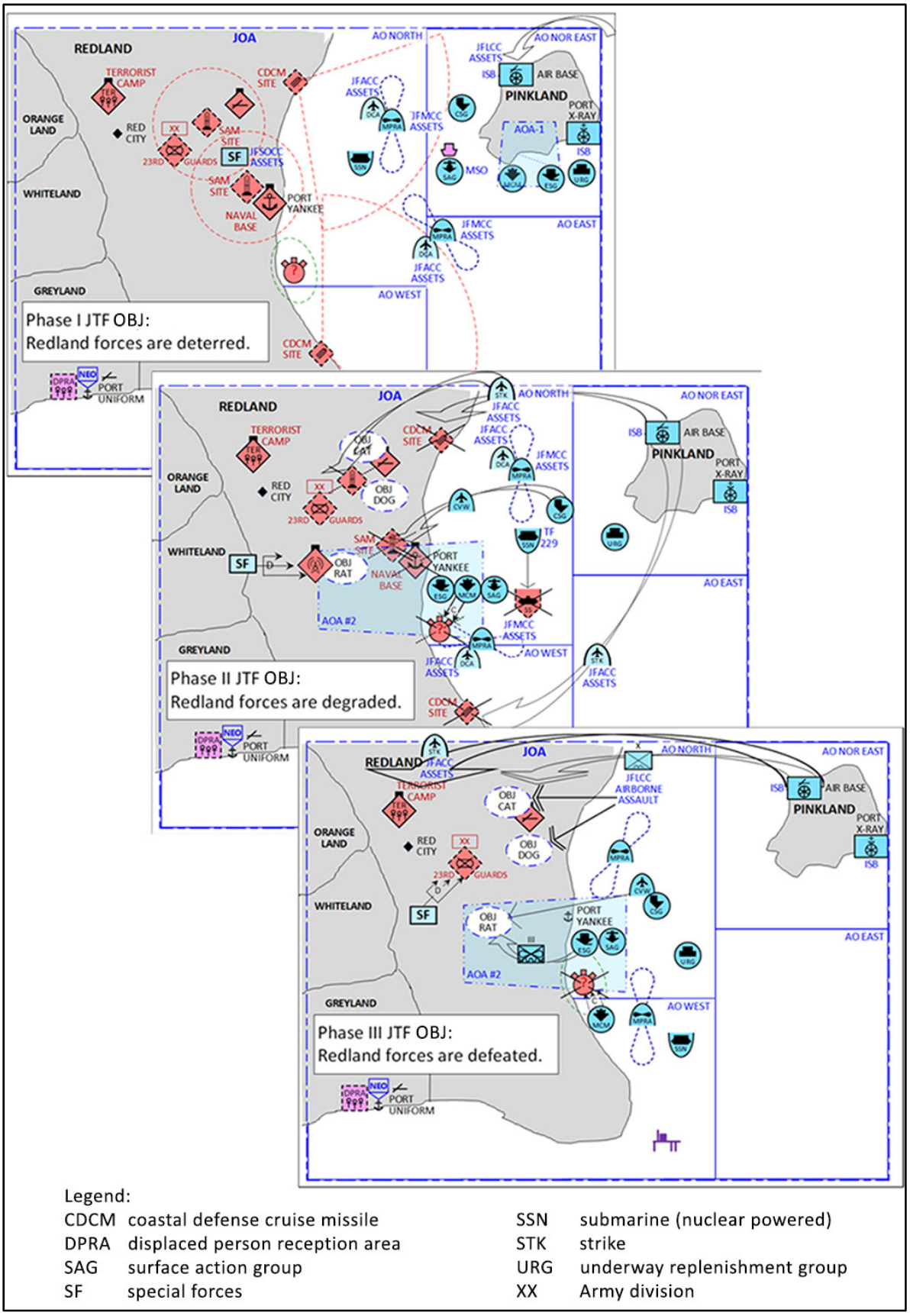


Figure 3-2. Joint Task Force Concept of Operations Sketch Examples (FREEDOM ASSURANCE) (Phases I–III)

Phase V (enable civil authority). *Operation FA, phase V, begins by enabling a legitimate Redland government to assume control of its sovereign territory and all JTF stabilization tasks are transferred.* The JTF supports the Department of State (DOS) and O/O transition JTF stabilization OPS to Redland or a Redland-designated international organization and redeploys JTF forces. JFLCC (ME) forms and trains new Redland security force, conducts joint security patrols, and O/O redeploys JFLCC forces. JFACC (SE) continues to provide CAS and supports redeployment through strategic airlift, and O/O redeploys JFACC forces. *JFMCC (SE) transitions port security to new Redland maritime/coast guard forces, transfers stabilization tasks, and O/O, redeploys JFMCC forces.* JFSOCC (SH) conducts SSE, and O/O redeploys JFSOCC assets. *Phase V ends when a legitimate Redland government has control of its sovereign territory, an international organization has assumed responsibility for remaining stabilization OPS, and all JTF forces have redeployed.*

3.3.1.2 Joint Force Maritime Component Commander Common Framework for Operation FREEDOM ASSURANCE

To discern the common framework for JFMCC COAs, the JFMCC's planning team examines the JTF CONOPS to identify the instructions that pertain to the JFMCC. The goal is to determine the main characteristics to be shared by each JFMCC COA to support the JTF CONOPS. In this example, JFMCC planners highlight the fundamental elements of the JFMCC's common framework by phase: the JFMCC's OBJ, role (ME, SE, SH, or SUS), and essential task(s) that drive development of JFMCC COAs. The example JFMCC common framework below is for phase I, which was derived from the JTF CONOPS for FA, described in 3.3.1.1.

Note

This is the starting point for the JFMCC's COA development. The JFMCC's common framework informs the JFMCC commander and staff's work to generate COA options.

Example: Operation FA, phase I, commences on C-day with JTF BLUE SWORD deploying forces into the JOA to defend Pinkland and deter Redland aggression. Hostilities have not been declared. JFMCC, the JTF phase I ME deploys forces to establish a maritime presence in the JOA to deter Redland aggression. Phase I ends when Redland agrees to diplomatic resolution or commences hostilities, and JFMCC forces have completed deployment to the JOA for FOPS.

The key point of emphasis for JFMCC's COA generation options is underscoring the common framework that must be developed between the JTF and the components prior to executing OPS. Examination of the JTF CONOPS enables the JFMCC to derive its phase I OBJ—Redland maritime forces are deterred; and highlight the JFMCC role in the JTF's phase I construct/scheme: JFMCC, the JTF phase I ME. And it emphasizes the JFMCC's essential task for this phase: maritime security operations (MSO) in the JOA IOT deter Redland maritime force aggression. The information needed to develop the framework is extracted from the JTF OPORD and the JFMCC's analysis of the order during mission analysis.

This methodology is repeated for each phase of the JTF CONOPS and serves as the foundation for generating COA options.

3.3.2 Analyze Relative Combat Power

RCPA compares friendly and enemy tangible (quantitative) and intangible (qualitative) factors that allow each to generate combat power. Combat power is the total means of destructive or disruptive force that a military unit or formation can apply against the opponent at a given time. It is created by combining the elements of operational functions (C2, intelligence, fires, movement and maneuver, sustainment, protection)[13] and leadership. The goal is to generate overwhelming combat power to accomplish the mission at least possible cost.

[13] At time of publication, Navy doctrine recognizes six operational functions and has not addressed the seventh joint function of information described in JP 3-0. Without approved Navy doctrine regarding how operational level of warfare commanders (i.e., NFC/NCC/MCC) will implement the new joint function of Information, the NPP continues to address the Navy-approved six operational functions.

Determining, analyzing, and comparing each force's strengths and weaknesses provides planners valuable insights needed for generating COAs. This analysis can also be useful to those charged with monitoring and directing the operation during its execution. Appendix D provides details on a methodology to conduct an RCPA and a matrix of comparison considerations, both of which have value to maritime planners. The matrix should be tailored for the specific planning effort or operation.

RCPA is primarily a subjective endeavor. By comparing each force's size, capabilities, strengths, and weaknesses, planners can gain insights into:

1. Friendly capabilities pertaining to the operation.

2. The types of OPS possible from friendly and enemy perspectives.

3. How, when, and where friendly and enemy forces may be vulnerable.

4. What additional resources may be required to execute the mission.

5. How to best allocate existing resources (organize friendly forces) to gain and maintain a relative combat power advantage.

As COAs are developed, planners will find it useful to consider whether or not friendly forces possess sufficient combat power at the right time, and in the required geographic locations, to accomplish the mission. During COA analysis and COA comparison, planners should remain cognizant of whether or not the COA ensures that friendly forces can generate the required amount of combat power to accomplish the assigned tasks with acceptable risk. This can be done by considering the RCPA insights during COA analysis and by including RCPA insights during COA comparison.

Maritime staffs involved in execution—specifically, those that monitor and direct—must also be keenly aware of whether or not friendly forces retain the relative combat power advantage required to accomplish the mission. For stability and civil support OPS, operational staffs might consider determining relative combat power by comparing available resources to civil support tasks. This is commonly known as troop-to-task analysis. This provides insight as to what options are available and whether or not additional resources are required. During such OPS, the elements of sustainment, movement and maneuver, nonlethal effects, and information operations (IO) may dominate. In all cases, planners should pay particular attention to analysis of critical capabilities, critical requirements, and critical vulnerabilities for friendly and enemy forces rather than simply relying on a purely numerical force comparison.

3.3.3 Generate Course of Action Options

Using the common framework and RCPA insights, the planning team generates options for COA development. The planning team applies critical thought, imagination, and creativity to produce the greatest range of options within the boundaries established by the common framework. The planning team must be unbiased and open-minded in evaluating proposed options. Team members identify valid COAs based on their particular areas of expertise. If one team member identifies information that might affect the development of valid COA(s), the member should share it immediately, eliminating wasted time and effort.

When formulating COA(s), the planners translate the commander's decisive, supporting, shaping, and sustaining actions into MEs, SEs, SHs, and SUSs. The planning team develops different distinguishable COAs by using different timing and/or sequencing of actions, establishing different intermediate OBJs, dividing phases into stages/steps/parts, and altering the array and synchronization of assigned forces and actions. These differences are applied in the framework of each COA as it is developed.

As different COAs are developed, the planning team must ensure that each COA shares the same ultimate OBJ, essential, and specified tasks and limitations identified earlier. The planning team should ensure all elements of the naval force are assigned appropriate tasks, areas of operation, command relationships, and resources. While these elements are presented sequentially, they actually have interrelated cause and effect connections that must be considered as a COA is developed.

A valid COA should be capable of defeating all feasible enemy COAs. In a totally unconstrained environment, the goal is to develop several valid (friendly) COAs. Since there is rarely enough time to do this, the commander's guidance may specify a number of COAs for the planning team to develop. In developing COAs, staff and planning team members should determine the doctrinal requirements for each type of operation they are considering, including doctrinal tasks to be assigned to subordinate units.

3.3.3.1 Establish the Course of Action's Framework

Using the common framework and identified tasks from mission analysis as the starting point, planners clearly define the commander's OBJ and prioritize the associated tasks as decisive, supporting, shaping, or sustaining actions. The task identified as the essential task in the common framework becomes the basis of the decisive action. Annotate the decisive action on the planning map and clearly state the task and purpose for the decisive action. For the remaining tasks, prioritize them as supporting, shaping, or sustaining actions (in that sequence and as described in 3.3.1), annotate them on the planning map and clearly state their priority, task and purpose.

New tasks will likely emerge as the COA framework becomes more detailed. These tasks are also prioritized and annotated on the planning map. Planners should also record timing considerations such as the conditions for beginning or ending the operation or phase, current assumptions and limitations, and other pertinent information that was initially developed during mission analysis. All COAs must accomplish the operation's specified and essential tasks within imposed limitations and nest within the HHQ common framework's phasing construct.

3.3.3.2 Array Initial Forces

The planning team arranges generic forces to accomplish major actions while applying the RCPA deliberations to each action. Array initial forces against the decisive action first, then follow with the supporting, shaping and sustaining actions. Place generic platform icons on the map to represent the type capabilities required for each action. It enables an accurate determination of the forces needed to accomplish essential tasks, identifies enemy-friendly relative combat power comparisons at specific locations, and conceptualizes responsibility geographically (for future control measures). Generic forces are used to avoid fixation on specific units and to allow the staff to build a solid conceptual plan first that will be populated with specific units in a later step.

Include task and purpose assignments when arraying generic forces. Determine what is necessary to accomplish the OBJ and assign the task and purpose required for each grouping of forces accomplishing each action. Ensure each task has a purpose to support the COA. Begin with the decisive action by stating the purpose and task(s) already identified and follow with the supporting, shaping, and sustaining action(s) until all applicable tasks are assigned. Valuable resources during this stage are the UJTL and the supporting UNTL to provide overall descriptions that can be applied to multiple levels of warfare. By reviewing these task lists, planning teams and staffs can ensure the tasks (with requisite purpose) are properly stated and designated for naval forces. Record the task and purpose for each action; they will become part of the COA statement in a subsequent step.

3.3.3.3 Convert Generic Units to Specific Units

The initial array of forces identifies generic units that possess required capabilities without regard to a specific unit, type, task organization, or other intangible aspect. During this step, unit types are converted from generic to specific by assigning available forces a task and purpose for each action. The force conducting the decisive action is designated as the ME. The forces conducting the supporting actions (that directly support the decisive action) and shaping actions (that set conditions for the success of subsequent OPS) become the SEs and SHs. The force conducting the sustaining action becomes the SUS.

Note

There is only one ME at any given time during a phase, but the ME can change in a phase if different decisive actions are assigned to different forces and are timed to occur in sequence. Also, if a decisive action applies to significantly displaced geographical areas that prevents accomplishment by a single force (e.g., obtain sea

control throughout the JOA), the forces (e.g., TF) assigned to accomplish the decisive action in the most critical area will be the ME and those other forces carrying out the same task in other areas are designated as SEs.

While assigning specific units to tasks, it may become apparent that more forces are required to accomplish the OBJ. If additional capabilities are identified as essential to the proposed COA(s), then those RFFs/RFCs shall be highlighted and briefed to the commander for approval. The additional forces ensure the capacity exists within the JFMCC to meet the force requirements identified through the COA development process. Likewise, the planning team should include the maritime support requirements to adjacent forces, and any support required from other components or forces applicable to each COA.

3.3.3.4 Task Organize

Subordinate forces are now organized to accomplish their assigned tasks and the commander's OBJ. The task organization captures how the planning team intends to structure and resource the force to conduct OPS. Through task organization, commanders establish command or support relationships and allocate resources to weight the ME. Additionally, it is used to establish appropriate C2 relationships (3.3.5). Task organizations should extend no more than two echelons down IOT address mission assignments thoroughly. Proper task organization ensures that subordinate forces are properly organized, sized, and resourced to accomplish their assigned tasks and contribute to accomplishing the mission.

3.3.3.5 Determine Control Measures

Planners now determine the control measures, under the commander's authority, best suited to C2 the operation. Control measures regulate forces and/or operational functions and are essential to coordinating subordinate actions. Control measures assign responsibilities, require synchronization between forces, impose restrictions, or establish guidelines to regulate freedom of action. Control measures assigned to the major subordinate commands provide flexibility to accomplish their assigned tasks and protect their force from enemy action. Good control measures foster freedom of action, decision making, and individual initiative.

3.3.3.6 Synchronize and Sequence Actions

The planning team synchronizes and sequences actions, deliberately creating complementary and reinforcing effects. During the development of the COA(s), the planning team makes sequencing and synchronization decisions to arrange the forces' actions in terms of time, space, and purpose. Additionally, the planners determine the anticipated duration of the operation, highlighting the phase begin and end criteria. Other considerations to address are when and under what conditions the ME may shift, when the ME is committed, and when success may be exploited. The planning team depicts the synchronization of actions across time and space in the COA sketch and statement.

Sequencing and synchronization decisions are recorded in the synchronization matrix and will be refined during COA analysis. The planning team further evolves the tasks and purposes to provide specifics about actions that are to occur. Tasks are recorded in the synchronization matrix to clearly convey the elements of who, what, when, where and why such that the force/unit does not have to guess what it is being told to do or try to achieve.

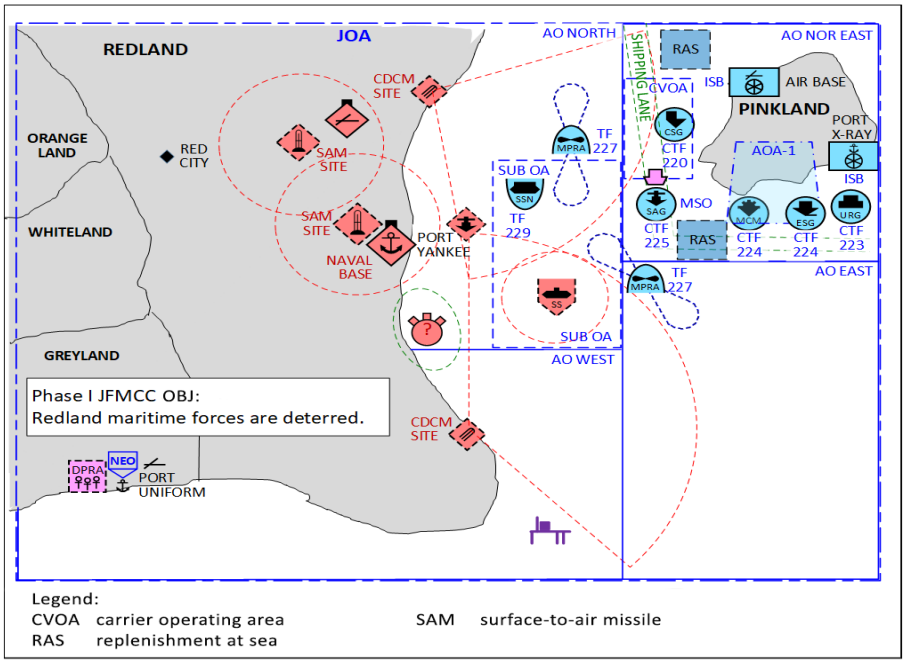
3.3.3.7 Commander's Input and Refinement

When possible, the planning team leader reviews the initial COA(s) with the commander, or their designated representative, to ensure they conform to initial intent and the COA development planning guidance. This is an opportunity for the commander to comment and make midcourse corrections before the planning team spends precious time on potential COA(s) that the commander does not approve.

3.3.4 Prepare Course of Action Sketch and Statement

The COA sketch and statement (Figure 3-3) broadly articulate the plan by providing a graphic and textual representation of the COA derived in 3.3.3. The sketch and statement is a proven and effective method for conveying understanding among commanders and staff. The COA statement forms the basis for the CONOPS developed during COA comparison and decision, and recorded in the execution paragraph of the upcoming directive. The COA sketch and statement clearly describe the timing to begin the operation or phase, the scheme of maneuver, and how the unit (JTF or component) will accomplish the OBJ (intermediate or ultimate). It describes how the COA achieves the commander's vision of decisive, supporting, shaping, and sustaining actions within the context of the framework. The COA sketch and statement include the type of operation, the subordinate units' task and purpose, and the form of maneuver, and they create a unique task organization for the operation.

Operation FA, phase I, commences on C-day with the JTF deploying forces into the JOA. Hostilities have not been declared. The JTF phase I OBJ is Redland military forces are deterred. JFMCC (ME) deploys forces to establish a maritime presence in the JOA IOT deter Redland aggression. JFACC (SE) conducts DCA in the JOA IOT protect JTF forces. JFLCC (SH) deploys forces to the ISB in Pinkland IOT support follow-on JTF offensive OPS in Redland. JFSOCC (SH) conducts SR in Redland IOT set conditions for future JTF OPS.



On D-day (On or about C+24) JFMCC, the JTF ME, conducts MSO in JOA IOT deter Redland maritime forces.

ME: Commander, task force (CTF) 225 (SAG) conducts maritime security patrols in AO NOR EAST ISO Pinkland IOT deter Redland maritime force aggression.

SE: CTF 227 (MPRA) conducts airborne surveillance and reconnaissance in the JOA IOT determine Redland maritime disposition and intentions.

SE: CTF 229 (SUBS) provide SUW, antisubmarine warfare, and ISR support in SUB AO IOT protect JFMCC forces and enable FOPS.

SH: CTF 224 (ESG) conduct amphibious rehearsal in AOA 1 IOT prepare for offensive OPS in Redland.

SH: CTF 220 (CSG) conduct integrated air and missile defense in AO NOR EAST IOT protect forces deploying to Pinkland.

SUS: CTF 223 provide LOG support IOT extend operational reach of JFMCC forces.

Other support required. Pinkland provides landside security for ISB (forward logistics site (FLS) and AOA). JFACC assumes security/protection requirements at the ISB/FLS.

Phase I ends when Redland agrees to a diplomatic resolution or commences hostilities, and JTF forces have completed deployment to the JOA for FOPS.

Figure 3-3. Joint Force Maritime Component Commander Course of Action Sketch and Statement, FREEDOM ASSURANCE, Phase I

Additionally, the COA statement identifies how SEs, SHs, and SUSs relate to the ME (to include a reserve if designated). Lastly, the COA sketch identifies necessary control measures, and the COA statement concludes with ending criteria. Ending criteria should be described in terms of enemy and friendly forces, and the OE. If the OPLAN or OPORD is phased, the sketch and statement for each phase follows this format.

The sketch visually depicts the key elements of the commander's scheme of maneuver. The elements and actions identified on the sketch are explained in the COA statement. Figures 3-3, 3-4, and 3-5 provide example JFMCC COA sketches and statements for phases I through III based on the example JTF CONOPS described in 3.3.1.1 and illustrated in Figure 3-2.

3.3.4.1 Course of Action Sketch

The COA sketch provides a picture of the COA's scheme of maneuver. A COA sketch includes arrayed forces and control measures to depict the primary actions taken by each subordinate unit illustrating their effort to accomplish the OBJ. A graphic control measure is a symbol used on maps/charts and displays to regulate forces and warfighting functions. Graphic control measures are prescriptive. They may include symbols for maneuver boundaries (operating areas (OPAREAs)) within the maritime AO, fire support coordination measures, some airspace control measures, AD areas, and specific theater warfare measures. Commanders establish them to regulate maneuver, movement, airspace use, fires, and other aspects of OPS. Examples of these control measures follow:

1. Establish unit or command (TF/TG) boundaries within the maritime AO or unit OPAREAs within the defined JOA.

2. Annotate unit (TF/TG) deployment/employment using mission graphics and unit symbols.

3. Use control graphics (waterspace management (WSM) diagrams, sea lines of communication (SLOCs), forward bases, joint AD commander/regional and sector AD commander, etc.) to direct/depict responsibility for these activities.

4. Use mission graphics to depict timing/sequencing/synchronization of events.

5. Designate the MEs, SEs, SHs, and SUSs to graphically depict the actions and relationships between these efforts.

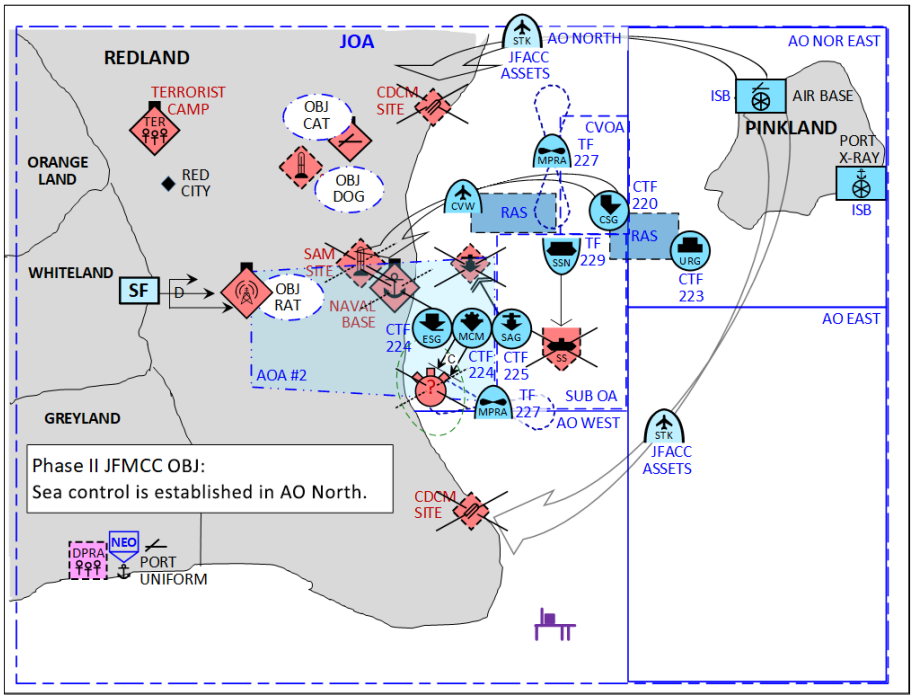
6. Annotate known or expected locations (e.g., ships and subs in port, at sea, or unlocated).

Planners can enhance the sketch with explanatory features to help orient the commander and staff. The sketch may be in any medium; what it portrays is more important than its form. The use of operational symbols and graphics convey specific instruction to the subordinate units for each of the first three phases of the operation. Appendix K describes symbology, control measure, and mission graphic conventions used in COAs sketches.

3.3.4.2 Course of Action Statement

The COA statement accompanies the COA sketch and describes how the forces will accomplish the OBJ. The COA statement concisely expresses the commander's CONOPS and governs the development of supporting plans or annexes. Planners develop a COA statement by summarizing the information determined in the generated COA options. Timing and coordination requirements between/among units are addressed by describing these interactions and the control measures used to depict the operation on the COA sketch. Planners develop COA statements using the same process regardless of the operation. A COA statement is expressed in clear, concise terms and contain these basic characteristics: timing/condition instructions for beginning and ending the operation/phase, HHQ CONOPS (MEs, SEs, SHs, and SUSs), COA framework (MEs, SEs, SHs, and SUSs,), and other support required from other commands, components, host nation (HN), coalition forces, U.S. agencies, and other organizations.

Operation FA, phase II, begins when hostilities commence and the JTF forces are in position to conduct offensive OPS in the JOA. The JTF phase II OBJ is Redland military forces are degraded. JTF forces conduct offensive OPS in the JOA IOT degrade Redland forces. JFMCC (ME), conducts offensive OPS in the JOA IOT gain sea control to set the conditions for amphibious OPS. JFACC (SE) conduct strike operations in the JOA IOT degrade Redland capabilities. JFSOCC (SE) conducts DA at OBJ RAT IOT degrade Redland C2. JFLCC (SH) stages forces in Pinkland IOT conduct phase III forcible entry OPS.



JFMCC, the JTF ME, conducts offensive OPS in the JOA IOT gain sea control in AO NORTH.

ME: CTF 225 (SAG) attacks Redland forces IOT establish sea control in AOA 2.

SE: CTF 227 (MPRA) conducts airborne surveillance and reconnaissance in AOs NORTH, and WEST IOT determine Redland maritime disposition and target Redland maritime forces.

SE: CTF 229 (SUBS) interdicts Redland maritime forces ISO CTF 225 IOT degrade Redland naval forces in AO NORTH.

SE: CTF 220 (CSG) conducts strike OPS in AOA 2 and AO NORTH IOT degrade Redland maritime forces.

SH: CTF 224 (ESG) O/O, conduct MCM OPS in AOA 2 to set the conditions for amphibious OPS.

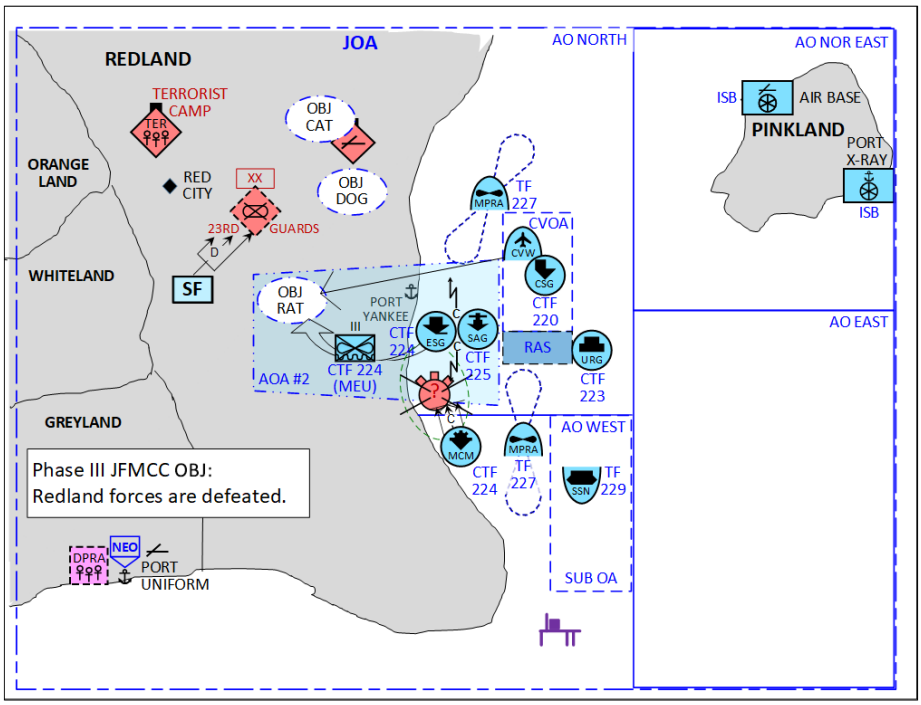
SE: CTF 223 provide LOG support IOT extend operational reach of JFMCC forces.

Other support required. Pinkland agrees to provide landside security for ISBs (air base and Port X-Ray). JFACC conducts strike and OCA in the JOA IOT degrade Redland sea denial capabilities.

Phase II ends when Redland forces are degraded in AO NORTH, sea control is established in AOA 2, and CTF 224 is in position to commence amphibious operations.

Figure 3-4. Joint Force Maritime Component Commander Course of Action Sketch and Statement FREEDOM ASSURANCE, Phase II

Operation FA, Phase III, begins when Redland forces in AO NORTH (AOA 2/OBJ RAT) are degraded. JTF forces are conducting simultaneous offensive operation in Redland. The JTF phase III OBJ is Redland military forces are defeated. JFLCC (ME) conducts airborne forcible entry into Redland IOT destroy Redland forces at OBJs DOG and CAT. JFACC (SE) conducts joint fire support in the JOA IOT degrade Redland and terrorist forces. JFMCC (SE) conducts amphibious operation in AOA 2 IOT fix Redland forces in OBJ RAT. JFSOCC (SH) conducts DA in the JOA IOT disrupt Redland reinforcements.



JFMCC (SE) conducts amphibious OPS in AOA 2 IOT fix Redland forces at OBJ RAT.

ME: CTF 224 (ESG) conducts amphibious assault at AOA 2 IOT fix Redland forces at OBJ RAT.

SE: CTF 225 (SAG) covers CTF 224 IOT enable the amphibious operation in AOA 2.

SE: CTF 229 (SUBS) O/O conducts surveillance and reconnaissance in AO WEST IOT monitor the OE.

SE: CTF 220 (CSG) conducts CAS in AOA 2 IOT support CTF 224 amphibious operation.

SH: CTF 227 (MPRA) conducts airborne reconnaissance and surveillance in the JOA IOT to determine remaining Redland capabilities.

SE: CTF 223 (underway replenishment group and numbered fleet commander) provide LOG support IOT extend operational reach of JFMCC forces.

Other support required. Pinkland agrees to provide landside security for ISBs (air base and Port X-Ray). JFSOCC conducts DA in OBJ RAT IOT fix Redland forces.

Phase III ends when Redland forces are defeated, JTF forces have consolidated in Redland, and JTF forces are postured for stability OPS.

Figure 3-5. Joint Force Maritime Component Commander Course of Action Sketch and Statement FREEDOM ASSURANCE, Phase III

A COA statement is typically produced by phase and should be simple, clear, and complete. See the sample methodology for COA statement development below:

1. Timing/conditions—beginning instructions. The COA statement starts with the timing/conditions instructions describing the actions which must be met to begin the operation or phase. Key aspects of time and/or conditions are the prerequisites that must exist to begin the operation/phase. Some things to consider are the minimum forces deemed necessary and the time when the force must be in position to conduct assigned tasks to commence OPS. These instructions are directed in the HHQ plan/order or derived from other instructions given to the commander.

Example: Operation FA, phase I, commences on C-day with the JTF deploying forces into the JOA. Hostilities have not been declared. The JTF phase I OBJ is Redland military forces are deterred.

2. HHQ CONOPS. Establish context for the COA by summarizing the HHQ CONOPS (MEs, SEs, SHs, and SUSs). Highlight the HHQ instructions to each of its subordinates (what actions are to be undertaken). Ensure these instructions address key component support relationships. Again, these instructions are provided in the HHQ plan/order or derived from other instructions given to the commander.

Example: JFMCC, the JTF phase I ME, deploys forces to establish a maritime presence in the JOA IOT deter Redland aggression. JFACC (SE) conducts DCA in the JOA IOT protect JTF forces. JFLCC (SH) deploys forces to the ISB in Pinkland IOT support follow-on JTF offensive OPS in Redland. JFSOCC (SH) conducts SR in Redland IOT set conditions for future JTF OPS.

3. COA description. Describe the generated COA for the operation or specified phase. Outline the roles assigned to specific units (MEs, SEs, SHs, and SUSs), emphasizing the instructions to each subordinate. Ensure these instructions address key support relationships. Begin with the ME; then address the SEs, SHs, and SUSs and their actions to accomplish the OBJ. Specify the actions undertaken by each subordinate (task and purpose). Again, these instructions are taken from the work done previously when generating COA options and from the COA sketch prepared earlier.

Example: On D-day (On or about C+24) JFMCC, the JTF ME, conducts MSO in the JOA IOT deter Redland maritime forces.

ME: Commander, task force (CTF) 225 (SAG) conducts security patrols in AO NOR EAST ISO Pinkland IOT deter Redland maritime force aggression.

SE: CTF 227 maritime patrol and reconnaissance aircraft (MPRA) conducts airborne surveillance and reconnaissance in the JOA IOT determine Redland maritime disposition and intentions.

SE: CTF 229 (SUBS) provide SUW, antisubmarine warfare (ASW), and intelligence, surveillance, and reconnaissance (ISR) support in the SUB operational area (OA) IOT protect JFMCC forces and enable FOPS.

SH: CTF 224 (ESG) conduct amphibious rehearsal in AOA 1 IOT prepare for offensive OPS in Redland.

SH: CTF 220 (CSG) conduct integrated air and missile defense (IAMD) in AO NOR EAST IOT protect forces deploying to Pinkland.

SUS: CTF 223 provide logistics (LOG) support IOT extend operational reach of JFMCC forces.

4. Other support required. Address the additional support required from adjacent commanders and partners to accomplish the operation.

Example: Pinkland provides landside security for ISB (FLS and AOA). JFACC assumes security/protection requirements at the ISB/FLS. JFLCC occupies position within Pinkland IOT prepare for combat operation against Redland. JFSOCC conducts SR IOT set conditions for future JTF OPS.

5. Timing/conditions—ending instructions. The COA statement ends with the timing/conditions instructions describing the actions that must be met to end the phase/operation. Key aspects of time and/or conditions are the prerequisites that must exist to end the operation/phase. Conditions should be described in terms of the enemy force, friendly force, and environment.

Example: Phase I ends when Redland agrees to a diplomatic resolution or commences hostilities, and JTF forces have completed deployment to the JOA for FOPS.

3.3.5 Recommend Command Relationships

Planners establish preliminary C2 arrangements and task organizations among major subordinates for each COA. This structure should consider the types of units to be assigned to a HQ or component and the span of control. C2 arrangements take into account the entire assigned maritime force organization. Planners also consider the special C2 requirements of OPS that include amphibious landings, support to other components, etc.

Each COA should propose a C2 arrangement for the assigned forces and propose initial TF and TG organizations, including OPCON, TACON, and support relationships,[14] as well as those relationships unique to Navy tactical OPS, such as designation of officer(s) in tactical command and support situation(s) by phase. This provides a baseline for comparison and refinement in later steps. It also enables review of force apportionment and dual tasking or capabilities in each phase and highlights the need for OPCON, TACON, or support relationships adjustments by phase. These arrangements should also consider types of capabilities/tasks retained by the commander, as well as capabilities assigned to subordinate TFs. Span of control and decision authorities for both the commander and subordinate commanders need to be considered in making C2 arrangements.

Some considerations include:

1. The mission(s)
2. Subordinate forces
3. The geography of the maritime AO
4. The commander's C2 capability
5. The subordinate staffs' planning capacity and C2 capability.

Organization options are: functional, domain, geographic, platform, or a combination. Refer to NWP 3-32, Maritime Operations at the Operational Level of War for detailed information on C2. Figure 3-6 provides an example C2 organization for the depicted JFMCC COAs with a hypothetical multinational maritime force (MMF) conducting cooperative OPS.

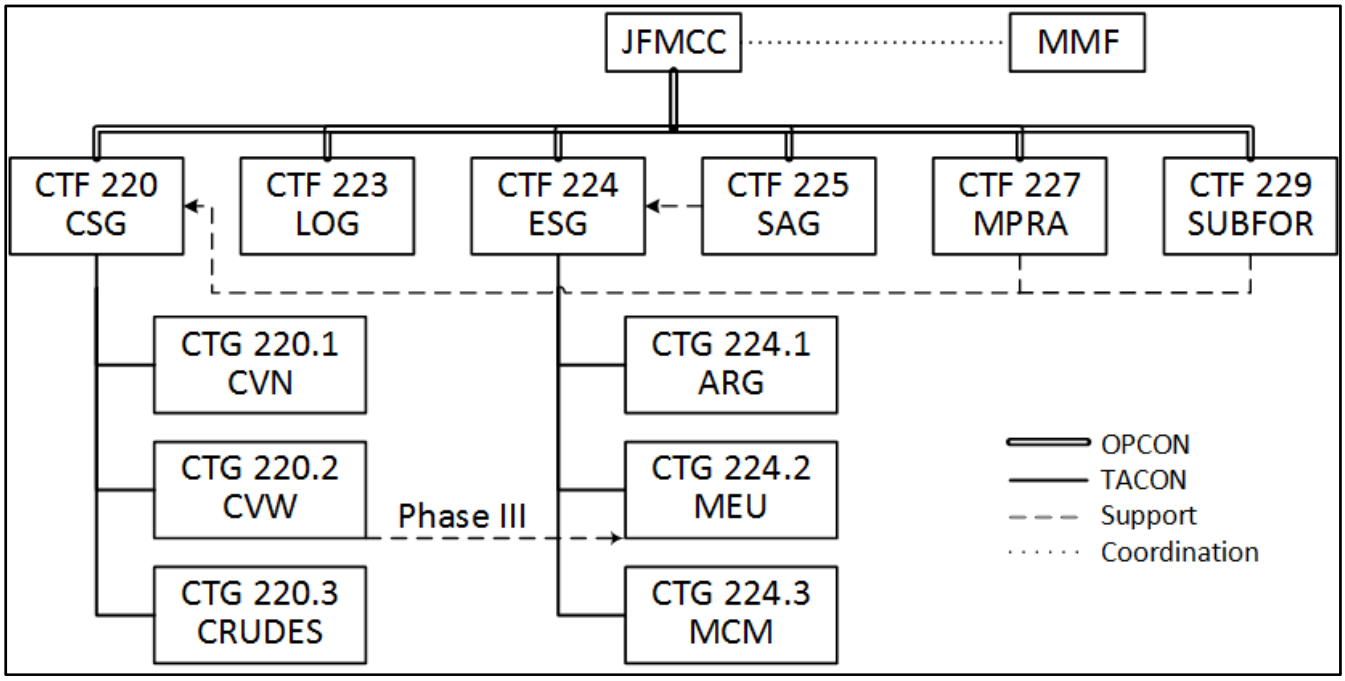


Figure 3-6. Command Relationships Example

[14] Refer to JP 1, JP-3-31, and NWP 3-32 for detailed information on support relationships: close, direct, mutual, and general support.

3.3.6 Authorities/Rules of Engagement Review

During COA development, planners should be familiar with the basic principles of self-defense, the law of the sea, the law of armed conflict (LOAC), and how these laws and policies apply to existing ROE. Planners should then identify those supplemental ROE required for mission accomplishment. Planners review existing U.S. ROE and consider whether or not they are sufficient to execute the COA. Planners should focus squarely on what they desire subordinate tactical forces to do and who in the chain of command will be able to authorize those actions.

Working with the assigned staff judge advocate, planners consider what, if any, supplemental ROE or other authorities are required to execute the COA. They should also identify what, if any, clarifications might be required with respect to the forces' ability to use force. At this stage in the planning process planners are not yet requesting additional ROE from an entity outside the planning team. Rather, planners should be able to articulate and justify what specific ROE are required to accomplish the mission and identify the risk to mission and forces that would result if the supplemental ROE were not obtained. IOT optimally employ forces, planners should also become familiar with coalition ROE. Coalition ROE will determine whether those partners can use force and, if so, when and how they can use force to conduct the mission.

3.3.7 Risk Assessment

The commander and staff continue the risk assessment and mitigation process in COA development, focusing on three steps:

1. Identify threats.

2. Assess threats.

3. Develop controls and make risk decisions.

Note

The refined risk assessment developed here may be different for each COA. See Appendix E for more information on risk assessment and mitigation.

3.3.8 Test for Validity

The staff reviews each proposed COA for validity. This test should address suitability, feasibility, acceptability, distinguishability, and completeness.

1. Suitability (adequacy). The COA must accomplish the mission and comply with the commander's guidance. However, the commander may modify guidance at any time. When the guidance changes, the staff records and coordinates the new guidance and reevaluates each COA to ensure that it complies with the change.

2. Feasibility. The force must have the capability to accomplish the mission in terms of available time, space, and resources. If requests for additional capability are essential for the COA(s) to pass the validity test and are not approved, then it is likely that the COA will become invalid.

3. Acceptability. The tactical or operational advantage gained by executing the COA must justify the cost in resources, friendly losses, time, position, and opportunity. This examines cost and risk assessment and is largely subjective.

4. Distinguishability. Each COA must differ significantly from any others. Significant differences may result from use of reserves, different task organizations, day or night OPS, sequential versus simultaneous OPS, or a different scheme of maneuver. This criterion is also largely subjective.

5. Completeness. Each COA must include the following: major OPS and tasks to be performed; major forces required; concepts for deployment, employment, and sustainment; time estimates for achieving OBJs; and the

desired military end state and mission success criteria. The order from HHQ normally provides the "what, when, where, and why" for the force as a whole. The who in the COA does not specify the designation of units; it arrays units by type (e.g., TFs/TGs or particular platform/capability). Designation of specific units within a C2 arrangement is refined later in the NPP.

3.3.9 Develop Course of Action Analysis Guidance

The planning team prepares COA analysis (wargaming) guidance for commander approval during the COA brief. Commander's war-gaming guidance may include:

1. Which friendly COAs to war-game against which enemy COAs (mostly likely or most dangerous or both).

2. Critical events (essential tasks), area of interest, LOOs/LOEs and/or decision points to be analyzed.

3. War-game method to use.

3.3.10 Develop Evaluation Criteria

The planning team should also develop initial evaluation criteria recommendations for use in COA comparison and decision (after wargaming). Evaluation criteria include the commander's governing factors but may also include other criteria developed by the staff. Evaluation criteria do not stand alone; each should have a clearly defined definition to minimize confusion. Many of the evaluation criteria lead directly to operation assessments conducted during actual execution. Evaluation criteria should:

1. Recommend modifications to governing factors.

2. Reflect the criteria for success established during the mission analysis.

3. Provide a reasonable basis for comparing the relative merits of the COAs under consideration.

4. Focus on the OBJs and DPs identified in the mission analysis.

5. Be quantifiable and measurable.

3.3.11 Prepare Course of Action Briefing

At this stage of the process, the planning team may propose (or the commander may require) a briefing on the COAs developed and retained. The purpose of this briefing is to gain the naval commander's approval of the COAs to be further analyzed, to receive guidance on how COAs are to be compared and evaluated, or to receive guidance for the revision of briefed COAs or the development of additional COAs. Figure 3-7 provides an example of information found in a COA briefing.

The COA briefing is another place where a collaborative session may facilitate subordinate planning. The COA briefing includes:

1. Updated IPOE.

2. Enemy COA(s) (event templates).

3. The restated mission.

4. The commander's and the higher commander's intent (two echelons above).

Briefer	Subject
N-3/N-5	Purpose of briefing.
N-2	Updated intelligence estimate and other IPOE products with potential enemy COAs.
N-3/N-5	HHQ mission, HHQ intent, JFMCC mission, JFMCC intent, JFMCC planning guidance.
N-3	RCPA.
OPT leaders	For each COA, the OPT presents: • Proposed task organization. • COA sketch and narrative that describe MEs, SEs, SHs, and SUSs of each phase. • Facts and assumptions that affect the COA. • Imposed limitations (constraints and restraints). • Authorities. • Risk. • Resource and capability shortfalls.
N-3/N-5	Proposed COA analysis guidance, governing factors, evaluation criteria, war-game method, and the enemy COAs to be considered.
N-3/N-5	Confirm commander's COA analysis guidance.

Figure 3-7. Example of a Course of Action Brief

5. The COA statement and sketch.

6. The rationale for each COA, including:

   a. Considerations that might affect enemy COAs.

   b. MEs, SEs, SHs, and SUSs in each phase.

   c. Deductions resulting from an RCPA (See Appendix D).

   d. Reasons units are arrayed as shown on the sketch.

   e. Reasons the staff used the selected control measures.

   f. Updated facts and assumptions.

After the briefing, the commander gives any additional guidance and may direct which enemy COAs should be used in COA analysis (wargaming), such as the most dangerous, most likely, or both. If the commander rejects all COAs, the planning team begins again. If one or more of the COAs is accepted, staff members develop evaluation criteria IOT begin COA analysis.

3.4 OUTPUTS

At the end of COA development, the following will have been produced:

1. Approved COAs.

2. Refined enemy COAs.

3. COA analysis (wargaming) guidance.

4. Initial evaluation criteria (for COA comparison).

5. Refined commander's intent.

6. Refined staff estimates—During the initial stage of COA development, staff sections should refine their initial staff estimates and share them with adjacent and subordinate commands to enhance common understanding of the situation. Once the planning team has identified draft COAs, the staff sections change their focus from information gathering to developing concepts of support for these COAs. See Appendix I for further discussion on staff estimates.

7. Refined risk assessment.

8. Recommended RFFs and RFCs.

9. Recommended standing ROE and supplemental ROE changes or additions.

10. Initial, refined decision tools: synchronization matrix, DST/DSM, risk analysis matrix, and CCIRs.

3.5 KEY POINTS

1. A COA is a broadly stated potential solution open to the commander, which if adopted would result in the accomplishment of the mission.

2. A COA shall meet validity tests of suitability, feasibility, acceptability, distinguishability and, most important, be complete with respect to who, what, where, when, why, and broadly how.

3. COAs continue to be refined after they are developed as facts and assumptions change. Refined COAs produce updated sketches, statements, decision tools, and other outputs.

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CHAPTER 4
Course of Action Analysis (Wargaming)

The war with Japan had been enacted in the game rooms at the War College by so many people and in so many ways that nothing that happened during the war was a surprise—absolutely nothing except the kamikaze tactics towards the end of the war. We had not visualized those.

Admiral Chester Nimitz, Naval War College lecture, (1960)

4.1 INTRODUCTION

COA analysis closely examines potential COAs to confirm their validity and refine them to account for potential enemy reactions. The commander and staff analyze each COA separately according to the commander's war-gaming guidance. COA analysis improves the plan by examining and refining friendly actions in light of enemy capabilities and potential reactions. The planning team analyzes the probable effect that enemy actions have on each friendly COA. This helps the commander, staff, and subordinates gain a common understanding of friendly and enemy COAs and allows them to determine the advantages and disadvantages of each COA that form a basis for COA comparison and the commander's decision. Based upon time available, the planning team analyzes each COA against the most likely and the most dangerous enemy COA (or critical events in noncombat OPS) identified during the IPOE process.

Wargaming is the means used to conduct COA analysis. War games are representations of conflict or competition in which people make decisions and respond to the consequences of those decisions. War games help visualize the flow of the operation given friendly and enemy capabilities, strengths and dispositions, as well as other aspects of the OE. The inputs, process, and outputs of wargaming are modified to fit the specific mission and OE, and this construct applies across the range of military OPS (Figure 4-1). The specific methods and approaches can vary greatly. In any case, each COA must be scrutinized with a healthy respect for the enemy's capability to prevent friendly force success.

Preferred methods of wargaming consist of those based on critical events (essential tasks), phases of the operation, or discrete amounts of time depending upon war-gaming guidance using an action, reaction, and counteraction method of force interaction. Other methods include the belt, box, and avenue in depth methods.

There are several approaches to the process that can be used depending on time, resources, and expertise available. A basic approach involves developing a narrative that visualizes and describes actions, reactions, and counteractions of friendly and enemy forces across space, time, and purpose. Sketches and notes can be further developed with respect to critical events covering broad areas using sequential numbers to drive the process. In a manual war game, the planning team uses maps or charts of the maritime OA to systematically explore a COA. The planning team may use discrete time periods, entire phases, or individual tasks to segment the game into manageable parts. Modeling and simulation can assist wargaming but are separate and distinct analytic tools that are not the same as war games.

Regardless of the method and approach, COA analysis war games use an iterative process of friendly action, enemy reaction, and friendly counteraction.

Wargaming produces refined COAs that can succeed against enemy reactions to a greater extent than the original COAs and helps synchronize the actions of the force in time, space and purpose. Refined products include validated COAs, a refined synchronization matrix, refined decision support tools, identified branches and sequels for further development, updated evaluation criteria, refined resource shortfalls, updated authorities, updated facts and assumptions, and refined staff estimates.

A planning team leader guide to organizing and conducting a manual war game is provided in Appendix F.

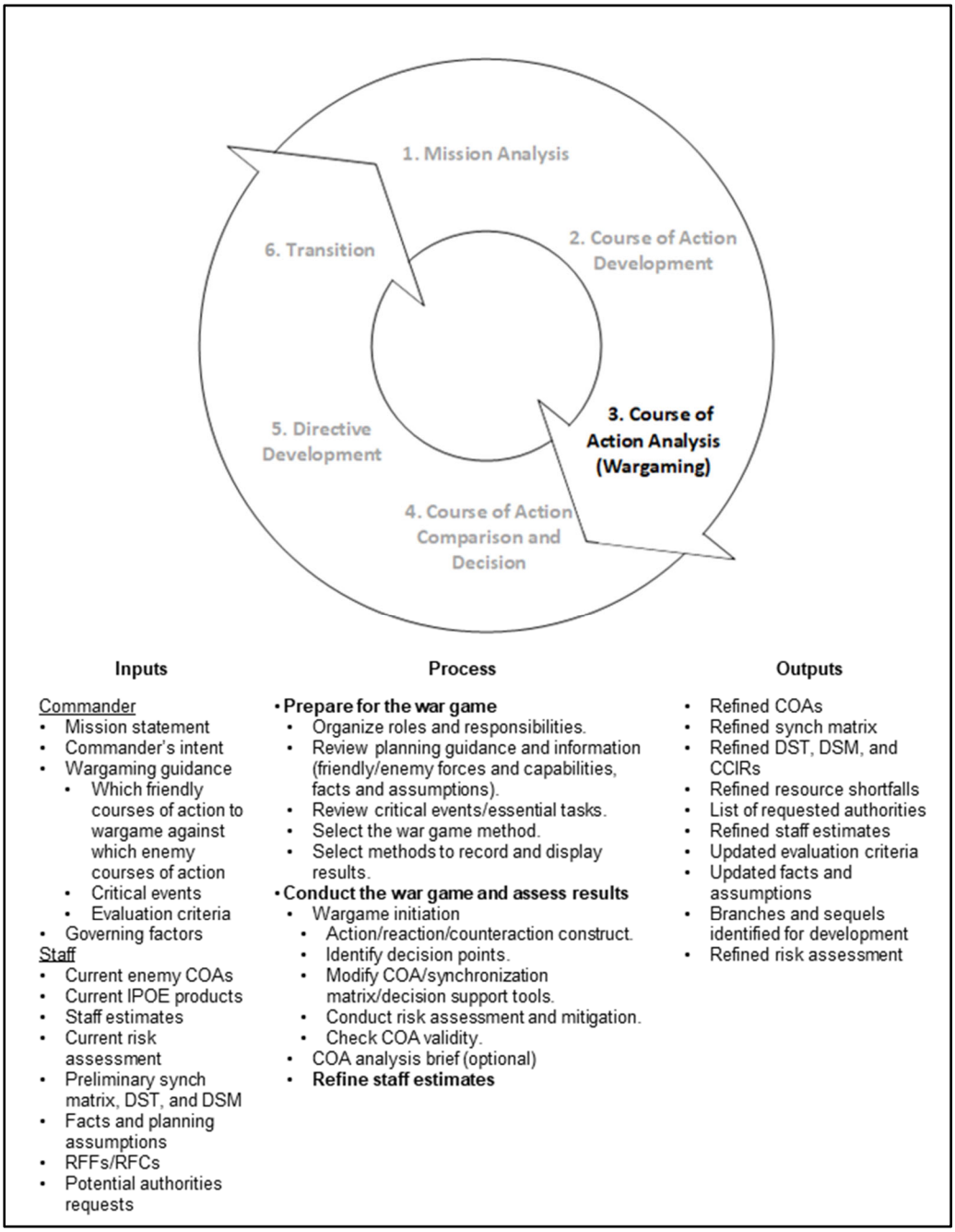


Figure 4-1. Course of Action Analysis (Wargaming)

4.2 INPUTS

The inputs for COA analysis are the results of the planning effort to this point in the process (see Figure 4-1).

1. Mission statement and commander's intent. The approved mission statement and commander's intent are brought forward to COA analysis.

2. Commander's war-gaming guidance. Guidance may include all or a portion of the following:

   a. Which friendly COAs to war-game against which enemy COAs (mostly likely, most dangerous, or both).

b. Critical events (essential tasks) and/or decision points to be considered.

c. Evaluation criteria that the commander deems critical to mission accomplishment.

3. Governing factors. Factors that the commander deems critical to mission success and drive, in part, COA development are another input to COA analysis. These factors are part of the validity test for approved COAs. The planning team may recommend adjusting, refining, deleting, or adding criteria to the commander as a result of analysis. These factors become part of evaluation criteria.

4. Current enemy COAs. Ongoing intelligence efforts may result in refinements to enemy COAs.

5. Current IPOE products. Ongoing intelligence efforts may yield updates to IPOE products in a manner that impacts planning.

6. Staff estimates. Throughout the planning process, the staff persists in developing estimates of supportability of the COAs based on their particular operational function or activity, the environment, the mission, and intent. They develop and use these estimates in a reciprocal manner to both contribute to the planning process, as well as to develop supporting annexes to the directive based on what was learned during the process. During COA analysis, staff functional representatives leverage their developing estimates to make meaningful contributions to wargaming.

7. Risk assessment and mitigation. Planning team efforts continue to assess risk within the framework of approved COAs. The planning team continues to identify new risks and mitigate known risks during wargaming.

8. Other products and information. Several other developing products and subprocesses are brought forward from COA development for COA analysis. These can include: updated facts and planning assumptions, requests for capability, and potential authorities.

4.3 PROCESS

COA analysis is a disciplined process with rules and steps that attempt to visualize the flow of the operation. Each war-game turn uses an action, reaction, and counteraction construct that provides a common frame of reference and drives the tempo of the war game. The commander indicates what aspects of the COA are to be examined and tested; that is, what friendly COAs the planning team and staff war-games against specific enemy COAs. The purpose is to develop a sound basis for determining valid COAs, contribute to a common understanding of the environment and problem, identify potential branches and sequels, and make modifications to the COA as required. Planning teams prepare for the war game, conduct the war game, assess results, and update staff estimates. Multiple substeps contribute to each step.

The planning team lead uses an agenda as a means to provide a source of purpose and direction for the time available. The number of war games scheduled are in accordance with (IAW) the commander's guidance. All participants should have a copy of the current synchronization matrix, the COA to be war-gamed, and the task organization. Functional planners should have their estimates available for reference. The red cell (which role-plays enemy COAs) should have access to all products, to include design (if used) and COG products, friendly COAs, task organizations, synchronization matrices, and decision products. The venue to conduct the war game should be large enough to accommodate required information displays, all war-game players, and technical support. The planning team posts all applicable products for easy reference within the war-gaming area to gain efficiencies during the process to include the war-game worksheet conspicuously posted for all to see.

4.3.1 Organize Roles and Responsibilities

Members of the planning team have distinct war-game roles and responsibilities. The size and composition of the planning team differ with mission, guidance, time available, and other variables. Some individuals may play dual

roles. The following general roles are for a manual war game and serve as a guideline. Detailed responsibilities are provided in Appendix F.

1. Planning team lead. This role functions as the unbiased controller who directs the war game to ensure that it achieves specified goals within the time available. This person also arbitrates impasses to allow the war game to continue. The planning team lead may be supported by a facilitator who executes the war-game agenda and guides dialogue and debate, and recorders who ensure that war-game records are captured and transcribed into planning products.

2. COA representatives. These individuals describe their COAs to the remainder of the war game team, coordinate counteractions to respond to envisioned enemy reactions, and ensure that COA modifications are documented.

3. HHQ and adjacent commander representatives. These roles provide information and feedback concerning unity of effort and synchronization from the perspective of the HHQ and adjacent commands. They help identify authority shortfalls and clarify command and support relationships. When the planning team lacks the manpower to fill all of these roles, adjacent commander roles can be combined.

4. Subordinate command representatives. These roles provide information and feedback concerning employment of subordinate forces during the war game. They explain their tasks and purposes, how their forces contribute to the higher COA, and what counteractions they may take in response to enemy reactions, and they help to identify resource and authority shortfalls. Subordinate command representatives serve as the movement and maneuver representatives of their respective TFs during the war-game turns. When the planning team lacks the manpower to fill all of these roles, subordinate command role playing is done by operational function representatives.

5. Operational function planning representatives. These individuals provide information and feedback on the overall schemes of maneuver and fires and other operational functions[15] that pertain to the COA. They ensure that each COA adequately addresses the operational functions that are relevant to the envisioned operation.

6. Intelligence representatives. Intelligence representatives play two supporting roles during wargaming: planner and analyst. The intelligence planner, supported by a collection management representative, represents IO from a functional perspective while the intelligence analyst represents the enemy within the red cell.

7. Red cell. The red cell is a composite cross-functional team from varied operational backgrounds and specialties. It emulates a thinking enemy during the war game by reacting to friendly force actions. The red cell's OBJ is to expose friendly critical vulnerabilities or gaps in their plan, improving feasibility and acceptability of the COAs through rigorous pursuit of perceived enemy OBJs against each friendly COA. The red cell combines operational experience with known enemy capabilities, requirements, vulnerabilities, doctrine, current disposition, and most likely and most dangerous COAs to provide a credible thinking enemy. The red cell must understand all friendly COAs as well as friendly and enemy COGs to leverage friendly vulnerabilities in attacking the friendly COG either directly or indirectly. The red cell must understand the rules established by the planning team lead and remain within those boundaries. Additionally, the red cell must understand the enemy capabilities and limitations and present reactions that fall within those capabilities, or the war game will lose realism.

During the war game, the red cell employs the directed enemy COAs against the friendly COAs to pursue perceived enemy OBJs and friendly vulnerabilities. This effort can result in potential COA modifications during counteraction, ranging from adjusted or additional friendly force tasks and purposes, additional authorities, and anticipated decision points, to additional coordination, control measure refinement, coordination gaps, resource shortfalls, C2 adjustments or reallocation of assets, and other potential operational function area adjustments.

[15] At time of publication, Navy doctrine recognizes six operational functions and has not addressed the joint function of information described in JP 3-0. Without approved Navy doctrine regarding how operational level of warfare commanders (i.e., NFC/NCC/MCC) will implement the new joint function of Information, the NPP continues to address the Navy-approved six operational functions.

Note

In addition to the red cell, other colored cells may be formed by the planning team lead to represent HN, third-party, or civilian aspects of the environment. During execution, these perspectives are captured as additional information on the synchronization matrix.

8. Special staff. In addition to the staff representatives listed above, the war game may include other staff personnel such as the legal advisor, a public affairs (PA) representative, contracting officer, chaplain, engineer, medical representative, space representative, electromagnetic warfare (EW) representative, and other subject matter experts as required.

4.3.2 Review Planning Guidance and Information

1. Review friendly forces and their capabilities. The planning team considers all units that can be committed to the operation, paying attention to task organization, command, and support relationships. Although the task organization may vary by COA to include assumptions made about additional capabilities necessary to continue planning, the friendly force list remains constant for all COAs. The force list reflects all units that may be employed. Ideally, a maneuver role player considers employment of their force two force levels down as they review capabilities and conduct war-game turns. For example, analysis of a JFMCC COA means that TF-level role players and the maneuver representative will consider employment of subordinate TGs and task units. An informal review of the RCPA product developed during COA development is also appropriate.

2. Review enemy forces and their capabilities. The planning team considers all enemy units that can be committed to the operation, paying attention to capabilities and potential employment of the enemy force. An informal review of enemy COG products developed during IPOE and RCPA products developed during COA development is appropriate.

3. Review facts and assumptions. The commander and planning team review facts and assumptions to determine continued validity. The review allows the team to think through impacts of assumptions as they relate to the COAs and the consequences regarding them. For example, while planning to conduct an opposed amphibious operation, the planning team may assume that the two minesweepers on the present force list are inadequate to clear lanes to the beach. The planning team may make the assumption that to continue planning, three additional minesweepers are required to protect the force. The consequence of not receiving these additional assets impacts the COA significantly with regards to risk to force and mission.

4.3.3 Review Critical Events/Essential Tasks

The planning team reviews and confirms critical events in each approved COA. Critical events are frequently essential tasks, or a series of critical tasks, conducted over a period of time. This may be expanded to important tasks within phase(s) of an operation or over a period of discrete time (e.g., D-day through D+5). Critical events directly influence mission accomplishment. They trigger significant actions or decisions. They are complicated actions or potential friction points requiring detailed analysis. As an example, it may include actions to gain local sea control, conduct an amphibious landing, and seize initial OBJs. Critical events are taken into consideration when war-gaming different phases or in discrete portions of time (e.g., D+1 to D+5).

Pursuant to reviewing critical events, the planning team also reviews associated decision points.[16] Decision points are frequently linked to a critical event at a DP, or perhaps an intermediate OBJ. They are associated with a CCIR, options (branches and sequels), locations, the time or phases that the decision is applicable to, and conditions (criteria) that may inform the decision. The fidelity of this information increases as planning progresses.

PIRs, a type of CCIR, are requirements stated as a priority for intelligence support that the commander and staff need to understand the enemy or other aspects of the OE. They are often associated with a named area of interest (NAI). FFIRs, another type of CCIR, describe information the commander and staff need to understand the status

[16] **decision point**—A point in space and time when the commander or staff anticipates making a key decision concerning a specific course of action. (DOD Dictionary. Source: JP 5-0.).

of friendly forces and supporting capabilities. Although a decision point may require a decision by the commander, it does not dictate the decision; only that the commander must make one at an expected place and/or time that bears on mission success.

4.3.4 Select the War-game Method

There are distinct methods the planning team can use to conduct the war game, which are illustrated in Figure 4-2.

OPERATIONAL LEVELS OF WAR CONSIDERATIONS:
At the high tactical-low operational level of war, these may be more conducive to wargaming:

By phase. Significant actions by phase (by subordinate unit and functional areas).

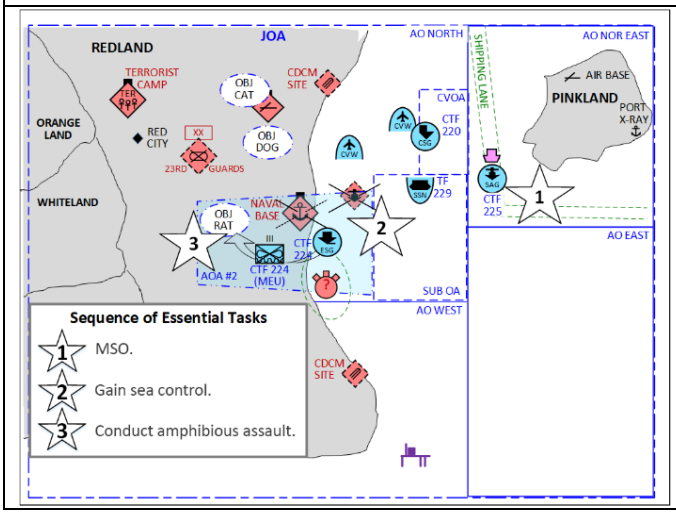
By deliberate time analysis. *Day by day or discrete blocks of time.*

IE: D-day D+1 D+2 D+3
or
D-day to D +10 D+10 to D+20

By critical event/sequence of essential tasks. Specific critical events that capture the essence of the COA.
IE: Deploy, control portions of operational area, conduct amphibious operations, etc.

Sequence of Essential Task/Critical Event
- War-games critical events in sequence
- Frequently associated with phases

Belt*
- Analyzes sequence of actions (all forces) in each belt.
- Preferred when focusing on all forces effecting an event during each specified time period.



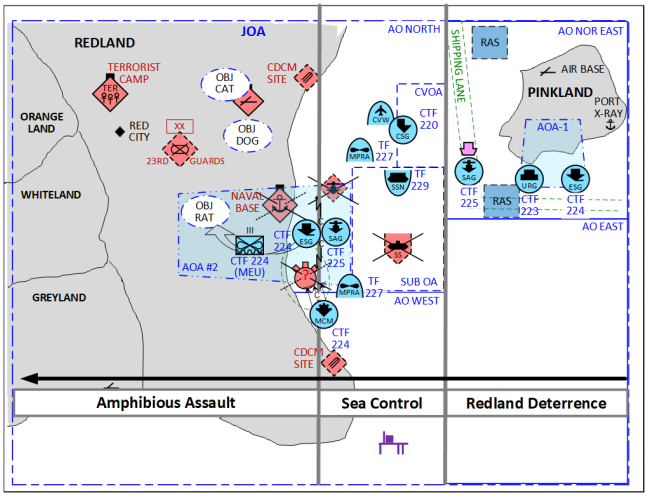


Figure 4-2. War-game Methods (Sheet 1 of 2)

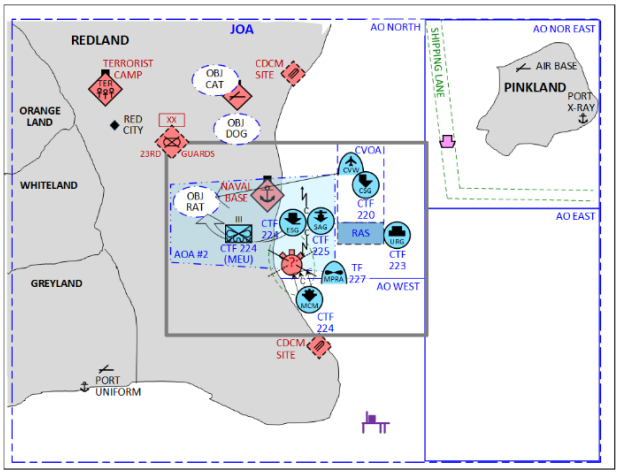
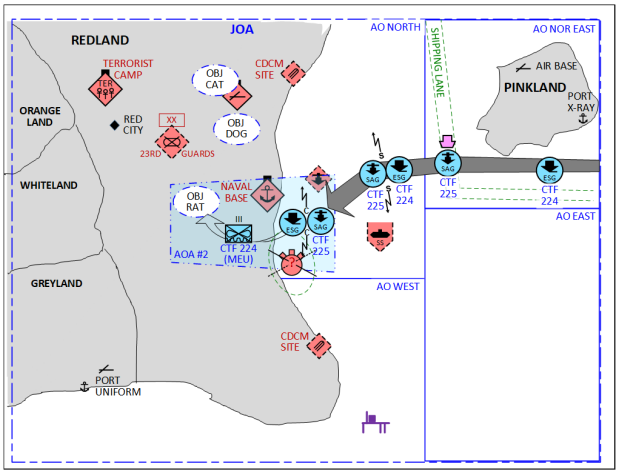
BOX*	AVENUE IN DEPTH*
- Focuses on a critical event within a defined area. - Preferred when detailed analysis of a critical area is required.	- Focuses on one avenue of approach at a time. - Preferred when space inhibits mutual support.
	
*These methods may not be ideal to advance planning at the operational level of warfare.	

Figure 4-2. War-game Methods (Sheet 2 of 2)

4.3.4.1 Phase Method

This method aligns each war-game turn to a phase in the operation. Critical events that occur during the phase are gamed within the appropriate turn. When time is available, this may be the most comprehensive of manual war games. Each turn in the phase has action, reaction, and counteraction played throughout the maritime AO. This method can be time consuming, but it does capture a detailed analysis.

4.3.4.2 Deliberate Time Analysis

This method partitions discrete portions of time (i.e., D+0 to D+4, D+5 to D+9, etc.) to conduct war-game turns. Each portion of time is a war-game turn accounting for action, reaction, and counteraction. This method also results in thorough analysis. An example of this type of turn would be D-day through D+2 of a COA involving amphibious OPS. In such situations, deliberate time analysis may be advantageous.

4.3.4.3 Critical Event/Sequence of Essential Tasks

This method focuses on specific critical events in sequence that capture the essence of the COA. A critical event is an event that directly influences mission accomplishment. It is usually tied to an essential task and the component ME. Critical events trigger actions requiring detailed analysis. Each critical event is war-gamed by action, reaction, and counteraction. The difference between this method and the phasing method is that the phasing method accounts for all significant maneuver in the phase by all subordinate TFs, while the critical event method may not address all TF actions throughout the maritime AO. Each war-game turn is focused only on the critical event occurring in that particular area of the maritime AO and the forces and functions being employed as a result of the event.

4.3.4.4 Belt Method

This method divides the maritime AO into a series of predetermined belts that span the AO. This technique is based on the sequential analysis of events in each belt; that is, events are expected to occur more or less simultaneously in that belt. It focuses simultaneously on all forces affecting particular events in the belt. Several

critical events can occur in each belt. Using this method divides the maritime AO into not more than three sequential belts that can be war-gamed by belt, by war-game turn, across the breadth and depth of the maritime AO. This technique is often more appropriate for analyzing plans at the tactical level of warfare.

4.3.4.5 Box Method

This method is a detailed analysis of an area, such as an AOA or a portion of the maritime OBJ area in which the maritime component is striving to achieve a degree of sea control to enable other actions. The planning team designates the area and focuses on the critical events within that area. Several critical events could occur in the area. The planning team assumes friendly TFs not engaged in the critical events in the box can continue to execute their assigned tasks in their portion of the maritime AO. These forces are not part of the war-game turn.

4.3.4.6 Avenue in Depth Method

This method focuses on one approach at a time, beginning with the ME executing the decisive action, then SEs that may be supporting the ME or shaping the environment. This technique may be conducive to war-game turns that relate to a series of actions by forces operating independently toward a specific OBJ.

4.3.5 Select Methods to Record and Display Results

The planning team records the results of war-game turns to synchronize each COA across time, space, purpose, functions, and activities (e.g., IO and OPS assessment). There are several techniques to capture information and display results.

4.3.5.1 War-game Worksheet

This worksheet (usually one or more large whiteboards) functions as the primary collection point for actions, reactions, and counteractions. It provides a visual reference during the war game that is visible to all participants and lends contextual reference to the progress of each war-game turn. Figure 4-3 illustrates the layout of a war-game worksheet.

4.3.5.2 Synchronization Matrix

The synchronization matrix, initiated earlier in the planning process, is carried forward to COA analysis and refined to become the basis of ensuing directives. Results captured in war-game worksheets are transcribed into the synchronization matrix (which serves as the source of tasks in the directive to be developed later). Other information can be added to the matrix such as other entities in the OE (i.e., HN, other government, or nongovernment agencies), authorities, resource shortfalls, or coordinating instructions. Each synchronization matrix is unique to the nature of the environment and mission. Figure 4-4 and Appendix G illustrate the layout of a synchronization matrix.

Note

Planning teams should exercise caution when using the synchronization matrix as the primary means to capture war-game results when it is not used ICW a war-game worksheet. The war-game worksheet records information about enemy reactions and possible friendly counteractions, while the synchronization matrix records the final decisions about the friendly actions to be taken.

	Action	Reaction	Counteraction	Remarks
Phase Beginning Conditions				
Commander, joint task force (HHQ)				
CFLCC				
Combined force special operations component commander (CFSOCC)				
Combined force air component commander (CFACC)				
Combined force maritime component commander (CFMCC) HQ				
Phase ending conditions				
CFMCC phase objective				
Subordinates				
CTF XXX ( )				
CTF XXX ( )				
CTF XXX ( )				
CTF XXX ( )				
CTF XXX ( )				
CTF XXX ( )				
CTF XXX ( )				
Coordinating instructions				
Operational functions				
C2				
INTEL				
Movement and Maneuver				
Fires				
Sustainment				
Protection				
IO				
Authorities required				
Decision points				
Branches				
Risk				

The source of information for this column comes from the current synchronization matrix.

This column is used to post enemy reaction notes.

This column is used to post friendly counteractions to be translated into the synchronization matrix by the role player posting the note.

Figure 4-3. War-game Worksheet

	Phase __	Phase __	Phase __ Stage A	Phase __ Stage B	Phase __
Phase Beginning Conditions					
CJTF (HHQ)					
CFMCC HQ					
CFLCC					
CFSOCC					
CFACC					
Phase Ending Conditions					
Phase Objective					
Subordinates					
CTF XXX ( )					
CTF XXX ( )					
CTF XXX ( )					
CTF XXX ( )					
CTF XXX ( )					
CTF XXX ( )					
CTF XXX ( )					
Coordinating Instructions					
Operational Functions					
C2					
INTEL					
M&M					
FIRES					
SUSTAINMENT					
PROTECTION					
IO					
AUTHORITIES REQUIRED					
DECISION POINTS					
BRANCHES					
RISK					

Figure 4-4. Synchronization Matrix

4.3.5.3 Decision Support Products

Initially developed earlier in the process, the planning team leverages the DST and DSM to not only acknowledge and discern known decision points during war-game turns, but to capture emerging decision points, their related CCIRs, the time and space relevant to the potential decision, potential options (to include branches and sequels), and the necessary supporting conditions (criteria) that help recognize that a decision is required and feasible. See Figure 4-5 and Appendix G.

4.3.5.4 Time-constrained Technique

This technique uses a graphic portrayal of the AO and a series of sequential notes that are, by themselves, individual worksheets. Participants place sequential numerals on the graphic in the areas to be war-gamed, and capture the resulting information on a series of notes that correspond to the numerals on the graphic. The note worksheet contains the sequential number of the event to be war-gamed, uses the action, reaction, counteraction construct, and can contain any additional required information, such as assets required to conduct the action at the critical event, a time reference for the event, decision points and CCIRs that correspond to the event, and references to control measures. The planning team can add to or delete any number of areas on the note worksheet that make sense for the mission. Operational functions and activities are considered, and their results are captured by functional planning representatives for developing staff estimates. This method is broader in nature than the synchronization matrix and can be used to condense the war game when time is a constraint. An example using the time-constrained technique to advantage can be made by reference to an amphibious assault whereby, first, a degree of sea control must be achieved; secondly, obstacles are identified and reduced; thirdly, the assault occurs; and lastly, initial OBJs areas are seized. See Figures 4-6 and 4-7.

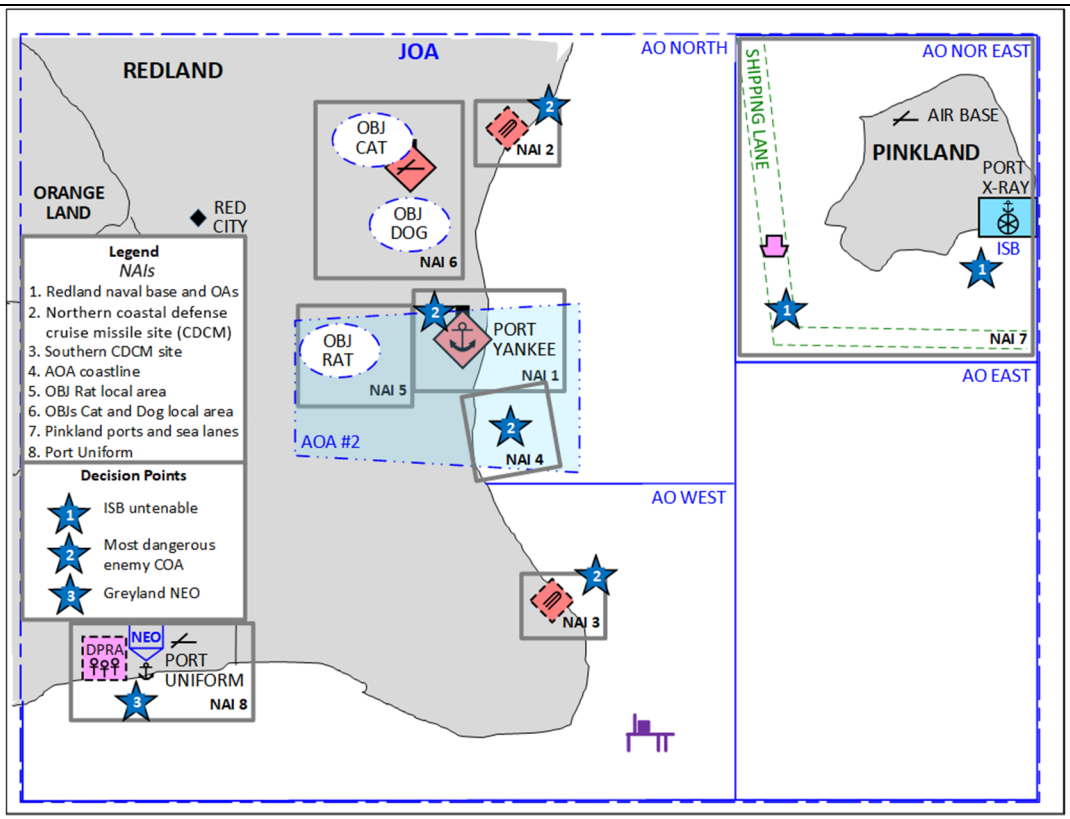
4.3.5.5 Additional War-game Records

War-game participants should be ready to capture additional information as it emerges from war-game turns, including:

1. Strengths (advantages) and weaknesses (disadvantages) of each COA based on its merits
2. Additional or refined evaluation criteria
3. Risk identification, assessment, and mitigation measures
4. COA sketch (control measures) and statement modifications
5. Task organization changes
6. Residual gaps in coordination or synchronization
7. Facts and assumptions
8. Authorities
9. Other issues that do not fit into any other product
10. Information pertinent to an optional briefing.

4.3.6 Conduct the War Game and Assess the Results

A COA analysis war game consists of a number of turns which consist of three moves: friendly action, enemy reaction, and friendly counteraction. This construct provides a common frame of reference and drives the tempo of the war game. The war-game method selected determines which events are examined, the sequence, and the level of detail. Appendix F provides a detailed guide for conducting a COA analysis war game.

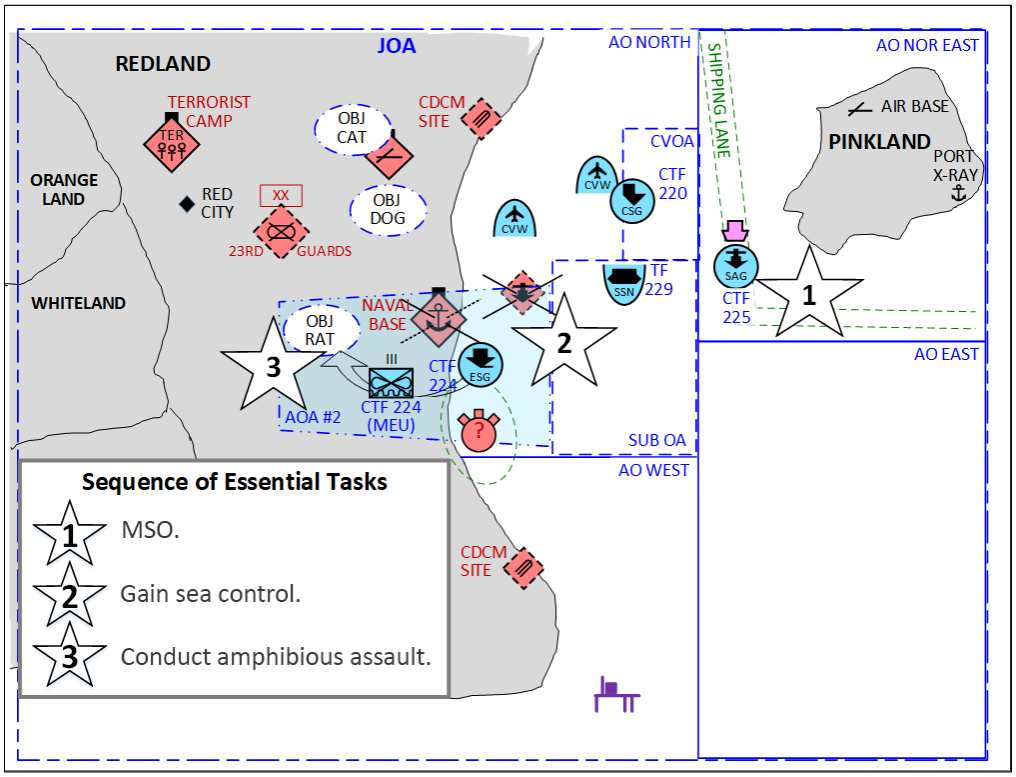


Decision Pts
1 –
2 –
3 –
4 –
5 –
6 –
7 –
8 –

NAIs
1 –
2 –
3 –
4 –
5 –
6 –
7 –
8 –
9 –
10 –
11 –
12 –
13 –
14 –

Decision Point	Timing	CCIR		Decision Options		Decision Support Criteria	Location (NAI)
Decision Point 1			1A: 1B: 1C:		1A: 1B: 1C:		
Decision Point 2			2A: 2B: 2C:		2A: 2B: 2C:		
Decision Point 3			3A: 3B: 3C:		3A: 3B: 3C:		
Decision Point 4			4A: 4B: 4C:		4A: 4B: 4C:		
Decision Point 5			5A: 5B: 5C:		5A: 5B: 5C:		
Decision Point 6			6A: 6B: 6C:		6A: 6B: 6C:		
Decision Point 7			7A: 7B: 7C:		7A: 7B: 7C:		
Decision Point 8			8A: 8B: 8C:		8A: 8B: 8C:		

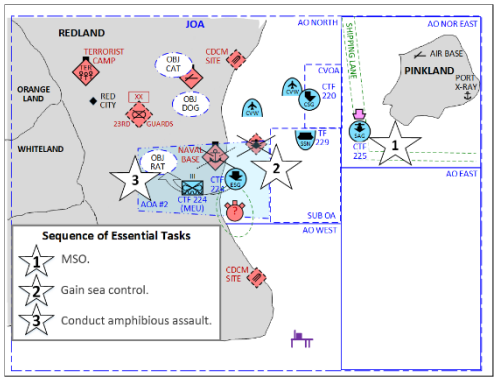
Figure 4-5. Decision Support Template and Matrix (DST and DSM)



Sequence #	Essential Task (Critical Event)	Action	Reaction	Counteraction	Forces/Assets	Remarks
1	MSO.	Conduct maritime security patrols in AO NOR EAST IOT deter Redland maritime force aggression.	Redland attempts to lay mines in shipping lanes near Pinkland.	National sensors monitor Redland naval base mine storage depot. TF 227 shadows Redland minelaying units in AO NORTH. TF 225 marks Redland minelaying units in AO NOR EAST.	National sensors, TF 225, TF 227.	Ensure ROE designate minelaying in international waters as a hostile act.
2	Sea control.					
3	Amphibious assault.					

Figure 4-6. Time-constrained Technique #1

1	Maritime Security OPS
Action	Conduct maritime security patrols in AO NOR EAST IOT deter Redland maritime force aggression.
Reaction	Redland attempts to lay mines in shipping lanes near Pinkland.
Counteraction	National sensors monitor Redland naval base mine storage depot. TF 227 shadows Redland minelaying units in AO NORTH. TF 225 marks Redland minelaying units in AO NOR EAST.
Assets	National sensors, TF 225, TF 227.
Time	D-10 to D-day.
Decisions	Neutralize Redland minelaying units.
CCIR	Mining activity by Redland navy units.
Control Measures	AO NOR EAST
Remarks	Ensure ROE designate minelaying in international waters as a hostile act.



3	Amphibious Assault
Action	
Reaction	
Counteraction	
Assets	
Time	
Decisions	
CCIR	
Control Measures	
Remarks	

2	Sea Control
Action	
Reaction	
Counteraction	
Assets	
Time	
Decisions	
CCIR	
Control Measures	
Remarks	

Figure 4-7. Time-constrained Technique #2

Note

COA analysis war games are not about one side winning or losing. They focus on exposing friendly COA flaws and improving friendly COAs.

The planning team lead uses an agenda as a means to provide a source of purpose and direction for the time available. The number of war games scheduled is IAW the commander's guidance. All participants should have a copy of the current synchronization matrix, the COA to be war-gamed, and the task organization. Functional planners should have their estimates available for reference. The red cell should have access to all products, to include design and COG products, friendly COAs, task organizations, synchronization matrices, and decision products.

4.3.6.1 War-game Initiation

Once the war game is called to order, the planning team lead or a facilitator establishes expectations and explains how the game will be conducted. A friendly force role player or COA representative explains the COA to be analyzed by presenting the COA sketch and statement, task organization, and decision support products. This player also highlights important considerations concerning the COA: mission, intent, facts, assumptions, risks, and current authorities. A red cell representative explains the enemy's perceived mission and intent, and enemy COAs relevant to the war game. Recorders are at the ready with electronic products displayed for refinement (synchronization matrix, decision support products, etc.).

Note

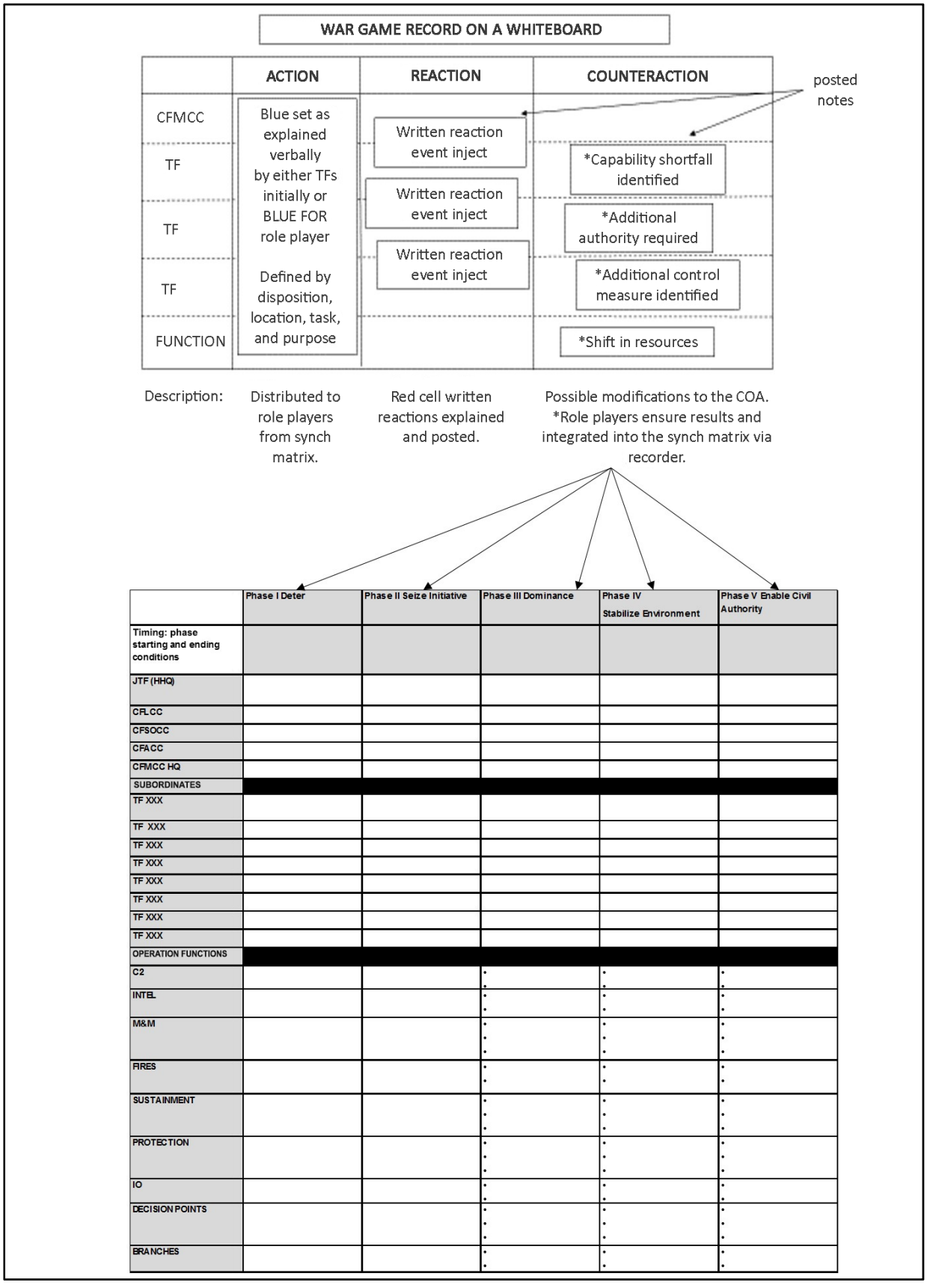
A sequence similar to this should initiate every new war game.

4.3.6.2 Nominal War-game Turn

The facilitator guides each war game turn using the action, reaction, counteraction construct (see Figure 4-8). Each war-game turn normally has these three moves. But, if necessary, portions of the war game turn may be extended beyond the three moves, especially if significant second- and third-order impacts emerge. The planning team lead decides if extending a war-game turn is required.

1. Action. Based on the war-game method used, the planning team lead or facilitator guides presentation of the actions to be analyzed in a manner that mirrors the organization of the synchronization matrix. The COA representative provides an overview of the COA that describes the starting and ending conditions of the turn, critical events, and other information relevant to the turn. All role players should present their actions in the same sequence: the decisive action first, followed by supporting, shaping, and sustaining actions in that sequence. Actions should be described in terms of task and purpose. Role players should explain how their forces are organized, their command relationships, authorities, and orientation within the maritime AO. Sequencing of actions should be explained (such as a TF conducting strike OPS as a shaping action prior to the decisive action). Subordinate command role players (referred to as TF representatives) explain actions to be taken two echelons below their level, but information to be recorded is limited to one echelon below their level. Operational function planning representatives explain details applicable to the war-game turn that support friendly actions from their more holistic functional perspective. Red cell members ask any relevant questions that lend clarity to friendly intentions before the turn is paused so that the red cell can form their reactions.

2. Reaction. During the reaction move, the red cell explains enemy OBJs and actions to pursue and describes responses to friendly actions. These responses can be in pursuit of enemy OBJs or actions to take advantage of friendly vulnerabilities. Red cell members post their responses on the war-game worksheet for display. TF and operational function planning representatives ensure they understand red cell reactions prior to the end of this step. The facilitator pauses the war-game turn so that friendly TF and operational function representatives can prepare counteractions to red cell reactions.



	Phase I Deter	Phase II Seize Initiative	Phase III Dominance	Phase IV Stabilize Environment	Phase V Enable Civil Authority
Timing: phase starting and ending conditions					
JTF (HHQ)					
CFLCC					
CFSOCC					
CFACC					
CFMCC HQ					
SUBORDINATES					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
OPERATION FUNCTIONS					
C2			•	•	•
INTEL			•	•	•
M&M			•	•	•
FIRES			•	•	•
SUSTAINMENT			•	•	•
PROTECTION			•	•	•
IO			•	•	•
DECISION POINTS			•	•	•
BRANCHES			•	•	•

Figure 4-8. Record Counteraction Outcomes in Synchronization Matrix

Note

In considering combat losses, potential losses during reaction or counteraction moves of a turn should focus on improvements to the COA and not accounting for potential losses. This effort could consider adjusting the scheme of maneuver, additional tasks and purposes, task organization adjustments, identifying capability shortfalls, improving risk, control measure adjustment, or mitigating vulnerabilities to prevent or mitigate losses. It should not be subjective speculation of numbers and type of losses on either side, as this can introduce bias and detract from improving the COA. The impact of potential losses is addressed later as branch or sequel plans.

3. Counteraction. Following the same sequence used during the action move, TF role players explain their present disposition and orientation, and describe adjustments to be made as a result of enemy reactions with respect to location, orientation, tasks and purposes, risk, unresolved gaps in coordination, capability shortfalls, and adjustments in priorities. Adjustments are posted to the war-game worksheet and include changes to location, orientation, tasks, and purposes. Role players also identify changes to risk assessment, unresolved gaps in coordination, capability shortfalls, and adjustments in priorities. Each operational function planning representative responds with any adjustments from their estimates to support friendly action. A friendly COA representative then guides the group to readdress and record any refinements in assumptions, decision points, modifications to the COA sketch or statement, task organization, command or support relationships, evaluation criteria, unresolved issues, gaps in coordination, adjustments to operational risk, authorities, or issues not belonging to any category. Identified advantages and disadvantages of the COA are also recorded. The war-game turn concludes with transcribing the war-game worksheet counteractions into the synchronization matrix and other products.

Note

During counteraction, decision points may emerge based on the response to red cell reactions. These decision points should be captured in the synchronization matrix, DST, and DSM along with related identified PIRs/FFIRs, timing, and supporting criteria associated with the decision point. Additionally, capture the conditions that existed when the decision point emerged to populate the synchronization matrix and subsequently the DSM. Do not attempt to fully provide feasible and suitable options (branches, sequels or lesser adjustments) during this time. If obvious options present themselves, note them in the synchronization matrix, DST, and DSM as appropriate.

4.3.6.3 Decision Points

A decision point is an event or a location in the AO where the commander anticipates making a key decision during execution. As decision points emerge during war-game turns, a recorder annotates the synchronization matrix and decision support products. Decision points do not dictate the substance of the decision; only that a decision must be made because the event is expected to affect friendly COAs. A recorder captures associated CCIRs for the decision points, the time the decision is applicable (for example, phases I through II if using a phasing method), any options discussed (branch or sequel discussions for potential future planning efforts), location, and any applicable supporting criteria (conditions under which the decision point can be made).

4.3.6.4 Conduct Risk Assessment and Mitigation

The planning team continually considers risk assessment and mitigation. It identifies and assesses threats, and then it develops controls and makes risk decisions. The team continually assesses the risk to mission and force, seeking to balance the potential cost versus the operational benefit. Risks that can be mitigated, but still retain residual risk, are subject to additional scrutiny in relation to the mission and OBJs. Assumptions carried through COA analysis have the potential to become additional risk. All COAs carry risk. It is a matter of what degree of risk becomes acceptable within the boundaries of the COA and commander's guidance.

4.3.6.5 Other Considerations

The planning team considers all possible forces, including templated enemy forces outside the maritime AO that could react to influence the operation. It continually evaluates the need for branches to the plan based on anticipated decisions focused on enemy actions. It considers all capabilities of friendly and enemy forces, the ranges and capabilities of respective weapons systems, friendly deployment considerations, and operational timelines.

4.3.6.6 Record Results

Several products are carried throughout the planning process and are used to record the results of wargaming. Modifications may be made to the COA sketch, statement, and task organization. Adjustments and refinements are recorded in the synchronization matrix and decision support products. Resource shortfalls and recommended authorities are captured. Emerging or adjusted evaluation criteria are noted. Additional assumptions are recorded. Any strengths and weaknesses of the COA, based on its merits, are noted and forwarded for comparison. Operational function planning representatives record information to take back to their staff section to further develop estimates.

4.3.6.7 Assessing Results

The aim of analysis is to test and improve the COAs. As analysis occurs, COA validity must be retained. To that end, planning teams use the same validity tests as used in COA development. Each COA is continually assessed based on the qualities the commander wants the COA to possess (the governing factors and evaluation criteria). The planning team should continually assess the COA's validity in terms of feasibility, suitability, acceptability, and completeness. Can the force accomplish the mission within the established time, space, and resource limitations? Can the force accomplish the mission within the commander's guidance? Have the potential cost and risk been balanced against the potential advantages gained? Is the COA complete? That is, does it incorporate OBJs, desired effects, tasks and purposes, forces required, concepts for deployment, employment and sustainment, and the end state conditions?

4.3.7 Course of Action Analysis Brief

A COA analysis brief is an optional brief. It may be required as part of commander's guidance or fleet operating procedures. If required, a recorder gathers insights as the result of wargaming, that is, refinements to assumptions, decision points, COA merits (strengths and weaknesses), command or support relationships, evaluation criteria, unresolved issues, gaps in coordination, operational residual risk, or authorities. Figure 4-9 provides an example outline of a COA analysis brief.

4.3.8 Refine Staff Estimates

Functional planners continue to update their estimates by capturing key functional support and execution issues to rectify as a result of war game turns. They identify resource and capability issues, position and disposition issues, or coordination gaps, and then return to their staff sections to continue the process of resolving these issues IOT better support the COA. Issues unique to particular operational functions are expected to be resolved and noted in their estimates and eventual annexes.

4.4 OUTPUTS

The outputs for COA analysis are the result of the planning effort to this point in the process from design (if performed) through wargaming. These outputs contribute to COA comparison and decision, CONOPS development, and directive development. The results of the war game help to understand the strengths and weaknesses of each COA.

Briefer	Subject
Planning team representative	• Purpose • Planning guidance for wargaming • Orientation to the joint and maritime areas of operation.
Intelligence representative	• Intelligence update • Enemy COAs.
Planning team representative	For each COA analyzed: • COA overview: ◦ Sketch and statement ◦ Task organization ◦ War-game method used.
Selected OPT representatives	• War game outcomes for the COA: ◦ Strengths and weaknesses ◦ Sketch and statement modifications ◦ Task organization modifications ◦ Supported and supporting relationship modifications ◦ Authorities required ◦ Additional assumptions required to advance planning ◦ Risk consideration modifications ◦ Coordination and synchronization adjustments ◦ Decision support adjustments ◦ Evaluation criteria (governing factor) adjustments.
Planning Team Representative	Recommendations for COA comparison

Figure 4-9. Example of a Course of Action Analysis Brief

4.4.1 Refined Courses of Action

COA analysis refines the COA sketches, statements, task organizations, and other COA development outputs. Each COA should remain suitable, feasible, distinguishable, acceptable, and complete. If any qualities of the COA become questionable, the commander can direct modification to the COA, discard the COA, combine aspects of the COA with other COAs, or direct the planning team to redevelop the COA and war-game it again. The COA should retain the qualities of the evaluation criteria that were developed in COA development. Strengths and weaknesses are identified and accompany each COA.

4.4.2 Refined Synchronization Matrix

Vital information from the war game, including changes to tasks and purposes, functional perspectives, risk, resource shifts, shifts of the main or SEs, support relationships, and coordination gaps are captured in the synchronization matrix.

4.4.3 Refined Decision Products (Decision Support Template and Matrix)

Each COA's DST and DSM (that include decision points and the CCIR related to them) is further developed during this step of the planning process. Decision support modifications that are captured during wargaming are reviewed for content and fidelity, resulting in increased specificity. At this point in the planning process, decision points should relate to specific DPs, OBJs, CCIRs, possible timing to the decision (which phases or time periods would the decision most likely occur), most likely execution or adjustment options (these can be proposed execution branch or sequel plans or other adjustment options), the likely conditions that will or must exist to facilitate the decision, and, if geographically focused, a location in the form of an NAI.

4.4.4 Refined Resource Shortfalls

Capability shortfalls that may exist are listed. These shortfalls, if not filled, may result in increased risk to accomplishing tasks for intended purposes. If a capability shortfall is related to an essential task, this could put the COA at risk of being inadequate and infeasible.

4.4.5 Authorities

A list of requested authorities may exist that, if not approved, could result in increased risk to the COA. These authorities were carried through from COA development and adjusted during war-game turns ISO task execution and OBJ achievement. For example, an authority may have been identified in a phase during wargaming that allows the maritime component the ability to engage enemy forces in neutral waters once hostilities have commenced. Failure to grant such an authority can put friendly forces at increased risk.

4.4.6 Staff Estimates

Information gleaned from the war game informs staff functional perspectives, which are incorporated into developing formal staff estimates. These staff estimates are eventually converted into annexes and appendixes that support the chosen COA. Intelligence event matrices and PIRs are refined. The scheme of fires is refined and target development and prioritization, as well as capabilities analysis, are performed. Information-related capabilities are updated and added to the IO functional synchronization matrix. The concept of logistics support (COLS) is refined and validated. Protection priorities are addressed and refined. OPS assessment indicators are refined and coordinated with intelligence and fires representatives. These staff efforts ensure the linkage of functional support to each COA.

4.4.7 Evaluation Criteria

As a result of the war game, evaluation criteria recommendations are made. This may include additions, deletions, or adjustments to the governing factors. These may be presented to the commander in either an optional brief or informally prior to commencing COA comparison and decision. Planners should exercise caution when recommending evaluation criteria at this point so as to not introduce bias during COA comparison.

4.4.8 Facts and Assumptions

Throughout the planning process, the facts and assumptions may change based on the current and ever-evolving OE. If facts change that significantly influence the COA, that calculus should be incorporated into any subsequent information or decision brief to the commander. The staff endeavors to consistently discern assumptions and learn when events confirm or deny that the assumption is a fact (or not). New assumptions made as the result of wargaming are necessary only to continue planning. The aim is to have as few assumptions as feasible remaining before execution of an operation, and to retain all assumptions as they pertain to the development of an operational plan.

4.4.9 Branches and Sequels

As a result of the war game, identified branches and/or sequels may require development. Branches are contingency options built into the original COA used for changing the mission, orientation, or direction of movement of a force to aid success of the operation based on anticipated events, opportunities, or disruptions caused by enemy actions and reactions. Sequels are the subsequent operation or phase that are based on the possible outcomes of the current operation or phase.

4.4.10 Other Products

Operational risk matrices are updated to reflect risk mitigation and residual risk.

4.5 KEY POINTS

1. The purpose of COA analysis is to develop a sound basis for determining valid COAs that accomplish the assigned OBJs, contribute to a common understanding of the environment and problem, identify potential branches and sequels, and make modifications to the COA as required.

2. Wargaming is a primary means to conduct COA analysis. It is a disciplined process, with rules and steps that attempt to visualize the flow of the OPS, resulting in refined COAs and draft planning tools for continued development after COA comparison and decision is completed.

3. The valid COA should remain suitable, feasible, acceptable, distinguishable, and complete. If any of the qualities of the COA become questionable, the commander can direct modification to the COA, discard the COA, combine aspects of the COA with other COAs, or direct the planning team to redevelop the COA and war-game it again.

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CHAPTER 5
Course of Action Comparison and Decision

When all is said and done the greatest quality required in commanders is 'decision'. . .

Viscount Montgomery of Alamein, Memoirs

5.1 INTRODUCTION

During COA comparison and decision, the commander evaluates each friendly COA against established evaluation criteria, compares them with each other, and selects the COA that best accomplishes the mission. The selected COA is then expanded and translated into a clear, concise CONOPS that describes in broad terms how and when the force and its subordinate tactical organizations will accomplish assigned tasks (see Figure 5-1).

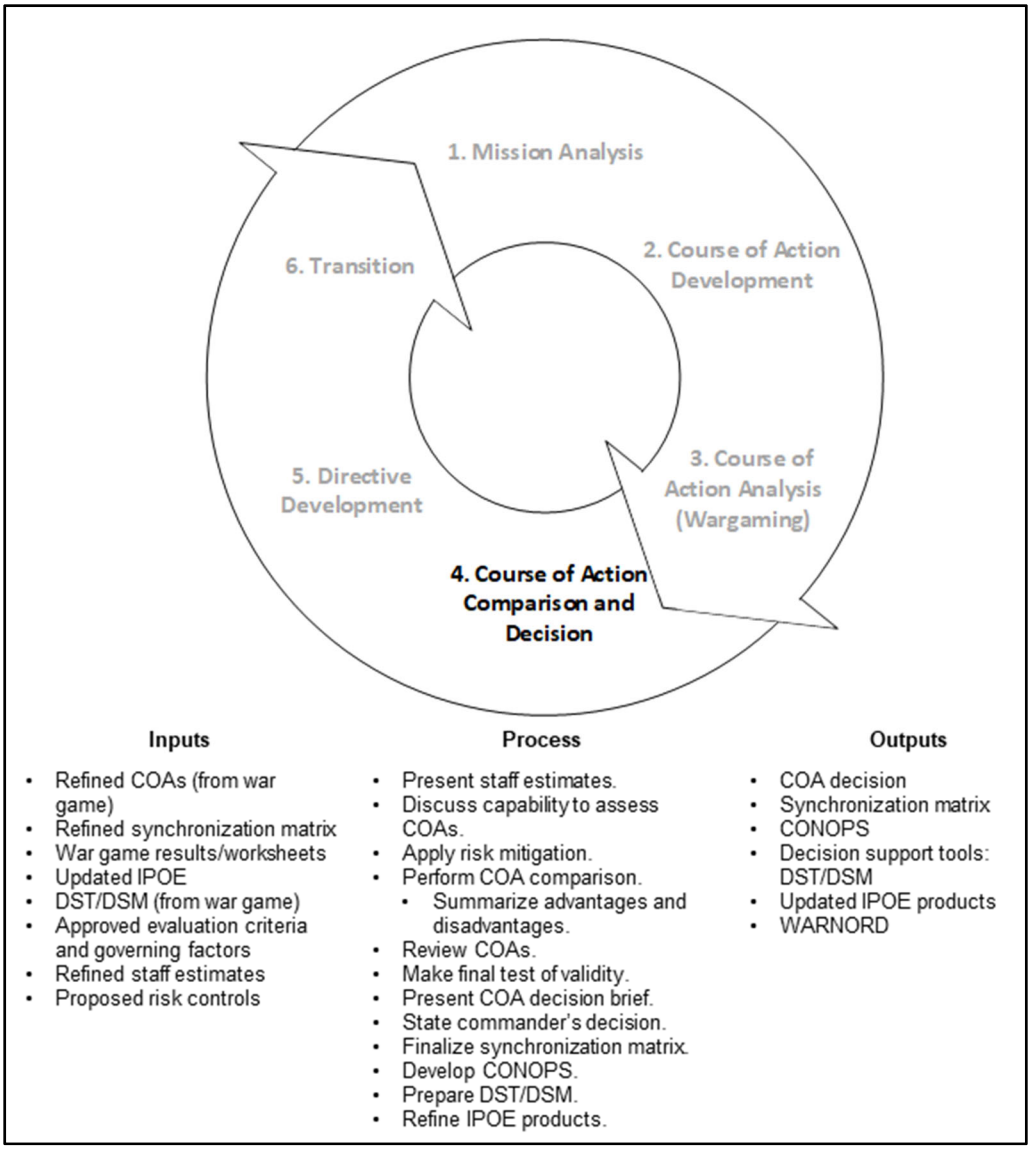


Figure 5-1. Course of Action Comparison and Decision

5.2 INPUTS

To help ensure that the COA comparison is thorough, the primary required inputs during this step include:

1. Refined COA sketches and statements.

2. War game results, worksheets, and refined synchronization matrix.

3. Updated IPOE products.

4. DST and matrix (from COA analysis).

5. Approved evaluation criteria to include governing factors and an understanding of their relative importance to the commander.

6. Refined staff estimates.

7. Proposed risk controls.

5.3 PROCESS

The COA comparison and decision process highlights the differences, advantages, disadvantages, and risks of each COA, and ends with the commander selecting the COA that best accomplishes the mission. The planning team and staff should have a candid discussion on the merits of each COA, assess each against the evaluation criteria, and compare each to provide the commander with the information needed to select the COA with the best tradeoff of advantages, disadvantages, and risks to best accomplish the mission. Executing this evaluation and comparison step requires that everyone involved in this process understands what the evaluation criteria are and how they are defined.

5.3.1 Present Staff Estimates

It is important to have staff input prior to the COA comparison step, to ensure that a detailed comparison can be made. Based on the analysis of the war game, each staff section should identify the strengths and weaknesses of each COA from their functional area perspective. Refined staff estimates should facilitate the commander's decision making by assessing the supportability of each COA.

5.3.2 Discuss Capability to Assess Courses of Action

At this point in the process, assessment plans for the various COAs are relatively mature. As part of the COA comparison and decision step, the assessment plans for each COA should also be compared to ascertain the effort required or ease of executing each respective assessment plan. Details concerning operation assessment can be found in Appendix H.

COA assessment plans can have varying levels of complexity in execution. Some COAs may lend themselves easily to the development of measures and assignment of resources for gathering data and evidence of progress. Others may present challenges. The ease of developing an assessment plan for a certain COA or the predicted ease of executing the assessment plan, however, should not impact the pursuit of developing or deciding to use that COA.

5.3.3 Apply Risk Mitigation

Step 3 of the risk mitigation process, develop controls and make risk decisions, is discussed in Appendix E. A final review of risk assessment and mitigation is required at this point to enable the commander to make sound risk decisions based on a comprehensive understanding of the threats to mission and forces.

5.3.4 Perform Course of Action Comparison

In the procedure for comparing COAs, the goal must be to assist the commander in reaching a sound decision. This step is subjective by its nature. The best method for accomplishing this is to highlight the advantages and disadvantages of each COA in a nonquantitative manner that is based on the commander's evaluation criteria and governing factors. A COA comparison provides the commander with enough detail to fully appreciate the differences between the COAs and make a sound decision. The most common techniques use a COA comparison matrix to facilitate the comparison discussion.

Note

Staffs often focus too much on the comparison methodology, wasting precious planning time on esoteric discussions about the process. COA comparisons are inherently subjective. Regardless of the comparison matrix or technique used, the most important aspect of the process is the discussion itself, and the resulting greater understanding of the relative differences between the COAs' advantages, disadvantages, and risks. The comparison technique used is only a tool to enable the comparison; it is not in itself the purpose of this step.

A COA comparison matrix is simply a staff planning decision aid and should be viewed as such. The goal of each comparison matrix must be to assist the commander, staff, and planners in highlighting strengths, weaknesses, advantages, and disadvantages of potential solutions that will lead to a sound decision. Each comparison matrix helps to compare COAs. The value of such matrices is that they allow the commander and staff to systematically review each COA against the evaluation criteria that the commander approved. The greatest utility of this comparison approach is the discussion and insights the staff and planners gain on the attributes of each COA relative to specified evaluation criteria.

5.3.4.1 Descriptive Comparison of Advantages and Disadvantages

An effective method of comparison is to develop a matrix summarizing the attributes of each COA in a manner that clearly describes the advantages and disadvantages of the considered COAs. This is perhaps the most important comparison step, and the planning team should capture the results in a matrix similar to the one depicted in Figure 5-2. Advantages and disadvantages are based on the commander's evaluation criteria and governing factors, even though they are not specifically listed. As the advantages and disadvantages are listed, the staff can begin to see where tradeoffs or modifications should be made to a COA. These should be included on the matrix, as well.

COA	Advantages	Disadvantages	Modifications
#1	Moderate support to JTF ME. Strong C2.	Sustainment difficult. Weak show of force.	Improve positioning to increase sustainment responsiveness. Adjust organization to enable better show of force.
#2	Strong support to JTF ME.		
#3	Very effective show of force.	Weak support to JTF ME.	Change force flow and positioning to enable more robust support to JTF ME.

Figure 5-2. Advantages Versus Disadvantages Matrix Example

5.3.4.2 Comparison of Evaluation Criteria

An alternative method of comparison not only reflects the advantages and disadvantages of each COA but explicitly does so in relation to specific evaluation criteria. This enables the staff to present a more nuanced, and perhaps more detailed, advantages and disadvantages comparison to the commander. Figure 5-3 illustrates this method.

	Evaluation Criteria 1: Effectiveness of Show of Force [A. Vital]	**Evaluation Criteria 2: Support to JTF ME [B. Important]**	**Evaluation Criteria 3: Minimize Risk to Sustainability [C. Necessary]**	**Modifications**
COA #1	Advantages: Disadvantages: Weak show of force.	Advantages: Moderate support to JTF ME. Strong C2. Disadvantages:	Advantages: Disadvantages: Sustainment difficult.	Improve positioning to increase sustainment responsiveness. Adjust organization to enable better show of force.
COA #2	Advantages: Disadvantages:	Advantages: Strong support to JTF ME. Disadvantages:	Advantages: Disadvantages:	
COA #3	Advantages: Very effective show of force. Disadvantages:	Advantages: Disadvantages: Weak support to JTF ME.	Advantages: Disadvantages:	Change force flow and positioning to enable more robust support to JTF main effort.

Figure 5-3. Matrix of Advantages Versus Disadvantages by Evaluation Criteria

5.3.5 Review Course of Action

Upon completion of the advantages and disadvantages summary, the COAs should be reviewed to determine whether any of the disadvantages of a COA can be overcome (without losing COA distinguishability). For instance, analysis of a COA could reveal that naval forces are placed at significant risk by an adversary's mine threat. However, the COA could potentially work better if strikes were used to destroy the mines or their mine-laying platforms or if a more robust ISR plan were implemented to determine whether mines had been laid. If any changes are made to a COA and time permits, the planning team should war-game the COA again to ensure that no new problems have been introduced. To maintain an unbiased approach in COA selection, any action proposed to overcome a disadvantage in one COA should be applied to all of the COAs, where appropriate.

5.3.6 Make Final Test for Validity

If the recommended COAs are modified from their original form as a result of the analysis in the previous step, the planning team applies a final validity test. During this final check, the planners once again verify the COAs for suitability, feasibility, acceptability, distinguishability, and completeness. If it is determined that any of the COAs fall short of the required criteria, they should be reworked to meet the requirements and analyzed again.

5.3.7 Present the Course of Action Decision Briefing

After completing its analysis and comparisons, the planning team identifies a preferred COA and makes a recommendation in a briefing to the commander. Subordinate commanders may be present but are not required, although their participation—either in person or through collaborative systems—enhances the process. The COA decision brief format includes:

1. The intent of HHQ.

2. The approved mission statement.

3. The status of friendly forces.

4. An updated IPOE.

5. Enemy COAs used in wargaming.

6. Friendly COAs, including assumptions used in planning, results of staff estimates, advantages and disadvantages (including risk) of each COA (with decision matrix table showing COA comparison), and feasibility and acceptability estimates.

7. Recommended COA.

The results of the COA comparison are briefed to the commander for final approval on a COA. Figure 5-4 provides the contents and sequence for the COA brief.

Briefer	Subject
N5	HHQ's intent Restated mission
N3	Status of own forces
N2	Updated intelligence estimate Weather analysis Enemy situation
N2	Enemy COAs used in wargaming
N3	Own COAs
N3, N2	Assumptions used in planning
N1, N4, N6	Results of staff estimate
N5	Advantages and disadvantages (including risk) of each COA (with decision matrix or table showing COA comparison) Recommended COA (may differ from other staff)
COS	Recommended COA

Figure 5-4. Sample Decision Briefing

5.3.8 State the Commander's Decision

The commander weighs the relative merits of the various COAs and selects the one that best accomplishes the mission. The commander has many choices after the decision briefing and may choose to:

1. Select a COA without modification.

2. Select a COA with modification.

3. Select a new COA by combining elements of multiple COAs.

4. Select none of the COAs and have the planning team start over with mission analysis and COA development as required.

5. Other than the first choice, any modified or new COA selected should be analyzed fully (to the maximum extent that planning time allows), to include wargaming. The commander may need to rely heavily on the planning team's professional judgment and experience. However, the ultimate decision is the commander's alone. The commander disseminates the decision through a clear and concise statement that translates the selected COA into a broad statement of what the force as a whole is to do. The commander may amplify the statement with other elements of the mission as appropriate. Each of these elements should be explained in writing in relation to the physical environment in which the expected action is to take place. The wording of the decision is not bound by rigid form.

Note

Observe two general rules in wording the commander's decision: Express it in terms of what is to be accomplished, and use simple language so that the meaning is unmistakable.

5.3.9 Finalize the Synchronization Matrix

The synchronization matrix initiated during mission analysis and refined during COA development and analysis is finalized to incorporate the commander's decision and used as the basis for preparing the CONOPS and operation directives. This internal staff planning tool is used in much the same manner as the war-gaming matrix. The synchronization matrix is an exceptional tool for synchronizing all units and functions. It provides the foundation for CONOPS development and OPORD construction. See Appendix G for more information and a recommended format.

5.3.10 Develop the Concept of Operations

The CONOPS includes refined information from planning products to be incorporated into the directive (OPORD, etc.), as well as annexes and appendixes. Additional documents from planning, such as DST/DSM, graphics, risk matrix, etc., are transitioned for use in execution. These additional documents may not be fully included in the directive but are considered part of the CONOPS.

Using the synchronization matrix, the approved COA sketch and statement, and any other current planning products, the planning team expands the approved COA into a fully integrated and developed CONOPS. This is a written summary of the approved COA and forms the backbone of the directive developed by the command. It directs how and when subordinate units are to work together and employ forces and capabilities to accomplish the mission. The CONOPS clearly and concisely expresses what the commander intends to accomplish and how it will be done using available resources. It describes how the actions of subordinate and supporting forces will be integrated, synchronized, and phased to accomplish the mission and may include potential branches and sequels.

The CONOPS describes the conditions required for the operation or phase(s) to begin and provides the sequence of actions to achieve the end state. The commander should also use the CONOPS to establish priorities of effort and amplify key points that may affect the force as a whole, such as critical aspects of the operational functions, risk tolerance and mitigation, ROE, strategic communication, and IO.

The staff writes or graphically portrays the CONOPS in sufficient detail so subordinate and supporting commanders understand their mission, tasks, and other requirements and can develop their supporting plans. During CONOPS development, the commander determines the best arrangement of simultaneous and sequential actions and activities to accomplish the assigned mission consistent with the approved COA and with the resources and authorities available.

The CONOPS should:

1. Provide for the application, sequencing, synchronization, and integration of forces and capabilities in time, space, and purpose (including those of joint, multinational, and/or interagency organizations as appropriate). This includes a description of the beginning and ending conditions for each phase, stage, step, or part as required. Phase beginning criteria are the same as the phase ending criteria of the previous phase.

2. Describe OBJs, desired effects to be created, tasks to be performed, major forces to be employed, and the central approach the commander intends to take to accomplish the mission, normally expressed in terms of MEs, SEs, SHs, and SUSs.

3. Relate the command's OBJs, desired effects, and actions to those of the next higher command and other organizations as necessary. It enables synchronization of the actions of subordinate and supporting forces to external activities conducted by higher and adjacent commands.

4. Describe when, where, and under what conditions the supported commander intends to conduct OPS and give or refuse battle, if required.

5. Describe all supporting concepts (concept of intelligence, concept of sustainment, etc.) and schemes (scheme of maneuver, scheme of fires, etc.) necessary for successful execution.

6. Describe the task organization and command relationships of subordinate forces.

The CONOPS includes:

1. A narrative description of the operation and its OBJs, typically derived from the COA narrative. The narrative forms the basis of the CONOPS paragraph included in the directive (described in Chapter 6).

2. All necessary graphics (OPAREAs, control measures, etc.) to enable execution.

3. The most current versions of the synchronization matrix, DSM, DST, and other planning products needed for execution.

4. The task organization and command relationships of subordinate and supporting forces.

5. All necessary supporting concepts and schemes to include: scheme of fires, concept of sustainment, and concept of protection.

6. Risk assessment matrix, including all risk controls developed.

5.3.11 Prepare Decision Support Tools

The DST/DSM are simple tools that link decision points, CCIRs, and decision options in a single template and matrix. The options vary by phase or circumstance. When a CCIR-related event occurs, this tool enables watchstanders to present decision makers with a sensible recommendation on short notice. Appendix G provides an example of a DST/DSM.

5.3.12 Refine the Intelligence Preparation of the Operational Environment

As the staff moves through the COA comparison process, it should continually use and refine the IPOE as necessary.

5.4 OUTPUTS

The output of COA comparison and decision provides the basis of directive development. The standard products of this step consist of the synchronization matrix, CONOPS, and decision support tools (DST and/or DSM). Additional outputs may include updated IPOE products and a WARNORD.

5.5 KEY POINTS

1. Keep the chain of command informed. Lower echelons of the planning team should keep their principal staff informed and, in turn, that staff should keep the major subordinate commanders informed of the planning progress.

2. Members conducting COA comparison should be the same as those who were involved in COA analysis; otherwise, much time is wasted making new planners fully aware of the COAs.

3. Do not discard the COAs that were not selected. There may be elements of them that can be used as branch plans during subsequent planning, such as a military deception plan to support the selected COA.

4. The CONOPS is the most critical product developed during planning. Sufficient time and attention should be afforded to the development of what will be the core of the OPORD.

CHAPTER 6
Directive Development

> It cannot be too often repeated that in modern war, and especially in modern naval war, the chief factor in achieving triumph is what has been done in the way of thorough preparation and training before the beginning of the war.
>
> *President Theodore Roosevelt*
> *Graduation address at the United States Naval Academy (1902)*

6.1 INTRODUCTION

The purpose of the directive development step of the NPP is to translate the commander's decisions from previous steps into oral, written, or graphic communication sufficient to guide implementation and promote initiative by subordinates. The directive, once completed, becomes the principal means by which the commander expresses the decision, intent, and guidance as shown in Figure 6-1.

During directive development, the planning team translates the CONOPS into a clear, concise, and authoritative directive. This directive is then back-briefed to the higher commander and crosswalked to other Service or functional components to ensure that it is synchronized, understood, and meets the higher commander's intent.

Directives not only direct and guide the actions of subordinate units, but also provide information to facilitate coordination among organizations outside the command. Directives synchronize supporting and subordinate units' activities in time, space, and purpose to achieve OBJs and accomplish missions; they also account for those military and civilian organizations involved in the operation. The detail provided in a directive depends on several factors, including the cohesion and experience of subordinate units and complexity of the operation. Directives should be written to direct forces one echelon down and be understood two echelons down.

Note

For example, a CTF order should task the subordinate commander, task groups (CTGs) and be understood by the subordinate commander, task units (CTUs). Effective directives encourage subordinates' initiative by providing the what and why of tasks to subordinate units, and leave the how to perform the tasks to those subordinates.

6.1.1 Types of Directives

The directive development step produces either a plan or order, depending on the nature of the planning.

1. A plan is prepared in a deliberate manner in anticipation of OPS and normally serves as the basis for an order. The procedure for producing a plan closely mirrors the preparation of an order. There are several types of plans that serve a variety of planning requirements. For more information about the different types of plans, see Appendix J.

2. An order is a written or oral communication that directs actions and focuses a subordinate's tasks and activities toward accomplishing the mission. Orders are promulgated at all levels of command. For more information about the different types of orders, see Appendix J.

Other operational directives provide amplification of plans and orders. These are generally promulgated in message format. These include OPGENs, OPTASKs, and daily intentions messages (DIMs) or similar instruction depending of the procedures of a given fleet. See NWP 3-56 for additional information.

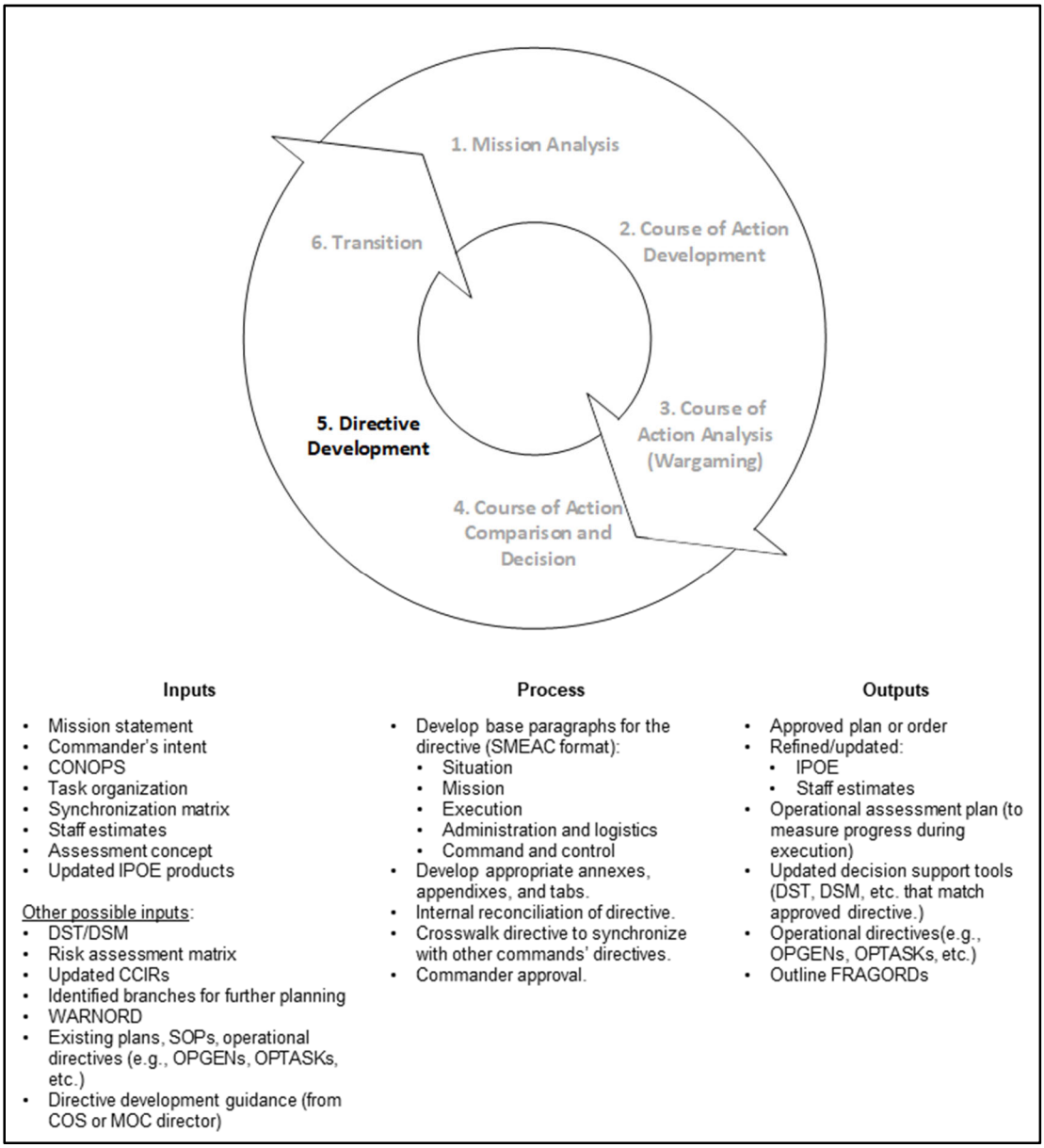


Figure 6-1. Directive Development

Note

When operating in a joint environment and referencing OPGENs, OPTASKS, and DIMs in a plan or order, staffs should be aware that these directives are unique to the Navy. Consider dissemination of these directives as an appendix or annex to improve visibility to the JFC and other components. For a more comprehensive discussion applicable at the operational and tactical levels of war, see NWP 3-56. There are also relevant discussions in JP 3-32, NWP 3-32, Navy Tactics, Techniques, and Procedures (NTTP) 3-32.1, Maritime Operations Center, and NIMS.

The commander, COS, MOC director, or plans director, as appropriate, may direct directive development. Directives contain only critical or new information, not routine matters normally found in SOPs. Thus, those SOPs outlined in existing OPGENs and OPTASKs should be referenced rather than repeated in the plan or order. A good directive is judged on its usefulness, not its length.

In the previous planning steps of the NPP, the planning team integrated the commander's selected COA with the staff estimates and planning support tools (developed in parallel) into a CONOPS that becomes the centerpiece of the directive. The need for a refined CONOPS cannot be overstated. In crisis situations, where time is of the essence and the OPS are of limited complexity, it is not uncommon for maritime commanders at the fleet and CTF levels to simply promulgate the CONOPS and direct execution of the CONOPS through an execute order (EXORD) or DIM. However, in situations where time is available or the OPS are complex, orders are the best way to direct OPS.

6.1.2 Writing Guidelines for Directives

When writing directives, words matter. Writers should remain consistent in their use of approved (doctrinal) terminology, particularly tactical tasks. For example, "conduct sea control operations" can refer to many very different tactical actions, from sinking civilian merchant vessels (suppressing enemy commerce) to protecting civilian merchant vessels (protecting vital sea lanes). Planners are responsible for ensuring the proper task is given to subordinate commanders with sufficient specificity to execute and enough latitude for tactical commanders to employ their forces as they see fit. Inappropriate terminology can lead to unintended consequences, including mission creep, gaps, or redundancies. In total, this results in a lack of control of the force. Directives writers should use the UJTL and UNTL as an effective way to articulate tasks, since the terminology is recognized and understood and tactical units train to many of them as they hone their unit's skills. When a task cannot be identified in an appropriate task list, directives writers should use plain language to clearly express the required task.

A well-written directive possesses the following important characteristics that help assure understanding of the directive and the accomplishment of the mission:

1. Clarity. Write in simple, understandable English and use proper military (doctrinal) terminology. Avoid jargon, and eliminate every opportunity for misunderstanding the commander's exact, intended meaning. The UJTL and UNTL are useful tools for providing a common, doctrine-based lexicon for assigning and describing tasks at all levels of command. Ensure contents of the directive are in the appropriate paragraphs. For example, all tasks must be included in the tasks and coordinating instructions paragraphs or appropriate annexes. Tasking forces in the CONOPS paragraph without an appropriate task in the tasks paragraph leads to confusion, ambiguity, and delays in execution.

2. Brevity. Be brief, clear, and concise with short words, sentences, and paragraphs. Avoid superfluous words and unnecessary details, but do not sacrifice clarity and completeness in the interest of brevity alone. Use acronyms unless clarity is hindered. Do not include material covered in SOPs and OPGENs/OPTASKs. Refer to those SOPs and OPGENs/OPTASKs instead. State all major tasks of subordinates precisely and include the task's purpose. The task and purpose should be stated in a manner that allows each subordinate sufficient latitude to exercise initiative. Short sentences are more easily and quickly understood than longer ones.

3. Authoritativeness. Unmistakably state what the commander wants the unit, and its subordinate units, to do and why. Authoritative expression through the commander's intent is reflected in orders, using direct language. IOT assure authoritative expression, writers use third person, present tense throughout the directive, with the possible exception of the situation paragraph and commander's intent paragraph.

4. Positiveness. State tasks and instructions in the affirmative, using the imperative mood. For example, "LOG ships remain in designated secure OPAREAs" instead of "LOG ships will not accompany strike groups in forward OPAREAs." The affirmative form of expression should be used throughout all directives because positiveness of expression suggests the commander's firmness of purpose.

5. Simplicity. Use language that is simple and direct to reduce misunderstanding and confusion. This requires that all elements be reduced to their simplest forms. Directives writers are responsible for explicitly describing the expectations for subordinate and supporting forces. The reader should never have to infer tasks, OBJs, or guidance from a directive.

6. Meaningful. Ensure tasks are clearly written to sufficient detail with appropriate purposes. Avoid hollow expressions such as “as soon as possible (ASAP)”. Ensure tasks are clearly written to sufficient detail with appropriate purposes. Vague tasks that are open to the interpretation of a subordinate force are a result of insufficient planning and lazy thinking.

7. Unambiguous. Articulate the flexibility of execution embodied in the directive’s CONOPS without creating ambiguity about the commander’s chosen CONOPS. The directive should identify decision points, and proposed options at those decision points, to build flexibility.

8. Factual. Only include necessary and relevant assumptions. Review assumptions made during planning to confirm or disprove them and ensure that only absolutely necessary and appropriate assumptions remain when starting development of the directive. If an assumption does not directly affect the CONOPS, consideration should be given to dropping it from the directive. Assumptions carried into execution constitute risks to the mission or force.

9. Timeliness. Send plans and orders to subordinates in time to allow them to collaborate, plan, and prepare their own actions. WARNORDs, planning orders (PLANORDs), and close communication between staffs provide subordinates with the most current information so that they can begin their preparation before receipt of the final plan or order. Concurrent planning saves time.

10. Completeness. The directive must contain all the information necessary to coordinate and execute the forthcoming action. It should also provide control measures that are complete, understandable, and that maximize the subordinate commander’s initiative. Only those details or methods of execution necessary to ensure that actions of the subordinate units concerned are synchronized with the CONOPS for the force as a whole should be prescribed.

6.2 INPUTS

The mission statement, commander’s intent, CONOPS, initial task organization, synchronization matrix, staff estimates, and assessment concept are the required inputs for orders development. Other inputs may include:

1. Updated intelligence and IPOE products. Intelligence products and IPOE should be updated continually throughout the planning process. Before beginning directive development, the planning team needs to ensure that all products are current and that the approved CONOPS is still relevant to the current situation. The most current intelligence product and IPOE will be used in the directive itself. If the situation has changed dramatically enough to bring the relevancy of the CONOPS into question, planners must consider if it is advisable to continue with the approved plan or go back to the commander for approval to replan portions of the CONOPS.

2. Planning support tools. Review all planner support tools such as the synchronization matrix, DSM, staff estimates, risk matrix, and the COA sketch and narrative.

3. Updated CCIRs. Review all of the CCIRs and ensure they have a DSM entry associated with them.

4. Identified branches for further planning. Any potential branch plans previously identified should be reviewed. A DSM for each potential branch should be constructed if not already produced.

5. WARNORD. Any previously developed WARNORDs can be used to help flesh out the directive. Additionally, changes from the WARNORD need to be identified to subordinate and supporting commands during crosswalk and transition.

6. Existing plans, standing operating procedures, orders, and other operational directives (e.g., OPGENs, OPTASKs, etc.). Review all relevant guidance from HHQ to ensure that their concerns are addressed.

7. Staff estimates. During planning, staff estimates provided critical information to support each step of the NPP. These staff estimates will provide key inputs into the directive. For example, the LOG estimate can serve as the foundation for paragraph 4 (administration and LOG) of the OPORD. Figure I-2 of Appendix I clearly demonstrates the relationship between staff estimates and OPORD development. Well-developed and refined staff estimates greatly assist in OPORD development. See Appendix I for more information.

8. COS or MOC director orders development guidance. Each command may have its own standards for directive development. Local guidance should be followed to ensure continuity within a given command.

6.3 THE DIRECTIVE PROCESS

Directives most frequently use the standard five-paragraph format, commonly referred to as SMEAC format that is briefly described below and elaborated upon in Appendix J. In complex OPS, much of the information required in the order is contained or amplified in the appropriate annexes and appendixes. However, the essential form of the commander's CONOPS, including the commander's intent, C2, task organization, and essential tasks and OBJs will be contained in the body of the order. The five basic paragraphs for all plans and orders are:

1. Situation. This paragraph provides the commander's summary of the general situation and ensures that subordinates understand the background for planned OPS. It often contains subparagraphs describing enemy and friendly forces relevant to the commander, OE, and task organization, as well as HHQ's guidance. The situation paragraph should be limited to those concerns that are relevant to the command issuing the directive and not a simple regurgitation of the situation paragraph produced by a senior command. Additionally, the situation paragraph is not just an intelligence summary, although much of the information is derived from intelligence products. It should clearly explain the relevant factors of the current situation that are likely to affect decision making by commands tasked with executing the directive.

2. Mission. The commander's mission is inserted exactly as previously approved. This is derived from the mission analysis step and contains those tasks deemed essential to accomplish the mission.

3. Execution. This paragraph is the centerpiece of the plan or order and describes how the commander intends to accomplish the mission. The execution paragraph typically contains four sub-paragraphs: commander's intent, CONOPS, tasks, and coordinating instructions. The CONOPS paragraph contains a general explanation of how the operation will unfold, including a description of any identified phases, and a description of all supporting concepts for the operation. The tasks paragraph is where the commander directly tasks all forces responsible for execution of the operation that the commander has the authority to task. Coordinating Instructions contain all information necessary for subordinates to coordinate the execution of their tasks.

4. Administration and LOG. This paragraph describes the execution of LOG, personnel, PA, civil affairs (CA), and medical services.

5. C2. This paragraph specifies command relationships, succession of command, and the overall plan for communications and control.

6.3.1 Develop Appropriate Annexes, Appendixes, and Tabs

To keep the directive as simple and understandable as possible, details and amplifying information are placed in annexes. Individual staff sections prepare appropriate annexes and supporting appendixes and tabs. These supporting documents are developed in coordination with higher, adjacent, and subordinate organizations and use the CONOPS and staff estimates as the basis for supporting documents. Typical annexes include detailed plans for intelligence and OPS (such as targeting, movement, undersea warfare, etc.), instructions necessary for C2 (such as the TF organizations), and other information requiring detailed explanation and not covered in the base directive. Annexes may be referenced in the appropriate part of the base order and should not include matters covered in SOPs.

Whereas joint doctrine identifies specific annexes typically used in joint OPLANs and CONPLANs, for a maritime supporting plan (MARSUPPLAN), usually only those annexes that are required to provide clarity, details, and amplifying information for the MARSUPPLAN are developed. See CJCSM 3130 (series) for more detail.

Lessons Learned

- While it is not necessary to develop all annexes, it is important to identify quickly those to be developed and the responsible points of contact within each staff section. Additionally, a specific staff member/action officer should be designated to collect and review each specified annex for clarity, completeness, and consistency.

- When using annexes, avoid placing major tasks to subordinates in an annex that are not reflected in the base order/plan. This burying of tasks could result in a miscommunication between higher- and lower-echelon commands.

Note

Directive development includes a two-step quality control process to ensure alignment and completeness—reconciliation and crosswalk.

6.3.2 Reconciling a Directive

Reconciliation is an internal process in which the planning team conducts a detailed review of the entire directive. This reconciliation ensures that the base directive and all the annexes, appendixes, etc., are accurate, coherent, complete, and in agreement. Reconciliation compares commander's intent, the mission, and the CCIRs against the CONOPS and the supporting functional concepts (intelligence, LOG, etc.). It identifies discrepancies or gaps in the planning. If discrepancies or gaps are found, the planning team takes corrective action. Specifically, the planning team compares assigned tasks of the base order with the primary annexes to ensure linkage. The synchronization matrix and DSM can be expanded to depict accurately the linkage and alignment of all parts to the directive. Additionally, other products developed to support the base directive, such as the time-phased force and deployment data (TPFDD), should be examined to ensure they support the overall CONOPS. The planning team should also check the coordinating instructions to ensure completion and appropriateness as well as ensure the collection plan supports the CCIRs.

The reconciliation process can be done a number of ways: formal, in-person meetings, electronic staffing, secure VTCs, and other collaborative tools. It is critical that all staff sections, cross functional teams, and special staff officers have an opportunity to weigh in on the consistency and correctness of the directive.

6.3.3 Crosswalk of a Directive

The crosswalk step is a review of the directive to ensure it is consistent with senior, adjacent, and subordinate commands. It identifies discrepancies or gaps in planning and, if any are found, enables the planning team to take corrective action. During crosswalk, the planning team compares the plan or order with the orders of higher-echelon commanders and components or subordinates to maximize synchronization and unity of effort and to ensure that the higher echelon commander's intent is met. The crosswalk process can be done through telephone, VTC, scheduled meetings and briefings, or a combination of methods. This process is the culmination of a feedback process that has been occurring continuously during the planning process. The planning team should be in regular contact with the planning teams of the senior commander, the other components, and subordinates. Operation phasing, timing, and critical event decision points, as well as operational concepts, are compared among the components and subordinate staffs. Orders crosswalk is the process of conducting the same detailed review that was done in reconciliation but it is executed with higher, adjacent, and subordinate staff representatives. It helps ensure process and organizational alignment among directives (e.g., OPGENs, OPTASKs, and the plan or order) to determine if and where modifications are required.

Lesson Learned

New technologies in collaborative planning systems are extremely useful for both collaborative and parallel planning for United States (U.S.) military participants. However, not all organizations involved in the planning process may have access to these systems (e.g., other United States Government (USG) agencies and multinational components).

6.3.4 Commander Approves and Issues Directive

The final action in directive development is the approval of the plan or order by the commander and issuing the directive. Although the commander does not have to sign every annex or appendix, it is critical that the commander review and sign the base plan or order. Once approved, the order must be transmitted to any command responsible for its execution. Plans are distributed based on the initial planning guidance provided by the senior command's planning directive.

6.4 OUTPUTS

The main output of directive development is an approved plan or order. Other outputs may include refined intelligence and IPOE products, planning support tools, other operational directives (e.g., OPGENs, OPTASKs, etc.) and outline fragmentary orders (FRAGORDs) for branches to the operation. In addition, staff estimates and operational assessment should be updated for accuracy. Staff estimates evolve from planning aids to staff tools that support execution of the operation. The assessment guidance (measures, etc.) is finalized as an integrated means to evaluate the command's achievement of assigned tasks/OBJs.

6.4.1 Operation Assessment

Operation assessment, which evolves during the progress of planning, is solidified during this step. During the reconciliation, crosswalk, and back brief, the staff ensures that operation assessment is nested with the CONOPS and that feedback mechanisms are in place to support the command's ability to measure its progress. While the operation assessment is dynamic and will adjust during the operation as more information becomes available or the situation changes, this is an important framework for the command to establish before the commencement of OPS.

6.4.2 Decision Support Tools

Most decision support tools are incorporated into the final directive. For example, the synchronization matrix forms the basis for the tasks subparagraph in the execution paragraph. However, these products are also used in the transition step and can be valuable to commands responsible for the execution of an order or training and maintenance associated with a plan. Planners must ensure that the latest version of each support tools is current and matches the written directive.

6.5 KEY POINTS

1. Regularly scheduled meetings to share information among participants are critical, especially in a crisis situation.

2. When possible, all commanders should attend a single meeting/VTC for back briefs. Component commander feedback pays big dividends in cross-component understanding.

3. Crosswalks in multinational environments are absolutely essential due to different interpretations of languages and unique doctrines. The crosswalk should include Annex A, a task organization of the various plans/orders with the force structure listed in the TPFDD.

4. Centralized distribution of orders ensures that information is current and has been properly disseminated. Collectively, the OPORD and ensuing FRAGORDs shall be the single source of authority. Briefings and slides may cause problems in dissemination of information because if not written in the plan, an order will not be executed.

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CHAPTER 7
Transition

Lord Nelson did not win at Trafalgar because he had a great plan, although his plan was great. He won because his subordinate commanders thoroughly understood that plan and their place in it well in advance of planned execution. You must be prepared to take action . . . when certain conditions are met; you cannot anticipate minute-by-minute guidance . . .

Vice Admiral Henry C. Mustin III, United States Navy
Commander, Second Fleet
Fighting Instructions, 1986

7.1 INTRODUCTION

Transition promotes a shared understanding of a plan as it shifts into execution or across planning horizons. Transition is conducted internally to increase a staff's understanding of a plan and externally to foster integration and synchronization with HHQ, adjacent components, and subordinate commanders. Figure 7-1 illustrates its role in the NPP. Figure 7-2 depicts the internal and external aspects of the transition step. Transition may not include the handover of an entire plan. Responsibilities for monitoring execution, plan maintenance, branch and sequel planning, and product upkeep should be specifically assigned during the transition step. Commanders, planners, and staffs will use a variety of techniques to convey a plan during the transition step.

7.2 INPUTS

For transition to occur, an approved directive must exist. This, along with additional staff products, forms the input for transition. These products include the synchronization matrix, DSM, a summary of risk, the ROE, IPOE, the assessment plan, and staff estimates.

7.3 PROCESS

The transition of a plan can be as simple as a brief within the MOC or as complex as a full force rehearsal. Figure 7-3 depicts types of transition events. It is possible that more than one transition event will be required to advance understanding of a plan by subordinates and adjacent components. Careful consideration must be given to the location of transition events. The selected location should maximize opportunity for attendance. A location that provides secure VTC may offer increased involvement of subordinates, partners, components, and other liaisons.

7.3.1 Prepare for Transition

1. Organize for transition.

   a. Determine the intended audience and focus (internal/external) for the transition event(s).

   b. Determine the type of transition event(s): brief, ROC, simulation, and partial or full force rehearsal.

   c. Develop a timeline.

   d. Determine the location and time of the transition event(s).

   e. Determine any technological requirements.

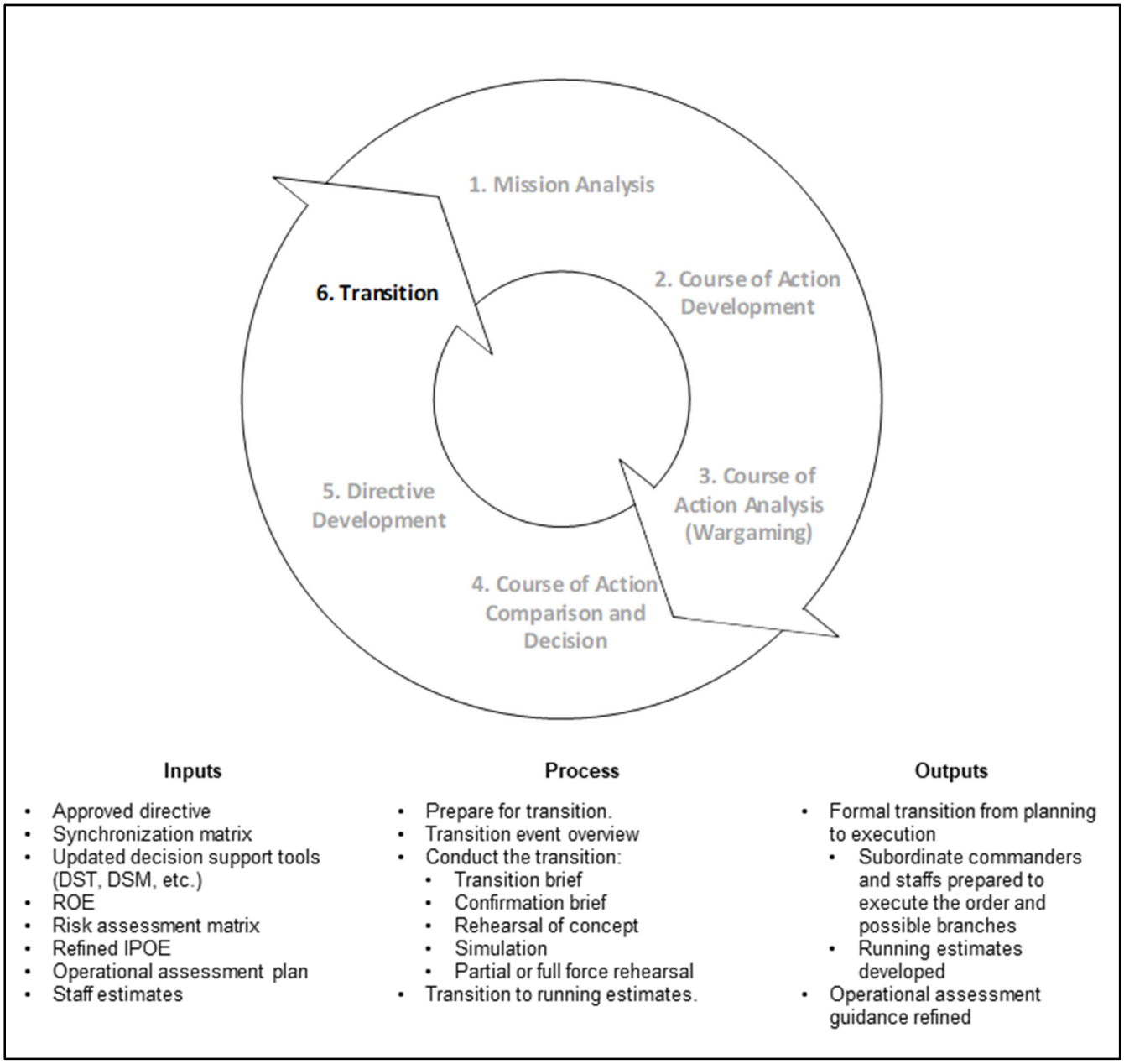


Figure 7-1. Transition

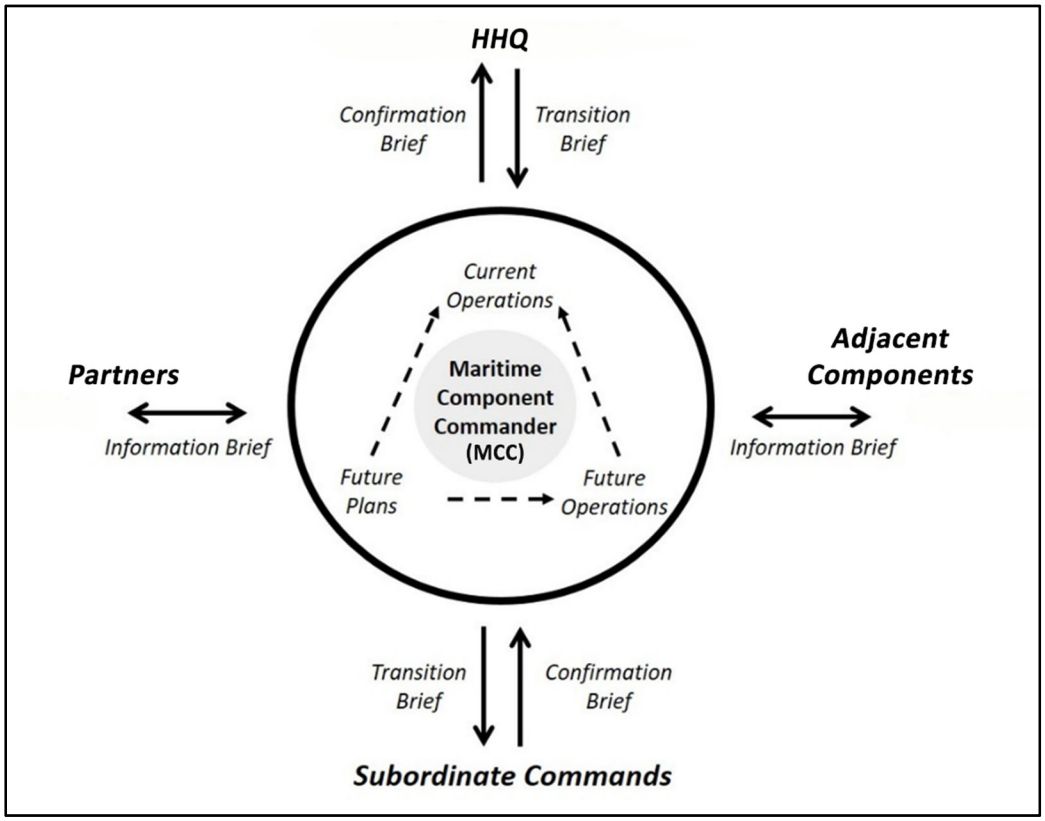


Figure 7-2. Transition Overview

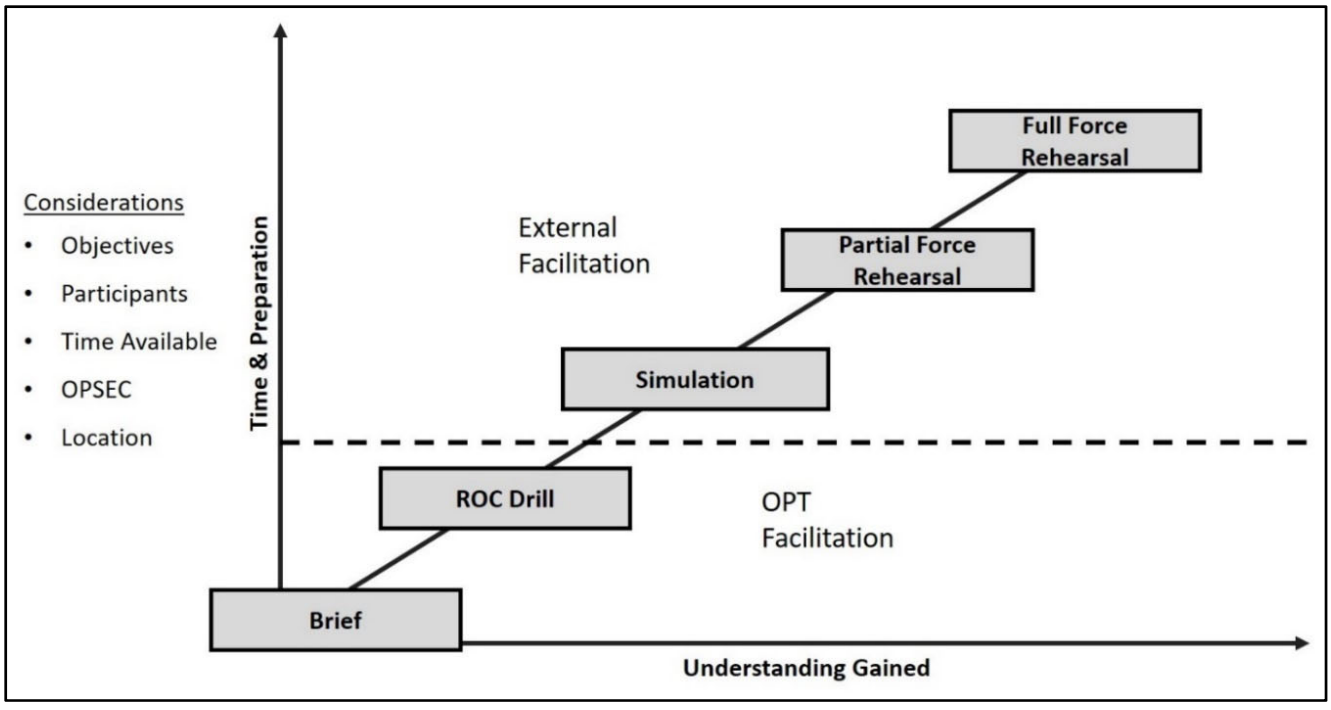


Figure 7-3. Transition Events

f. Determine any security requirements.

g. Prepare the briefing location, training aids, and technology.

2. Establish roles and responsibilities within the OPT.

a. Designate the transition facilitator (typically the OPT lead).

b. Assign briefers/role players (CTF/LNO/red cell).

c. Assign an information manager.

d. Assign a timekeeper (typically the OPT deputy).

e. Assign individual responsibility for product development/refinement (military directive/synchronization matrix/decision support products/brief sections/read-ahead).

f. Determine any personnel exchanges related to the transition.

g. Confirm post-transition points of contact between participating organizations.

7.3.2 Transition Event Overview

1. Brief. A brief is the least resource- and time-intensive method to communicate a plan across echelons of command.

2. ROC. This technique is used by the commander and staff to facilitate understanding of a plan throughout all levels of the command and to gain insight into seams and points of friction within a plan. ROC drills may be conducted internally or externally and may be used in lieu of a confirmation brief. The ROC drill may be as simple as gathering the OPT around a chart for a walk-through of a CONOPS or as complex as a multiple component rehearsal of an entire plan.

3. Simulation. Simulations are used by commanders to gain detailed insight into the decision making, CONOPS, and risk assessment tied to a plan's execution. A simulation is time and resource-intensive, requiring dedicated support personnel and technology. For these reasons, simulations are often conducted outside of the command HQ.

4. Partial force rehearsal. A partial force rehearsal uses a portion the maritime force (commanders, staffs and hardware units) to rehearse a CONOPS. Partial force rehearsals may be combined with a simulation to increase the level of participation and complexity.

5. Full force rehearsal. A full force rehearsal is the most comprehensive but also the most resource-intensive. This technique involves all participants (commanders, staffs, and hardware units) rehearsing a CONOPS. Full force rehearsals may be conducted within the actual OPAREA or in similar areas.

7.3.3 Transition Brief

The transition brief is the most common transition event. The transition brief provides an overview of the mission, commander's intent, task organization, and adversary and friendly situation. The briefing ensures that all actions necessary to implement the order are known and understood by the staff section monitoring and directing the execution of the order, by the subordinates executing the order, and by supporting components. The transition brief may be provided internally to the staff or externally to subordinates and adjacent components. Figure 7-4 provides a general outline for a transition brief. Time available dictates the level of detail in the transition briefing. Orders and supporting materials should be transmitted as early as possible.

- Joint OPAREA/maritime AO orientation
- Current situation (enemy/friendly)
- Review HHQ mission statement and intent
- Review JFMCC mission statement and intent
- Facts and assumptions
- Limitations
- Task organization (by phase if applicable)
- CONOPS by phase
- CCIRs
- Authorities
- Coordinating instructions
- Risk summary
- Decision support tool
- Other transition activities

Figure 7-4. Example Transition or Confirmation Brief Outline

7.3.4 Confirmation Brief

A confirmation brief is a transition event delivered by a subordinate commander to their HHQ. The confirmation brief demonstrates the nesting of a subordinate's concept with that of HHQ. The confirmation brief allows the superior commander to identify discrepancies between the order and the subordinate's plans. The superior commander often uses the confirmation brief as a means to formally approve a subordinate's mission for execution.

7.3.5 Rehearsals

A rehearsal is a series of briefings, guided discussions, and walk-throughs used to advance understanding of a plan. Rehearsals may be used for internal or external transition events to include confirmation briefs. They improve the ability of the commander and staff to supervise OPS, exercise operational decision making, and increase the situational awareness of the subordinate commanders and staffs. The commander should identify and prioritize the events to be rehearsed and designate participants. In time-constrained situations, rehearsals may be abbreviated to focus on the most critical events within an operation.

7.3.5.1 Rehearsal Roles

1. Facilitator. The rehearsal facilitator is responsible for the organization and conduct of the rehearsal. Typically, this individual has participated in planning and is normally the OPT leader. The facilitator should also have a clear understanding of the commander's intent and rehearsal guidance. The facilitator keeps the rehearsal on track by adhering to the agenda and rehearsal timeline.

2. Red cell. A red cell is established if there is a war-gaming component to the rehearsal. The red cell portrays enemy COAs and, possibly, other events (weather, natural disaster) that may impact the mission. Members of the red cell must possess expertise in enemy activities. The red cell should be an independent group of participants and not dual-hatted to represent both friendly and adversary forces and capabilities.

3. Role players/briefers. LNOs or OPT members can be used to role-play the activities of subordinate commands, adjacent components, and other actors during ROC drills. The role players must understand the details of their respective commanders' CONOPS, as well as the operating capabilities and limitations of that organization.

4. Recorder. A recorder should be identified to capture commander's comments, points of coordination, and any other items that require further action. Effective techniques for the recorder include posting on a whiteboard, butcher block paper, or projecting notes to a screen.

7.3.5.2 Rehearsal Considerations

1. Prepare a script. A script is used as a tool to control the scope and tempo of the rehearsal. A well-prepared script will dictate the flow and timing of the rehearsal, appropriately scope the content, and ensure that the equities of all participants are considered. Figure 7-5 provides an outline for development of a script.

Introduction/welcome.
- Introduce participants.
- Administrative remarks: Facility overview (as necessary), Read-ahead/information package.
- Rehearsal overview: Describe the OBJs, agenda, and rules associated with the rehearsal.

Conduct the rehearsal.
- Review adversary/friendly/neutral (if applicable) situations.
- Area orientation: area of Interest, JOA, maritime AO, DPs, NAIs, and control measures.
- Red overview: force lay-down and enemy COAs,
- HHQ overview: mission statement, commander's intent, and task organization.
- Maritime component commander (MCC) overview: mission statement, commander's intent, task organization, and CONOPS overview.
- MCC ROC (by critical event, phase, stage, step): MCC phase overview, MCC task organization (changes), initial force laydown, MCC tasking by TF (start with ME), component coordination requirements, functional overview, risk, decision point review, phase end transition criteria, questions (by phase), and reaction/counteraction events (if the rehearsal includes a war-gaming component).

Conclude the rehearsal.
- Questions and discussion.
- Review decisions and follow-on actions.
- Closing remarks.

Figure 7-5. Sample Rehearsal Agenda

2. Issue rehearsal instructions. A directive should be developed by the staff to provide specific information about the rehearsal. Rehearsal instructions should include time, location, required attendees, security requirements, and points of contact.

3. Assemble resources and support personnel. Conducting a rehearsal can be time- and resource-intensive. Preparing for a rehearsal will require some pre-event production. This may include the development of training aids, map preparation, technology setup and testing, database construction, and generation of virtual forces. Facilitators must consider the time and personnel required to prepare for the rehearsal.

4. Develop reference material. This will provide essential information to the audience and participants. The contents of reference material is situationally dependent but could include task organization, description of the OPAREA and associated control measures, decision points, and ROE.

7.4 OUTPUTS

The outputs of the transition step include:

1. Formal transition of a plan or order from planning to execution. For example, from FOPS to COPS.

2. Formal transition of a plan or order across planning horizons. For example, from FUPLANS to FOPS.

3. Increased understanding of a plan or order across the MOC.

4. TF commanders understand their responsibilities within the MCC's plan.

5. A clear understanding of support relationships between the MCC and adjacent component commanders.

6. A clear understanding of support relationships between subordinate TF commanders.

7. Confirmation that the maritime CONOPS corresponds with that of HHQ.

8. Assignment, within the MOC, of any additional planning requirements and maintenance of key planning products (DSM, synchronization matrix, risk matrix, etc.).

7.5 KEY POINTS

1. The transition step of the NPP promotes a shared understanding of a plan as it shifts into execution or across planning horizons.

2. For transition to occur, an approved directive must exist.

3. Transition may not always include the handover of the entire plan. Responsibilities for monitoring execution, plan maintenance, branch and sequel planning, and product upkeep should be specifically assigned during the transition step.

4. A confirmation brief is delivered by a subordinate commander to their superior commander. A common practice is to use an ROC drill in lieu of a confirmation brief.

5. Commanders, planners, and staffs will use a variety of briefs and rehearsals to convey the plan during the transition step. The level of understanding increases with time available to conduct the transition. As the complexity of the transition increases, additional preparation time and resources are required.

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APPENDIX A

Planning in a Time-constrained Environment

A.1 DECISION MAKING IN A TIME-CONSTRAINED ENVIRONMENT

This appendix recommends techniques and procedures for compressing planning timelines to develop flexible, tactically sound, fully integrated, and synchronized plans that balance ends, ways, means, and risk. The NPP provides a sound procedural process that can be adapted to varying situations and differing timelines. Before a planning staff can support decision making in a time-constrained environment, it must master Navy planning. Organization and preparation are the keys to success in a time-constrained environment. The techniques and procedures of this appendix are not exhaustive, but they have proven useful to planning staffs.

A.2 GENERAL CONSIDERATIONS

The NPP is flexible and adaptable enough to support both contingency and time-constrained planning. The process facilitates development of creative solutions in the timeframe allotted. The requirement of the planner remains the creation of a plan that balances ends, ways, means, and risk. The general considerations that follow assume an understanding of Navy planning as outlined in the main body of this publication. Figure A-1 offers some general insight regarding time to plan.

1. Involve the commander throughout time-constrained planning. Increased participation of the commander provides the planning team with direction and guidance that reduces time spent on unproductive efforts.

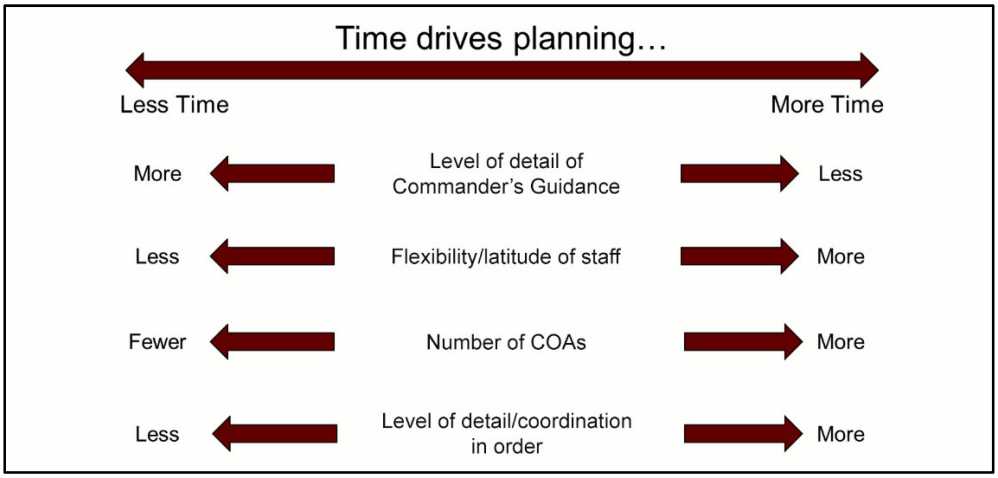


Figure A-1. Time and Planning Considerations

2. Insist on experienced staff members, LNOs, and additional planners. Experience matters in time-constrained planning. The ability to use the NPP in a time-constrained environment requires planning experience and practice.

3. Create a plan to plan and start with the end in mind. Take the time to establish a planning timeline and organize the planning team. The plan to plan should include planning team assignments, as well as a detailed timeline to include all deadlines for briefs, orders, and transition events. Figure A-2 provides a notional allocation of planning time. Note that this is an example and not prescriptive.

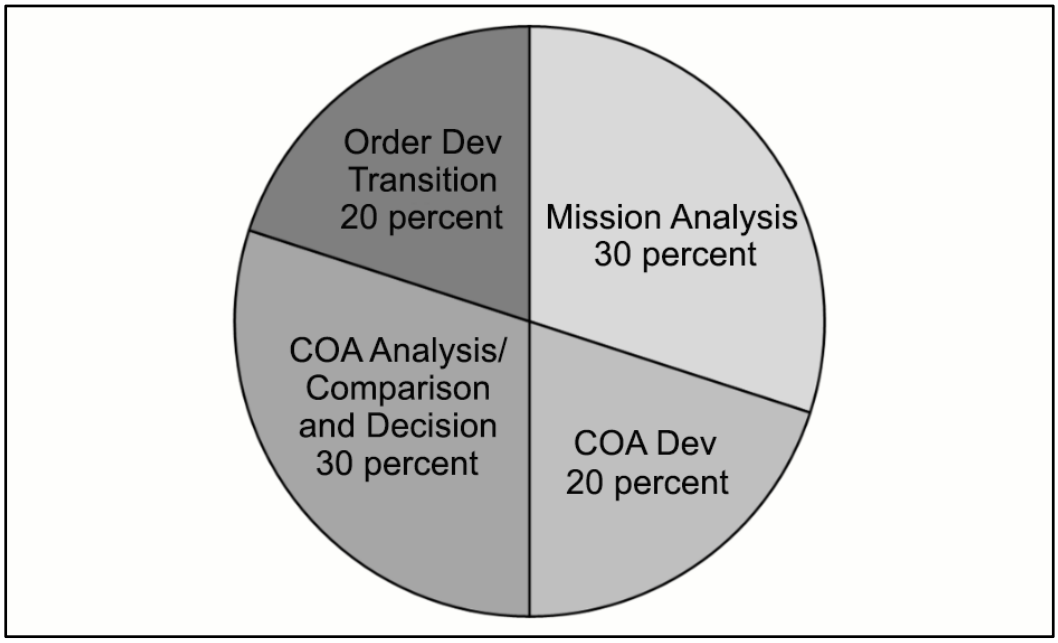


Figure A-2. Notional Allocation of Planning Time

4. Apply the NPP with judgment. Skipping steps invites risk to the planning process. Some planning steps and substeps can be compressed or conducted concurrently to save time. For example, directive development can begin at the outset of planning and completed in stride throughout the planning process.

5. Leverage parallel and collaborative planning with subordinates, adjacent components, HHQ, and other organizations (nongovernmental organization (NGO), inter-agency, etc.). While this applies to all planning, information sharing is especially critical when conducting time-constrained planning. Ensure open lines of communication and use verbal WARNORDs to initiate subordinate planning.

6. Use existing analysis, planning products, and guidance. This includes existing staff estimates, intelligence analysis, conceptual and detailed planning products, and commander's guidance.

7. Deliver the required directive on time. A good plan delivered today is likely more useful to a commander than a perfect plan delivered next week. Planning always suffers from information gaps, obstacles to communication, and resource constraints. Time-constrained planning magnifies these frustrations. The commander and planner must be willing to accept less detail and more ambiguity during time-constrained planning.

A.3 THE COMMANDER'S ROLE

In a time-constrained environment, commanders rely on their experience, intuition, creativity, and situational awareness to decide how to adjust the NPP, giving specific guidance to focus the process and save time. The commander may elect to have the staff spend most of its time developing, refining, and war-gaming a single COA rather than developing multiple COAs. When possible, the commander consults with subordinate commanders before making a decision. Subordinate commanders are closer to the fight and can more accurately portray the adversary's situation and that of their own units. Additionally, consulting with subordinates will give them insight into the upcoming operation and allows them to conduct parallel planning.

A.3.1 Design Methodology in a Time-constrained Environment

Design methodology by the commander, staff principals, and planners can quickly frame an initial approach, establish operational OBJs, and initiate commander's guidance to drive planning when time is constrained. It

offers early engagement with the commander and can result in identification of a directed COA. Use of design methodology hastens planning. Appendix B describes design methodology in detail. A general outline for conducting design methodology in a time-constrained environment follows.

1. Understand the operational direction.

   a. Determine ends, ways, means, and risk articulated by higher authorities.

   b. Determine maritime OBJs and the desired end state.

2. Understand the OE.

   a. Analyze the spatial features within the maritime environment.

   b. Analyze friendly and enemy actors influencing the maritime environment. This is a useful analytical model when time is constrained is to address key actors, their functions, relationships, and resulting tensions.

   c. Describe the current state of the maritime environment in terms of friendly and enemy conditions.

   d. Describe the friendly and enemy desired end state and identify the resulting points of friction.

3. Define the problem.

   a. Identify the key tensions, from the analysis in step two, that must change to achieve the friendly end state.

   b. Craft a problem statement that describes the key tensions, disputed areas, and vulnerabilities within the maritime environment that impede achievement of the friendly desired end state. Highlight any opportunities that may be exploited to achieve the friendly desired end state.

4. Develop an operational approach.

   a. Develop lines of effort or OPS that broadly frame the ways to achieve the desired end state.

   b. Include maritime OBJs and DPs in the operational approach.

   c. Craft initial commander's intent and guidance to commence detailed planning.

A.3.2 Commander's Guidance

A key way to save time is in the issuance of the commander's initial planning guidance. Detailed guidance keeps the staff focused by establishing parameters within which to work. This type of detailed guidance limits the staff's flexibility and initiative to save time, but it allows the staff more time to synchronize the COA during the war-game session. The elements of the commander's initial planning guidance may be the same as described in the full planning process, but the guidance is more detailed and directive. The commander can provide detailed information outlining what is expected in each COA developed, including tentative task organization and scheme of maneuver. The commander may also determine which adversary COAs should be war-gamed against friendly COAs as well as the branches or sequels to be incorporated in each COA. Once the guidance is issued, the WARNORD is immediately sent to subordinates.

Once the commander's initial planning guidance is received, the COS or MOC director, as appropriate, establishes an organization and timeline for accomplishment of this planning effort. Whether the use of an established working group (WG), OPT, crisis action team, or other crisis planning body is to be used, leadership must resource their planning requirements, streamline their periodic discourse with the commander, and enforce their product delivery timeline to enable successful crisis planning.

A.4 THE STAFF'S ROLE

The staff should use every opportunity to conduct parallel planning with the unit's HHQ. Parallel planning can save significant time, but if not carefully managed, it can also waste time. The majority of time spent conducting parallel planning should be spent developing the foundation of the plan such as mission analysis.

A.4.1 The Importance of Staff Estimates Increases as Time Decreases

Decision making in a time-constrained environment often takes place after a unit has entered into the AO and has begun to execute OPS. This means that the IPOE and some portion of the staff estimates should already exist. Detailed planning before OPS provides the basis for information the commander and staff will need to make knowledgeable decisions as OPS continue. Staff members must keep all required estimates up-to-date so that when planning time is limited, they can provide accurate, up-to-date assessments quickly and move directly into COA development. When time is short, the commander and staff use as much of the previously analyzed information and products from earlier decisions as possible. Although some of these products may change significantly, such as the IPOE that is continuously updated, many remain the same or require little change.

A.4.2 Intelligence Preparation of the Operational Environment

The IPOE process requires constant attention. Many delays during mission analysis can be traced to the IPOE. The Navy intelligence staff officer (N2) must quickly update the IPOE based on the new mission and changed situation. This is critical to allow reconnaissance assets to deploy early and collect information needed to adjust the initial plan. Adversary event templates must be as complete as possible prior to the mission analysis briefing. Because they are the basis for wargaming, they must be constantly updated as new information becomes available.

A.5 THE NAVY PLANNING PROCESS IN A TIME-CONSTRAINED ENVIRONMENT

Commanders and planners must be willing to accept less detail in terms of information and product refinement when planning in a time-constrained environment. While the NPP steps remain the same, they can be compressed or completed concurrently IOT save time.

A.5.1 Mission Analysis

There are several techniques that the planning team can use to save time. The first, mentioned earlier, is that the commander can provide specific planning guidance to drive the analytical work of the planning team.

Another successful technique to generate momentum is to divide the OPT into breakout groups to address mission analysis substeps separately. When using breakout groups it is important to schedule time to brief the analytical work back to the planning team. Outputs from the breakout groups are synthesized into coherent mission analysis output for the commander. Figure A-3 offers one way to organize breakout groups during missional analysis. The use of breakout groups is situationally dependent.

Using existing analysis from conceptual planning, staff estimates and previous planning will save the planning team considerable time. The ability for a planning team to quickly process and assimilate information is critical in a time-constrained environment. The identification of an information manager and rules for dissemination of information must be established up front to avoid delays in planning.

The information obtained during mission analysis must be communicated to the commander, staff, and subordinates in an efficient manner. When time is constrained, the requirement to continue planning may force a less formalized briefing format. This may take place as a desktop brief to the commander or a walk-through of mission analysis products within the planning team spaces. External communication of the analysis, usually delivered via a written product, may be condensed to a verbal WARNORD via phone or video teleconference.

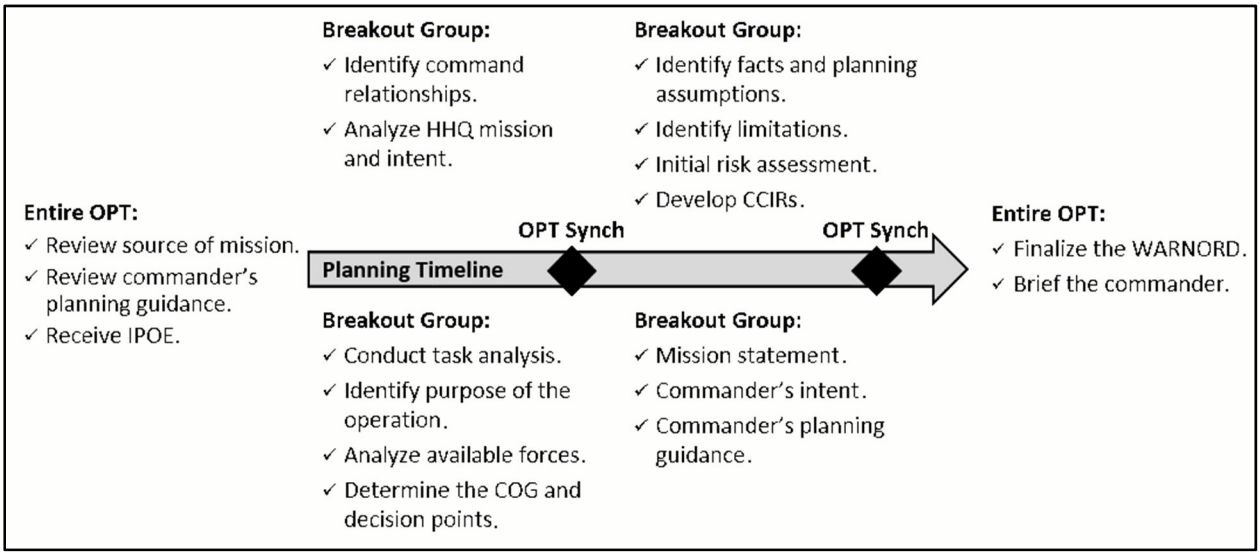


Figure A-3. Notional Breakout Groups for Mission Analysis (Not Prescriptive)

A.5.2 Course of Action Development

During COA development, time is saved by increasing the commander's involvement and the issuing of detailed commander's guidance. Commander's guidance should limit the number of COAs to be developed and may specify development of a single COA. If multiple COAs are directed, the planning team can subdivide and develop COAs concurrently.

The use of preplanned responses and SOP of subordinate forces will also speed COA development. The Navy CWC doctrine includes preplanned responses and command relationships that enable mission command. The use of CWC and related warfare area operation tasking orders facilitate planning and decentralized execution. For more information on the CWC doctrine, review NWP 3-56.

When time is severely limited, the process can be expedited by establishing a small planning team with direct commander involvement developing a single COA against the adversary's most likely COA. The commander determines which staff officers are critical to assist in this phase, depending on the type of operation being planned. The commander may also integrate subordinate commanders.

A.5.3 Course of Action Analysis (Wargaming)

The requirement to test a COA against possible enemy responses does not change in a time-constrained environment. The depth of analysis will depend on time and resources available. The commander and planning team can save time by conducting a less formalized, compressed war-gaming process discussed in Chapter 4. Compressing COA analysis allows the commander to determine preferred COAs and reject unsuitable COAs. This may include early selection of a single COA for further analysis. However, in some situations, time may only be available to conduct the compressed war game.

The commander's involvement can save significant time by allowing the staff to focus on only the most essential aspects of the war game. The commander can supervise the war game and be prepared to make decisions, provide guidance, delete unsatisfactory concepts, and assist in keeping the staff focused.

The method for conducting COA analysis in a time-constrained environment is situationally dependent. However, time can be saved by focusing on specific critical events, a sequence of essential tasks, or a specific geographic area as described in Chapter 4. Expand the analysis to supporting, shaping, and sustaining actions as time and resources permit.

The commander and staff war-game the COAs to ensure that all elements are fully integrated and synchronized. An early decision to limit the number of COAs war-gamed or to develop only one COA saves the greatest amount of time. It is best to war-game against all feasible enemy COAs. However, the commander can save additional time by having the staff war-game against a smaller number of or single enemy COA that represents a broad spectrum of possible enemy reactions.

The use of recorders is particularly important. These recorders must capture coordinating instructions, subunit instructions, and information required to synchronize the operation during the war-gaming process. If this occurs, a portion of the order is written before the planning process is complete. The location used for the war game must be prepared and configured by the time the staff is ready to conduct the war game. Charts and boards must be cleaned and prepared for use, and supporting intelligence products should be in position.

When only one COA is developed, the purpose of the COA analysis is to verify, refine, synchronize, and integrate the commander's COA and to recommend modifications as necessary. However, the analysis should follow the formal war-gaming process as much as time allows to help the commander visualize the outcome and identify potential branches and sequels.

A.5.4 Course of Action Comparison and Decision

If the commander decides to war-game only one COA or chooses one during the war game, the COA must still be evaluated against the established evaluation criteria. If multiple COAs have been war-gamed and the commander has not made a decision, the staff should conduct the COA comparison detailed in Chapter 5. Limiting the evaluation criteria/governing factors is the only significant shortcut in this step.

If the commander has observed and participated in the planning process, the decision may be rapidly apparent, and the commander can make an on-the-spot decision. If the commander has not participated in the process to this point or has not made a decision, a decision briefing will still be required. Good charts that clearly display the strengths and weaknesses of each COA and COA sketches assist the commander in visualizing and distinguishing between each COA. The staff should ensure that the COAs are complete with tentative task organizations, COA statements, and tasks for each subordinate unit. Limiting the COA briefing to only the most critical points can also save time.

If only one COA was developed, no decision is required unless the developed COA becomes unsuitable, infeasible, or unacceptable. If this occurs, another COA should be developed. Once the decision is made, and with HHQ concurrence as applicable, the WARNORD is immediately sent to subordinates.

A.5.5 Directive Development

In a time-constrained environment, a best practice is to develop the order throughout the planning process. Many elements of the five-paragraph order are available early in the NPP, including the situation and mission statement. The execution, administrative and LOG, and C2 paragraphs can be developed between COA development and COA comparison and decision, and refined and finalized during directive development. However, the staff must complete a final reconciliation and crosswalk to ensure clear instructions to subordinate units.

A.5.6 Transition

Lead planners must provide adequate time to communicate the maritime plan. The purpose of the transition step is to ensure the situational awareness of the commanders and staffs responsible for the execution of a plan. In a time-constrained environment, the transition step may offer an optimal opportunity to confirm coordination and synchronization across echelons and may offer an opportunity to rehearse the plan.

APPENDIX B
Design Methodology

B.1 GENERAL

Operational design is a methodology to aid commanders and planners in organizing and understanding the OE. Although conceptually linked to operational design, the term design methodology refers to a process tailored for maritime or Navy component OPS. Design methodology is a nonlinear, iterative process used to assist in the conceptualization and articulation of an operational approach, i.e., a framework for solving a problem. Commanders may direct the planning team to use design methodology for any problem, but it is most useful when faced with a complex, ill-structured, or unfamiliar situation. Design methodology directly supports the conceptual aspect of planning by assisting the commander in visualizing the OE and the nature of the problem. Conceptual thinking helps understand the problem and establishes a framework solving it. Although design methodology is a deliberate process initiated to understand a complex, and/or ill-structured problem set, significant changes to the OE or the operational guidance may cause planners to revisit design methodology.

Today's naval commanders plan against a myriad of problems across the range of military OPS. In the context of highly interconnected, multifaceted environments, solutions are often hard to determine. Design methodology provides a process to assist commanders and staffs in visualizing and articulating initial planning guidance for solving these complex problems, which, by definition, are novel and require creative thought to develop nonobvious and original solutions.

Design methodology sets the conditions for the commanders, their staff, and planners to apply critical and creative thinking toward demanding planning situations. Critical thinking is characterized by actively and skillfully conceptualizing, applying, analyzing, synthesizing, and evaluating information to reach a conclusion. It is disciplined thinking that is clear, rational, open-minded, and informed by evidence. Creative thinking involves thinking in new and innovative ways.

B.2 ORGANIZING FOR DESIGN METHODOLOGY

When faced with complex and unfamiliar situations, it is often beneficial for the commander to establish a design team to determine the direction to be provided to the OPT in the commander's initial planning guidance. A design team consists of select subject matter experts from across the staff and may include representatives from the OPT. Experts from outside the OPT bring unique insights to understanding the operational problem. Including OPT members reduces the risk of creating a disconnect between the thinking that produces the initial planning guidance and the detailed planning done by the full OPT. A cross-section of the staff should be represented ISO the design team's efforts. In some situations, obtaining SMEs from external organizations may be desirable.

The commander is the linchpin of the design team. The commander's judgment and decisions are required to effectively and efficiently guide the staff through the process. The more complex and ill-structured the problem, the greater the need for commander involvement. Since design methodology provides the context for the commander's eventual guidance for detailed planning, it is imperative that the commander participates in the discourse to set the conditions for developing the commander's understanding. Commanders draw on design methodology to mitigate complexity and uncertainty, leveraging their knowledge, experience, judgment, and intuition to generate a clearer understanding of the conditions needed to focus effort and achieve success. Design methodology supports the exercise of command, providing a broad perspective that deepens understanding and facilitates visualization.

In addition to possessing critical and creative thinking skills, members of the design team should possess baseline knowledge of design methodology and be familiar with the OE. Design methodology will assist the commander and OPT in gaining insights into the operating environment, problem, and approaches to resolve the problem.

Note

Design methodology and its application as described in this appendix provide one example of how a commander and staff may conduct conceptual thinking. Applying design methodology as a rubric, rather than a rote procedure, provides direction for solving complex problems without prescribing an inflexible process.

B.3 APPLYING DESIGN METHODOLOGY

By definition, all complex, ill-structured, and unfamiliar problems are unique and require individually-tailored solutions. Design methodology provides commanders and planners a repeatable process to develop usable solutions in the face of complex operational problems. Design teams should tailor the methodology to fit the needs of the organization and the uniqueness of the problem being addressed. This appendix describes a technique that can be used as a point of departure for a maritime planning team. Generally, design methodology consists of four components that should be considered when attempting to understand a complex situation (see Figure B-1). The components to consider are:

1. Understand the operational direction.

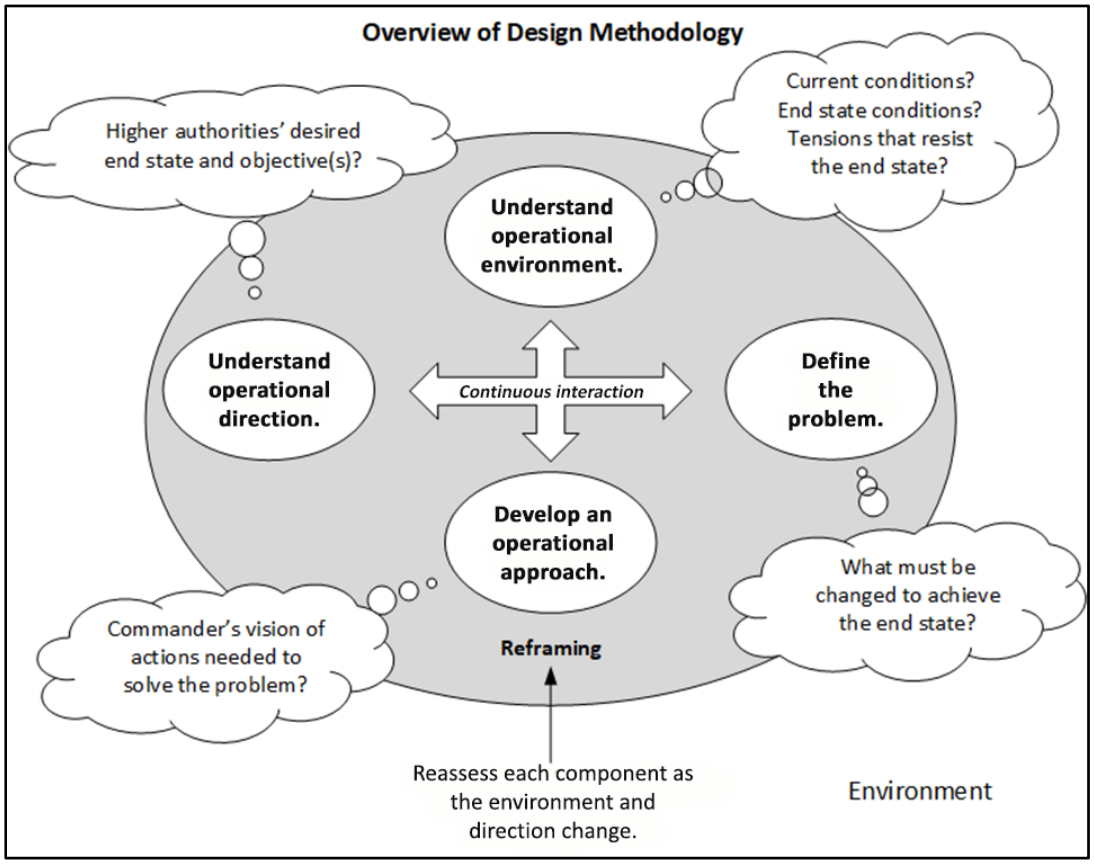


Figure B-1. Overview of Design Methodology

2. Understand the OE.

3. Define the problem.

4. Develop an operational approach.

While working through the components of design methodology, the design team should be aware of the requirement to "reframe." Reframing recognizes and accommodates change to the operational direction or environment, or to the design team's understanding of the operational problem. It provides a mechanism for adjustments to the initial facts, assumptions, and hypotheses through iteration and reexamination of prior work. Reframing captures learning continuously, providing space for the design team to develop and update different aspects of the operational approach as learning occurs. As new information is discovered, the design team must be prepared to revisit previous components of the design methodology process to ensure that the operational approach remains valid throughout planning.

B.3.1 Understand the Operational Direction

The operational direction provided by or determined from the HHQ serves as the foundation for design methodology (see Figure B-2). Strategic guidance that is relevant to the situation is traditionally included in the operational direction from HHQ. If the strategic issues are not properly addressed, the design team may choose to include an analysis of strategic issues while developing an understanding of the operational direction. The design team must exercise caution to avoid second guessing their HHQ or interpreting strategic issues in a manner that puts their efforts out of synch with their HHQ. Any disparities identified by the design team should be resolved with HHQ as soon as feasible.

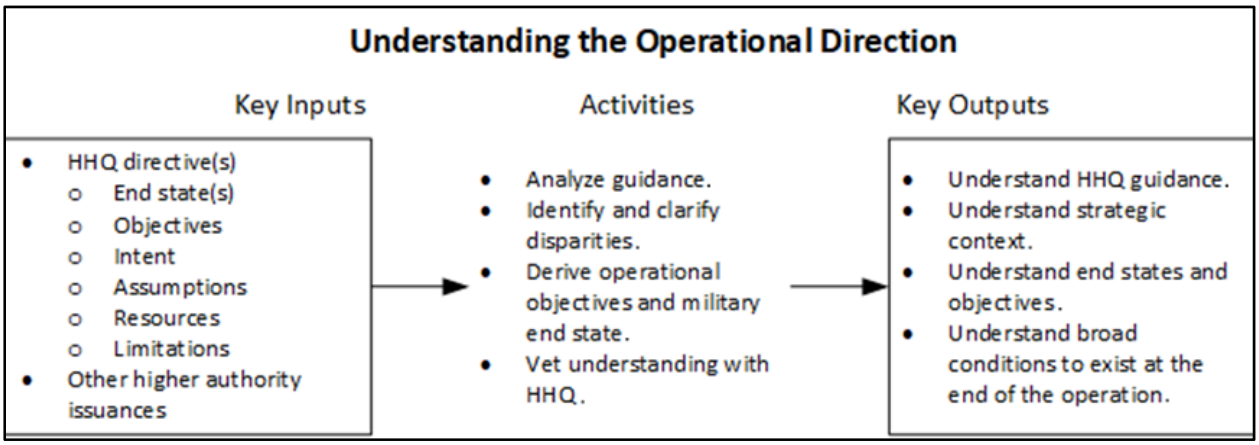


Figure B-2. Understanding the Operational Direction

Note

The strategic environment and strategic-level considerations are contemplated when developing the operational direction. Additionally, the commander and staff define the direction in view of the military end state and OBJ(s). Finally, the commander and staff record their planning assumptions, limitations, etc. from the start of design methodology to inform conceptual thinking and detailed planning.

B.3.1.1 Inputs

Direction from the HHQ initiates planning and could come in a number of formats, from written orders to verbal direction from the HHQ commander. The HHQ guidance defines what constitutes success (ends), available resources (means), and approaches (ways) to accomplish the operational OBJ for the HHQ. The design team examines the HHQ's end state, OBJs, assumptions, and intent and determines how their portion of the operational problem relates to the HHQ's direction. Design methodology facilitates initial development of OBJs for use and validation during the NPP. The desired end state and OBJ(s) are critical to gaining a thorough understanding of the operational direction and should be prominently displayed to remind the design team of the purpose of their efforts as they attempt to answer the question: To what end? As planning progresses and deeper understanding occurs, the commander and staff frequently revisit the HHQ stated end state and OBJs to ensure that the ends are achievable and properly framed.

Additionally, the HHQ guidance often provides resources and limitations that will shape the commander's operational approach. Just as the planning team does during the NPP, the design team seeks to balance ends, ways, means, and risk. If, during design, the ways and means are assessed as less than optimal to accomplish the ends, the risk will expand accordingly. The commander and the design team should identify ways to mitigate those risks and apprise the HHQ of proposed modifications to the ends if the risks are assessed as too great.

B.3.1.2 Activities

The design team considers the HHQ guidance in detail to ensure a proper understanding of the guidance. If available, the design team should also consider the guidance two levels up to enable understanding of the overall direction and how the current operation fits into the larger concerns. Once the design team has analyzed higher authorities' guidance to form a thorough understanding of their stated OBJs and the military end state, it determines the commander's contribution to those aims, identifies the commander's OBJs, and describes the associated end states from the commander's perspective. If at any time there is a disparity in guidance or something that is unclear, the design team must attempt to seek clarity through discussion with HHQ or challenge their own research into the larger operational or strategic issues shaping the environment.

In the absence of well-articulated guidance from higher authority, the design team derives operational OBJs and the military end state from the existing guidance available. If possible, this guidance should be vetted through the HHQ to ensure a common understanding with HHQ of the operational direction.

Figure B-3 illustrates the types of information organized to understand operational direction.

Note

An OBJ is clearly defined, decisive, and attainable. OBJs and their supporting effects provide the basis for identifying tasks to be accomplished. Military OBJs specify what must be accomplished and provide the basis for describing desired effects. OBJs describe what must be achieved to reach or attain the end state. Achieving operational OBJs ties execution of tactical tasks to reaching the military end state. There are four primary considerations for an OBJ:

- An OBJ establishes a single desired result (a goal).
- An OBJ should link directly or indirectly to higher-level objectives or to the end state.
- An OBJ is specific and unambiguous.
- An OBJ does not infer ways and/or means—it is not written as a task.

B.3.1.3 Outputs

The outputs of understanding the operational direction are a thorough understanding by the design team of:

1. HHQ guidance
2. End states and OBJs
3. Broad conditions to exist at the end of the operation.

When time and resources permit, the key outputs should be captured in writing and made available to the design team and subsequent planners.

<table>
<tr><th colspan="2">Operation FA Example</th></tr>
<tr><td colspan="2">The design team examines JFC guidance to discern the desired military end state and OBJs. Guidance from CJCS, Secretary of Defense (SECDEF), and other higher authorities (although not shown) is also studied to ensure that strategic and nonmilitary end state and OBJs are also understood.</td></tr>
<tr><th>Source</th><th>Stated End State and Objectives</th></tr>
<tr><td>JFC mission statement
(2.4.5)</td><td>O/O JTF BLUE SWORD defeats the Redland 23rd Guards Div and destroys terrorist forces and their infrastructure in Redland IOT eliminate the terrorist base of OPS in the area, secure the territorial integrity of Pinkland</td></tr>
<tr><td>JFC commander’s intent (future state and OBJs)
(Figure 2-2)</td><td>PURPOSE: Enable a stable and secure environment in which Redland terrorist and conventional forces no longer threaten the government of Pinkland.
END STATE: The 23rd Guards Division is defeated, and terrorist forces and their camps in Redland are destroyed. Territorial integrity of Pinkland is secured, and Pinkland is prepared to assume responsibility for security OPS.</td></tr>
<tr><th>Source</th><th>Approach to Accomplish the Objective</th></tr>
<tr><td>JFC commander’s intent (ways to accomplish)
(Figure 2-2)</td><td>METHOD: There will be three primary focus areas: 1) neutralizing Redland C2, 2) destroying Redland conventional forces, and 3) eliminating terrorist groups and their infrastructure. We will respect the boundaries of neighboring neutral countries and minimize damage to civilian infrastructure within Redland. Operating within a joint organization construct, U.S. forces will cooperate with multinational partners and synchronize with Pinkland to ensure unity of effort. Maximize employment of all JTF capabilities, especially our precision and mobility, to dominate the OE. Speed and timing are essential to mission success.</td></tr>
<tr><td colspan="2">The design team synthesizes the stated end state and OBJs from higher authorities and their implications for the JFMCC to develop a thorough understanding of the operational direction. Guidance from the JFC is most relevant to JFMCC planning while guidance from other higher authorities provides additional context about what constitutes success and the approach to be taken. The developed understanding of the operational direction becomes the envisioned future state and OBJs in the JFMCC’s operational approach.</td></tr>
<tr><th>Desired End/Future State</th><th>Objectives</th></tr>
<tr><td>• Redland respects territorial boundaries.
• Redland military forces that threaten Pinkland are destroyed.
• Terrorists in Redland are eliminated.
• Pinkland can protect/control its territory.</td><td>• Redland anti-SLOC OPS are deterred.
• Cities and ISBs in Pinkland are protected.
• Redland forces are degraded.
• Sea control between ports X-Ray and Yankee is established.
• SPOD at Port Yankee is established.</td></tr>
</table>

Figure B-3. Operational Direction Example

B.3.1.4 Reframing

Reframing is an important aspect of design methodology as it records change throughout the process. When changes to the operational direction or OE occur, the design team reviews their understanding of the operational direction to determine if it needs to be updated to reflect the new situation. Reframing is considered when analyzing each component of the design methodology.

B.3.2 Understand the Operational Environment

Gaining an understanding of the OE is critical to successful application of design methodology (see Figure B-4). The OE is the composite of the conditions, circumstances, and influences that affect the employment of capabilities and bear on the decisions of the commander. It encompasses physical areas and factors of the air, land, maritime, cyber, and space domains; the electromagnetic spectrum; and the information environment. Contained in the environmental analysis are the adversary, friendly, and neutral actors that are relevant to a specific operation situation. Understanding the OE helps the design team to better identify the problem; anticipate potential outcomes; and understand the results of various friendly, adversary, and neutral actions and how these actions affect attaining the military end state.

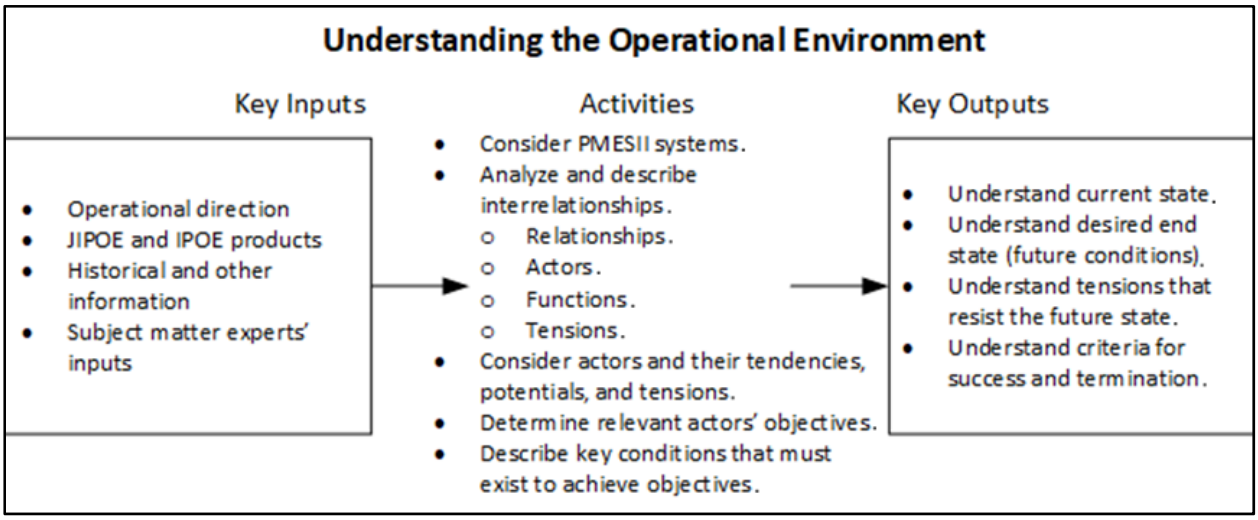


Figure B-4. Understand the Operational Environment

B.3.2.1 Inputs

1. The strategic and operational direction are carried over from the first component of design methodology. The understanding developed by the design team forms the context in which the OE is studied. There is an infinite amount of information available in any OE, and the design team uses their understanding of operational guidance to ensure the study of the OE focus on issues that are relevant to the current situation and of sufficient consequence to warrant study.

2. Joint and maritime IPOE products contain most of the information relevant to understanding the OE component of design methodology. See JP 2-01.3, Joint Intelligence Preparation of the Operational Environment, and NWP 2-0 for more information. Appendix D of JP 2-01.3 describes network analysis and other techniques that may be valuable for developing an understanding of the OE. The design team may choose to review additional, external information relevant to the current situation and environment. Documents such as those addressing the recent history and an analyses of these pertinent materials provided by other governmental agencies (DOS, other national intelligence agencies, coalition partners, and/or nongovernmental agencies) are useful in the environmental analysis.

B.3.2.2 Activities

1. During this component of design methodology, the design team seeks to describe both the current state of the OE and the desired future state of the OE when OPS conclude (desired military end state and derived OBJs) to aid in visualizing the problem and an approach to solving the problem. To develop this understanding, the design team studies the joint intelligence preparation of the operational environment (JIPOE), IPOE, and any additional information within the context of the operational direction to develop a thorough understanding of where differences exist between the current and desired conditions (see Figure B-5).

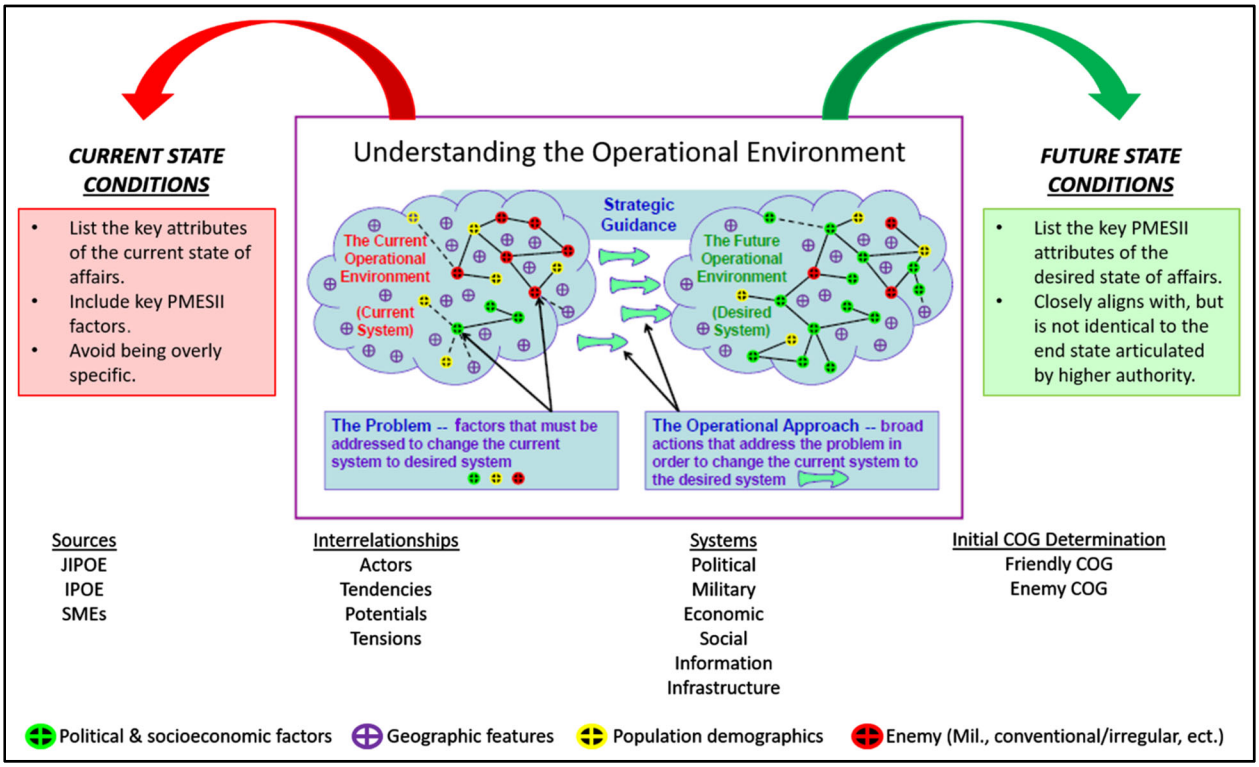


Figure B-5. Identifying and Depicting Relevant Relationships

2. Operational-Level Considerations:

a. In analyzing the current and future OE, the design team can use one or more analytical tools, such as a political, military, economic, social, information, and infrastructure model (PMESII) described in JP 2-01.3 to determine relationships and interdependencies relevant to the specific operation or campaign. Another technique is to use the rubric RAFT (relationships, actors, functions, and tensions) ICW PMESII. Combining the two helps to ensure that the design team addresses multiple aspects within the environment.

The RAFT rubric keys the commander and design team in on the key actors (enemy, friendly, and neutral) in their environment. An examination of the actors and their relationships within and between their systems provides insights as to how well they are functioning in their PMESII roles. The way they function leads to opportunities and/or challenges which cause tensions within and between their systems. Synthesizing RAFT and PMESII information is an effective way to gather, assess, and categorize vast amounts of information in a timely manner.

b. The size and scope of the analysis varies depending on particular aspects of the OE. For example, land-centric adversaries could possess substantial space-based capabilities that could influence the maritime environment.

c. Additional factors to consider include:

(1) Geographic features, and meteorological and oceanographic characteristics.

(2) Population demographics (ethnic groups, tribes, ideological factions, religious groups and sects, language dialects, age distribution, income groups, and public health issues).

(3) Social and cultural factors of adversaries, neutrals, and allies in the OE (beliefs, how and where they get their information, and types and locations of media outlets).

(4) Political and socioeconomic factors (economic system, political factions, and tribal factions).

(5) Infrastructure, such as transportation, energy, and information systems.

(6) Operational limitations such as ROE, rules for the use of force (RUF), or legal restrictions on military OPS as specified in U.S. law, international law, or HN agreements.

(7) All friendly, adversary, and enemy conventional, irregular, and paramilitary forces and their general capabilities and strategic OBJs (including all known and/or suspected chemical, biological, radiological, and nuclear threats and hazards).

(8) Environmental conditions (earthquakes, volcanic activity, pollution, and naturally occurring diseases).

(9) Location of toxic industrial materials in the area of interest that may produce chemical, biological, radiological, or nuclear hazards.

(10) Psychological characteristics of adversary decision making.

(11) All locations of foreign embassies, international organizations, and NGOs.

(12) Friendly and adversary military and commercial capabilities provided by assets in space, their current or potential use, and critical vulnerabilities.

(13) Knowledge of the capabilities and intent, COGs, and critical vulnerabilities of forces. The identification of COGs in design is based on the determination of initial OBJs for friendly and enemy forces. The COG is further refined and updated throughout the planning process.

(14) Financial networks that could impact the adversary's ability to sustain OPS.

d. To produce a holistic view of the relevant adversary, neutral, and friendly systems within a larger system that includes many external influences, analysis should describe how these systems interrelate. Most important to this, the holistic analysis is describing the relevant relationships within and between the various systems that directly or indirectly affect the problem at hand.

e. Tendencies and potentials. As the design team develops its understanding of the OE, several elements emerge during the analysis that are important to the design effort.

(1) The first is the interaction between actors, tendencies, potentials, and tensions. Actors are individuals, groups, nations, etc. that act to advance an interest. Tendencies indicate the inclination of an actor to think or behave in a certain manner. By identifying tendencies, the design team assesses the range of possible actions (potentials) an actor could take. Additionally, the identification of tensions allows the commander and the design team to recognize points of resistance or friction between actors that often occurs when differing interests or ends collide.

(2) The desired end state accounts for tendencies and potentials that exist among the relevant actors or other aspects of the OE.

Note

Think critically when addressing tendencies and potentials. Not all interactions and relationships support attaining the desired end state.

(3) The commander and the design team define and describe the levers to act upon to influence the relevant actors to conform to or move away from their tendencies and potential according to the desired end state.

f. Describe the key conditions that must exist in the future OE to achieve the OBJs. What in the environment must change to move from the current to the future state?

g. Determine the OBJs of relevant actors affecting the OE. These actors will have different sets of conditions for achieving their respective OBJs.

(1) Actors whose OBJs are in opposition can be expected to take actions to thwart friendly OBJs.

(2) Other actors, neutral or friendly, may not have an opposing mindset, but may have desired conditions (or unintended consequences of their actions) that oppose desired end state conditions.

(3) The analysis of the OE should identify where the contradictions between the competing sides, allies, partners, and neutrals lie and recognize the conflicts of interests.

B.3.2.3 Outputs

The commander and design team develop a thorough understanding of the conditions in the current OE (current state) and how they differ from the desired end state (future state). The design team considers the effects, both intended and unintended, of any military actions within the OE. The design team builds its understanding of the OE by addressing these questions:

1. What are the current OE conditions (current state)?

2. What are the desired future OE conditions (future state)?

3. What prevents the desired future state from occurring?

4. How is success defined (termination criteria)?

Figure B-6 provides an example how OE information can be organized.

B.3.2.4 Reframing

The OE constantly changes due to actions by friendly, adversary, and neutral forces as well as natural processes. Therefore, continuous evaluation of the OE is necessary to provide insights on those changes and how they affect accomplishment of the end state and the OBJ. When the OE has significantly changed, the design team may determine that the other components of the design methodology need to be reframed to account for the new environment. The decision to reframe should be deliberate because the results from reframing may cause the current CONOPS to be invalid and require replanning.

B.3.3 Define the Problem

Defining the problem identifies the differences between the current state and desired future state that must be resolved by the operational approach. Defining the problem begins with a review of the tendencies and potentials of the relevant actors and identifying the relationships and interactions among their respective desired conditions and OBJs. The problem statement articulates how the operational variables can be expected to resist or facilitate transformation and how inertia in the OE can be leveraged to ensure the desired conditions are achieved (see Figure B-7).

Operation FA Example

The JFMCC uses the PMESII model and RAFT rubric (or other useful analytical tools) to examine the OE, summarize the relevant relationships and current state conditions, describe conditions associated with the desired end (future) state, and identify tensions that prevent achieving the current state. The table below partially illustrates information considered to describe current and future state conditions.

	Actors	Relationships	Functions	Tensions
Political	Redland.	Adversarial.	Authoritarian.	Redland efforts to delegitimize the government of Pinkland and infringe upon freedom of navigation and other nations' exclusive economic zones (EEZs).
	Pinkland.		Democratic.	
	United States	Ally of Pinkland.	Support Pinkland sovereignty and guarantee freedom of navigation in international waters.	
Military	Redland.	Dominant regional military power.	Expand Redland influence. Coerce neighboring states.	Redland sustainment and support of terrorists.
	United States	Ally of Pinkland.	Provides military support to Pinkland.	
Economic	Pinkland.	Region's main global trading partner.	Dominates regional economy.	Pinkland economic influence within the region.
Social	Terrorists.	Pinkland ethnic minority. Redland ethnic majority.	Destabilize Pinkland economy/government and create an independent state within Pinkland.	Terrorist attacks on foreign nationals within Pinkland.
Information	Redland.	Support of terrorist aims.	Foment discord within Pinkland.	Misinformation ISO terrorist activities.
Infrastructure	Terrorists.	Disrupt Pinkland economic infrastructure.	Terrorize Pinkland citizens and foreign nationals.	Attacks on Pinkland and foreign-owned facilities.

Summary of current state conditions:

Pinkland government is being undermined by Redland-sponsored terrorism. Pinkland is unable, militarily, to thwart terrorist activities and secure its SLOCs from the threat posed by terrorists who are supported by Redland's superior military forces. Pinkland appealed to the international community for political and military assistance to counter the Redland threat. The United States has agreed to send military forces to deter and if necessary, defeat the Redland threat IOT preserve legitimacy of Pinkland government and compel the Redland regime to cease activities that threaten freedom of navigation in international waters and infringe on other nations' EEZs.

Desired future state conditions:

Redland regime ceases all threats against Pinkland and commercial shipping and agrees to respect customary international law concerning uses of the oceans. Redland ends its activities as a state sponsor of terrorism and disbands the terrorist training camps within Redland. Pinkland maritime forces are able to enforce their nation's legitimate sovereign rights within its contiguous and economic exclusion zones. Should hostilities commence, Redland maritime forces and maritime-based terrorists are destroyed and cannot conduct organized OPS.

Figure B-6. Operational Environment Example (Sheet 1 of 2)

Opposing end states:

The Redland regime, through coercion and/or the use of force (conventional, irregular, and terrorism) seeks to interdict or sever SLOCs and launch terror attacks against Pinkland territory. Their OBJ is to degrade Pinkland's economic infrastructure, intrude upon its maritime economic resources, and destabilize its government.

Pinkland and the United States are aligned to prevent Redland military forces and terrorists from interdicting Pinkland's SLOCs and launching ballistic missile and terror attacks against Pinkland economic infrastructure. Their OBJ is to deter/defeat Redland military and terrorist forces; and assist Pinkland forces in securing its SLOCs and enforcing its sovereign rights at sea. The JFMCC focus is to prevent sea denial and defeat ballistic missile attacks by Redland forces throughout the maritime environment.

Figure B-6. Operational Environment Example (Sheet 2 of 2)

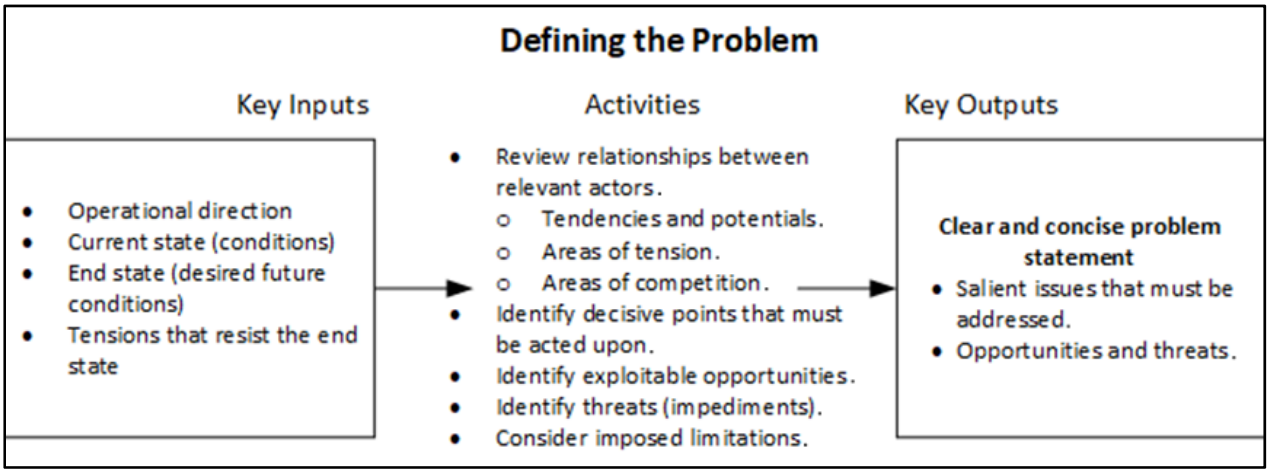


Figure B-7. Defining the Problem

B.3.3.1 Inputs

The commander and the design team review the current and future state assertions developed during the previous components. Additionally, the design team reviews the desired end states (military, political, etc.) and ensures that the identified desired conditions align with the desired end states and conforms to the operational direction. Proper problem definition is essential to resolving a complex problem.[17]

B.3.3.2 Activities

1. The commander and the design team determine which systems, components of systems, conditions, actors, or other concerns are preventing movement within the OE from the current state to the future state to achieve the desired conditions and end states. The disparity between the desired future state and current conditions forms the basis of the problem definition.

   a. The problem statement identifies the areas for action to transform existing conditions (current state) toward the desired end state (future state). Defining the problem begins with a review of the tendencies and potentials of the relevant actors and identifying the relationships and interaction among their desired end states and OBJs. It extends beyond analyzing interactions and relationships in the OE (Figure B-8) and identifies areas of tension and competition—as well as opportunities and challenges—that commanders

[17] "The existence of a discrepancy representing a wicked (complex) problem can be explained in numerous ways. The choice of explanation determines the nature of the problem's resolution." Rittel, Horst W. J.; Webber, Melvin M. (1973). "Dilemmas in a General Theory of Planning."

must address to transform the current state to attain the desired future state. The commander and the design team identifies the tension by analyzing the context of the relevant actors' tendencies and potentials within the OE.

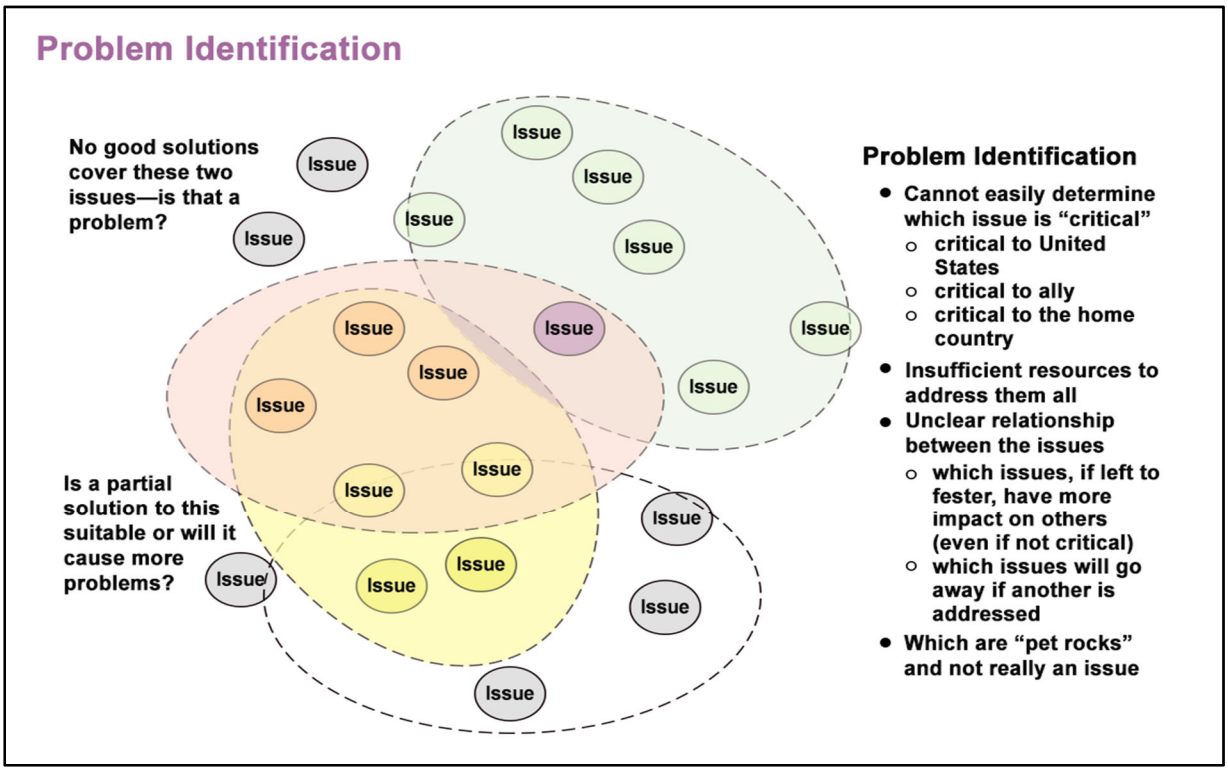


Figure B-8. Problem Identification[18]

b. Critical to defining the problem is determining what needs to be acted on to reconcile the differences between existing (current state) and desired conditions (future state). Some of the conditions are critical to success, some are not. Some may be achieved as a secondary or tertiary result of another condition. In identifying the problem, the planning team identifies the tensions between the desired conditions and identifies the areas of tension that merit further consideration as areas of possible intervention.

c. The commander and the design team must identify and articulate:

(1) Tensions between current and future state conditions desired at the end state.

(2) Elements within the OE which must change or remain the same to attain desired end states.

(3) Opportunities and threats that either can be exploited or will impede the command from attaining the desired end state.

(4) Operational limitations imposed by HHQ or external agencies.

d. A concise problem statement is used to clearly define the problem or problem set to solve. It considers how tension and competition affect the OE by identifying how to transform the current conditions to the desired end state—before adversaries begin to transform current conditions to their desired end state. The statement broadly describes the requirements for transformation, anticipating changes in the OE while identifying critical transitions.

[18] JP 5-0, Joint Planning, 01 December 2020, Fig. IV-3, p. IV-12.

B.3.3.3 Outputs

The critical output of this component is a clear and concise problem statement. The problem statement articulates how the operational variables can be expected to resist or facilitate transformation and how inertia in the OE can be leveraged to ensure the desired future state conditions are achieved. The problem statement is a written narrative which briefly describes the existing current state conditions which must change to move from the current to future state conditions and achieve the desired end state. The problem statement identifies the salient issues that prevent or impede attainment of the desired end state. The statement should include the following considerations that may be separately articulated, if desired:

1. Tensions between current and future state conditions preventing the achievement of the desired end state.

2. Element within the OE that must be changed to move from current to future state conditions.

3. Opportunities and threats to be acted upon to achieve the desired end state.

4. Operational limitations which may prevent movement from the current to future state conditions and achievement of the desired end state.

Figure B-9 provides an example problem statement.

Operation FA Example
The JFMCC problem statement below addresses the tensions identified earlier in the current state preventing movement to the future state according to the JTF/JFMCC end state and OBJ.
Problem statement: Redland-sponsored terrorists pose a threat to U.S. interests and Pinkland security. Redland armed forces operate in an aggressive manner that violates international law. Pinkland lacks the military power to counter Redland armed forces that sustain the terrorists and threaten its sovereignty. U.S. must eliminate terrorists based in Redland, deter or neutralize Redland forces that threaten Pinkland, and compel Redland to respect sovereignty of Pinkland. Over time, Pinkland must develop the capabilities needed to defend its sovereign territory.

Figure B-9. Problem Statement Example

B.3.3.4 Reframing

Just as the OE is constantly changing, the commander and the design team's understanding of the problem is constantly changing. As the understanding develops, the design team reframes the problem statement to reflect the new understanding. The team should always be on the alert for changes significant enough to warrant redesign of the entire operation.

B.3.4 Develop an Operational Approach

The operational approach is a commander's description of the broad actions the force can take to achieve OBJs ISO HHQ OBJs or attain a desired military end state (see Figure B-10). It is the commander's visualization of how the operation should transform current conditions into the desired conditions—the commander's envisioned OE at the conclusion of OPS. Using the developed understanding of the tensions that resist transformation of current state conditions to the desired future state conditions, the commander and the design team develop the broad operational approach for transforming current conditions into desired conditions. The operational approach underpins the operation and the detailed planning that follows.

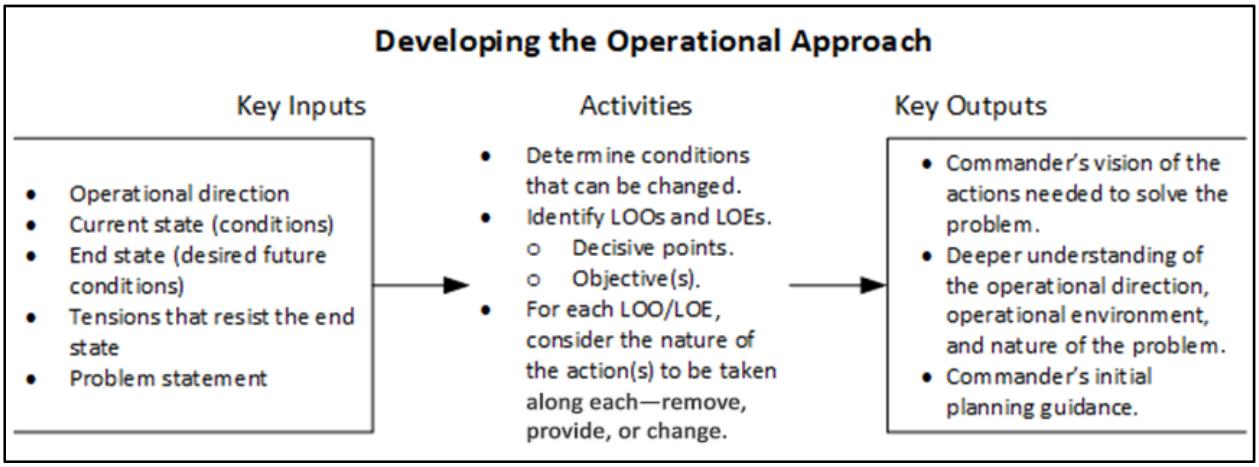


Figure B-10. Develop an Operational Approach

B.3.4.1 Inputs

The inputs to this component of design methodology are developed by the other components: the desired conditions at end state, tensions in the OE, conditions that need to be changed, threats and opportunities for change, and the definition of the problem. Most important is the understanding gained while developing these products.

B.3.4.2 Activities

The commander and the design team review the previous work and determine which of the tensions, threats, opportunities, or conditions can be acted against by military forces to change the current state conditions in the OE to the desired future state conditions. The commander and the design team determine which conditions can be changed and what actions should be taken to enact those changes. Typically, the operational approach is described in terms of LOEs and LOOs that result in the desired changes identified by the commander and the design team in previous components. Each LOO or LOE identifies key actions that lead to overall operational success. Figure B-11 illustrates depiction of an operational approach.

The rubric "remove, provide, or change" can be used to summarize the nature of actions to be taken at DPs along each LOE or LOO to achieve its OBJ. The commander and the design team determine what must be removed, provided or changed along each LOE or LOO. For example, the "degrade" LOO/LOE of Figure B-12 will likely consist of offensive OPS to neutralize, destroy, disrupt, and otherwise degrade enemy forces to "remove" enemy capabilities.

The operational approach reflects the ways of employing military forces to achieve the ends determined from understanding the operational direction. The direction provides the broad conditions of the OE that will exist at the conclusion of the operation that are acceptable to higher authorities. It incorporates all previous aspects of design methodology: direction, environment, and problem. The approach's composition provides insight into the commander's and design team's newly acquired understanding.

B.3.4.3 Output

The operational approach captures the commander's vision of how the operation will unfold. It incorporates all other components of design methodology–direction, environment, and problem–and describes the broad actions the force can take to achieve the OBJ and end state. The operational approach shares the commander's and design team's insights and understanding with the remainder of the staff and OPT. Figure B-11 and Figure B-12 illustrate a possible depiction of an operational approach.

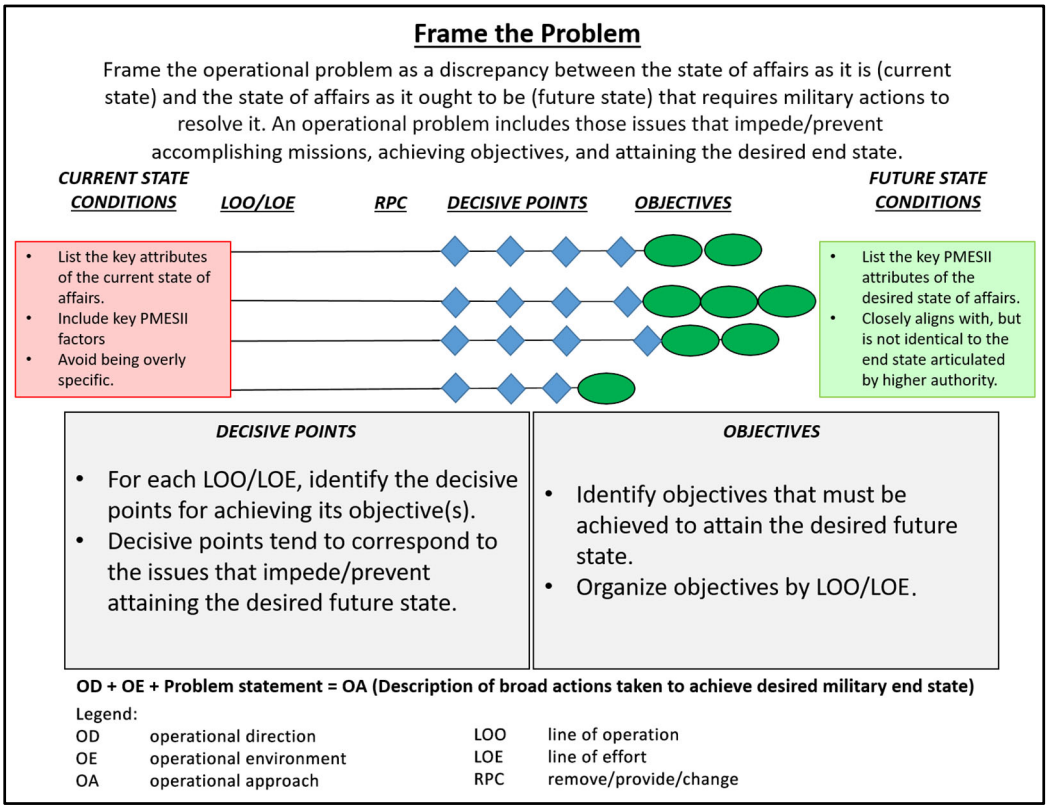


Figure B-11. Depicting an Operational Approach

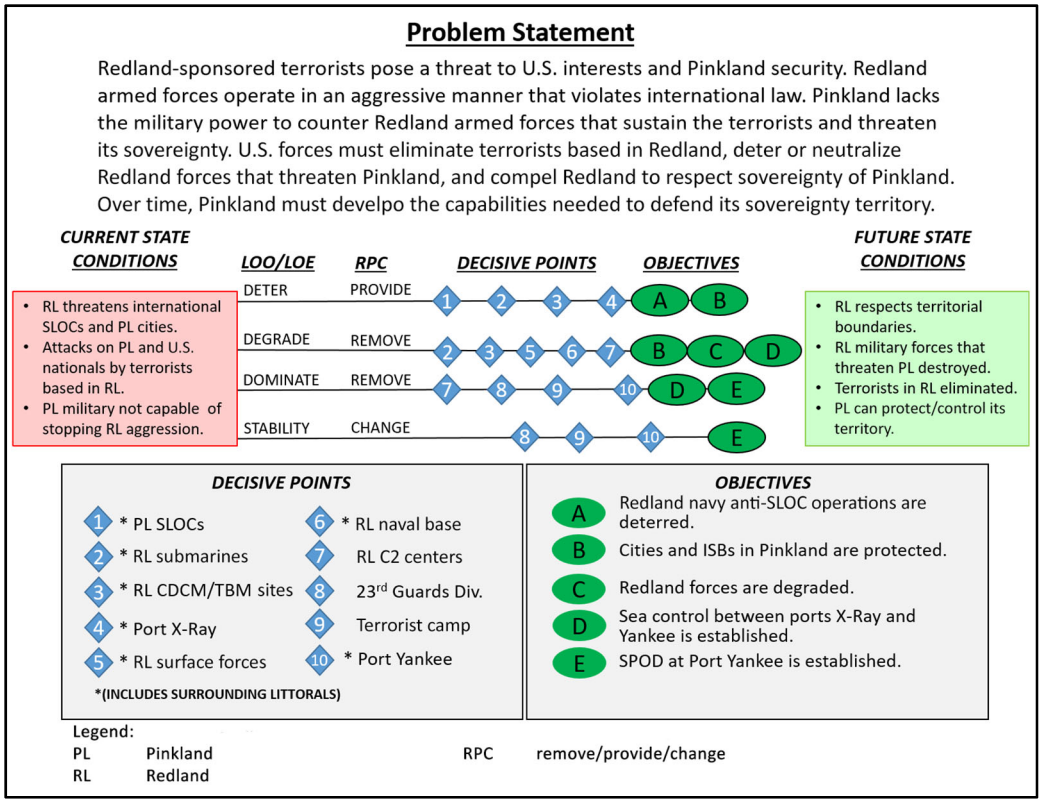


Figure B-12. Operational Approach Example

B.4 INCORPORATING DESIGN METHODOLOGY INTO PLANNING GUIDANCE

Commanders share the understanding developed through design methodology via the commander's initial planning guidance to the staff. They then update that guidance as the operation develops to adapt the operational approach to changes in direction, the OE, or a changed problem.

The format for the commander's planning guidance varies based on the personality of the commander and the level of command, but should adequately describe the logic to the commander's understanding of the OE, the methodology for reaching the understanding of the problem, and a coherent description of the operational approach. It should include the following elements:

1. Description of the OE. Some combination of graphics showing key relationships and tensions and a narrative describing the OE (current and future state; tensions in the OE) will help convey the commander's understanding to the staff

2. Definition of the problem to be solved. A narrative problem statement that includes a timeframe to solve the problem and embodies understanding of the operational direction.

3. Description of the operational approach. A combination of a narrative describing OBJs, DPs, and potential LOEs and LOOs, with a summary of limitations (constraints and restraints) and risk (what can be accepted and what cannot be accepted).

4. Commander's initial intent. The commander's initial intent ties together the improved understanding developed by design methodology to concisely describe the purpose of the OPS, the general method (operational approach), and desired military end state to focus the efforts of the OPT. It may also describe the operational risks associated with the operation, including where the commander will and will not accept risk. Chapter 2 provides more information on commander's intent.

B.5 KEY POINTS

1. This description of design methodology offers one approach for addressing complex and ill-structured problems.

2. The commander's participation in the discourse is critical to achieving the outcome: initial planning guidance.

3. Design methodology benefits from employing a design team that consists of both subject matter experts and OPT members.

4. Dialog amongst the commander and the design team develops deeper understanding.

5. Each component of design methodology is reframed as circumstances change to ensure that the operational approach remains valid.

6. Design methodology facilitates initial development of OBJs for use and validation during the NPP.

7. **Planning teams should be alert for the pitfall of falling into a design loop.** This happens when the planning team continues to improve its conceptual model–admiring the problem–at the expense of beginning detailed planning. A balance must be sought.

APPENDIX C
Center of Gravity Analysis

C.1 CENTERS OF GRAVITY

One of the most important tasks confronting a staff during planning is identifying and analyzing the COGs of friendly and enemy forces. The COG is the source of power or strength that enables a military force to achieve its objective and is what an opposing force can orient its actions against that will lead to enemy failure. COGs are determined by their impact on the military end state. Success requires protecting the friendly COG while defeating the enemy COG.

COGs can exist at different levels. At the strategic level, a COG could be a military force, an alliance, political or military leaders, a set of critical capabilities or functions, or national will. At the operational level, a COG is often associated with the threat's military capabilities such as a powerful element of the armed forces but could include other capabilities in the OE.

COGs may change in time as the strategic environment or OE changes due to shifts in friendly, neutral, or threat diplomatic, information, military and/or economic/commercial conditions or objectives. A force's will to fight, for example, may increase or decrease throughout the course of an operation with dramatic impact on the joint force's success. Similarly, victories or defeats may cause adversaries or enemies to reassess objectives or strategies that can alter the COG.

COGs exist in an adversarial context involving a clash of moral wills and/or physical strengths. COGs do not exist in a strategic or operational vacuum; they are formed out of the relationships between adversaries and enemies. They define COGs through their unique view of the threats in the strategic environment and OE as well the requirements to develop and maintain power relative to their own objectives. Commanders, therefore, must not only consider their threat's COGs but they must also identify and protect their own.

The COG construct is useful as an analytical tool to help commanders analyze friendly and adversary or enemy sources of strength as well as weaknesses and vulnerabilities. A faulty conclusion resulting from a poor or hasty analysis can have serious consequences; such as the inability to achieve strategic and operational objectives at an acceptable cost.

The identification of COGs is not a static N-2 process conducted once during JIPOE. Planners must continually analyze and refine COGs due to actions taken by friendly forces and the threat's reactions to those actions.

The determination and analysis of COGs (friendly and threat), as well as understanding the relationships between them, is a key step in operational design. Intelligence analysts determine potential options for the threat COG by identifying what needs to be weakened or destroyed to defeat the enemy. The N2, in conjunction with other operational planners, then determine if the proposed COGs are truly critical to the enemy's strategy. This analysis is a linchpin in the planning effort. Planners in all sections on the staff conduct similar analysis to identify friendly COGs and their strengths and weaknesses. Once COGs have been approved, commanders and their staffs determine how to attack enemy COGs while protecting friendly COGs. Understanding the relationship among COGs facilitates greater precision in thought and expression in operational design. Figure C-1 illustrates the identification and deconstruction of a COG.

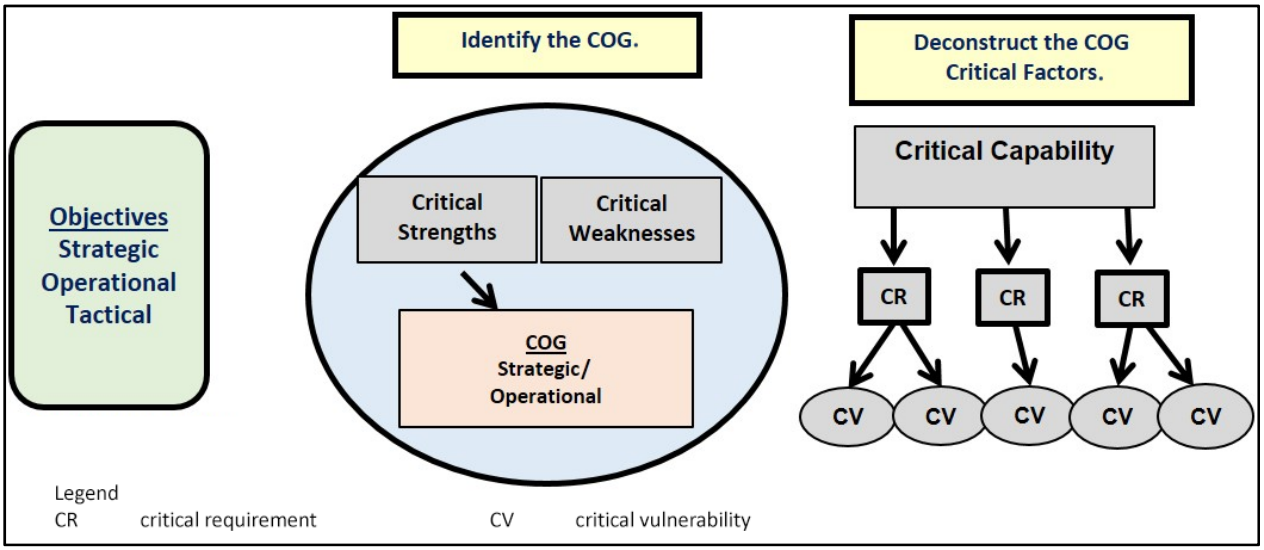


Figure C-1. Center of Gravity Identification and Deconstruction

C.2 IDENTIFY THE OBJECTIVE(S)[19]

One must know and understand the OBJ(s) before determining the COG(s). If this portion of the analysis is flawed, then the error infects the remainder of the process. The planner should first determine the strategic end state and then the operational and intermediate OBJs, ensuring that they all nest. An objective is clearly defined (i.e., specific, measurable, achievable, relevant, and time-bound), decisive, and attainable goal toward which an operation is directed. Military OBJs specify what must be accomplished and provide the basis for describing desired effects. OBJs describe what must be achieved to reach or attain the end state. Achieving operational OBJs ties execution of tactical tasks to reaching the military end state.

There are four primary considerations for an OBJ.

1. An OBJ establishes a single desired result (a goal).

2. An OBJ should link directly or indirectly to higher-level OBJs or to the end state.

3. An OBJ is specific and unambiguous.

4. An OBJ does not infer ways and/or means—it is not written as a task.

C.3 IDENTIFY CRITICAL STRENGTHS AND WEAKNESSES

Critical strengths and weaknesses are those attributes considered crucial for the accomplishment of the OBJ. These strengths and weaknesses (that describe the environment in relationship to the OBJ) should be identified and classified as either sufficient (critical strength) or insufficient (critical weakness). Critical strengths and critical weaknesses can be military or nonmilitary, and quantifiable (tangible) or unquantifiable (intangible). Critical strengths and weaknesses are present at each level of war; they require constant attention because they are relative and subject to changes resulting from the actions of one's forces or of the enemy's actions. It is important while conducting the analysis for this step that planners maintain a sharp eye on the OBJs identified in the first step—each OBJ has critical strengths and weaknesses that are unique to that OBJ.

[19] JP 5-0, Joint Operational Planning, provides a more in-depth review of operational art and the relationship of objectives and COGs to the overall operational design.

When determining critical strengths, the question to be asked is: "Which military or nonmilitary capabilities are considered essential to accomplish one's or the enemy's military OBJ(s)?" The most important, or the one determined necessary to accomplish the OBJ among the critical strengths is the COG.

For example, instead of simply listing an enemy critical strength as naval forces, specify which military force or capability of those naval forces into clear-cut critical strengths such as subs or antiship cruise missile (ASCM) equipped surface combatants.

When determining critical weaknesses, the question to be asked is: "What are the military or nonmilitary capabilities considered essential to the accomplishment of one's or the enemy's military OBJs but, in terms of quantity, quality, or both, are insufficient or inadequate to perform their intended functions?"

For example, a strategic critical weakness, such as a strategic leader having a tenuous communications link to fielded forces, may also create an operational critical weakness for fielded forces unable to reliably communicate with their higher command.

The answers to these two questions will produce a range of critical strengths and critical weaknesses associated with specific levels of war. One should note that, like the close relationship expected to be found between strategic and operational OBJs, there will undoubtedly be some critical strengths and critical weaknesses that have a similar close relationship between the strategic and operational levels of war.

The more discrete that planners are in the determination of critical strengths and weaknesses, the more focused the resultant COAs can be. At the operational level of war, particularly in a force-on-force scenario, the operational COG will typically be a physical force. Therefore, when listing critical strengths planners should be as discrete as possible.

C.4 IDENTIFY THE CENTERS OF GRAVITY

The importance of the COG concept in the NPP is that it is directly linked to COAs development. While COGs are critical strengths that actually accomplish OBJs at specific levels of war, COAs should be focused on defeating the enemy COG(s) and protecting the friendly COG(s) that have been identified.

Note

> Planners should note that COGs can also be considered similarly for natural disaster/epidemic phenomena. A good example would be malaria/yellow fever mitigation actions during the building of the Panama Canal. Mosquitoes were determined to be the carriers of the disease and thus can be considered malaria/yellow fever's COG. Once the mosquitoes were nearly eradicated in the Canal Zone, deaths from malaria/yellow fever were tremendously reduced to negligible, thus allowing canal construction to proceed.

The COGs at each level of war are found among the listed critical strengths identified at that level of war. While all of the identified strengths are critical, the planner must deduce which among those strengths identified rise(s) above all others in importance in accomplishing the OBJ (that is, those tangible and intangible elements of combat power that would accomplish the assigned OBJs)—this critical strength is the COG. A method to do this somewhat analytically is to take each critical strength and specifically ask the question: "Does this critical strength accomplish the OBJ?" If the answer is that it does not accomplish the OBJ but only assists in accomplishing the OBJ, it is probably a critical capability or critical requirement but not the COG.

Before solidifying COGs into the plan, planners should analyze and test the validity of the COGs. The defeat, destruction, neutralization, or substantial weakening of a valid COG should cause either the friendly or enemy force (depending on which COG analysis is being conducted) to change its COA or prevent the achievement of its OBJs. If analysis and/or war game demonstrates that this does not occur, then planners have likely misidentified the COG, and they must revise their COG and subsequent analysis. The conclusions, while critically important to the planning process itself, must be tempered with continuous evaluations because derived COGs and critical

vulnerabilities are subject to change at any time during the campaign or operation. Accordingly, commanders and their subordinates should be alert to circumstances during execution that may cause derived COGs and critical vulnerabilities to change and adjust friendly plans and OPS accordingly. For example:

1. If the enemy has an operational OBJ to seize island X and the planners have identified some enemy critical strengths as: carrier air power, sustainment forces, and the landing force, the planners should ask: "Can carrier air power seize island X?" and "Can sustainment forces seize island X?" The answer to each is obviously no, but carrier air power and sustainment forces are instrumental in providing air superiority over the island and keeping the landing force operational. Therefore, neither carrier air power nor sustainment forces are the enemy COG but each may prove to be a critical capability or critical requirement to achieving the OBJ.

2. Next ask: "Can the landing forces seize island X?" The answer is yes, which makes the landing forces the enemy COG. This does not diminish the importance of the other critical strengths; however, it forces the planner to examine closely the relationships of the various critical strengths to one another and the OBJ.

C.5 DECONSTRUCTING THE CENTER OF GRAVITY

Once planners have identified the likely threat COG, they need to identify the best method to attack or weaken it. This process forms the core of COA development and assists with the identification of missions and tasks.

Direct attacks against adversary COGs resulting in their neutralization or destruction provide the most direct path to victory. Since direct attacks against COGs mean attacking an opponent's strength, commanders and staffs must determine if friendly forces possess the power to attack with acceptable risk. Commanders normally attack COGs directly when they have superior forces, a qualitative advantage in leadership, and/or technological superiority over enemy weapon systems.

In the event a direct attack is not a reasonable solution, commanders should consider an indirect approach until conditions are established that permit successful direct attacks. At the operational level, the most common indirect method of defeating an enemy's COGs is to conduct a series of attacks against selected aspects of the enemy's combat power. Indirect methods of attacking the enemy's COGs (through critical vulnerabilities) could entail reducing the enemy's operational reach, isolating the force from its C2, and destroying or suppressing key protection functions such as air defense. Whenever applicable, commanders should consider developing simultaneous and/or synchronized action with both direct and indirect approaches. In this manner, the enemy or adversary's derived vulnerabilities can offer indirect pathways to gain leverage over its COGs.

Identifying the missions and tasks can be accomplished by analyzing and deconstructing the COG within a framework of three components: critical capabilities, critical requirements, and critical vulnerabilities. In general, a command must possess sufficient operational reach and combat power or other relevant capabilities to take advantage of an enemy's critical vulnerabilities while protecting friendly critical capabilities within the operational reach of an enemy.

C.5.1 Identify Critical Capabilities

A critical capability is a means considered to be a crucial enabler for a COG to function as such and is essential to the accomplishment of the specified or assumed OBJ(s). If the COG is a physical force (as is often the case at the operational level), the commander and staff may wish to begin their examination of critical capabilities by reviewing the integration of the joint/operational functions: C2, intelligence, movement and maneuver, fires, sustainment, and protection (joint doctrine includes information as a separate function). Moreover, these capabilities often are located within the critical strengths and weaknesses previously identified. The planner should be alert for two major considerations. First, although a capability might be a critical strength, if it bears no relationship to the identified COG, it cannot be considered a critical capability. The second consideration is that although some capability may be perceived as a critical weakness, if it is an essential enabler for the enemy COG, then it is a critical capability, albeit weak in nature.

For example, a critical capability for an operational COG to accomplish its mission might be its ability to direct forces (C2)—its ability to receive direction as well as communicate directives to subordinates. The fact that this capability has been deduced to be a weakness does not diminish its importance to the COG for accomplishment of its assigned mission. The planners should exploit or mitigate the weakness during COA development.

C.5.2 Identify Critical Requirements

Once critical capabilities are identified, the next step is for the staff to identify those essential conditions, resources, and means for a critical capability to be fully operational. These are the critical requirements that support each of the critical capabilities. This is essentially a detailed view of what comprises a critical capability. An example of this would be using the ability to C2 as a critical capability, the critical requirements might be tangible elements such as: communication nodes, antennas, frequency bands, individual command posts (CPs), spare parts, bandwidth, specific satellites, etc. It may also include intangibles such as commander's perceptions and morale.

C.5.3 Identify Critical Vulnerabilities

A critical vulnerability is an aspect of a critical requirement that is deficient or vulnerable to direct or indirect attack that will create decisive or significant effects. The planner should contemplate those critical capabilities and their supporting critical requirements in this regard, keeping in mind that these weaknesses shall bear a direct relationship to a COG and its supporting critical capabilities for it to be assessed as a critical vulnerability. Striking a weakness that bears no such relationship is simply a measure taken to harvest low-hanging fruit that offers no decisive benefit. The planner should also take this opportunity to consider the previously assembled lists of critical strengths and critical weaknesses to determine if there are any with a close relationship to the COG that were not captured in the previous critical capability/critical requirement steps.

While the planner first seeks critical weaknesses within the critical capabilities and supporting critical requirements as implied by the definition, there might be opportunities found in critical strengths that provide decisive or significant results disproportionate to the military resources applied.

For example, the integrated air defense system (IADS) that is protecting an operational COG. While this critical capability might be assessed as a strength, its neutralization and the subsequent opening of the COG to direct attack may be assessed by the commander as more favorable in regard to the amount of resources and time expended to achieve the desired effects.

C.6 IDENTIFY DECISIVE POINTS

A DP is key terrain, key event, critical factor, or function that, when acted upon, enables commanders to gain a marked advantage over an enemy or contribute materially to achieving success. See also center of gravity. As with all previous steps, the value of a DP is directly related to its relationship to a COG and its OBJ. A DP is neutral in nature; that is, it is by definition as important to both enemy and friendly commanders. Planners should remember that DPs might not always be physical in nature. They could be functions or key events such as gaining maritime superiority or support of the local populace. In noncombat situations, such as a disaster response, a DP could be accessing stagnant bodies of water where mosquitoes breed IOT treat or drain them to prevent the spread of disease. Although DPs are usually not COGs, they are the keys to attacking or protecting them.

An example of this would be if an aerial port of debarkation (POD)/SPOD complex is a DP for a friendly commander, enabling that commander to project the COG through it on the way to the OBJ. The enemy commander will also assess the complex as a threat to the enemy COG and should attempt to deny the friendly force commander control of the DP. In both cases, this DP, if within the capability of the force, will undoubtedly become an OBJ or task assigned to both enemy and friendly subordinate commands.

C.7 TASKS TO TACTICAL ORGANIZATIONS

Vulnerabilities identified in the previous step will often translate into tasks for the tactical-level commander. Critical vulnerabilities are related to a desired effect: the exposure of the operational COG. Additionally, critical

capabilities may identify tasks to subordinates when the capability is not inherent in the COG. Critical vulnerabilities may include a force's LOG hubs or LOG ships. A task to a tactical commander may include strike OPS against those LOG assets.

C.8 ENEMY CENTER OF GRAVITY ANALYSIS/DECONSTRUCTION EXAMPLE

Figure C-2 provides an example of COG identification and deconstruction.

Enemy OBJ: sea denial. Deny JTF BLUE SWORD the use of territorial and adjacent seas out to maximum over-the-horizon range for amphibious OPS and sustained carrier-based air support.

Critical strengths.

1. Surface combatants—three frigates, five corvettes, eight fast attack craft (missile), 11 coastal patrol craft.
2. Subs—four Kilo subs.
3. Coastal defenses—three CDCM launchers (mobile) and two CDCM launchers (fixed) (30 missiles total).
4. Land-based aircraft—30 MiG 29s, Su 27s, and 45 MiG 25s/27s.
5. Sea mines—450 (contact and influence).
6. Forces—disciplined forces/high morale and loyalty to leadership/home field advantage.

Critical weaknesses.

1. C2—centralized, limited satellite communications (SATCOM) and encryption.
2. No maritime patrol aircraft.
3. Sustainment—limited antiship missile inventory, no underway replenishment (UNREP) capability.
4. Sea mines—still in storage locations (example of how a strength can be a weakness).
5. Lack of joint interoperability.

Enemy operational COG: Redland Kilo subs.

Critical Factors

1. Critical capabilities.

a. C2—ability to plan/task organize and direct subs against enemy ships.

b. Intelligence—ability to detect enemy ship movements/analyze information on ship movements to determine where and when subs should be employed against which targets.

c. Movement and maneuver—ability to manage water space/maneuver sub forces/provide indications and warnings (I&W)/direct/provide WSM to deconflict OPS against enemy shipping.

d. Fires—ability to provide complementary sea denial fires by channeling blue forces into optimal kill areas

e. Sustainment—ability to resupply/repair/replenish subs.

f. Protection—ability to mitigate JTF BLUE SWORD force ASW capabilities/ability to safeguard sub bases.

2. Critical requirements.

a. C2—CP locations/communications links (physical and electromagnetic) and the sub's ability to transmit/receive with these links.

b. Intelligence—CP/fusion centers/human intelligence (HUMINT)/coastal surveillance radars/signals intelligence/C2 structure and communications links for passing information on JTF BLUE SWORD ship locations.

c. Movement and maneuver—suitable water space and WSM to support sub OPS/bases of OPS support/trained crews.

d. Fires—antiship mobile platforms and missile storage locations/surface fleet/C2 structure/ISR capabilities to detect JTF BLUE SWORD forces/airfields to support aviation OPS/sea mines and minelayers/mine storage facilities.

e. Sustainment—storage and repair facilities/sub tenders/ports/lines of communications between storage sites and ports/C2 Infrastructure.

f. Protection—use of OPSEC/adequate I&W of JTF BLUE SWORD movements/local air superiority/deception OPS.

Figure C-2. Center of Gravity Identification and Deconstruction Example (Sheet 1 of 2)

3. Critical vulnerabilities.

a. C2—CPs/communications links (physical and electromagnetic).

b. Intelligence—CPs/fusion centers/coastal surveillance radars.

c. Movement and maneuver—bases of support.

d. Fires—mine storage facilities/C2 structure.

e. Sustainment—storage and repair facilities/ports/sub tenders.

f. Protection—use of OPSEC/local air superiority.

4. Decisive points. Key terrain, key event, critical factor, or function that, when acted upon, enables commanders to gain a marked advantage over an enemy or contribute materially to achieving success.

a. Geographic—ports/airfields/coastal surveillance radars.

b. Specific key event—establishing maritime superiority/identification of sub movement.

c. Critical factor or function—ability to communicate with subs/ability to locate and track subs.

Figure C-2. Center of Gravity Identification and Deconstruction Example (Sheet 2 of 2)

C.9 CENTER OF GRAVITY ANALYSIS WORKSHEETS

Figures C-3 and C-4 provide example worksheets for COG analysis.

Strategic OBJs:	
Operational OBJs:	
Critical Strengths:	**Critical Weaknesses:**
Operational COG: •	

Figure C-3. Center of Gravity Worksheet

Critical Factors		
Critical Capabilities:	**Critical Requirements:**	**Critical Vulnerabilities:**
Decisive Points: Key terrain, key event, critical factor, or function that, when acted upon, enables commanders to gain a marked advantage over an enemy or contribute materially to achieving success. • • •		

Figure C-4. Center of Gravity Deconstruction Worksheet

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APPENDIX D
Analyzing Relative Combat Power

D.1 ANALYZING RELATIVE COMBAT POWER

RCPA compares those friendly and enemy tangible and intangible factors that allow each force to generate combat power to achieve their OBJs. The most important information derived from an RCPA is an understanding of which factors provide each force an advantage. This understanding coupled with knowledge gained from IPOE and mission analysis serve as a guide for COA development by determining how, when, and where to generate overwhelming friendly force combat power to accomplish the mission while mitigating the risks posed by those areas in which the enemy has the advantage. RCPA helps provide an understanding of the aspects of the problem that guide or limit potential solutions.

Maritime operational planners should conduct an RCPA as part of generating COAs. Doing so provides planners with a deeper understanding of friendly and enemy force numbers, capabilities, strengths, and weaknesses relative to each other at a given time and in a particular geographic location. When conducting an RCPA, a numerical comparison of major air, surface, and subsurface platforms must be balanced by comparing the actual capabilities of what are often multimission platforms. Other intangible factors such as will to fight, training, the presence or absence of an alliance/coalition, leadership, morale, discipline, soundness of doctrine, and combat readiness must also be compared when conducting an RCPA.

Many of the inputs required to conduct an RCPA are generated prior to the COA development step in the NPP. For example, several tangible and intangible factors related to time, space, and force identified during IPOE should be compared when conducting an RCPA. Similarly, factors identified while conducting the enemy COG analysis and deconstruction should also be considered when conducting an RCPA. During mission analysis, the results of both the friendly COG analysis as well as the "analysis of friendly forces and assets" should also contribute to the RCPA.

When conducting an RCPA, planners should consider whether friendly forces achieve and/or maintain a sufficient combat power advantage to act upon a DP previously identified during enemy and friendly COG analysis. Additionally, such an analysis and comparison should help planners consider how best to mitigate friendly critical vulnerabilities and protect the friendly COG.

RCPA provides planners with insights that help them determine force array in relation to the enemy, location and requirement for control measures, and location and requirement for ISR. These insights and discussions should be captured and used throughout COA development.

D.2 METHODS TO CONDUCT A RELATIVE COMBAT POWER ANALYSIS

The RCPA method used depends upon the time and resources available as well as the scope and scale of the planning problem (e.g., a campaign or major operation versus a tactical engagement or battle).

D.2.1 Mathematical Force Ratios

Mathematical force ratios are the most common method and can be especially helpful for a specific enemy that has been studied and analyzed for years. Mathematical forces ratios require time, resources, and in-depth analysis for development. Development of mathematical force ratios for maritime forces can be especially challenging due to the multimission nature of many naval platforms, the qualitative difference among various nations' platforms and their tactical employment, the time and resources required to analyze weapons systems' kill probability and effectiveness of defensive systems, and the characteristics of the maritime domain itself. As mathematical force ratios take time to develop, they may be more appropriate for use in contingency planning where plans are

focused on specific enemies and are developed and refined over a long time. Examples of mathematical force ratios are the force ratio tables developed during the Cold War (U.S. versus Soviet forces). Tangible factors (e.g., weapons system capabilities, force numbers, etc.) and intangible factors (e.g., training, doctrine, C2 organization, etc.) were studied, analyzed, and distilled into force ratio tables that expressed force equivalents (e.g., X number of U.S. force/unit type = Y number Soviet force/unit type).

D.2.2 Planning Considerations Method

When planning against an enemy for which no force ratios have been developed and/or time is limited (e.g., time-constrained planning) naval planners may benefit by focusing on the subjective comparison of friendly and enemy tangible and intangible factors that reveal relevant information for developing COAs. This subjective RCPA method provides planners with perspective in assessing whether the forces applied to a particular task or OBJ are adequate in force and capabilities in comparison to those of the enemy.

The focus of this method is not the worksheet produced (Figures D-1 and D-2), but rather the rich dialogue that takes place through analysis and comparison. The worksheet is a tailorable tool that focuses the dialogue and concisely captures key planning considerations to be applied as different COAs are developed. The worksheet should be tailored to account for friendly and enemy tangible and intangible capabilities and qualities that are relevant to the planning problem. For example, Figures D-1 and D-2 do not address special operations forces (SOF) capabilities. If friendly or enemy SOF capabilities are relevant to the planning problem, the planners should add this to the worksheet. Moreover, the worksheet could be tailored for deeper analysis of a specific area by planners focused in a specific mission area. For instance, IO planners may tailor the worksheet to analyze friendly and enemy capabilities through information-related capabilities (IRCs) as part of the IO order of battle. Cyber planners may tailor the worksheet to analyze friendly and enemy capabilities in relation to cyber terrain, cyber DPs, and offensive/defensive capabilities.

In this method, a key output from comparing the quantitative and qualitative tangible/intangible factors of engaging forces is the planning considerations and crucial insights that apply to the actual COA(s) developed. In addition to providing insights that help determine friendly force array and TF composition in relation to the enemy, these considerations may guide task sequencing and how or when tasks are accomplished, as well as identify needed capabilities, foreclose certain options, or dictate friendly OPAREAs. The purpose of the planning considerations and insights derived through RCPA is to serve as guides for application by the planning team in the development of COAs.

D.2.2.1 Preparation

When preparing to conduct an RCPA, planners might find the following helpful:

1. When developing IPOE products, the raw number-type data, such as quantities, can be captured in advance of conducting the RCPA.

2. Impressions of capabilities, particularly the enemy's, established during the RCPA process can be long lasting, even past planning and into execution. If a capability is discussed, ensure that it is a factually based discussion, rather than one based on conjecture. If an item cannot be researched adequately, capture it as a subjective estimate, or even an assumption, rather than building awareness around data that may in fact be incorrect.

3. Planners should be prepared to discuss:

   a. Friendly and enemy doctrine and how it affects their respective actions/equipment

   b. Friendly and enemy weapons capabilities and limitations for class/type/model/series of military equipment

   c. Friendly and enemy functional capabilities, issues, and impacts

   d. Friendly and enemy training issues and impacts

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
Aircraft carriers				
Cruisers				
Destroyers				
Frigates				
Corvettes				
Fast attack craft (ASCM equipped)				
Patrol boats/ Fast attack craft				
Minelayers				
MCM platforms				
Mines				
Subs (nuclear-powered) (SSNs)				
Subs (diesel/electric)				
Amphibious ships				
Marines				
Naval air (strike)				
Naval air (fighter)				
Naval air (EW/airborne early warning (AEW))				
Ship-based helicopters				
Unmanned aerial vehicles (UAVs)				
MPRA/ISR				
LOG ships				
Other Forces or Capabilities That Support or Impact Maritime OPS				
Air force				
Ground force				
SOF				
Ballistic missiles				

Figure D-1. Relative Combat Power Analysis Worksheet (Sheet 1 of 3)

Intangible Factors	Enemy Force Quality	Friendly Force Quality
Doctrine		
Training		
Experience		
Warfare Area Capability		
SUW		
AMD		
ASW		
Amphibious		
MIW		
EW		
Cyber		
Operational Factor—Time		
Operational Factor—Space		
Operational Factor—Force		
Operational Functions		
C2		
Intelligence		
Fires		
Movement and maneuver		
Protection		
Sustainment		

Figure D-1. Relative Combat Power Analysis Worksheet (Sheet 2 of 3)

Advantage/ Disadvantage Summary	Enemy Force	Friendly Force
Advantages		
Disadvantages		
Key points/Elements Derived From Enemy Force and Friendly Force Advantages/Disadvantages for Application in COA development		

Figure D-1. Relative Combat Power Analysis Worksheet (Sheet 3 of 3)

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
Aircraft carriers.	0.	No navy tactical air (TACAIR). Reliant upon land-based (air force) TACAIR for strike and defense of Redland.	One (U.S.); Nimitz class.	Significant advantage in strike and standoff capability. Single carrier limits ability to conduct 24/7 flight operations.
Cruisers.	0.	No surface combatants larger than a frigate. Significantly limited blue water capabilities.	Two (U.S.); Ticonderoga class.	One-BMD capable. Relative to guided missile destroyer (DDG) BMD capabilities, guided missile cruiser (CG) better used for AD mission.
Destroyers.	0.	No surface combatants larger than a frigate. Significantly limited blue water capabilities.	Five (U.S.); Arleigh Burke class DDGs.	Three-BMD capable. Based on Redland short-range ballistic missile threat, three DDGs available for BMD msn will allow for unit rotation and continuous protection of bases in Pinkland.

Figure D-2. Relative Combat Power Analysis Example (Sheet 1 of 12)

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
Frigates.	Three.	Multipurpose guided missile frigate (FFG); ASUW: C802A ASCM, 76-millimeter gun; ASW: sonar (hull and tactical towed-array sonar (TACTAS)), torpedoes and two triple launchers; AAW: limited to short range 8–10 nautical miles point defense; frigates are Redland's high-end blue water capability. Lack over-the-horizon targeting (OTHT) capability to effectively employ ASCMs beyond 15–20 nautical miles. One embarked helo used for daytime ISR. Limited ASW training. Well trained and equipped for minelaying, easily carry 25–30 mines. No UNREP capability, endurance (3–4 weeks).	Three (Pinkland).	Multipurpose FFG; ASUW: Exocet, 76-millimeter gun; ASW: sonar (hull and TACTAS), torpedoes and two triple launchers; AAW: phased array radar, medium range (25 nautical miles) surface-to-air missile (SAM); area AD capable. Embarked helo used for day/night ISR, OTHT, and ASW. Well trained (all warfare areas), well equipped crews—high readiness. Routinely exercise with United States and U.S. allies—can be easily integrated into TF OPS. UNREP capable. Must ensure compatible crypto.
Corvettes.	Five.	Primarily used as littoral patrol craft in the littorals. ASUW: 76-millimeter gun; ASW: TACTAS and rocket-thrown bomb unit 6000 (RGB 60 depth charge). AD limited to short range man-portable air defenses (MANPADs) (SA 24) 3 nautical miles range. Well trained and equipped for minelaying—can carry 10–15 mines. Limited endurance (2 weeks).	Three (Pinkland). Three littoral combat ships (LCSs) (U.S.).	Pinkland corvettes- ASUW platforms with Exocet and 76-millimeter gun; short range (8 nautical miles) point defense SAM; no ASW capability. Limited endurance (2 weeks). UNREP capable, but can only UNREP with United States Navy (USN) using astern refueling method. USN LCS-SUW package: Naval Strike Missile, Hellfire missiles and 30-millimeter guns. Sea rolling air frame missile for AD. No ASW capability. UNREP capable but tied to shore-based maintenance/support back in Pinkland.

Figure D-2. Relative Combat Power Analysis Example (Sheet 2 of 12)

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
Fast attack craft (ASCM equipped).	Eight.	Littoral ASUW platform with C802A ASCMs and 30-millimeter gun. Lack OTHT capability to effectively employ ASCM beyond 15–20 nautical miles. No ASW capability. AD limited to short-range MANPADs (SA 24/SA 14). Well trained and equipped for minelaying—can carry up to five mines. Limited endurance (4 days).	0.	Redland has the advantage in terms of platforms optimized for littoral OPS. Blue force task organization must maximize ISR, OTHT and standoff/long-range. weapons capabilities across the force.
Patrol boats/fast attack craft.	11.	Littoral patrol platform. One 25-militmeter gun and various crew-served weapons. Short-range MANPAD (SA 24/SA 14) for AD. Well trained in swarm tactics. Trained and equipped for minelaying; can carry two–three mines. Limited endurance 2–3 days.	0.	Redland has the advantage in terms of platforms optimized for littoral OPS. Pinkland corvettes and LCS with SUW package provide significant antiswarm boat capabilities-integrating corvettes/LCS with cruisers/destroyers among the different TFs will provide good layered defense.
Minelayers.	All ship and subs.	Redland has significant advantage in minelaying. The number of potential minelayers, their readiness, and number of mines present the most formidable threat to opposing naval forces.	P-8s and subs.	Air superiority required/air threats must be mitigated prior to P-8 employment for minelaying.
MCM platforms.	Two minehunters, coastal.	Well trained and equipped for minelaying. Primarily employed for minefield maintenance. MCM training and capability unknown.	One MCM (U.S.), Two LCSs (U.S.).	Small capability relative to the Redland minelaying threat. Additional MCM capability required (RFF). MCM very slow and both MCM and LCS (MIW package) have limited self-defense capabilities. Consider targeting Redland's minelaying capabilities at the start of shaping fires IOT neutralize the threat.
Mines.	450.	Mix of contact and influence. Best weapon in inventory to keep enemy navy out. Fixed mine storage sites are easily attacked. Must disperse mines and minelayers early to avoid enemy attack. Can be used to channel approaching enemy naval forces into areas for attack by other weapons systems.	75 contact/ influence mix (Pinkland); 50 influence (U.S.).	Pinkland MIW capabilities not available for use against Redland (for defense/defensive mining of Pinkland waters only). U.S. influence mines could be used to deny Redland access to/from their naval base.

Figure D-2. Relative Combat Power Analysis Example (Sheet 3 of 12)

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
SSNs.	0.	Lack long-range/high-endurance sub capability—but not necessarily needed for defense of Redland.	Two (U.S.); one LA class, one VA-class.	Disadvantage in numbers and ability to operate in littoral waters. Environment ideally suited for Kilo-class sub employment. Consider ROE authorization for employment of active search capabilities early in the COA.
Subs (diesel/electric).	Four Kilo class.	Environment is ideal for Kilo sub employment in defense of Redland. Mines and fixed CDCMs can be used to channel enemy naval forces into weapons range of waiting Kilo subs. Endurance and snorkeling requirement for extended OPS are vulnerabilities.	0.	Neutralize/destroy Redland subs at the start of the employment of lethal fires. Consider additional ASW capabilities—are Pinkland subs available for OPS?
Amphibious ships.	0.	Redland lacks amphibious warfare capability to project power from the sea to threaten Pinkland.	Four (U.S. ESG); one LHA, one LPD, and two LSDs.	Embarked landing craft: two landing craft, utility, four landing craft, air cushion; Significant advantage in power projection from the sea. Credible deception force to force Redland to spread its forces to counter potential threat. Maritime superiority must be established prior to amphibious OPS in the littorals.
Marines.	0.	Redland lacks ability to project power from the sea to threaten Pinkland.	MEU.	ACE: six F35s, four AH-1Z Cobras, three UH-1Y Venoms, four CH-53 Super Stallions, 12 MV-22 Ospreys; ground combat element: BN landing team; four M1A Abrams tanks, 15 amphibious assault vehicles, 16 light armored vehicles, six 155-millimeter howitzers (T); combat LOG BN: 92 HMMWVs, 30 7-ton trucks. Employment of Marine maneuver and fires in raids, demonstrations, and feints to keep Redland forces off balance.

Figure D-2. Relative Combat Power Analysis Example (Sheet 4 of 12)

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
Naval air (strike).	0.	Reliant upon land-based air force fourth-generation aircraft TACAIR for strike (offensive OPS). Limited/no joint OPS capability/experience between the Redland navy and air force.	36 strike fighter squadron (VFAs) (U.S.); CVW mix of F/A-18s and F-35s.	CVW has significant advantage in training, readiness and integration of air wing capabilities. Achieve overmatch wherever we can bring Navy TACAIR against Redland surface forces.
Naval air (fighter).	0.	Reliant upon land-based air force third-generation aircraft for defense of Redland.	Same 36 VFAs (U.S.).	Limited to one carrier operating tempo (OPTEMPO). Coordinate with JFACC to synchronize/integrate air OPS IOT achieve two-carrier effect. Consider deception to lure out and defeat Redland strike aircraft prior to engaging Redland Air Force third-generation aircraft (sequential approach).
Naval air (EW).	0.	Four shore-based ISR aircraft have EM spectrum surveillance capability only no EA. Four fixed EW sites along coast for surveillance (limited range due to antennae/receivers 25 nautical miles).	Four (U.S.); EA-18 Growlers.	EW advantage for degrading Redland IADS and fixed EW sites.
Naval air (AEW).	0.	Reliant upon coastal radar systems (range 25 nautical miles) and GCI.	Four (U.S.); E-2D Hawkeyes.	Significant AEW advantage especially when combined with fleet tactical link system. Limitation: single carrier OPTEMPO—coordinate with JFACC for synchronization/integration with Airborne Warning and Control System.
Ship-based helicopters.	Five.	One per frigate; two at Navy base for training and rotation. Helo capable of ASUW and ASW however crew training and proficiency is limited and poor. Used primarily as an organic ISR asset during daytime OPS; has limited/no nighttime sensor capabilities.	16 (U.S.); mix of SH-60S and SH-60R; 5 (Pinkland); NH90.	Significant shipboard helo advantage for ISR, OTHT, SUW, and ASW OPS. Significant Redland MANPAD threat requires employment of standoff sensors and weapons capabilities.
UAVs.	Two.	Shore-based, limited range out to 25 nautical miles from shore for up to 4 hours; sensors limited to daytime OPS.	Five Scan Eagles (U.S.) Two Fire Scouts (U.S.).	Day and night sensor packages. Excellent endurance for ISR and OTHT. Develop task organization to disseminate capability across the different TFs (e.g., CSG, ESG, and SAGs).

Figure D-2. Relative Combat Power Analysis Example (Sheet 5 of 12)

Tangible Factors	Enemy Quantity	Significance for Enemy Force	Friendly Quantity	Significance for Friendly Force
Naval Order of Battle				
MPRA/ISR.	0 MPRA, four ISR.	Limited range out 100 nautical miles from shore for up to 5 hours; E/O sensors for daytime; nighttime sensors limited to radar and EM surveillance.	One P-8 (U.S.).	Shortfall in relation to ASW threat. RFF for additional MPRA capability for ISR, ASW and SUW support. Shore-based in Pinkland, requires protection from Redland Air Force and shore-based SAMs.
LOG ships.	0.	Redland navy ships are reliant upon access to/from shore bases for sustainment. Naval base is the main hub with full support capabilities. Other Redland ports can be used for resupply of food and fuel. Use of other Redland ports for rearming and repairing Redland navy ships is extremely limited. Access to/from the Redland naval base is essential for OPS.	Four (U.S.): two fleet replenishment oilers (MSC) (T-AOs), one fast combat support ship (MSC) (T-AOE), one T-AKE.	Good mix of UNREP/replenishment at sea (RAS) capability to maximize force time at sea. Access to ISB in Pinkland is essential. Coordinate with N4 and log CTF to establish UNREP/RAS locations in rear areas such that CTFs can screen/protect replenishment sites.

Figure D-2. Relative Combat Power Analysis Example (Sheet 6 of 12)

Other Forces or Capabilities That Support or Impact Maritime OPS				
Air force.	30 fourth-generation, mix of MiG 29 and Su 27 aircraft, 45 third-generation; mix of MiG 25 and MiG 27 aircraft.	Lacks joint OPS experience with Redland navy. Reliant upon shore-based GCI with shore-based sensor capability limited to 25 nautical miles.	12 F-15s (Pinkland), 12 F-16 E/Fs (Pinkland).	Pinkland air forces, while well trained, equipped, and ready, are limited to defense of Pinkland out to 50 nautical miles from Pinkland's shore.
Ground force.	Two fixed CDCM sites. Three mobile CDCM transporter, erector, launchers; 20 C802 CDCMs; five coastal artillery sites.	While CDCMs have range out to 65 nautical miles, Redland lack of OTHT capability limits effective employment out to 25–30 nautical miles. Coastal artillery range of 12 nautical miles, rudimentary capability to coordinate with Navy units for spotting/targeting.	Eight Exocet missiles in two mobile four-cell launchers (Pinkland).	Exocet Mobile Coastal Defense System. Excellent capability, well trained force able to coordinate OTHT with naval units. System used only for Pinkland defense (not deployable).
IADS.	Two SAM sites co-located with search/ targeting radars.	Use Russian S-125 system (SA-3 Goa) at fixed sites. One SAM site each at naval base and air base. Colocated fixed radar systems at each SAM site can feed SAM launchers at either location. Radars: P-15 Flat Face target acquisition radar 135 nautical miles range, SNR-125 Low Blow tracking, fire control and guidance radar 28 nautical miles range, PRV-11 Side Net height finder radar 9 nautical miles range/ 105,000 feet. maximum height. Good early-warning range but shorter (medium range) engagement capability.	Aegis system with Standard missiles (SMs) 2 and 3 (U.S.). Phased array radar with Aster 15 missiles (Pinkland frigates).	Integrated/linked systems. Pinkland systems compatible for tactical link data exchange but not cooperative engagement.
Ballistic missiles.	Five mobile SCUD launchers; 10 missiles.	Weapons can reach bases in Pinkland. Level of training and proficiency fair/good based upon observed missile tests 3 years ago.	0.	Strike limited to TLAMs (aboard CG, DDG, and subs) and CVW strike aircraft.

Figure D-2. Relative Combat Power Analysis Example (Sheet 7 of 12)

Intangible Factors	Enemy Force Quality	Friendly Force Quality
Doctrine.	Defense and sea denial. Operate from a central position with home field advantage. Defend using layered area defense capabilities: ASCM-equipped aircraft and ships outside the littorals followed by ASCM-equipped fast attack craft and layered area denial capabilities: CDCMs and defensive mine belts closer to shore. Sea denial by seeking to conduct defensive first strike to inflict heavy/high-profile damage and loss of life to defeat opposition's national support/will to fight.	Deter and if necessary, compel/defeat. Minimize collateral damage and maintain multinational support to maintain legitimacy. Expeditionary oriented sea-based force. Maximizes operational reach and advantages in fires and maneuver to operate from the exterior position. Deters through the nonlethal fires and maneuver. Compels/defeats through sea control OPS and strike/amphibious OPS ISO joint force OBJs.
Training.	Enemy forces are a conscript force with only basic-level training. Lacks unit cohesion.	Highly trained using modern equipment and technology. Minimal turnover of forces/highly cohesive forces.
Experience.	Inexperienced in joint operations. Inexperienced operating outside of territorial areas.	Experienced in joint operations. Highly experienced in the employment of technology during OPS.
Warfare Area Capability		
ASUW.	**Good** Operating in familiar waters. OTHT and night OPS limitations prevent Redland from using weapons to full extent of capabilities. Redland capabilities optimized for layered defense of in the littoral environment.	**Very Good** Significant advantage in OTHT (day or night) and target discrimination. Traffic density in the littorals may limit employment.
AAW.	**Poor** Key vulnerability is Redland's short range SAMs/MANPADs for shipboard AD.	**Excellent** Long-range integrated and layered AD.
ASW.	**Poor to Fair** Redland has good ASW weapons. However, ASW training for surface forces has been extremely limited - sub force assessed to be better trained in ASW.	**Very Good** Excellent ASW capability and readiness in the force. Limitations based upon requirement to operate in the littorals and number of MPRA currently allocated to operation.
Amphibious.	**None**	**Excellent** Well equipped, highly trained force with excellent readiness. Redland layered defense of littorals poses significant challenge for surface-borne maneuver of amphibious forces.

Figure D-2. Relative Combat Power Analysis Example (Sheet 8 of 12)

Intangible Factors	Enemy Force Quality	Friendly Force Quality
MIW.	**Excellent** Large portion of Redland surface force is well trained and equipped to conduct offensive/defensive mining. Large inventory of mines.	**Poor** While the friendly force has capable MCM systems and well-trained personnel, severely limited due to significant shortfall in MCM capability relative to the threat. Small inventory of mines/mine kits.
EW.	**Poor** EW capability limited to EM spectrum surveillance. Limited range/coverage in the maritime domain.	**Excellent** Full range of EW capabilities dispersed across the force.
Cyber.	**Poor** No known dedicated military cyber force/capability. Internet heavily used for IO and strategic messaging. Threat from civilian hacktivists ISO Redland or contracted by Redland does exist.	**Excellent** Cyber support team available with significant capabilities. May be limitations in terms of effects due to enemy force infrastructure and organization.
Operational Factor–Time		
Ability to prepare.	Redland advantage: have had adequate time to reinforce, prepare defenses, and initiate IO as they initiated the crisis.	Limited time to prepare as friendly force is responding/reacting. Once friendly force is on station and initiates OPS, controlling tempo may shift advantage to friendly force.
Ability to mobilize.	**Fair/Poor** Redland has very limited ability to regenerate or mobilize new forces once combat OPS begin and existing forces are attrited. However, terrorists have been able to recruit/force more people to join their cause.	**Good** Significant force available if strategic level deems crisis requires additional forces/ capabilities. Long lines of communications (LOCs)—time required for mobilized forces to travel to the JOA.
Ability to deploy.	**Good** Redland operating in own home waters and can deploy forces quickly. Limitation is OPTEMPO and endurance—Redland forces have at-sea endurance limitations as their sustainment is all based ashore in port.	**Good** Pinkland ISB shortens SLOC requirements for forces in the JOA. The long LOCs from the continental United States (CONUS) to the JOA requires significant transit time for forces.
Ability to reconstitute.	**Poor** Redland has limited reserves and repair capability. Once combat OPS begin, Redland infrastructure lacks depth capability to rapidly repair/restore damages vessels.	**Good** Friendly force has significant potential reserves from which to draw and Pinkland has significant sustainment/repair capabilities.
Ability to regenerate.	**Poor** Redland has limited/no international support. Lack the capability to build new units once they are attrited in combat.	**Good** Friendly force has significant potential reserves from which to draw. Limitation is the time it takes for replacement forces/capabilities to arrive from CONUS.

Figure D-2. Relative Combat Power Analysis Example (Sheet 9 of 12)

Intangible Factors	Enemy Force Quality	Friendly Force Quality
Ability to maintain OPTEMPO.	**Poor** Degrades over time. With the finite force that Redland has, it is vulnerable to wearing down if it cannot control the pace of OPS and keep friendly forces from seizing the initiative.	**Good** Improves over time once all friendly forces are on station and sustainment capabilities are established across the LOCs. With the full force on station, the friendly force has sufficient capabilities to keep Redland forces off balance and in a reactionary posture.
Operational Factor–Space		
Open ocean (blue water) environment.	Redland forces at a disadvantage due to lack of UNREP capability and limited endurance. Lack OTHT capability to use weapons to their full capability.	Nearest support bases (FOB and ISB for United States) in Pinkland. Pinkland bases are within Redland SCUD range. Long LOCs back to CONUS—Redland could potentially strike LOG units with air force strike assets and subs.
Basing areas.	One major naval base (able to provide full sustainment functions); five other Redland ports from which Navy routinely operates (typically provide basic—i.e., fuel/food—sustainment. However, land lines of communications (LLOCs) among the ports and naval base could be used to move other supplies (ammunition, etc.).	Friendly force holds exterior position. Mobile LOG force and friendly force ISR, OTHT. Maneuver and fires advantages enables the force to use the open ocean as maneuver space from which to strike Redland forces from over the horizon.
Position.	Redland holds central position. While friendly force navy can intercept supplies bound for Redland, LLOCs and air lines of communications through neighboring neutral countries remain routes for resupply from external sources.	Critical infrastructure for Pinkland-loss/damage could impact Pinkland participation in—or support to—OPS against Redland. Consider force array/maneuver in the COA to keep friendly forces in position to interdict any Redland attempts to attack the gas-oil platform (GOPLAT) field.
GOPLATs in the NW sea near Pinkland.	Potential target for Redland to exploit—threaten Pinkland oil production, cause an environmental disaster, and make portion of sea space unusable.	International support—UNSCR calling for sanctions against Redland and action against terrorists. Concerted IO effort required to influence Redland populace to not support Redland military in protection/support of terrorists.
Popular support.	Redland civilians will assist/support Redland military OPS. Redland population heavily influenced through Redland government strategic messaging and propaganda. Support for terrorists not strong and limited to certain small groups/segments of Redland society.	LOOs into Redland predictable. Use deception to deny enemy to obtain I&W of friendly force. ME.
Redland bounded by three neutral countries.	Can be exploited by Redland—LOCs through neighboring neutrals enable bringing in supplies from external sources. Redland forces may hide in/along border regions. May use border area as route to move/maneuver forces.	Nearest support bases (FOB and ISB for U.S. forces) in Pinkland. Pinkland bases are within Redland SCUD range. Long LOCs back to CONUS—Redland could potentially strike LOG units with air force strike assets and subs.

Figure D-2. Relative Combat Power Analysis Example (Sheet 10 of 12)

Intangible Factors	Enemy Force Quality	Friendly Force Quality
Operational Factor–Force		
Organization.	Navy—well-defined C2 structure with strong leadership. Heavily reliant upon centralized C2—deployed force reliant upon direction and guidance from HQ. No real joint experience/ coordinated OPS. Terrorist lack maritime capability other than waterborne IEDs near shore/harbor.	Well-defined C2 structure. Forces trained to execute from mission type orders and apply initiative based upon commander's intent. Distributed planning and execution. CWC approach to battles/engagements. Well trained in joint OPS. Excellent U.S.-Pinkland interoperability.
Doctrine.	Rigidly adhere to Soviet model—especially C2. Target discrimination and LOAC not of great concern.	Excellent doctrine and application through tactics, techniques, and procedures, OPTASKs. Adherence to LOAC and ensuring target discrimination.
Combat readiness/ experience.	Navy—no combat experience against an enemy Navy. Combat readiness varies depending upon warfare area.	Experienced in a number of warfare areas, well trained, and high state of readiness.
Will to fight.	Good—naval forces are well disciplined and have high morale.	Very good—disciplined, mission=focused force with good morale.
Operational Functions		
C2.	Poor and vulnerable—centralized C2, deployed forces heavily reliant upon direction and guidance from HQ. Limited SATCOM and encryption capability. Communications rely on high frequency (HF)/very high frequency (VHF)/ultrahigh frequency (UHF) (line of sight).	Excellent—mission type orders, decentralized execution, and CWC. Full range of communications (extremely high frequency-VHF) with robust encryption capability.
Intelligence.	ISR—limited in range. Advantage in HUMINT: support from Redland civilian mariners in reporting enemy navy location and activities.	Limited HUMINT but significant ISR advantage with capabilities ranging from space to below the sea.
Fires.	Capabilities are organized in a layered fashion, but Redland forces not well trained in combined arms (integrated fires).	Robust systems for planning and executing integrated fires. Significant OTHT advantage.
Movement and maneuver.	Operating on interior lines and in home waters. Limited endurance and tied to shore-based LOG.	Operating on exterior lines and can maximize open ocean (blue water) beyond Redland ISR/sensor range for maneuver.
Protection.	Layers of capabilities arrayed to protect Redland littorals. Significant advantage in MIW.	Excellent capabilities and numbers to protect against surface and air threats. Significant shortfall in MCM compared to the threat. Highly capable ASW capabilities but number in relation to the enemy requires more capability. Pinkland bases within Redland SCUD range.
Sustainment.	Poor and vulnerable. Single Navy port is the main sustainment hub. No UNREP capability, so Redland must ensure access to/from port. No force regeneration capability.	Excellent ability to UNREP, sea base, and apply mobile LOG force. Availability of bases in Pinkland shorten SLOCs in JOA, but LOC back to CONUS very long.

Figure D-2. Relative Combat Power Analysis Example (Sheet 11 of 12)

Intangible Factors	Enemy Force Quality	Friendly Force Quality
Force Comparison		
Advantage/ Disadvantage Summary	**Enemy Force**	**Friendly Force**
Advantages.	• Minelaying and number of mines. • Forces/capabilities optimized for operating in the littoral environment. • Operating in home waters along short interior lines. • Bounded by three neutral countries • SCUD capability. • HUMINT, support from local mariners.	• ISR, AAW, strike, ASUW, amphib OPS, EW, and OTHT. • Open ocean (blue water) OPS and sustainment. • Maneuver, integrated fires, and endurance. • C2 systems and capabilities. • Excellent night OPS capability.
Disadvantages.	• ISR, EW, AAW (short range shipboard AD systems), OTHT, and amphibious OPS. • Forces tied to shore bases for sustainment (no UNREP capability). • Limited C2 systems (HF/UHF/VHF) will minimal encryption capability. • Poor night OPS capability. • Size of force in relation to area to be defended.	• MCM capability. • ASW capability in littoral environment. • Lack of local popular support/limited HUMINT.
Key Points/Elements Derived From Enemy Force and Friendly Force Advantages/Disadvantages for Application in COA Development		
1. Neutralize/destroy enemy minelayers, mine inventory and kilo subs early—target in first strike/at commencement of hostilities. 2. Use open ocean (blue water) for maneuver space, sea basing, and leverage OTHT to attrite enemy forces in the littorals. 3. Ensure distribution of UAV capabilities across the TFs to maximize force ISR and OTHT capabilities. 4. Ensure allocation/distribution of Pinkland corvettes and U.S. LCSs among appropriate TFs to protect against enemy swarm boat capability. 5. Overmatch achieved wherever friendly force TACAIR can be applied against enemy surface force. 6. Apply maneuver and deception to spread enemy out and preserve OPSEC of ME. 7. Leverage night OPS advantage—maneuver friendly force into littorals at night to close and strike/raid the enemy. 8. Coordinate with JFACC to achieve two-carrier effect and determine BMD requirements for protection of Pinkland bases. 9. Coordinate with Pinkland LNO regarding protection of GOPLATs in NW sea and availability of Pinkland navy subs for OPS. Propose RFF for additional MCM and ASW capabilities.		

Figure D-2. Relative Combat Power Analysis Example (Sheet 12 of 12)

D.2.2.2 Methodology Steps

The steps in the planning considerations RCPA method are:

1. Create a list of friendly and enemy tangible and intangible factors that should be analyzed and compared based on the nature of the mission (the factors included in Figures D-1 and D-2 are examples of initial lists that should be tailored to the specific planning situation.).

2. For the tangible factors, if it useful to list a number or quantity related to a particular factor, do so. However, do not just stop at listing the number/quantity and determine advantage/disadvantage based on which side has more. Be sure to address strengths and/or vulnerabilities, highlight weapons systems of concern, and highlight/capture key points as they are discovered vice waiting until later.

3. For the intangible factors, if it useful to concisely describe the quality of a particular factor, do so. Whenever possible capture the so-what of the factor–impact upon or the significance to enemy/friendly forces.

Note

The essence of operational art is the ability to balance the factors of time, space, and force. As such, as part of the RCPA and prior to COA development it is useful for planners to analyze, and possibly compare, certain critical factors related to time, space, and force that may have been identified during IPOE. An initial list of time, space, and force factors are included on the worksheets below for consideration.

4. Analyze and compare each tangible and intangible factor selected.

5. Summarize enemy/friendly force advantages/disadvantages.

6. List planning considerations and key insights that are to be applied during development of COA options and individual COAs.

Note

The planning considerations and key insights are key outputs of the RCPA process. The worksheet is merely a tool to focus the planners and organize on the relevant factors against which the enemy and friendly force are analyzed. The worksheet approach is of limited utility without actual discussion, analysis, and the ultimate distillation of the analysis into planning considerations and key insights.

See Figure D-1 for a worksheet of considerations for RCPA. This worksheet is an example and should be tailored to support the specific operational planning effort being undertaken. Figure D-2 is a filled-in example.

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APPENDIX E

Risk Assessment and Mitigation

> One of the most difficult things we have to do in war is to recognize the moment for making a decision…the information comes in degrees. Shall we make a decision now or shall we wait a little longer?
>
> *Adolf von Schell, Battle Leadership*

E.1 RISK

The planning process enables commanders to make informed decisions, solve problems, and accomplish missions in the face of an enemy, challenging environment, or other obstacles. During the process, the staff will develop various options that take advantage of opportunities to solve potential problems. Risk is inherent in any use of military force or routine military activity. Opportunity and risk have an inherent relationship that in many cases influences COA decisions. Greater opportunity may require greater risk. In the planning process, the commander and staff balance these opportunities against the risks. The staff will attempt to develop various risk mitigation measures. Risk discussed in relation to Navy planning is associated with the dangers that exist due to the presence of the enemy, the uncertainty of the enemy intentions, and the potential rewards or dangers of friendly force action in relation to mission accomplishment.

Risk is "the probability and consequence of an event causing harm to something valued."[20] Risk within Navy planning is aligned to the Joint Risk Analysis Methodology (JRAM) with the concept of risk relationship to potential gain. The JRAM uses a framework with three major components and four steps or activities (see Figure E-1) to comprehensively address risk. The three components are:

1. Risk appraisal—generation of knowledge and understanding

2. Risk management—decisions and actions to manage or mitigate risk

3. Risk communication—the exchange of risk perspectives across processes and among leadership.

The risk analysis process consists of four steps:

1. Problem framing—establishing the risk conventions and "Risk to what?"

2. Risk assessment—identifying and scaling threats; "Risk from what?"

3. Risk judgment—developing a risk profile ("How much risk?") and evaluating the risk—i.e., determining how much risk is acceptable.

4. Risk management—decisions and actions to accept or mitigate—"What should be done about the risk?"

[20] CJCSM 3105.1, Joint Risk Analysis Methodology (JRAM).

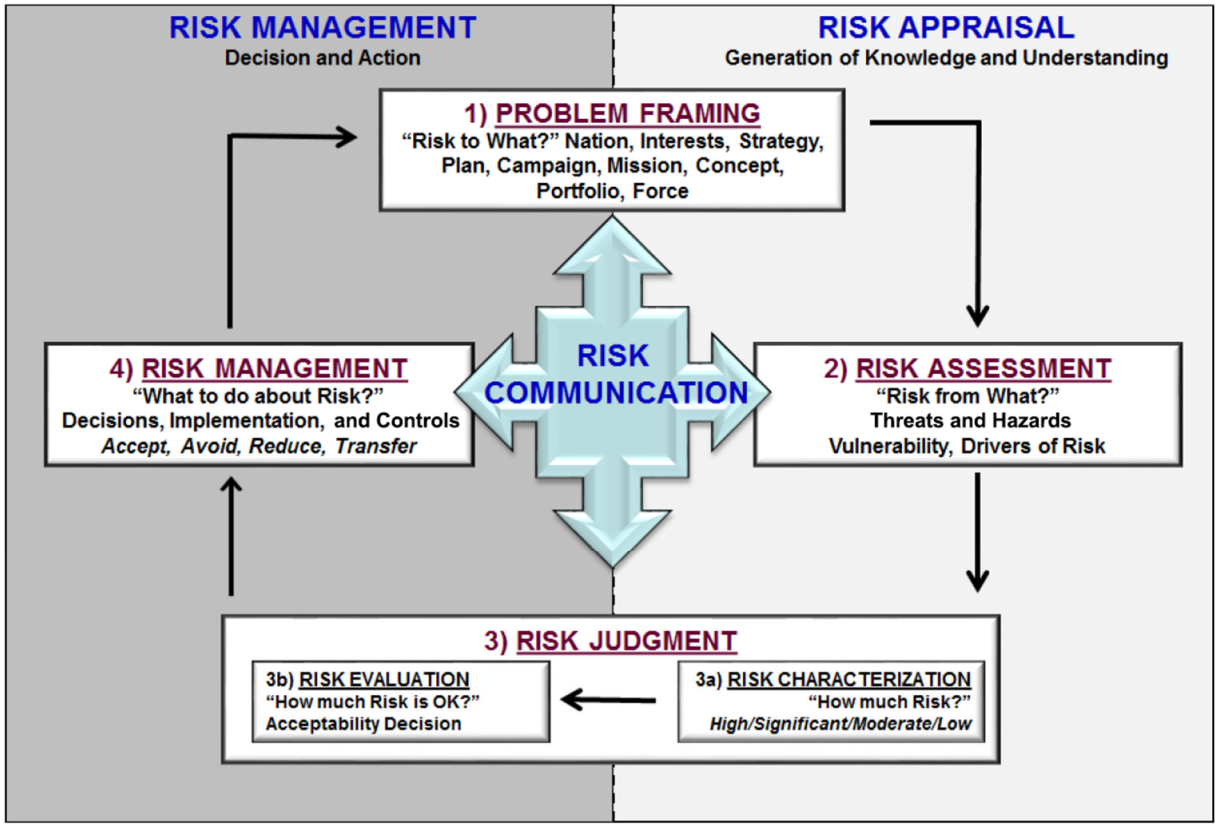


Figure E-1. Joint Risk Analysis Methodology Process[21]

The strategic, operational, and tactical levels of warfare view risk from different perspectives. At the national and theater strategic levels, the JRAM process enables the CJCS to make consistent, timely risk assessments and provide the best possible military advice on risk management ISO the national military strategy and other United States Code (U.S.C.) Title 10 responsibilities. Strategic risk assessment serves national, departmental, and military leaders as they set priorities and allocate resources to mitigate risks. At the operational level, commanders and their staffs work to integrate and synchronize the tactical actions of multiple subordinate commanders and staffs. Operational level risks are much more complex as they attempt to integrate the operational functions in multiple types of OPS, including single Service, joint, and multinational. At the tactical level, the time horizon tends to focus on upcoming potential tactical actions. Risk consideration outside military action is minimized apart from considerations such as collateral damage. This allows the tactical commander to evaluate in a very linear manner the balance of risk to force against task accomplishment.

One emerging trend is that risks have linkages and there may be cause and effect between each. However, risks depend on the perspective of the viewer (e.g., what may be assessed as high risk at the operational level may be considered as a low risk at the strategic level). Because of this, risk communication is critical between HHQ and subordinate commanders and their staffs, to ensure risk is effectively understood at all levels and proper commander's guidance is clearly comprehended throughout the force.

A second clear trend is the complexity of missions in the contemporary operating environment. These missions, while just as complex as previous missions, spring from different kinds of complexity that have caused issues for forces seeking to adapt. A major contributing cause of this complexity is the lack of capability and/or capacity of some civilian agencies and interagency bodies often to carry out American foreign policy.

[21] CJCSM 3105.01, Joint Risk Analysis.

From a planning perspective, risk falls into two broad categories:

1. Risk to mission (primary focus at operational level of war)

2. Risk to forces (primary focus at tactical level of war).

Based upon HHQ's input, direction, and guidance, the commander alone determines how and where to accept risk, but the staff plays a critical role in helping the commander identify the various risks and offering options for mitigation.

While risk cannot be totally eliminated, it can be mitigated, to some extent, by a systematic approach that weighs the costs—time, personnel, and resources—against the benefits to mission accomplishment. Commanders have always mitigated risk through intuition, by their past experiences, and by their judgment. Risk mitigation will not prevent losses, but properly applied, it allows the commander to take necessary and prudent risks without arbitrary restrictions while maximizing the application of military capabilities.

E.2 RISK AND OPPORTUNITY AS THEMES PERMEATING THE PLANNING PROCESS

Elements of operational art, such as risk, COG analysis and deconstruction, DPs, and CCIRs are developed and refined throughout the planning and IPOE processes. Choosing which enemy capability or vulnerability to attack provides a broad range of options for use in COA development and presents possible asymmetric opportunities. Likewise, friendly COG analysis provides clarity on capabilities that may be vulnerable to enemy attack. The friendly COG analysis forms the basis for threat identification in the mission analysis' initial risk assessment. As the planning process continues, the risk to friendly forces and overall mission across the developed friendly COAs are analyzed and measured in a cost/benefit comparison. As greater understanding is gained throughout the planning process, mitigation measures are developed to ensure the most effective use of military force in achieving OBJs. During execution, these mitigation measures are implemented and constantly reviewed and refined.

Commanders rely on the planning and execution processes carried out by their staffs to make informed and timely decisions under uncertainty. Risk at the operational and tactical levels is much more than the potential negative effects of enemy capabilities. Equally as important are factors from the strategic and OEs, friendly force readiness, capabilities and capacities, the quality and timeliness of decision making, and C2 capability. The operational commander—with staff support—develops a vision that contemplates the cumulative impact to achieve OBJs and accomplish missions. The relationship between opportunity and risk needs to be understood and balanced by the commander and staff. Anticipating and seizing an opportunity for decisive victory may include taking greater risk in time, space, and/or force.

During the planning process, there are four primary purposes for analyzing risk:

1. Provide the commander greater insight to develop the vision for mission accomplishment,

2. Identify and research knowns and unknowns to better inform and improve the appreciation of the environment,

3. Develop risk mitigation measures,

4. Improve decision making by the commander, staff, and force during execution.

Ultimately, risk consideration in planning transitions to decision making in execution. Commanders and their staffs that have a coherent appreciation of risk should gain much greater clarity on the when of decision making.

E.3 RISK METHODOLOGY

Our forces will be placed in an environment with risk. Ultimately, during Navy planning, the commander envisions making conscious and well-informed decisions on how to deal with threats. Accepting risk is a function of both risk assessment and risk mitigation. The approach to accepting risk entails the following actions: identify and assess threats (risk assessment) and mitigate risk; develop controls and make risk decisions; and supervise and evaluate.

E.3.1 Identify Threats

Identify threats to the mission and force. Consider all aspects of mission, enemy, terrain and weather, troops and support available, time available, and civilian considerations for current and future situations. Sources of information about threats include reconnaissance, intelligence, and experience/expertise of the commander and staff.

E.3.2 Assess Threats

Assess each threat to determine the risk potential based on probability and severity of the threat. Determining the risk is more an art than a science. Use historical data, intuitive analysis, and judgment to estimate the risk of each threat. Probability and severity levels are estimated based on the user's knowledge of probability of occurrence and the severity of consequences once the occurrence happens. The level of risk is assessed by a combination of the threat, its probability of occurring, and degree of severity.

Note

Ideally, the term probability would equate to statistically valid frequency data that has been collected and analyzed thoroughly. Realistically, this is not often possible; requiring the application of subjective professional judgment within the context of the operation to generate a written description of what is really subjective likelihood. Therefore, while the term probability is used as part of risk analysis in recognition that mathematically supportable probabilities are the goals, in most instances it is an approximated likelihood.

E.3.2.1 Risk Assessment Process and Matrix

Figure E-2 illustrates the three-step process for conducting a risk assessment, and Figure E-3 illustrates how risk is categorized by severity and probability of occurrence to assign a risk assessment code.

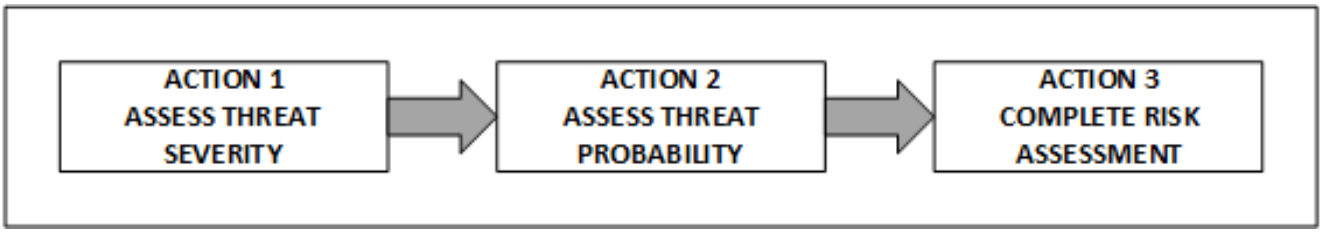


Figure E-2. Conducting a Risk Assessment

		Probability				
Severity		**Frequent A**	**Likely B**	**Occasional C**	**Seldom D**	**Unlikely E**
Catastrophic	I	E	E	H	H	M
Critical	II	E	H	H	M	L
Marginal	III	H	M	M	L	L
Negligible	IV	M	L	L	L	L
E–Extremely High		H–High		M–Moderate		L-Low

Figure E-3. Risk Assessment Codes

The risk assessment matrix combines severity and probability estimates to form a risk assessment for each threat. The risk assessment matrix should be utilized to evaluate the acceptability of a risk and the level at which the decision on acceptability will be made. The matrix may also be used to prioritize resources, to resolve risks, or to standardize threat notification or response actions. Severity, probability, and risk assessment should be recorded to serve as a record of the analysis for future use. This table is only a tool to summarize the risk assessment, understanding the rationale behind the assessment is necessary to fully put the matrix in context for the operation.

E.3.2.2 Risk Assessment Definitions

The following risk assessment definitions can be tailored to differentiate risk to mission or risk to forces.

1. E—extremely high risk: loss of ability to accomplish the mission if threats occur during mission.

2. H—high risk: significant degradation of ability to accomplish mission, inability to accomplish all parts of the mission, or inability to complete the mission to standard if threats occur during the mission.

3. M—moderate risk: expected degraded mission capabilities if threats occur during the mission.

4. L—low risk: expected risk has little or no impact on mission accomplishment.

E.3.2.3 Severity Categories

When developing risk assessments, the severity of potential risks should be considered (see Figure E-4).

Category	Definition
Catastrophic (I)	Loss of ability to accomplish the mission or mission failure. Death or permanent disability. Loss of major or mission-critical systems or equipment. Major property (facility) damage. Severe environmental damage. Mission-critical security failure. Unacceptable collateral damage.
Critical (II)	Significantly degraded mission capability, unit readiness, or personal disability. Extensive damage to equipment or systems. Significant damage to property or the environment. Security failure. Significant collateral damage.
Marginal (III)	Degraded mission capability or unit readiness. Minor damage to equipment or systems, property, or the environment. Injury or illness of personnel.
Negligible (IV)	Little or no adverse impact on mission capability. First aid or minor medical treatment. Slight equipment or system damage but fully functional and serviceable. Little or no property or environmental damage.

Figure E-4. Risk Severity Categories

E.3.2.4 Probability Categories

Figure E-5 identifies and defines elements of risk probability.

E.3.3 Develop Controls and Make Risk Decisions

For each threat, develop options that will mitigate or reduce the risk of the threat. Specify who, what, where, when, and how. Determine any residual risk and revise the evaluation of the level of risk remaining. The commander alone then decides whether or not to accept the level of residual risk. If the commander determines that the risk is too great to continue the mission or a COA, then the commander directs the development of additional measures to account for the risk, or the COA is modified or rejected.

Probability Definitions	
Element Exposed	**Definition**
Frequent (A) Occurs very often, continuously experienced	
Single Item	Occurs very often in service life. Expected to occur several times over duration of a specific mission or operation.
Fleet or inventory of items	Occurs continuously during a specific mission or operation, or over a service life.
Individual	Occurs very often. Expected to occur several times during mission or operation.
All personnel exposed	Occurs continuously during a specific mission or operation.
Likely (B) Occurs several times	
Single Item	Occurs several in service life. Expected to occur during a specific mission or operation.
Fleet or inventory of items	Occurs at a high rate but experienced intermittently (regular intervals, generally often).
Individual	Occurs several times. Expected to occur during a specific mission or operation.
All personnel exposed	Occurs at a high rate but experienced intermittently.
Occasional (C) Occurs sporadically	
Single Item	Occurs some time in service life. May occur about as often as not during a specific mission or operation.
Fleet or inventory of items	Occurs several times in service life.
Individual	Occurs over a period of time. May occur during a specific mission or operation but not often.
All personnel exposed	Occurs sporadically (irregularly, sparsely, or sometimes).
Seldom (D) Remotely possible; could occur at some time	
Single Item	Occurs in service life but is only remotely possible. Not expected to occur during a specific mission or operation.
Fleet or inventory of items	Occurs as isolated incidents. Possible to occur sometime in service life but rarely. Usually does not occur.
Individual	Occurs as an isolated incident. Remotely possible but not expected to occur during a specific mission or operation.
All personnel exposed	Occurs rarely within exposed population as isolated incidents.
Unlikely (E) Can assume will not occur but not impossible	
Single Item	Occurrence is not impossible but can assume will almost never occur in service life. Can assume will not occur during a specific mission or operation.
Fleet or inventory of items	Occurs very rarely (almost never or improbable). Incidents may occur over service life.
Individual	Occurrence is not impossible but may assume will not occur during a specific mission or operation.
All personnel exposed	Occurs very rarely but is not impossible.

Figure E-5. Risk Probability Definitions

The following actions may assist the staff with thinking through options to mitigate risk:

1. Adherence to the principles of joint OPS (mass, OBJ, offensive, security, economy of force, maneuver, unity of command, surprise, simplicity, restraint, perseverance, and legitimacy)

2. Requests for forces, capabilities, or support

3. Modifications to ROE

4. Modifications to command relationships

5. Concept and employment of fires

6. Concept and employment of IO activities

7. Concept and employment of protection

8. Employment of reserve forces

9. Modifications to scheme of maneuver

10. Effective operational assessments.

E.3.4 Implement Controls

Think through the threat: What information will provide indications that the risk is no longer acceptable? Ensure that subordinates and staff are informed of the importance of communicating the status of those indicators or controls and that they are included in the staff's operational assessment measures.

E.3.5 Supervise and Evaluate

In execution, monitor the status of the indicators and enact further options as warranted. After the operation, evaluate the effectiveness of each option in mitigating or eliminating risk.

E.4 APPLYING RISK MITIGATION

Risk mitigation requires a clear understanding of what constitutes unnecessary risk, when the benefits actually do outweigh costs, and guidance as to the command level to make those decisions. When a commander decides to accept risk, the decision should be coordinated with HHQ and other components, partners, and subordinate commands potentially affected; where and how the commander is willing to accept risk are detailed in each COA.

Figure E-6 identifies the risk mitigation steps that should be accomplished during each step of the planning process.

E.5 RISK ASSESSMENT FOR COMMANDERS AND THEIR STAFFS

Commanders seek to gain and maintain initiative, causing the enemy to make mistakes and providing friendly opportunities. The commander promotes calculated, disciplined risk-taking focused on winning rather than preventing defeat—often even when preventing defeat appears safer. Because uncertainty is a normal condition at any level, every decision incurs risk. Risk at the operational/operational-tactical level has characteristics that can be different from risk at lower tactical levels. For example, functional and Service component commands plan and operate as part of a joint force; JFMCC or NCC operational risk is closely linked to other components and the JFC's mission and OBJs at that level.

NPP Steps	Step 1: Identify Threats.	Step 2: Assess Threats.	Step 3: Develop Controls and Make Risk Decisions.	Step 4: Implement Controls.	Step 5: Supervise and Evaluate.
Mission analysis	X	X			
COA development	X	X	X		
COA analysis (wargaming)	X	X	X		
COA comparison and decision			X		
Directive development			X	X	
Transition	X	X	X	X	X

Figure E-6. Risk Mitigation Steps within the Navy Planning Process

Regardless of the level, risk is composed of two distinct, yet interrelated areas:

1. Risk established at the HHQ and delegated to its subordinates

2. Risk shared between adjacent organizations (functional or Service component commanders or shared between TF commanders).

Each of these risk areas needs to be analyzed as it relates to the mission and OBJs and, as discussed earlier, to friendly and enemy COGs. In the initial design of an operation, establishment of the broad operational approaches will provide better clarity on potential intersection of enemy threats and friendly LOOs. During mission analysis and COA development, the commander will provide more details on how risk is distributed to adjacent and subordinate organizations. The supported/supporting functional, Service component commanders or CTFs may be directed to mitigate or accept risk IOT comply with the commander's method to attack the enemy. A supporting commander may incur a higher risk to place the enemy at a disadvantage elsewhere. The supported commander may identify a requirement to the supporting commander that will result in accepted risk. The JFMCC would then conduct normal risk assessment and mitigation.

E.6 DISPLAYING RISK ASSESSMENT AND MITIGATION FOR THE COMMANDER

There are many options for displaying risk assessment and mitigation for a commander. Figure E-7 provides examples of initial risk assessments and Figure E-8 provides an example of how risk mitigation can be communicated. Staffs should appreciate that every commander may desire a different method for displaying risk—these examples are only provided as a starting point from which a staff can deviate.

Risk, therefore, is a factor that is present in nearly every aspect of military actions whether at the tactical or operational level. Operational commanders need to consider where risk is located and where risk needs to be accepted. The following is a checklist of JFMCC, NCC, numbered fleet commander (NFC), or designated TF commander risk areas that should be considered:

1. What risk has been accepted by HHQ that directly impacts the JFMCC, NCC, NFC, or designated TF commander?

2. Assumptions identified during planning that have not been validated, such that they will carry through execution, should be considered risks and mitigated accordingly.

Risk to Mission—Assessment			
Phase I	**Severity**	**Probability**	**Risk**
Terrorist attack(s) against Redland	Critical	Likely	High
Mining of AOA or Pinkland SLOCs	Catastrophic	Occasional	High
Preemptive attack by Redland conventional forces	Critical	Unlikely	Low
Phase II	**Severity**	**Probability**	**Risk**
Loss of CVN	Critical	Seldom	Moderate
Loss of LHA/LPD/LSD before amphibious assault	Catastrophic	Occasional	High
Loss of CG/DDG/SSN	Marginal	Occasional	Moderate
Risk to Force–Assessment			
Phase I	**Severity**	**Probability**	**Risk**
CDCM threat	Catastrophic	Unlikely	Moderate
Mine threat	Critical	Seldom	Moderate
Theater ballistic missile (TBM) threat to Pinkland ISB	Catastrophic	Seldom	High
Phase II	**Severity**	**Probability**	**Risk**
Coordinated CDCM and ASCM attack by Redland forces	Catastrophic	Frequent	Extremely high
Sub attack on high value units	Catastrophic	Likely	Extremely high
Mine threat during strait transit	Critical	Likely	High

Figure E-7. Example of Initial Risk Assessments

Risk to Mission–Mitigations					
Phase I	**Severity**	**Probability**	**Initial Risk**	**Mitigation Actions**	**Residual Risk**
Terrorist attack(s) against Redland	Critical	Likely	High	• Conduct ISR and MSO to identify and intercept potential terrorists.	Moderate
Mining of AOA or Pinkland SLOCs	Catastrophic	Occasional	High	• Monitor mine storage facilities. • Conduct ISR of potential minelayers.	Low
Preemptive attack by Redland conventional forces	Critical	Unlikely	Low	• Monitor all indicators of possible preemptive strike.	Low

Figure E-8. Example of Final Risk Assessments and Risk Mitigation (Sheet 1 of 2)

Risk to Mission–Mitigations					
Phase II	**Severity**	**Probability**	**Risk**		**Risk**
Loss of CVN	Critical	Seldom	Moderate	• CVN remains clear of known threats.	Low
Loss of LHA/ LPD/LSD before Amphib Assault	Catastrophic	Occasional	High	• Amphibious forces remain clear of AOA until threats neutralized.	Low
Loss of CG/ DDG/SSN	Marginal	Occasional	Moderate	• Neutralize/destroy surface, subsurface, and CDCM threats.	Moderate
Risk to Force–Mitigation					
Phase I	**Severity**	**Probability**	**Risk**	**Mitigation Actions**	**Risk**
CDCM threat	Catastrophic	Unlikely	Moderate	• Monitor CDCM sites. • Avoid potential Redland targeting platforms. • Avoid CDCM engagement arcs.	Low
Mine threat	Critical	Seldom	Moderate	• Monitor mine storage facilities. • Conduct ISR of potential minelayers. • Obtain ROE to neutralize/destroy vessels laying mines in international waters.	Low
TBM threat to Pinkland ISB	Catastrophic	Seldom	High	• Position IAMD sensors and units to defeat threat. • Disperse LOG footprint.	Low
Phase II	**Severity**	**Probability**	**Risk**		**Risk**
Coordinated CDCM and ASCM attack by Redland forces	Catastrophic	Frequent	Extremely High	• Neutralize/destroy Redland NAVFOR CDCM threats.	Low
Sub attack on high value units	Catastrophic	Likely	Extremely High	• Conduct ASW against deployed Redland navy subs. • Neutralize/destroy Redland naval base.	Moderate
Mine threat during Strait Transit	Critical	Likely	High	• Neutralize/destroy potential Redland navy minelayers. • Destroy Redland mine depot.	Moderate

Figure E-8. Example of Final Risk Assessments and Risk Mitigation (Sheet 2 of 2)

3. During parallel planning, what additional risks are introduced or identified during COA development, analysis, comparison, and the developed CONOPS?

4. As a supporting commander, what risk has been incurred to support a different functional component commander? Can these risks be mitigated?

5. As a supported commander, what risk is accepted in the JFMCC, NCC, NFC, or designated TF commander CONOPS that impacts the other functional component commanders? Is there a mitigation method available? What risk mitigation actions are the other functional component commanders providing?

6. As a higher commander, what risks are accepted that are passed to subordinate tactical commanders? How will these risks be mitigated?

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APPENDIX F
War-game Guide

F.1 PURPOSE

This guide provides a technique to execute and prepare a manual war game at the operational level of war. It provides the planning team lead a guide for directing planning teams and conducting effective war games.

F.2 GENERAL

This guide is organized into two sections:

1. **Section One—Organize the Effort:** Describes roles and responsibilities and provides checklists that aid in organizing people and the venue prior to war-game initiation (see Figure F-1).

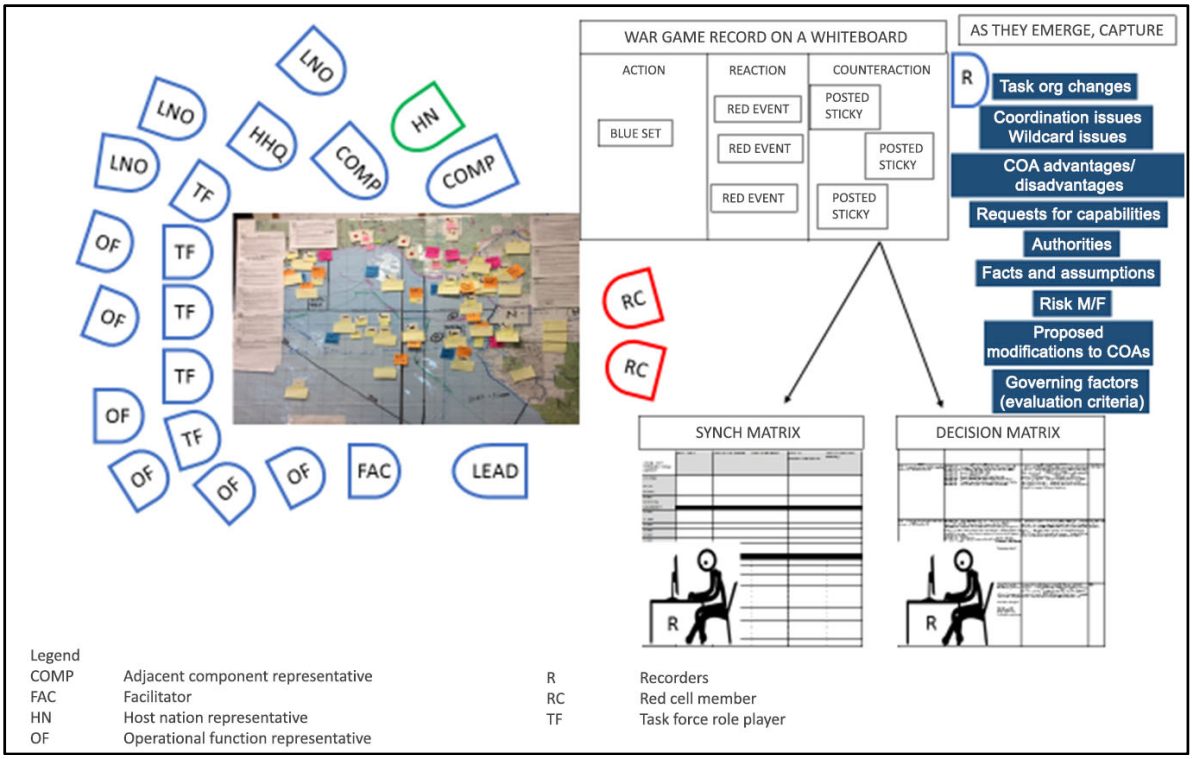


Figure F-1. Notional War-game Setup

2. **Section Two—Conduct the War Game:** Provides a notional script to initiate the war game and guide war-game turns.

SECTION ONE—ORGANIZE THE EFFORT

F.3 IDENTIFY ROLES AND RESPONSIBILITIES

The following information describes how different role players prepare and execute their roles in war games. War-game organization depends upon the nature and complexity of the COAs to be analyzed and the number of

participants available. Not all roles need to be filled, and some participants may fill multiple roles. Figure F-2 (which follows the descriptions of roles and responsibilities) depicts a notional worksheet to be used to assign roles and responsibilities for the war game. Figure F-3 depicts notional room ergonomics for setting physical conditions of war gaming.

War-game Assignments

Position	Names
Planning team lead	
Facilitator	
Blue COA representative	
Red cell representative	
Recorder: RFF/RFC	
Recorder: task organization	
Recorder: sketch and narrative	
Recorder: decision support products (DST/M)	
Recorder: risk	
Recorder: authorities	
Recorder: facts and assumptions	
Recorder: requests for information	
Recorder: war-game record (THEN synchronization matrix)	
Recorder: insights gained for information brief	
Recorder: evaluation criteria	
Recorder: coordination gaps and issues	
Recorder: information for COA analysis brief slides	
Higher and adjacent component role players	
TF XXX	
TF XXX	
TF XXX	
TF XXX	
TF XXX	
TF XXX	
TF XXX	
C2 Function	
Intelligence function	
Fires function	
Sustainment function	
Protection function	

Figure F-2. War-game Role Assignments Worksheet (Sheet 1 of 2)

Position	Names
IO function	
Operational assessment	
Special staff (judge advocate general, PA officer, etc.)	
Regional role player impacting the COA (as required)	
HN liaison	
Other partner nation liaison (as required)	
Other government agency liaison (as required)	
Subject matter expert (as required)	
Nongovernment entities (as required)	

Figure F-2. War-game Role Assignments Worksheet (Sheet 2 of 2)

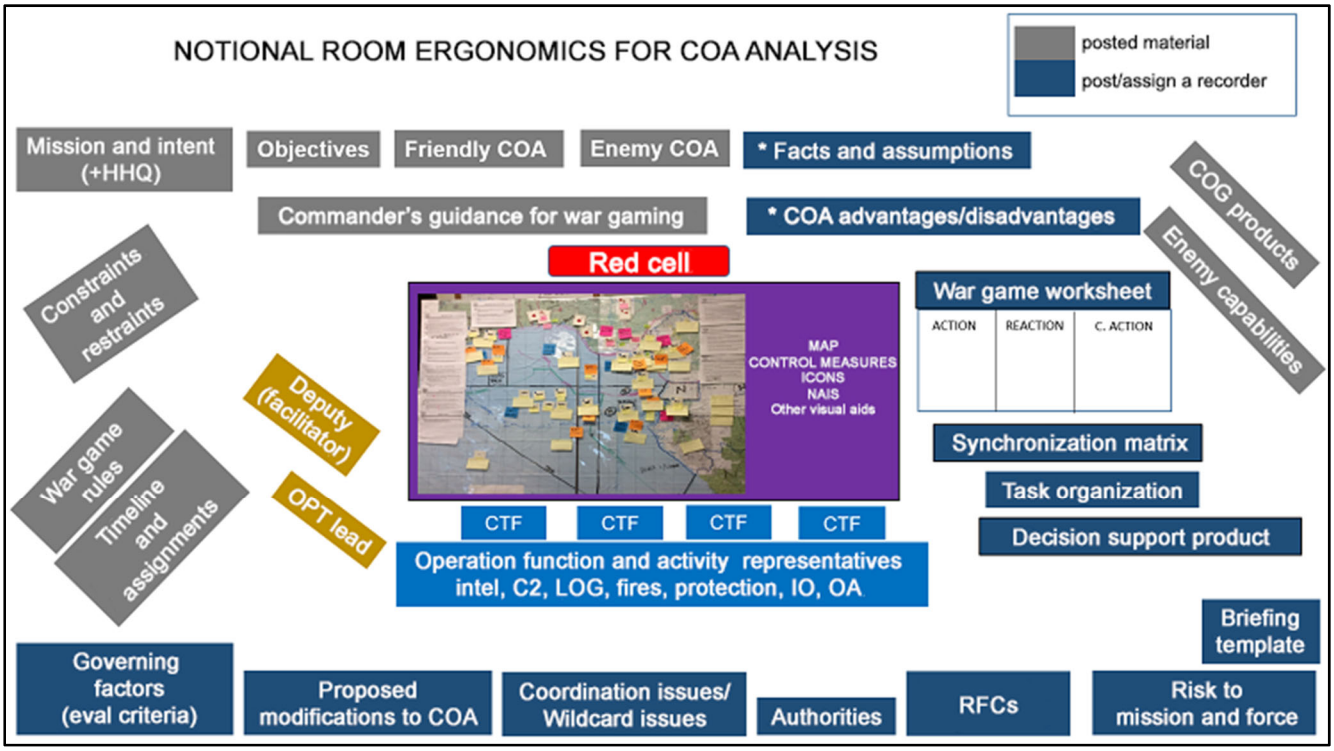


Figure F-3. Room Ergonomics

F.3.1 All Participants

1. Prepare.

   a. Know the mission, commander's intent and guidance, and OBJs.

   b. Know the war-game method and expectations.

   c. Know the timeline and agenda.

d. Know the rules of the war game and acceptable conduct.

e. Know the responsibilities and requirements of your role.

f. Know how to record outcomes for your position.

g. Know both the friendly and enemy COA being analyzed.

h. Know the outcomes of design, mission analysis, and COA development to this point in the process.

i. Know the task organization and command relationships of the force during every turn.

j. Obtain a copy of the COA sketch and statement, synchronization matrix, task organization, and decision support products.

2. Execute.

a. Remain OBJ and do not allow personality or a sense of what the commander wants influence participation.

b. Be anticipatory in your responses by maintaining awareness during turns.

c. Do not distract from the analysis being conducted.

d. Keep cogent notes—take note of advantages and disadvantages of each COA as they emerge.

e. Continuously assess feasibility, acceptability, and suitability of each COA.

f. Avoid drawing premature conclusions, and gather facts to support conclusions.

g. Ensure you record outcomes in your area of responsibility.

h. Maintain awareness during war-game turns so you can explain how your function contributes to and supports friendly actions and counteractions.

i. Record any adjustments, and ensure they are reconciled at some point with recorders.

F.3.2 Planning Team Lead

1. Prepare.

a. Clarify any ambiguities in the planning guidance.

b. Define COA analysis outcomes (specific expectations).

c. Prepare the agenda and timeline.

d. Develop the rules for the war game and individual conduct.

e. Select the war-game method.

f. Establish war-game recording requirements.

g. Assign roles. Break up like roles; that is, the intelligence functional representative should not be the MPRA TF representative.

h. Coordinate support from liaisons, staff, and subject matter experts as required.

i. Oversee room setup (ergonomics, information posted, recording tools, etc.)

j. Have current authorities and permissions available for refinement.

k. Know the current risks associated with each COA.

2. Execute.

a. Direct execution of the war game; reiterate war-gaming guidance; describe the process and its outcomes; maintain oversight of actions.

b. Encourage dialogue.

c. Anticipate and coordinate timely support from liaisons, staff, and other subject matter specialists.

d. Adjust the pace and tempo of war-game turns based on remaining time available.

e. Arbitrate any impasses; decide disputes to allow turn tempo to continue.

f. Be prepared to directly facilitate a challenging aspect of analysis.

g. Maintain awareness of the emergence of possible decision points during turns.

F.3.3 Facilitator

Note

The planning team lead carries out these functions when a facilitator is not specifically designated.

1. Prepare.

a. Become familiar with all aspects of the COA, mission, intent, visual information references, agenda, timeline, and authorities.

b. Help develop the rules for conducting the war game.

c. Assist with the agenda for war gaming.

d. Ensure the posting of visual informational references pertinent to the conduct of the war game (complete room ergonomics setup).

e. Coordinate with the planning lead to ensure that charts, maps with control measures, unit icons, and recording methods are in place.

f. Have current authorities and permissions available for refinement.

2. Execute.

a. Direct the actions of the planning team.

b. Facilitate the execution of war-game turns.

c. Execute the war-game agenda by timeline and turn, fully cognizant of the required outcomes.

d. Monitor time allotted for each turn and adjust pace and tempo, as necessary.

e. Enforce the rules for group and individual conduct.

f. Facilitate dialogue and debate.

g. Ensure required information is recorded to drive outcomes.

F.3.4 COA Representative

1. Prepare.

   a. Know the mission and commander's intent.

   b. Know the task organization and command and support relationships.

   c. Know the facts and assumptions.

   d. Know major risks to mission and force.

   e. Know current authorities and supplemental ROE.

   f. Prepare to orient all to the map board, current joint and maritime AOs, current control measures, major geographical points, and current NAIs on the map.

   g. Know the known decision points, if any, and associated CCIRs.

2. Execute.

   a. Be prepared to explain tasks during initiation and be able to do the same during counteraction (depending on how the effort is organized).

   b. Guide the group during action to address:

      (1) Facts and assumptions.

      (2) Decision points and PIR/FFIR adjustments.

      (3) Evaluation criteria.

      (4) Additional authorities and supplemental ROE.

      (5) Residual risk.

F.3.5 Recorders

Note

Assign a recorder for each recording tool. A recorder may be responsible for multiple recording tools.

1. Prepare.

   a. Know the requirements of your role.

b. Know recording formats IOT direct information or assist in recording information in all logic tools. Recording tools for each game include:

(1) Planning guidance (anticipate new guidance for COA comparison and decision).

(2) Synchronization matrix (specific details spanning forces and functions are noted).

(3) DST and DSM (that identify decision points, decision options, the associated timing, CCIRs, NAI, and supporting criteria).

(4) Evaluation criteria for COA comparison and decision.

(5) Mission (wording may evolve for clarity).

(6) Intent (wording may evolve for clarity).

(7) Task organization (may evolve as tasks or efforts change).

(8) Sketch (may change as decisive, shaping, and sustaining actions are gamed).

(9) Narrative (may be adjusted as the MEs, SEs, SHs, and SUSs possibly evolve).

(10) Authorities and permissions (record new authorities as identified).

(11) Limitations (as the need to remove or to add limitations is identified).

(12) Risk matrix (new risks and changes to residual risk are noted).

(13) Facts and assumptions (changes to facts and assumptions as noted).

(14) Gaps in capabilities (shortfalls as noted). RFF?

(15) RFIs (information gaps noted).

(16) Coordination gaps are noted.

(17) Wildcard information or issues that do not fit neatly into any other category.

(18) Advantages and disadvantages (insights on inherent strengths and weaknesses of the COA are noted).

(19) Insights gained as the result of war-gaming turns.

c. Prepare digital recording templates for any standard and specialized requirements in advance of the analysis.

d. Review formats for brief templates associate with the assigned role.

2. Execute.

a. Record with due diligence the information gleaned from dialogue and debate.

b. Do not skew information by putting your own spin on it.

c. Attain clarity from participants by what was meant and the appropriate wording to convey the meaning.

d. Translate recorded data into brief slides.

F.3.6 Red Cell

1. Prepare.

   a. Know IPOE outcomes.

   b. Know the enemy's capabilities, requirements, vulnerabilities, doctrine, current disposition, and most likely and most dangerous COAs to provide a credible thinking enemy; review and have enemy COG data available.

   c. Know the friendly capabilities and COAs used to support COA analysis; review and have friendly COG data available.

   d. Understand the war-game rules and remain within those boundaries.

   e. Anticipate your first reactions to the first war game as you will be familiar with initial friendly actions.

   f. Prepare to brief each COA and the reconnaissance and surveillance (R&S) efforts that support it.

   g. Based on the commander's guidance, prepare candidate reactions that anticipate apparent weaknesses or critical vulnerabilities (from COG analysis) in the friendly COAs or which contribute to achieving enemy OBJs within the enemy COA.

   h. Strive not to mirror friendly tactics, but think through your actions as the opposition to achieve your OBJs based on known information about enemy ways.

2. Execute.

   a. Brief the enemy situation and COA for each war game, and as necessary, rebrief for each turn.

   b. Employ the directed enemy COAs against the friendly COAs to pursue perceived enemy OBJs and friendly vulnerabilities; fight your force capabilities in reaction to friendly force actions in a manner consistent with the enemy COAs analyzed.

   c. Clarify rules (boundaries) regarding enemy reactions during games and turns.

   d. Calculate your reactions based on a short review of friendly actions, your OBJs, and both enemy and friendly vulnerabilities.

   e. Demonstrate, through your reactions, emerging gaps in friendly actions or synchronization, gaps in risks, unprotected vulnerabilities, and considerations in time and space that bring cogent critical thought to friendly counteractions.

   f. Ensure that reactions are recorded for accuracy and reconciliation with recorders.

F.3.7 Higher Headquarters and Adjacent Component Representative(s)

1. Prepare.

   a. Become familiar with their organization's current task organization, capabilities, the major operational plan, and actions pertinent to the COAs being analyzed.

   b. Understand their commands' CONOPS as it applies to each war-game turn.

   c. Know their MEs and SEs and those of their HHQ.

d. Know their command and support relationships and their purpose.

e. Know location of adjacent forces, if appropriate, and their main tasks and purposes.

2. Execute: Explain the information above when called upon during the turn.

F.3.8 Subordinate Command (Task Force) Representatives

Note

Throughout the description of the war-gaming process, TF refers to assigned forces and subordinate commands' roles in the COA.

1. Prepare.

a. Know the task organization, command level, and capabilities of their TF.

b. Be familiar with potential assigned tasks, associated purposes, and the location of those tasks supporting the particular COA.

c. Be familiar with the developed COA in terms of force, space, time, command and support relationships, MEs and SEs, risk, and enemy COAs.

d. Know the TF's assigned tasks and purposes as it relates to the turn.

e. Know the present and proposed authorities and permissions.

2. Execute.

a. Maintain awareness during war-game turns IOT explain the TF's tasks and associated purposes and their locations when called upon.

b. Be prepared to explain the TF's tasks and purposes and how the TF contributes to the COA.

c. Explain your TF actions and counteractions during war-game turns as required.

d. Serve as the TF's movement and maneuver representative during war-game turns.

e. Record any adjustments to tasks assigned to you, task organization, risk, resource shortfalls, and RFIs, and ensure they are reconciled at some point with recorders.

F.3.9 Operational Function Planning and Special Staff Representatives (General)

1. Prepare.

a. Be familiar with the developed COA in terms of force, space, command and support relationships, MEs and SEs to be war-gamed, and enemy COAs against which to war-game.

b. Bring your partially developed staff estimate for reference.

c. Know the current and anticipated authorities and permissions.

2. Execute.

a. Maintain awareness during war-game turns and be ready to explain how your function contributes to and supports friendly actions and counteractions.

b. Record any adjustments and ensure they are reconciled at some point with recorders.

c. Capture refined information on the synchronization matrix based on specific outcomes.

d. Record strengths and weaknesses of each COA as they pertain to your function.

F.3.10 Intelligence Planner

Note

See red cell for intelligence analyst role.

1. Prepare: Identify intelligence OPS and requirements that support the COA being war-gamed.
2. Execute.

   a. Provide the intelligence overview.

   b. Discuss intelligence OPS related to NAIs and target areas of interest that are associated with PIRs.

   c. Update IRs from war-gaming turns as decisions or additional information emerges.

F.3.11 Movement and Maneuver Planner

1. Prepare: Understand the overall scheme of maneuver and expected movements of subordinate forces.
2. Execute.

   a. Role-play friendly maritime forces.

   b. Capture emerging decision points, corresponding CCIRs, timing of decisions, location if required, conditions that bear on the decision, and potential branch plans and sequels.

   c. Record the COA's strengths and weaknesses.

F.3.12 Fires Planner

1. Prepare.

   a. Study the draft scheme of fires with target sets and related fires effects as a result of target development and prioritization which support maneuver aspects of the operation.

   b. Understand the planning guidance, HHQ targeting guidance, apportionment effort, target lists (e.g., time-sensitive targets), and each COA.

2. Execute.

   a. Assess each COA's ability to provide supporting fires.

   b. Capture emerging decision points, corresponding CCIRs, timing of decisions, location if required, conditions that bear on the decision, and potential branch plans and sequels.

   c. Adjust the scheme of fires as necessary including those based on refinements in movement and maneuver.

F.3.13 Sustainment Planner

1. Prepare.

   a. Ensure that personnel (N1) and LOG (N4) provide representatives to the war game.

   b. Develop a COLS[22] for each COA.

   c. Assess the personnel aspect of building and maintaining individual and unit manning requirements.

2. Execute.

   a. Estimate potential personnel losses and assess the adequacy of human resources support for the operation.

   b. Ensure mission execution is supported across the full spectrum of sustainment functions throughout all phases of an operation.

   c. Identify potential shortfalls, and recommend options to ensure units maintain adequate resources to accomplish their mission.

   d. Provide analysis of LOG feasibility; identify potential supply, transportation, and sustainment issues; and assess the LOG functions that must be conducted to support the COA.

F.3.14 Command and Control Planner

1. Prepare.

   a. Understand each COA's required authorities, task organization, command and support relationships, communications, task synchronization, coalition force integration, battlespace geometry, and airspace control and coordination.

   b. Understand the required coordination with international organizations, NGOs, and other governmental agencies.

2. Execute: Identify C2 functions which should be modified.

F.3.15 Protection Planner

1. Prepare.

   a. Determine the essential operational protection task(s), protection capability shortfalls, and priorities for employment of protection capabilities for each phase.

   b. Develop the proposed critical-asset list for each phase.

2. Execute.

   a. Observe TF representatives as they describe their force composition and assigned tasks.

   b. Ensures critical assets have protection.

   c. Discuss all remaining protection shortfalls (forces or authorities) with war-game participants to determine risk mitigation.

---

[22] The COLS states how an operation is sustained through all phases by identifying the key requirements, available resources, and sourcing options for resupply. The COLS explains how all key LOG functions will be synchronized in the operation. NTTP 5-01.4, Navy Planning, Logistics, describes how the COLS is developed.

F.3.16 Information Operations Planner

1. Prepare: Become fully aware of all IRCs available, potential target audiences, desired effects, and all associated IRC tasks and purposes (the IO representative will have an appreciation for how all IO tasks are sequenced and synchronized within each COA analyzed).

2. Execute.

   a. Describe in broad terms how IO actions are employed by addressing the timing, key tasks and purpose, target audience, and location(s) of those actions.

   b. Ensure that key IO tasks are captured in the synchronization matrix.

   c. Determine counteractions to enemy reactions in coordination with IRC representatives and make any necessary changes, refinement, or additional actions to improve IO's contribution to the COA.

F.3.17 Operation Assessment Planner

1. Prepare: Ensure understanding of the draft assessment collection plan, with draft measures of effectiveness (MOEs) and performance that measure force effectiveness and efficiency toward creating desired operational effects and OBJs.

2. Execute.

   a. Observe war-game moves to ascertain adjustments to effects, OBJs, and tasks IOT refine measures and collection actions that contribute to the assessment plan.

   b. Provide recommendations to the war-game group when task-effect-OBJ inconsistencies emerge.

F.3.18 Host Nation Representative (Optional)

1. Prepare.

   a. Know HN military capabilities and limitations.

   b. Know host national space.

   c. Know how the HN integrates with the friendly COA.

   d. Know the resources the HN may provide during the course of the war game.

2. Execute.

   a. Know how the HN may be able to contribute as part of friendly forces.

   b. Record any contributions and ensure they are reconciled at some point with recorders.

F.4 ORGANIZE THE ROOM

1. Prepare the room. The venue to conduct the war game should be large enough to accommodate required information displays, all war-game players, and technical support. Figure F-3 provides an example war-game room layout. Shipboard space limitations often play a major factor in the layout. Key considerations are:

   a. Ensure that all applicable planning guidance is readily accessible and visible for reference when needed.

b. Use a paper (physical) map or chart to facilitate dialogue and ensure common understanding.

c. Keep the recording method simple and flexible. Sticky notes tend to work best.

d. Have computer workstations available to permanently record observations, and update the plan and supporting products as decisions and follow-on actions emerge.

2. Prepare the agenda and timeline. The war-game agenda, based on commander's guidance and available time, should be posted and readily available. Each war game consists of a series of turns, and each turn normally consists of three moves: action (friendly), reaction (enemy), and counteraction (friendly, based on enemy reaction). Sufficient time should be allotted to every turn to ensure each war game can be completed. The timeline can include steps to initiate each war game, conduct turns of each war game, refine products, make modifications to the COAs, and prepare for an optional brief.

3. Gather tools and known products. Gather the necessary tools, materials, and data for war-game turns. Planning teams can war-game using maps and charts, computer simulations, and other tools that accurately reflect the AO. The planning team posts the COA on a map/chart (printed or electronic) with graphic control measures and necessary unit icons. Other data, information, and place holders that can either be posted or held in electronic form can include, but are not limited to:

a. A display of information as the result of mission analysis: higher and own mission and intent, OBJs, constraints and restraints, COG products, and friendly and enemy capabilities.

b. A display of information gleaned from COA development: friendly COAs and task organization, enemy COAs, updated facts and assumptions, updated CCIRs, recommended authorities, operational risk, commander's guidance for war gaming, capability shortfalls, and evaluation criteria.

c. Chart or map of JOA/OA (either paper or digital) with control measures and unit icons.

d. The COA synchronization matrix (electronic and paper).

e. The COA DST and DSM (electronic and paper).

f. A blank war-game worksheet prominently displayed that depicts columns for action, reaction, and counteraction.

g. The optional brief template (electronic and paper).

h. Coordination issues placeholder (electronic and paper).

i. Event template (if produced by the intelligence staff).

j. Updated staff estimates.

F.5 REVIEW COURSES OF ACTION, PLANNING GUIDANCE, AND SUPPORTING INFORMATION

1. Review friendly forces and their capabilities.

a. Consider all units that can be committed to the operation, paying attention to task organization and command and support relationships.

b. Review the RCPA (friendly force strengths and weaknesses).

2. Review enemy forces and their capabilities.

a. Consider all enemy units that can be committed to the operation, paying attention to capabilities and potential employment of the enemy force.

b. Review the enemy COGs.

c. Review the RCPA (enemy force strengths and weaknesses).

3. Review facts and assumptions.

a. Review facts and assumptions to determine continued validity.

b. Consider impacts of assumptions as they relate to the COAs and the consequences regarding them.

4. Review critical events/essential tasks.

a. Review and confirm critical events/essential tasks in each approved COA.

b. Review decision points[23] associated with each critical event/essential task.

c. Associate critical events/essential tasks to CCIRs, locations, timing/phases, and conditions (decision criteria).

F.6 SELECT THE WAR-GAME METHOD

The planning team lead selects the war-game method to be used based on the overall situation and circumstances. Chapter 4 describes war-game methods. Each technique is suited to different requirements, time available, and the situation.

F.7 SELECT METHODS TO RECORD AND DISPLAY RESULTS

The planning team records the results of war-game turns to synchronize each COA across time, space, purpose, functions, and activities (i.e., OPS assessment). Recorders are assigned as required to capture additional information as it emerges from war-game turns. There are several techniques to capture information and display results.

1. War-game worksheet.

a. Typically displayed on one or more large whiteboards.

b. Provides a visual reference during the war game that displays action, reaction, and counteraction columns visible to all participants in the room.

c. Functions as the primary collection point for actions, reactions, and counteractions.

d. Friendly action, enemy reaction, and friendly counteraction discussion outcomes are affixed to the worksheet for all to see.

e. Outcomes are transcribed into the synchronization matrix and decision support products (see Chapter 4).

2. Synchronization matrix.

a. Typically recorded in a computer spreadsheet.

b. Records results captured in war-game worksheets.

[23] **decision point**—A point in space and time when the commander or staff anticipates making a key decision concerning a specific course of action. (DOD Dictionary. Source: JP 5-0).

c. May include additional information, such as other entities in the OE (e.g., HN and other government or nongovernment agencies), authorities, resource shortfalls, or coordinating instructions.

d. Unique to the nature of the environment and mission.

3. Decision support products.

a. Establish links between decision points, CCIRs, and decision options.

b. DST: combined intelligence and OPS graphic that depicts geographic locations where CCIR-related events are expected to occur.

c. DSM: matrix associated with a DST that lists decision points and their corresponding CCIRs, decision options, decision support criteria, and associated geographic area(s).

4. Time-constrained technique.

a. Graphic portrayal of the AO and a series of sequential notes that are individual worksheets.

b. Broader in nature than the synchronization matrix.

c. Used to condense the war game when time is a constraint.

d. See Figures 4-6 and 4-7.

5. Additional war-game records (noninclusive).

a. Strengths (advantages) and weaknesses (disadvantages) of each COA based on its merits.

b. Additional or refined evaluation criteria.

c. Risk assessment.

d. COA sketch (control measures) and statement modifications.

e. Task organization changes.

f. Residual gaps in coordination or synchronization.

g. Facts and assumptions.

h. Authorities.

i. Other issues that do not fit into any other product.

j. Information pertinent to an optional briefing.

SECTION TWO—CONDUCT THE WAR GAME AND ASSESS RESULTS

F.8 INITIATE THE WAR GAME

This portion explains how to set the conditions for war-game turns. It should be repeated at the start of each new war game. It is not repeated for each turn within the war game.

1. Planning lead:

a. Call the war game to order.

b. Take roll call.

c. Review the timeline and agenda.

d. Explain the selected war-game method to be used.

2. Facilitator: Explain the war-game rules. At a minimum, these may include:

a. Take all direction from the facilitator.

b. Be prepared to change the tempo of war-game turns based on remaining time available.

c. Everyone is a recorder.

d. The lead is the deciding authority on all matters.

e. Be prepared to yield the floor during turn discussions within steps to gain different perspectives of improvement considerations and their impacts.

f. Stay professional.

Note

Each role player requires a copy of the current synchronization matrix and task organization regardless of the degree to which it has been developed. Each friendly force player should be familiar with the unit locations, tasks, purposes, and priorities unique to that player's role during the turn.

3. COA representative:

a. State the mission and commander's intent.

b. Orient participants to the JOA; explain the space that constitutes the current joint and maritime areas of operation; explain current control measures, major geographical points (NAIs), and corresponding DPs.

c. Provide an overview of the COA task organization for the first turn.

d. Explain current authorities/permissions.

e. Review facts and assumptions.

f. Review risks applicable to the turn.

g. Explain current decision points, if any, and associated PIRs/FFIRs (CCIRs) from the synchronization matrix.

h. Review current evaluation criteria.

4. Intelligence representative: Explain the friendly R&S plan.

Note

Reference geographical NAIs when explaining the friendly R&S plan.

5. Red cell representative: Explain the enemy COA considered and the enemy R&S plan.

F.9 CONDUCT A WAR-GAME TURN

Each war game normally has a set number of turns. Each turn consists of three moves unless special circumstances dictate additional moves (see Figure F-4.).

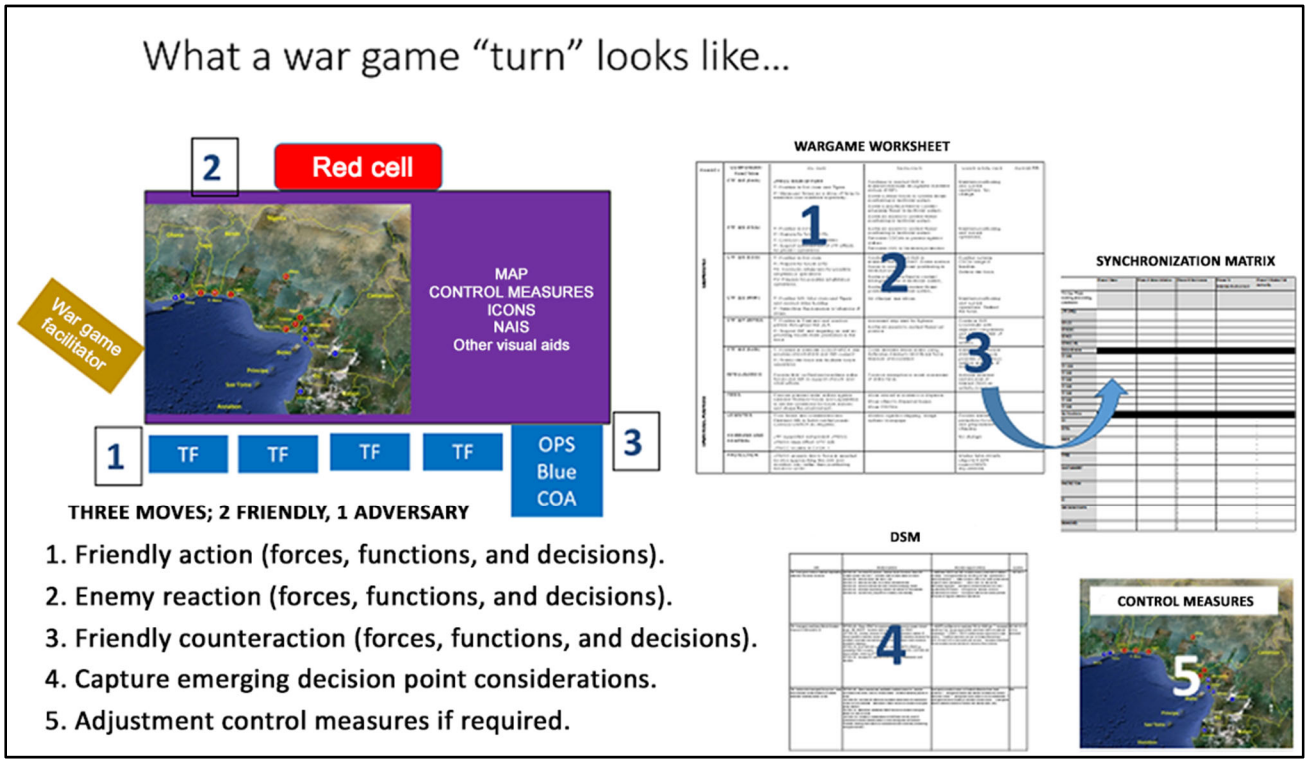


Figure F-4. Notional Moves in a War Game Turn

F.9.1 Friendly Action

1. Facilitator: Guide the group through friendly actions to be analyzed (as determined by the war-game method) and provide cues for each role player.

2. HHQ and adjacent command representative(s): State HHQ and adjacent component CONOPS and support relationships applicable for this turn. Each role player representing higher and adjacent components explains a condensed version of that component's CONOPS during the turn. Explain the main task and purpose of each component for the turn.

Note

Consider this order to present higher and adjacent CONOPS: HHQ, ME component, other components.

3. COA representative: State the commander's OBJ for the current war-game turn. State the starting and ending conditions (or time) of the turn based on the construct selected (e.g., critical event, time, phase).

4. Red cell representative: State the disposition of enemy forces across the JOA applicable to maritime OPS.

5. Task force representatives: State the following, starting with the ME, followed by each SE to the ME, then SHs, and then SUSs, in that order:

a. Explain how the TF is organized.

b. Explain command relationships. Who has OPCON or TACON of the TF's resources?

c. Indicate TF locations (OA(s)).

d. Explain assigned tasks and their related purposes germane to the particular turn per the synchronization matrix.

6. Operational function representatives: Each functional representative states their role and effort priorities for the current turn that support MEs and SEs.

Note

During subsequent war game turns, the planning lead may direct the COA representative to explain the friendly set (disposition, location, tasks, and purposes) in the interest of time. This does not relieve TF representatives of their responsibilities to role-play their TF during counteraction.

7. COA representative: Guide the group to explain:

a. Facts and assumptions.

b. Decision points applicable to this turn.

c. CCIRs applicable to this turn.

d. Current evaluation criteria.

e. Current and proposed authorities and supplemental ROE.

f. Applicable risks.

8. Red cell representative: Ask questions needed to clarify friendly disposition or intention.

9. Facilitator:

a. Announce an administrative pause to ensure that any information that requires updating can occur.

b. Announce the amount of time in minutes allocated to the red cell to develop reactions to friendly dispositions and actions.

c. Ensure that the war-game record captures information that must be incorporated into the synchronization matrix.

F.9.2 Enemy Reaction

1. Facilitator: Call the war game to order and guide the group to ensure that enemy reactions presented by the red cell are understood and consistent with enemy COAs to be considered.

Note

In considering combat losses, potential losses during reaction or counteraction moves of a turn should focus on improvements to the COA and not accounting for potential losses. This effort could consider adjusting the scheme of maneuver, additional tasks and purposes, task organization adjustments, identifying capability shortfalls, improving risk, control measure adjustment, or addressing vulnerabilities to prevent or

mitigate losses. It should not be subjective speculation of numbers and type of losses on either side, as this can introduce bias and detract from improving the COA. The impact of potential losses is addressed later as branches or sequels.

2. Red cell representative:

   a. Explains enemy OBJs and actions to pursue them.

   b. Explains enemy reactions to take advantage of critical friendly vulnerabilities.

   c. Ensures reactions are recorded and posted to the map and eventually the war-game record.

   d. Cages responses in terms of achieving their OBJs and/or acting on friendly critical vulnerabilities (as noted in COG analysis).

3. Facilitator: Announce an administrative pause for participants to form counteractions.

4. Task force representatives:

   a. Ensure understanding of red cell reactions before pausing to consider counteractions.

   b. Discern responses to enemy reactions.

   c. Couch results in terms of tangible outcomes. Gather as a group during the pause to discern counteractions to red cell reactions.

   d. Use critical and creative thinking as well as the implicit experience of the group.

Note

When developing tangible outcomes as a result of red reactions, consider:

- Location and orientation adjustments.
- Control measure adjustments.
- Tasks and purpose refinement.
- Risk adjustment (mitigate and translate to additional tasks).
- Identify coordination gaps.
- Identify capability shortfalls.
- Adjustments in priorities.

5. Facilitator: Ensure that the war-game record captures information that must be incorporated into the synchronization matrix.

F.9.3 Friendly Counteraction

Note

During counteraction, decision points may emerge based on the response to red cell reactions. These decision points should be captured in the synchronization matrix, along with related identified PIR/FFIR, timing, and supporting criteria associated with the decision point. If a DST/DSM recorder has been designated, ensure that the recorder captures the initial information. Additionally, capture the conditions that

existed when the decision point emerged to populate the synchronization matrix and subsequently the DSM. Do not attempt to fully provide feasible and suitable options (branches, sequels, or lesser adjustments) during this time. If obvious options present themselves, note them in the synchronization matrix, DST, and DSM as appropriate.

1. Facilitator: Call the war game to order and guide discussion of friendly counteractions to ensure that all enemy reactions are considered.

2. Task force representatives: Starting with the ME, followed by SEs, then SHs, and lastly SUSs, each TF role player explains and records tangible adjustments as they pertain to them as a result of group discourse.

   a. Location and orientation adjustments.

   b. Control measure adjustments.

   c. Tasks and purposes refinement.

   d. Risk adjustment (mitigate and translate to additional tasks).

   e. Identification of coordination gaps.

   f. Identification of capability shortfalls.

   g. Adjustments in priorities.

   h. Refinement of authorities.

3. Facilitator: Guide each operational function, in turn, to explain any adjustments to red cell reactions from a functional perspective to identify coordination gaps, potential resource adjustments, or capability shortfalls.

4. COA representative: Guide the group to readdress refinements to:

   a. Facts or assumptions.

   b. Decision points and PIR/FFIR adjustments.

   c. Strengths or weaknesses (advantages or disadvantages) of the COA that emerged.

   d. Modifications to the COA sketch (locations and control measures) or narrative.

   e. Task organization and command or support relationships.

   f. Evaluation criteria.

   g. Unresolved issues for resolution.

   h. Gaps in coordination not previously identified.

   i. Additional considerations for risk to mission or force.

   j. Additional authorities and permissions for supplemental ROE requests.

5. Facilitator:

   a. Ensure that the war-game record captures information that must be incorporated into the synchronization matrix.

   b. Announce a reset for the next game or turn. Record the appropriate information.

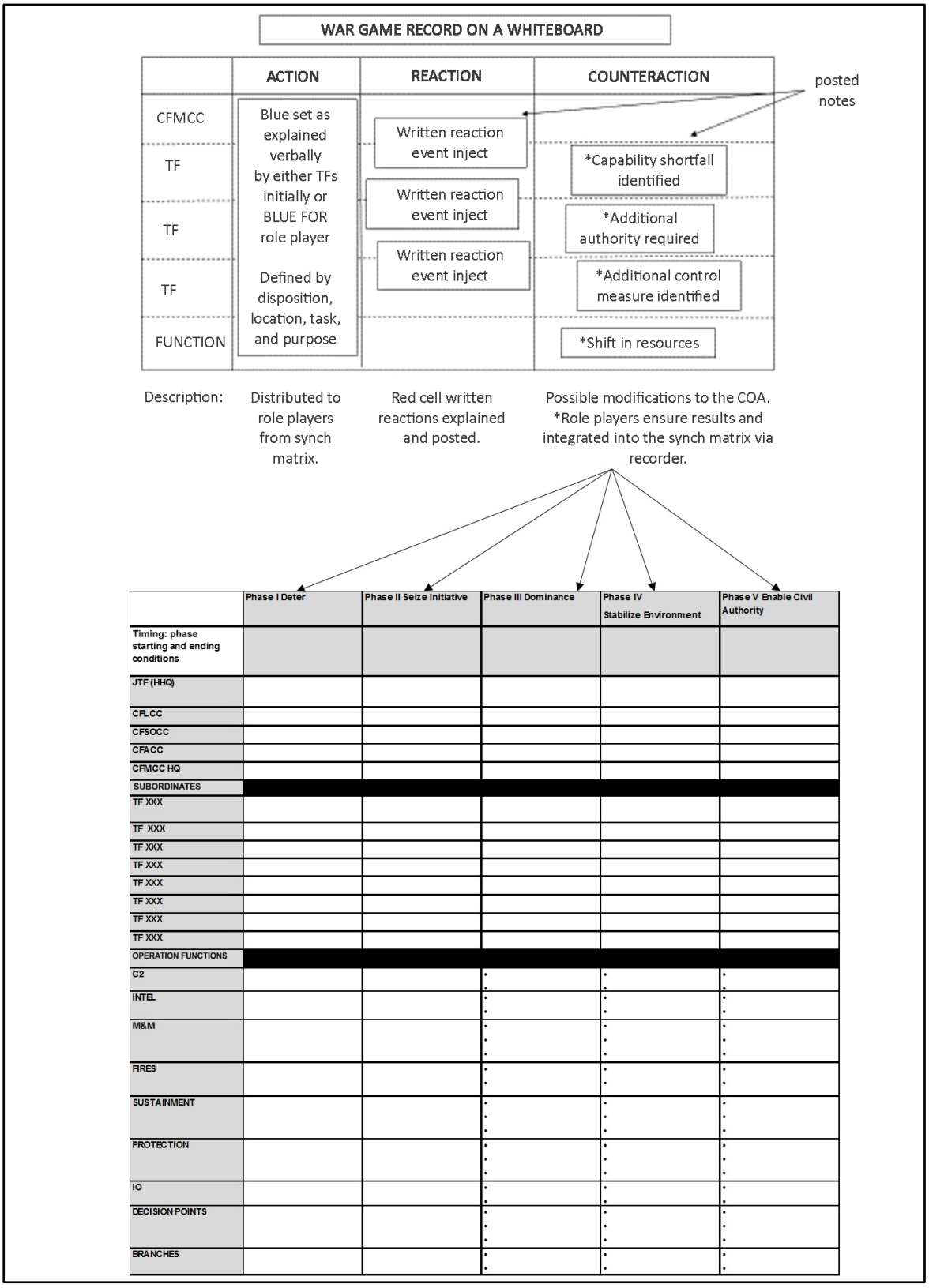
Note

After the turn has concluded, capture adjustments from the posted war-game record. The originator of recorded information personally explains each adjustment to the recorder incorporating that information into the synchronization matrix.

This constitutes one notional turn of a war game. To increase the tempo of subsequent turns, planning leads may truncate steps as feasible and adjust the pace or tempo to accommodate time restrictions. Also, the facilitator may appoint a single representative to explain the entire initial friendly force disposition at the start of the action move. The friendly force intelligence representative should brief the R&S setup at the start of each war-game turn.

F.10 TRANSFER WAR-GAME RECORD TO SYNCHRONIZATION MATRIX

As the war game concludes, the war-game record is transcribed into planning tools to produce refined products that support development of directives for the selected COA (see Figure F-5).



	Phase I Deter	Phase II Seize Initiative	Phase III Dominance	Phase IV Stabilize Environment	Phase V Enable Civil Authority
Timing: phase starting and ending conditions					
JTF (HHQ)					
CFLCC					
CFSOCC					
CFACC					
CFMCC HQ					
SUBORDINATES					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
TF XXX					
OPERATION FUNCTIONS					
C2			•	•	•
INTEL			•	•	•
M&M			•	•	•
FIRES			•	•	•
SUSTAINMENT			•	•	•
PROTECTION			•	•	•
IO			•	•	•
DECISION POINTS			•	•	•
BRANCHES			•	•	•

Figure F-5. Transfer War-game Record to Synchronization Matrix

APPENDIX G
Synchronization and Decision Tools

G.1 INTRODUCTION

This appendix describes planning tools that assist the staff throughout the planning process and into execution. The principal tools described are the synchronization matrix, DST, and DSM.

G.2 SYNCHRONIZATION MATRIX

The synchronization matrix is a staff planning and execution aid that depicts the operation throughout its phases in an easily understood format. When used properly, the synchronization matrix ensures that an operation is synchronized across time, space, and purpose in relation to its OBJs. Once completed, the synchronization matrix provides the staff with:

1. An understanding of the major actions being conducted by the HHQ, adjacent commands, and subordinate commands (e.g., the JFC, other components, and maritime TFs).

2. A listing of subordinate tasks to be conducted during the operation, and a means to refine the synchronization of events/actions throughout planning.

3. A portrayal of the key decision points and their linked CCIRs for each phase of the operation.

4. A clear display of operational functions and key supporting activities (LOG, IO, intelligence collection, etc.) that support the maritime component's scheme of maneuver.

5. A means to identify and prioritize branch planning requirements.

6. A portrayal of the maritime directive—a complete synchronization matrix makes development of execution paragraph 3 of the base directive much clearer.

7. A sense that the COA represented in the synchronization matrix is valid (i.e., suitable, acceptable, feasible, distinguishable, and complete).

Synchronization matrices are crucial planning and execution tools for all echelons of command. The methodology applies to JTF-level staffs to synchronize the actions and activities of their components, component-level staffs to synchronize the actions of their TFs, and task-force-level staffs to synchronize the actions of their subordinate tactical units.

G.2.1 Synchronization Matrix Development

Development of the synch matrix should begin early in the planning process. During mission analysis, the planning team should identify those responsible for the creation and maintenance of the synchronization matrix. During COA development, as the planning team designs the task organization, the synchronization matrix is used to record tasks assigned to subordinates and supporting commands. COA analysis is when the synchronization matrix starts to be refined through interaction between the friendly and enemy COAs. However, the full value of the matrix is realized after the commander has approved a COA, and the final sequencing of the operation has been established. Upon receipt of the commander's decision, the planning staff should refine the synchronization matrix for the approved COA. See Figure G-1 for a more detailed description of the process for constructing a synchronization matrix.

NPP Step	Information to Record
Mission Analysis	- Timing: Record phase beginning and phase ends from the HHQ order. - HHQ: Record the HHQ essential or key task for each phase.
COA Development	- Maritime component: Record the maritime component decisive action for each phase. - Adjacent commands: Record what adjacent commands (e.g., components) are doing ISO the maritime component. Do not record tasks assigned to the other commands unless they directly relate to the maritime component. - Subordinate HQs: Record tasks the maritime component is assigning to the subordinate TFs. Consider the use of the five W's when writing the tasks: - Who? Which TF? - What is the actual task to be executed? - Where will the task be executed? The JOA and maritime AO are usually too broad for tactical tasks assigned to subordinate TFs. Use the control measures (OPAREAs) developed ICW the applicable COA. - When? Tasks will be effective for execution either immediately when the phase begins (commence maritime patrols), at a specific point in time (on C+5, commence maritime patrols), at a yet undetermined point in time (O/O commence maritime patrols), or perhaps not at all (BPT commence maritime patrols). - Why? The purpose of a task. - Operational functions: OPT records input from staff cross-functional teams (intelligence, fires, sustainment, protection) ISO each COA if available. At a minimum, the OPT should be able to provide input for C2 (maritime component established supported/supporting relationships) and movement and maneuver (big picture of how the force is to be employed).
COA Analysis	- General: war-game actions and counteractions during a game turn will be recorded in a war-game worksheet. Transfer modifications or additional information as the result of the war game to the synchronization matrix. Modifications include additional or refined tasks, authorities, decision points, etc. Conduct this process as necessary for each phase, discrete time, or critical event. - Operational functions: Record the operational-level activities and priorities for each operational function for each phase. Input should be provided by cross-functional teams and support the maritime component scheme of maneuver. - Branch plans: Record any identified branch plans. - Timing: Refine the phase beginning and phase end by using maritime component specific conditions where applicable. These statements will be utilized in the CONOPS phase narratives.
COA Comparison and Decision	**Pre-decision brief:** - Majority of effort will be applied to refining the COAs and their individual synch matrices prior to the decision brief. Capture any additional tasks or changes directed by the commander during the brief. **Post-decision brief:** - Conduct final refinement of the synchronization matrix IOT simplify preparation of the plan or order. - Ensure accuracy of tasks being assigned to subordinates and those agreed to by adjacent commands. Pay particular attention to alignment of tasks with their purposes. These tasks will be directly copied into the maritime directive. See notes above in COA Development for guidance on writing tasks. - Record any additional decision points ISO the CONOPS. These decision points should be consistent with the content of the DST/DSM. - The DSM options column should be reviewed for potential BPT-type tasks to subordinates.

Figure G-1. Synchronization Matrix Development (Sheet 1 of 2)

NPP Step	Information to Record
Directive Development	- Transfer all tasks to subordinates and supporting commands from the synchronization matrix to the draft directive. - As the draft plan or order is edited, ensure that the synchronization matrix is updated IOT maintain the validity of the tool. - Compare the synchronization matrix with the DST/DSM to ensure both tools are consistent with each other.
Transition	- As the order (or a portion of the order) is transitioned to another internal staff section (COPS or FOPS), the planners who developed the synchronization matrix will transition responsibility for corresponding portions of the synchronization matrix.

Figure G-1. Synchronization Matrix Development (Sheet 2 of 2)

G.2.2 Elements of the Synchronization Matrix

The staff should record the following content in the synchronization matrix:

1. Timing (phase beginning). The starting conditions of the discrete phase or critical event being synchronized. During plan development, details concerning exact timing are unlikely to be available. In this case describe the beginning conditions for the applicable phase, stage, or step of the operation.

2. HHQ. Record what the HHQ (e.g., the JTF) is doing in each phase of the operation. Consider including the HHQ commander's essential tasks, OBJs, or activities (for example: phase I: Deploy joint forces into the JOA, establish C2, and posture for potential noncombatant evacuation operation (NEO)). Phase II: Establish air and maritime superiority in vicinity of country X IOT disrupt enemy offensive momentum.)

3. Adjacent commands. Record the main tasks that adjacent commands (e.g., components) are performing during each discrete time period, particularly those tasks they are performing ISO the maritime component. If there is another maritime force operating in the assigned battle space (e.g., HN or individual nation), create a separate line on the sync matrix and record what they are doing ISO the maritime component OPS. Identify the JFC's ME during each phase or time period.

4. Maritime component. Record what the maritime component is doing during each phase, discrete time, or critical event. This input should remain broad in nature and align with the force's decisive action for each discrete time period. Consider recording support provided by the maritime component to other adjacent commands. Also consider including any maritime OBJs for the time period in question.

5. Subordinate forces. Record tasks to be executed by subordinate forces. Information contained in these cells applies to a single echelon (e.g., for JFMCC's synchronization matrix, assign tasks to subordinate TFs but do not assign tasks to TGs). The tasks recorded here are the main source of tasks by phase which will be transferred to the tasks to subordinates section of the directive.

6. Timing (phase ending). Record the desired conditions at the conclusion of the phase or critical event.

7. Coordinating instructions. Record those tasks, purposes, or other important information that apply to two or more subordinate tasks forces.

8. Operational functions. Record the major operational level task(s) and or priorities for each function (intelligence, fires, LOG, C2, etc.) as identified by its supporting cross-functional team. Do not relist tasks to subordinates that are already recorded in prior rows of the synchronization matrix.

9. Authorities. Record any identified authorities required ISO execution.

10. Decision points. Record any identified decision points which may be encountered in each respective phase.

11. Branches. Record where and for what a branch plan might be required. This will assist in prioritizing additional planning requirements.

Staffs often find it most useful to create a synchronization matrix with more detail early in the timeline and less detail later in the timeline. This technique allows for detailed synchronization of the events that are near term (and better understood) and requires less detail for those phases that are further in the future (and less understood). Detailed examination of the earlier phases also allows for a sharper focus upon force closures—especially important if critical capabilities are not in place in the assigned AO (e.g., JOA, maritime AO, etc.), and their arrival supports a specific event. As planning continues throughout the operation, these later phases should be refined with greater detail. See Annex 1 to this appendix for an example of a JFMCC synch matrix.

G.2.3 Use of the Synchronization Matrix

Perhaps the most important function of the synch matrix is to ensure that tasks are done at the appropriate time and in the correct order (for instance, not employing LOG aircraft in the combat zone for resupply purposes without first obtaining air superiority). When completed, the matrix also allows the planners to look vertically and ensure all tasks identified in mission analysis and COA development are assigned without unnecessary duplication, as well as showing that all operational functions are adequately employed during each time/event/phase. By looking across each row, the matrix also allows the planners to identify gaps or seams between the time/event/phase transitions more readily.

Given proper attention throughout planning, the synchronization matrix provides the planning team an invaluable tool for writing a detailed CONOPS after COA selection. The synchronization matrix (with tasks and purposes identified for each subordinate force) comprises the tasks to subordinates section of the directive.

A good synchronization matrix also facilitates transitioning the order to execution. For example, when the order transitions to execution, the synchronization matrix can provide the COPS cell a detailed template (execution matrix) of how the operation was planned to flow, so it can follow through on subordinate tasks. It can also help the COPS cell recognize when the order is off course due to friendly, enemy, and environmental interaction.

G.3 DECISION SUPPORT TEMPLATES AND DECISION SUPPORT MATRICES

DSTs and DSMs provide a critical connection between planning and execution. The primary value of the DST/DSM is to capture the linkages between decision points, CCIRs, and planner-identified decision options when CCIR-related events occur IOT speed decision making during execution.

1. DST. A combined intelligence and OPS graphic based on the results of war gaming. The DST depicts the geographic locations of where CCIR-related events are expected to occur.

2. DSM. A matrix associated with a DST that has a column for CCIRs, a column for decision options if the CCIR-related event occurs, a column for decision support criteria supporting the CCIRs, and a column that identifies a geographic area related to the decision.

The DST/DSM complements the maritime CONOPS. The CONOPS explains what the commander wants to accomplish and describes the operational-level how of the friendly plan. The DST/DSM records decisions the commander should anticipate for operational level contingencies that would disrupt execution of the CONOPS should they occur. In essence, the DST/DSM captures advance thinking about what-if scenarios. In that regard, the DST and DSM are products of the planning process and tools that connect planning to execution. The purpose of the DST/DSM is to provide information that anticipates and links pre-identified decision points, CCIRs, and decision options that the commander may select during execution. The DST/DSM documents anticipated ways to address contingencies that may occur. This tool ensures that planners, who have detailed knowledge of the plan, record and pass on to COPS information requirements and potential decision options that are aligned with the commander's guidance and intent. For this reason, the DST/DSM is a critical document when transitioning the corresponding military directive from planning to execution (e.g., from FOPS to COPS).

G.3.1 Relationship between Objectives, Decision Points, Commander's Critical Information Requirements, and Decision Options

Identifying relevant decision points and their related information requirements can be a challenge. To ensure relevance, planners must consider decisions that must be made which are directly or indirectly related to important aspects of the plan. Tasks associated with the decisive action, MEs and SEs, or COGs are all sources of potential decision points. However, the most important consideration for determining decision points lies with the OBJs, both ultimate and intermediate, identified during planning.

Take for example an intermediate OBJ of "Landing force is established ashore" as depicted in Figure G-2. The key decision that must be made is whether to commence the amphibious assault. The planning team should determine what contingencies could prevent attainment of the OBJ. The contingencies focus on potential enemy action and friendly requirements and can be used as a starting point for establishing CCIRs. Realistic options can be determined for each CCIR, with supporting criteria to assist in recommending the best option. The key point is to focus on the most important decisions that the commander will need to make IOT execute and adjust the approved plan and not arbitrarily identify issues that may require a decision.

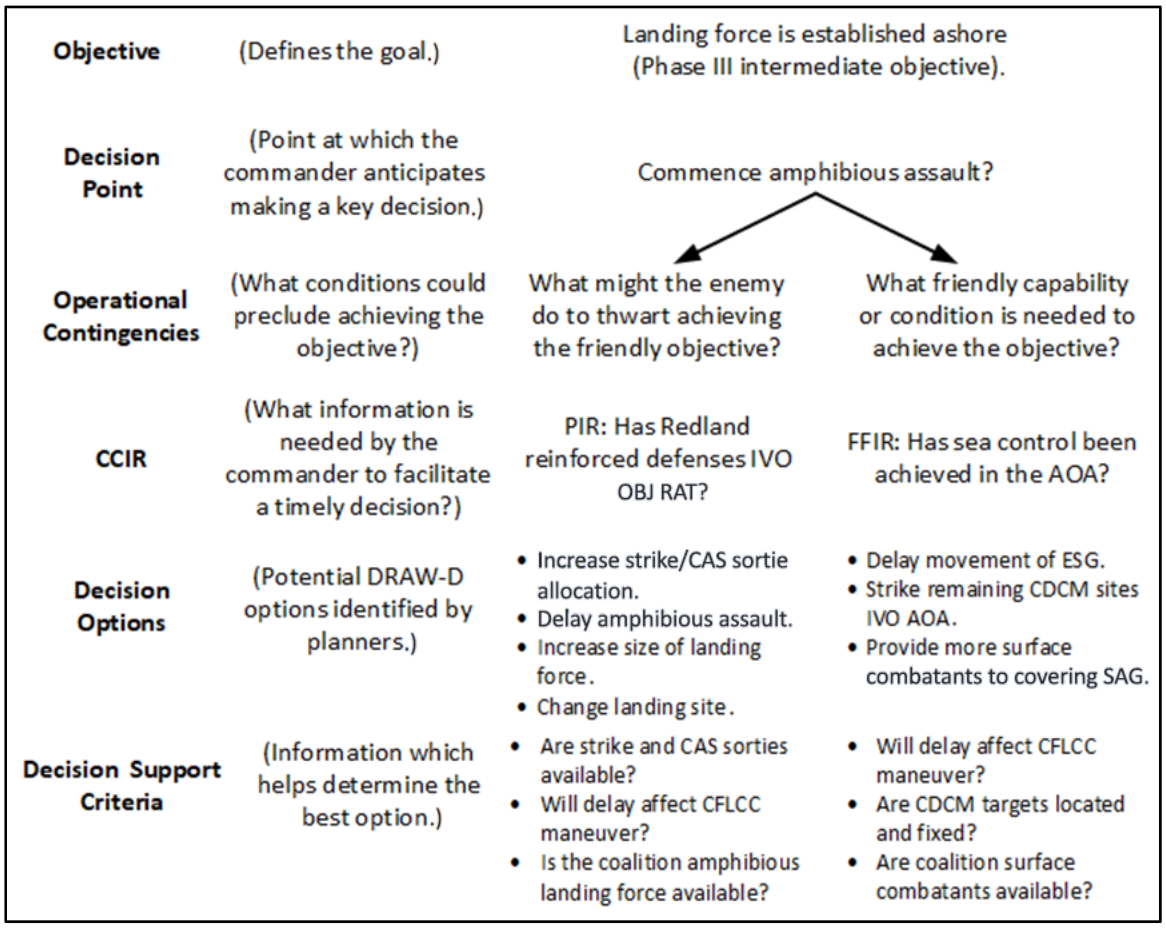


Figure G-2. Relationship between Objectives, Decision Points, Commander's Critical Information Requirements, and Decision Options

G.3.2 Decision Support Template and Decision Support Matrix Development

As with a synch matrix, DST/DSM construction should begin during mission analysis with identification of persons responsible for capturing the information and maintaining the products. Initial identification of CCIRs and decision points will begin as part of mission analysis. Decision points and decision options (by phase of the

operation) are identified and refined throughout the planning process. The war-gaming process during COA analysis is of particular benefit in determining and validating decision options. A separate DST/DSM for each phase of the operation may need to be developed depending on the complexity of the operation. Figure G-3 provides more information concerning DST/DSM development.

G.3.3 Use of Decision Support Template/Decision Support Matrix

DST/DSM products are valuable to planners because they help ensure clear association between decision points and CCIRs. This may assist in keeping CCIRs to the minimum required in the operation, as well as documenting when CCIRs change by phase of the operation. The DST/DSM also documents planner ideas about valid options when a CCIR-related event occurs, illustrating how the planners thought through potential options/solutions. This helps ensure that planners, who have the most detailed knowledge of the order's intricacies, identify decision options that are aligned with the commander's intent and guidance.

The DST/DSM is a critical document to be provided in transitioning an order from plans to execution (e.g., COPS and watch standers). The DST/DSM provides an immediate list of valid decision options to recommend in execution once a CCIR-related event occurs. Having these documents available will provide viable decision options to the commander sooner, thus increasing the speed of decision making and tempo of OPS. Figure G-4 provides an example DST. Figure G-5 provides an example of the accompanying DSM.

Note

Decision support criteria listed in the DSM may be aligned, color coded, or annotated to indicate which decision options they support.

Navy Planning Process Step	Operational Planning Team Actions	Decision Support Template/Decision Support Matrix Planner
Mission analysis	Identify candidate CCIRs to propose during the mission analysis brief. To spur discussion of potential CCIRs, the OPT should consider what enemy action or friendly event will prevent the maritime force from achieving the friendly OBJ or executing essential tasks. Keep this list short, but do include both PIRs and FFIRs.	Record initial CCIRs identified by the OPT in the template. It is not yet required to identify options or criteria.
COA development	OPT will develop a DST/DSM for each COA. The primary DST/DSM consideration is to continue the identification of potential decisions for the commander. Potential decision points/CCIRs can be refined by discussing contingencies that can impede the maritime component's ME from accomplishing its task.	Record new CCIR/decision points as they emerge. Record any options or criteria which are discussed, although they are not required yet.
COA analysis	During the war game, the OPT will identify additional CCIRs and decision points. The red cell should attempt to trigger known CCIRs during the war game if possible. Planners should discuss the conditions under which the decision point emerged, and if time allows, potential options and related criteria that will assist in recommending one option over another.	The DST/DSM will be used as a recording device for new decision points/CCIR, options, and criteria as that information emerges during the war games. A draft DST/DSM is an output of COA analysis.
COA comparison and decision	**During Decision Brief** OPT should verify all CCIRs are presented to the commander. **Post-decision Brief** Planners responsible for the synchronization matrix and DST/DSM need to ensure their products are aligned. Where applicable, options that are actually BPT-type tasks to subordinates or supporting commands should be recorded in the synch matrix for eventual inclusion in the base directive.	**Pre-decision Brief** Refine the draft DST/DSM for each COA by ensuring options are provided for each decision point/CCIR and continue work on the criteria column. The DST/DSM should be organized by phase, stage, or step if possible. **Post-decision Brief** Complete the remaining input for all DPs/CCIRs. In addition to scanning the options for potential tasks to subordinates, ensure that input for all columns is complete.
Directive development	Ensure alignment between the directive in development, DST/DSM, and synchronization matrix. CCIRs in DST/DSM should match those in coordinating instructions. Where applicable, options should be captured as tasks in the execution portion of the directive.	Crosswalk and reconcile the DST/DSM with the military directive to ensure CCIRs match between products. Fill in and socialize all columns with appropriate organizations.
Transition	DST/DSM is a primary product and discussion point during transition and should be addressed during transition briefs. COPS and subordinate commands need to understand what these products include and the thought process behind them.	Provide DST/DSM to COPS prior to the transition brief and be prepared to brief decision points during transition brief.

Figure G-3. Decision Support Template/Decision Support Matrix Development

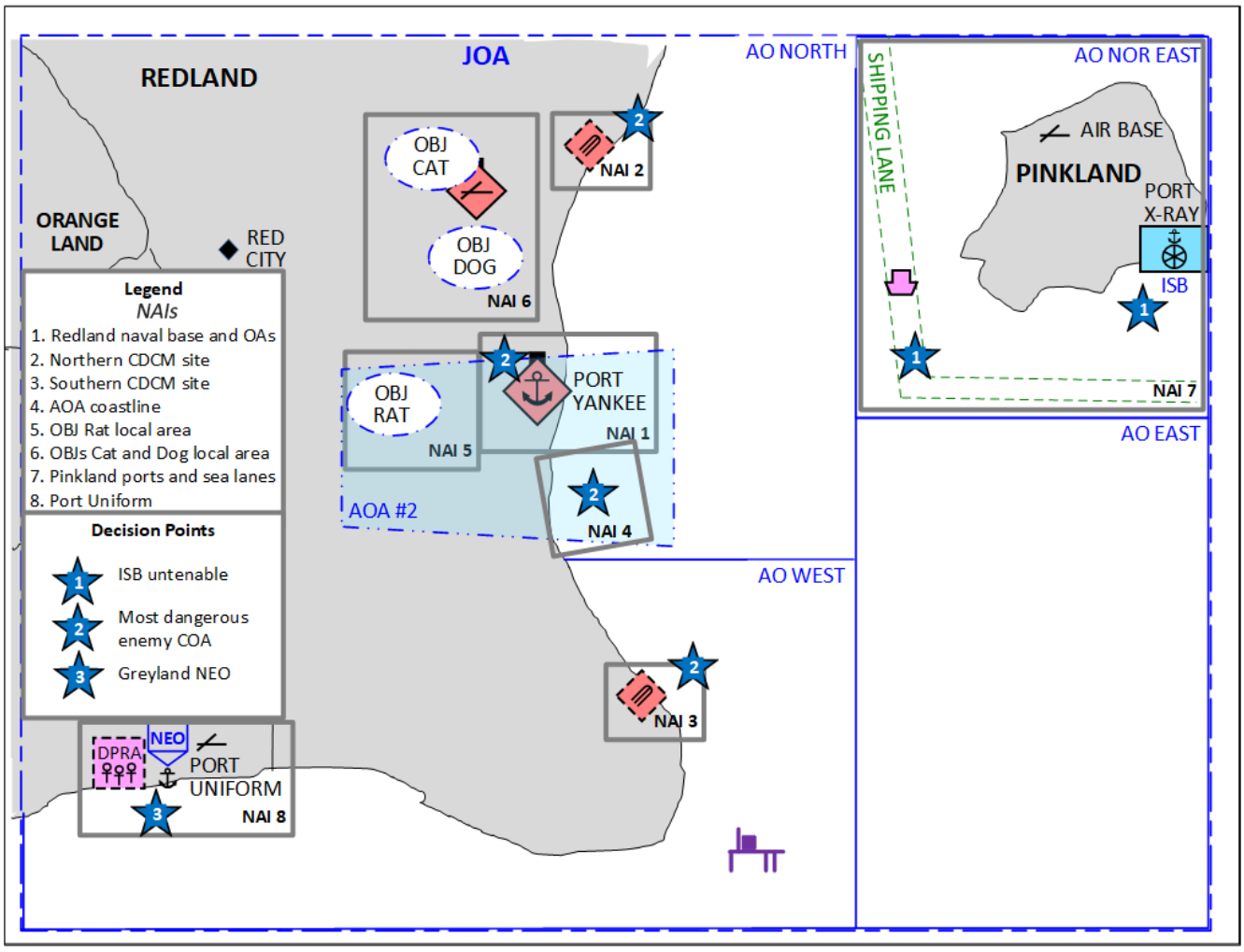


Figure G-4. Decision Support Template Example

Sample DSM–Operation Freedom Assurance Phase I				
Decision Point	**CCIR**	**Decision Options**	**Decision Support Criteria**	**Location**
1	ISBs and/or approaches to ISBs are untenable.	Option 1A: Shift ISB to alternate location. Option 1B: Redeploy IAMD, MCM, and force protection (FP) units to counter threats as required. Option 1C: Recommend early transition to Phase II to neutralize threats.	I&W of minelaying of Pinkland SLOCs. Terrorist attacks on or in vicinity of Pinkland air base and Port X-Ray. Ballistic missile attack(s) on Pinkland. Attacks on shipping in vicinity of Pinkland. Alternate ISB available.	NAI 7
2	Is Redland preparing for sea denial OPS?	Option 2A: Expedite transition to Phase II to neutralize threats. Option 2B: Delay phase III. Option 2C: Proceed with phase II as scheduled (and accept increased risk).	Redland deploys attack sub(s) (diesel-electric propulsion (SSK(s)). I&W of impending minelaying activities. Activation and dispersal of CDCM sites. Reinforcement of Port Yankee defenses. Minelaying and/or SSK OPS within or in vicinity of AOA 2. Availability and readiness of JFMCC forces. Availability of supporting JFACC forces.	NAIs 1, 2, 3, and 4
3	U.S. citizens in Greyland are threatened by civil unrest.	Option 3A: Deploy the forward command element. Option 3B: Reposition ESG assets. Option 3C: Request JFACC assistance.	Threat levels in Greyland. Availability of ESG. Availability JFACC forces. Security of Port Uniform airport.	NAI 8

Figure G-5. Decision Support Matrix Example

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ANNEX 1 TO APPENDIX G

Example Synchronization Matrix

The following pages provide a synchronization matrix example.

	Phase I: Deter	Phase II: Seize Initiative
Estimated Timeframe	C-day	
Phase beginning conditions	Commences on C-day. Hostilities have not been declared.	Hostilities have commenced, and JTF forces are in position to conduct offensive OPS in the JOA.
Phase ending conditions	Redland agrees to diplomatic resolution or commences hostilities. Maritime commerce avoids JOA. JFMCC forces have completed deployment to the JOA for defense of Pinkland.	Sea control is established in AO NORTH. CTF 224 is positioned to conduct amphibious OPS. Redland forces are degraded in AOA 2.
JTF HQ	Deploy joint forces into the JOA IOT protect Pinkland and deter Redland aggression.	Conduct offensive OPS in the JOA IOT degrade Redland forces.
JTF phase objective	Redland military forces are deterred from aggression.	Redland military forces are degraded.
JFLCC	JTF SH: - Deploy forces to Pinkland IOT support follow-on offensive OPS in Redland. - Assume land-side security at JTF ISBs.	JTF SH: - Stage forces in Pinkland IOT conduct phase III forcible entry operation. - Prepare for forcible entry OPS against Redland.
JFSOCC	JTF SH: - Conduct SR in Redland IOT set conditions for future JTF OPS.	JTF SE: - Conduct DA at OBJ RAT IOT degrade Redland C2. - Conduct surface observations in AOA 2 IOT prepare for follow-on amphibious assault.
JFACC	JTF SE: - Conduct DCA in the JOA IOT protect JTF forces. - Assumes security/protection requirements at ISBs/FLS. - Provide aerial refueling support IOT extend TF 220 operational reach. - Designated area AD commander.	JTF SE: - Conduct strike OPS in the JOA IOT degrade Redland capability. - Conduct OCA in Land AO to degrade Redland air forces.
JFMCC	JTF ME: - Deploy forces to establish a maritime presence in the JOA IOT deter Redland aggression. - Conduct maritime stability OPS in the JOA IOT deter Redland maritime forces.	JTF ME: - Conduct offensive OPS in the JOA IOT gain sea control in AO NORTH.

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 1 of 11)

	Phase III: Dominate	Phase IV: Stabilize Environment
Estimated Timeframe		
Phase beginning conditions	Sea control is established in AO NORTH. CTF 224 is positioned to conduct amphibious OPS. Redland forces in AO NORTH (AOA 2/OBJ RAT) are degraded.	Redland forces are defeated. JTF forces have consolidated in Redland. JFMCC forces are postured for stability OPS.
Phase ending conditions	Redland forces are defeated. JTF forces have consolidated in Redland. JTF forces are postured for stability OPS.	Redland forces do not pose a threat to Pinkland. Pinkland maritime forces are able to conduct stability OPS within the Pinkland EEZ. Commercial maritime activity has returned to precrisis levels.
JTF HQ	Conduct simultaneous offensive OPS to defeat Redland forces.	Conduct stability OPS throughout the JOA IOT set conditions for relief by Pinkland and other designated forces.
JTF phase objective	Redland military forces are defeated.	International partners are able to provide regional security.
JFLCC	JTF ME: - Conduct airborne forcible entry into Redland IOT destroy Redland military forces at OBJs DOG and CAT.	JTF ME: - Conduct counterinsurgency and counterterrorism OPS in Redland IOT stabilize security environment. - Recommend timeline priority for redeployment of JFLCC attached forces. - BPT conduct foreign humanitarian assistance (FHA) IOT relieve suffering in Greyland.
JFSOCC	JTF SE: - Conduct DA against high-value targets IOT disrupt Redland reinforcements.	JTF SE: - Conduct foreign internal defense with Pinkland security forces IOT enable transfer of security missions. - Recommend timeline and priority for redeployment of JFSOCC attached forces. - BPT conduct DA in Redland against terrorist IOT prevent resurgence.
JFACC	JTF SE: - Conduct joint fire support IOT degrade Redland forces in JOA.	JTF SE: - Protect JFLCC and JFSOCC forces in Redland IOT ensure freedom of maneuver. - Recommend timeline and priority for redeployment of JFACC attached forces. - BPT conduct strike OPS to prevent resurgence of Redland aggression.
JFMCC	JTF SE: - Conduct amphibious OPS in AOA 2 IOT fix Redland forces at OBJ RAT.	JTF SE: - ICW Pinkland naval forces, conduct stability OPS in the maritime AO IOT prevent terrorist exploitation. - BPT conduct reconstruction at Port Yankee IOT ensure rapid opening of SPOD.

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 2 of 11)

	Phase V: Enable Civil Authority
Estimated Timeframe	
Phase beginning conditions	Redland forces do not pose a threat to Pinkland. Pinkland maritime forces are able to conduct stability OPS within the Pinkland EEZ. Commercial maritime activity has returned to precrisis levels.
Phase ending conditions	Pinkland or other designated organizations have assumed responsibility for JFMCC missions. JFMCC forces have redeployed.
JTF HQ	Coordinate transition of JTF missions with Pinkland MOD IOT ensure a seamless continuation of security activities.
JTF phase objective	JTF forces are redeployed.
JFLCC	JTF ME: - Conduct relief in place with designated multinational force (MNF) IOT ensure continuation of security missions. - O/O, redeploy JFLCC attached forces.
JFSOCC	JTF SE: - In coordination with JFLCC, conduct relief in place with designated MNF IOT ensure continuation of security missions. - O/O, redeploy JFSOCC attached forces.
JFACC	JTF SE: - Conduct relief in place with Pinkland's air force IOT ensure continuation of security missions. - O/O, redeploy JFACC attached forces.
JFMCC	JTF SE: - Conduct relief in place with Pinkland Navy IOT ensure continuation of MSO. - O/O, redeploy JFMCC attached forces.

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 3 of 11)

	Phase I: Deter	Phase II: Seize Initiative
JFMCC Phase Objective	Redland maritime forces are deterred.	Sea control is established in AO NORTH.
Subordinates		
CTF 220 (CSG)	JFMCC SH: - Conduct IAMD in AO NOR EAST IOT protect deploying forces. Priority of effort is CTF 224 forces conducting amphibious rehearsal. - BPT conduct aerial show of force to deter Redland aggression.	JFMCC SE: - Conduct strike OPS in the JOA IOT degrade Redland maritime forces. - Protect CTF 224 forces in AO NORTH to preserve combat power for phase III.
CTF 223 (LOG)	JFMCC SUS: - Conduct LOG IOT extend operational reach of JFMCC forces. - Conduct intratheater airlift in the JOA IOT support JFMCC deployment.	JFMCC SUS: - Conduct LOG IOT extend operational reach of JFMCC forces.
CTF 224 (ESG)	JFMCC SH: - Conduct amphibious rehearsal in AOA 1 IOT prepare for offensive OPS in Redland. - BPT conduct defensive mine clearance IOT ensure access to ISBs and SLOCs.	JFMCC SH: - O/O, conduct MCM in AOA 2 to set the conditions for amphibious OPS.
CTF 225 (SAG)	JFMCC ME: - Conduct patrols in AO NOR EAST IOT deter Redland maritime aggression.	JFMCC ME: - Attack Redland forces IOT gain local sea control in AO NORTH. - Conduct strike OPS IOT degrade Redland defensive capability.
CTF 227 (MPRA)	JFMCC SE: - Conduct ISR in the JOA IOT determine Redland maritime disposition and intentions.	JFMCC SE: - Conduct ISR in AOs NORTH and WEST IOT determine Redland maritime disposition and intentions. - Conduct armed reconnaissance in AOs NORTH and WEST IOT degrade Redland maritime capabilities.
CTF 229 (SUB)	JFMCC SE: - Collect target information in SUB OA IOT hold Redland subs at risk.	JFMCC SE: - Interdict Redland maritime forces ISO CTF 225 to establish sea control in AO NORTH.
Pinkland Forces	JFMCC SH: - Conduct patrols in AO NOR EAST IOT protect Pinkland ISBs. - BPT conduct defensive mine clearance of Pinkland ISBs IOT ensure FA force flow.	JFMCC SE: - O/O, escort CTF 224 to AOA 2 IOT prevent Redland interference.
Coordinating Instructions	PIR 1—What is the current disposition and intent of Redland mine laying and CDCM forces? PIR 2—What is the threat to forces establishing ISB in Pinkland? FFIR 1—Has JFMCC IAMD capability been degraded?	PIR 1—What is the disposition of Redland maritime forces in AOs WEST and CENTRAL? PIR 2—What is the disposition of Redland ground forces in vicinity of OBJ RAT? FFIR 1—Is sea control established in AO CENTRAL? FFIR 2—Is CTF 220 capable of conducting flight OPS? .

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 4 of 11)

	Phase III: Dominate	Phase IV: Stabilize Environment
JFMCC Phase Objective	Redland maritime forces are deterred.	Maritime AO is secure.
Subordinates		
CTF 220 (CSG)	JFMCC SE: - Conduct CAS in AOA 2 IOT support CTF 224 amphibious operation.	JFMCC SE: - Conduct combat air patrol (CAP)/surface combat air patrol (SUCAP) IOT protect CTF 225 and CTF 224 forces in AOs CENTRAL and WEST. - BPT conduct strike OPS IOT prevent resurgence of Redland aggression.
CTF 223 (LOG)	JFMCC SUS: - Conduct replenishment at sea IOT extend operational reach of JFMCC forces.	JFMCC SUS: - Conduct replenishment at sea IOT extend operational reach of JFMCC forces.
CTF 224 (ESG)	JFMCC ME: - Conduct amphibious assault at AOA 2 IOT fix Redland forces at OBJ RAT.	JFMCC SE: - Conduct counterterrorism OPS in vicinity of OBJ RAT IOT secure area in preparation for turnover to Redland forces. - BPT conduct FHA in Greyland IOT relieve human suffering. - BPT conduct mine clearance OPS in AO WEST IOT ensure access to commercial shipping. - BPT conduct port reconstruction at Port Yankee IOT ensure rapid opening of SPOD.
CTF 225 (SAG)	JFMCC SE: - Cover CTF 224 IOT enable amphibious OPS in AOA 2.	JFMCC ME: - Conduct MIO in AO CENTRAL and WEST IOT deny freedom of movement to suspected terrorists.
CTF 227 (MPRA)	JFMCC SH: - Conduct ISR in the JOA IOT determine remaining Redland capability. - Conduct armed reconnaissance in AOs NORTH and WEST IOT prevent Redland interference with amphibious OPS.	JFMCC ME: - Conduct MIO in AOs CENTRAL and WEST IOT deny freedom of movement to suspected terrorists.
CTF 229 (SUB)	JFMCC SE: - Attack Redland naval forces IOT expand sea control in the maritime AO.	JFMCC SE: - Conduct surveillance of Redland maritime forces to determine disposition and intentions.
Pinkland Forces	JFMCC SH: - Protect CTF 223 forces in AO NORTH IOT prevent disruption of combat LOG.	- Conduct MIO in AOs NE IOT provide secure operating environment.
Coordinating Instructions	PIR 1—What is the remaining Redland maritime capability? PIR 2—Are Redland ground forces reinforcing in vicinity of OBJ RAT? FFIR 1—Are conditions set to commence amphibious assault? FFIR 2—Is CTF 224 capable of conducting amphibious OPS?	PIR 1—Are terrorists using the maritime domain to conduct attacks against JTF or Redland forces? FFIR 1—Are international partners prepared to assume security missions?

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 5 of 11)

	Phase V: Enable Civil Authority
JFMCC Phase Objective	Maritime OPS are turned over to the Pinkland navy.
Subordinates	
CTF 220 (CSG)	JFMCC SE: - O/O, redeploy attached forces.
CTF 223 (LOG)	JFMCC SUS: - In coordination with CTF 224 and CTF 225, redeploy attached forces.
CTF 224 (ESG)	JFMCC SE: - O/O, redeploy attached forces.
CTF 225 (SAG)	JFMCC ME: - Conduct relief in place with Pinkland navy IOT ensure continuation of MSO. - O/O, redeploy attached forces.
CTF 227 (MPRA)	JFMCC ME: - Conduct relief in place with the Pinkland navy IOT ensure continuation of MSO. - O/O, redeploy attached forces.
CTF 229 (SUB)	JFMCC SE: - O/O, redeploy attached forces.
Pinkland Forces	- Assume sole responsibility for maritime stabilization OPS in Pinkland EEZ.
Coordinating Instructions	

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 6 of 11)

	Phase I: Deter
Operational Functions and Other Staff Activities	
C2	- JFMCC is the supported component for initial ISB protection and for deterrence OPS in the MAO (JFACC is supporting). - JFACC is the supported component for IAMD (JFMCC is supporting). - CTF 225 is the supported TF for deterrence OPS in AO NOR EAST (227 and 229 supporting). - CTF 220 is the supported TF for ISB protection.
INTELLIGENCE	- Priority for this phase is to discern Redland maritime force disposition and intent with regard to Pinkland. Secondary priority is to identify Redland ability to reinforce OBJ RAT.
Movement and Maneuver	- Deploy surface and air forces into AO NOR EAST to deter Redland aggression and protect Pinkland ISBs. - Deploy sub forces in AO NORTH to collect target information.
Fires	Fires effect: Redland maritime forces remain out of Pinkland EEZ. - Nonlethal fires to support JFMCC deployment. - Deliberate lethal fires are prohibited.
Sustainment	- Establish JFMCC FLS. - Establish RAS procedures and priorities.
Protection	- Conduct IAMD of Pinkland ISBs until relieved by JFACC IOT enable JTF deployment. - OPSEC: sub OAs and disposition; potential beach landing sites. - Priority for protection: ISBs, LOG ships, and CTF 225 forces in AO NOR EAST.
IO	- Strategic message: JFMCC is working with Pinkland naval forces to protect Pinkland sovereignty. - Effect: Redland is deterred from aggressive actions against Pinkland.
Authorities Required	- Board suspected mining and smuggling vessels in Pinkland EEZ. - Detain individuals and vessels conducting mining and smuggling. - Collective self-defense. - Use force, up to and including deadly, to stop in-situ mining in Pinkland territorial sea and international waters.
Decision Points	- Recommend transition to phase II.
Branches	- Situation stabilizes. - Conduct NEO of coalition citizens. - Pinkland mine clearance OPS.
Risk	

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 7 of 11)

	Phase II: Seize Initiative
Operational Functions and Other Staff Activities	
C2	- JFMCC is the supported component for establishing sea control in maritime AO (JFACC is supporting). - CTF 225 supported TF for establishing sea control in AO NORTH (TFs 220, 227, and 229 are supporting). - CTF 220 is the supported TF for strike OPS (TFs 225 and 229 are supporting).
INTELLIGENCE	- Priority for this phase is to assess remaining Redland maritime capabilities in AOs NORTH and WEST. Second priority is to locate Redland forces in vicinity of OBJ RAT.
Movement and Maneuver	- Maneuver forces to establish sea control in AO NORTH. - Posture amphibious forces to be prepared for OPS in AOA 2.
Fires	- Effect: Redland maritime forces unable to dispute sea control in AO NORTH. - Effect: Land reinforcements in vicinity of AOA 2 are degraded. - Focus: Underway maritime forces in vicinity of AO NORTH; Ground forces in vicinity of AOA 2.
Sustainment	CASEVAC/MEDEVAC of JFMCC personnel during offensive OPS - Coordinate reloading SMs and TLAMs at Pinkland FLS. - Priority: CTFs 225, 220, 227, and 224.
Protection	- Fratricide avoidance during offensive OPS in AO NORTH. - Personnel recovery and tactical recovery of aircraft and personnel to recover isolated personnel. - Priority for protection: ISBs, CTF 225 in AO NORTH, LOG ships.
IO	- Strategic message: JFMCC is supporting Pinkland's right to defend itself from Redland aggression. - EA and OCO to degrade Redland maritime C2. - Military information support operations (MISO) to transmit JTF themes and messages of FA legitimacy.
Authorities Required	- Declare Redland naval forces in MAO hostile. - Conduct strike OPS against military and dual-use targets in Redland. - Conduct electronic attack.
Decision Points	- Commence CTF 224 movement to AOA 2. - Recommend transition to phase III.
Branches	
Risk	

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 8 of 11)

	Phase III: Dominate
Operational Functions and Other Staff Activities	
C2	- JFLCC is the supported component for forcible entry OPS; JFMCC, JFSOCC, and JFACC are supporting. - JFMCC is the supported component for amphibious OPS in AOA 2; JFACC and JFSOCC are supporting. - CTF 224 is the supported TF for amphibious OPS; CTFs 220, 225, 227, and 229 are supporting.
INTELLIGENCE	- Priority of this phase is to assess remaining Redland capability in vicinity of AOA 2. Second priority is to determine remaining Redland maritime capabilities.
Movement and Maneuver	- Amphibious forces fix Redland ground forces at OBJ RAT. - Surface and sub forces interdict Redland forces to maintain sea control.
Fires	- Effect: Redland maritime forces are defeated. - Redland ground forces are fixed at OBJ RAT. - Focus: Underway maritime forces, armor/mechanized forces in vicinity of OBJ RAT.
Sustainment	- CASEVAC/MEDEVAC of JFMCC personnel during amphibious OPS. - Sustainment of forces ashore at OBJ RAT. - Priority: CTFs 224, 220, 225, 227, and 229.
Protection	- Protect amphibious forces from Redland interference. - Priority for protection: amphibious ships, ISBs, and CTF 225 LOG ships.
IO	- Strategic message: JFMCC actions are IAW UNSCR condemning Redland. - EA and OCO to degrade Redland ground and maritime C2. - MISO to transmit JTF FA themes and messages of legitimacy.
Authorities Required	- Enter the land, air, and sea area of Redland for purpose of conducting combat OPS. - Conduct indirect fires against Redland ground forces. - Erect barriers in Redland. - Require authorization to declare Redland forces within AOA 2 hostile.
Decision Points	- Reinforce CTF 224 at OBJ RAT. - Air sortie apportionment.
Branches	
Risk	

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 9 of 11)

	Phase IV: Stabilize Environment
Operational Functions and Other Staff Activities	
C2	- JFLCC is supported component for counterinsurgency and counterterrorism OPS in Redland, and FHA in Greyland; JFMCC, JFACC, and JFSOCC are supporting. - JFMCC is supported component for maritime stability OPS in MAO; JFACC and JFSOCC are supporting. - CTF 225 is the supported TF for MIO in AO CENTRAL/WEST; CTFs 220, 227, and 229 are supporting. - CTF 224 is the supported TF for FHA if requested.
INTELLIGENCE	- Priority: Assess whether Redland forces or terrorists are capable of conducting attacks from the maritime domain.
Movement and Maneuver	- Employ surface forces and MPRA to patrol coastal areas IOT deter and disrupt terrorist use of maritime domain. - Amphibious forces support JFLCC efforts on land.
Fires	- Effect: Redland maritime forces are unwilling to threaten Pinkland. - Effect: Terrorists are deterred from aggressive actions in the MAO. - Nonlethal fires. - Focus: Redland navy commanders, terrorist leaders along coast.
Sustainment	- CASEVAC/MEDEVAC of JFMCC forces operating in Redland and Greyland. - Transportation of HA supplies to Greyland. - Engineering support at Port Yankee. - Priority: CTFs 224, 225, and 220.
Protection	- FP of forces operating ashore in Redland and Greyland. - Mine clearance OPS in Redland territorial sea. - Priority for protection: forces ashore, amphibious forces, ISBs, and CTF 225 LOG ships.
IO	- Strategic message: JFMCC is working with international partners to ensure regional security. - PA to deliver FA themes and messages. - EA to disrupt terrorist communications.
Authorities Required	- Enter the land, air, and sea area of Redland for purpose of conducting combat OPS. - Conduct indirect fires against Redland ground forces. - Erect barriers in Redland.
Decision Points	
Branches	
Risk	

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 10 of 11)

	Phase V: Enable Civil Authority
Operational Functions and Other Staff Activities	
C2	- JTF and functional components relinquish OPCON of attached forces and disestablish. - Theater service components execute OPCON/TACON of forces IAW geographic combatant commander (GCC) deployment order (DEPORD).
INTELLIGENCE	- Priority: Assess ability of Pinkland to assume stabilization missions.
Movement and Maneuver	- Maritime forces redeploy per GCC DEPORD. - Theater service components resume activities and mission IAW theater campaign plan.
Fires	
Sustainment	- Disestablish FLS. - Determine LOG support requirements to enable Pinkland maritime OPS.
Protection	- Coordinate protection requirements with Pinkland and other multinational partners.
IO	- Strategic message: Pinkland, in coordination with international partners, can effectively patrol and regulate its EEZ. - PA to deliver GCC themes and messages.
Authorities Required	- Revert to phase I authorities. - Transport foreign civilians on military aircraft for the purpose of conducting HA missions. - Embark foreign nationals on Navy ships for the purpose of providing acute medical care during HA missions.
Decision Points	
Branches	
Risk	

Figure G-1-1. Joint Force Maritime Component Commander Synchronization Matrix Example (Sheet 11 of 11)

APPENDIX H
Operation Assessment

H.1 PURPOSE

This appendix provides an overview of operation assessment and approaches to developing an assessment plan and conducting operation assessment in execution. It complements processes described in JP 5-0; NTTP 5-01.3, Multi-Service Tactics, Techniques, and Procedures for Operation Assessment; NWP 3-32OPS; NTTP 3-32.1; the NATO Operations Assessment Handbook; and Office of the Chief of Naval Operations (OPNAV) M-3500.42A, Maritime Operations Center Standardization Manual.

H.2 OVERVIEW

Assessment is an integral part of the commander's decision cycle and is a cross-functional activity. Operation assessment occurs during planning and execution. Operation assessment enables measurement of progress toward achieving operational OBJs and supports decision making. Figure H-1 depicts how operation assessment spans planning and execution.

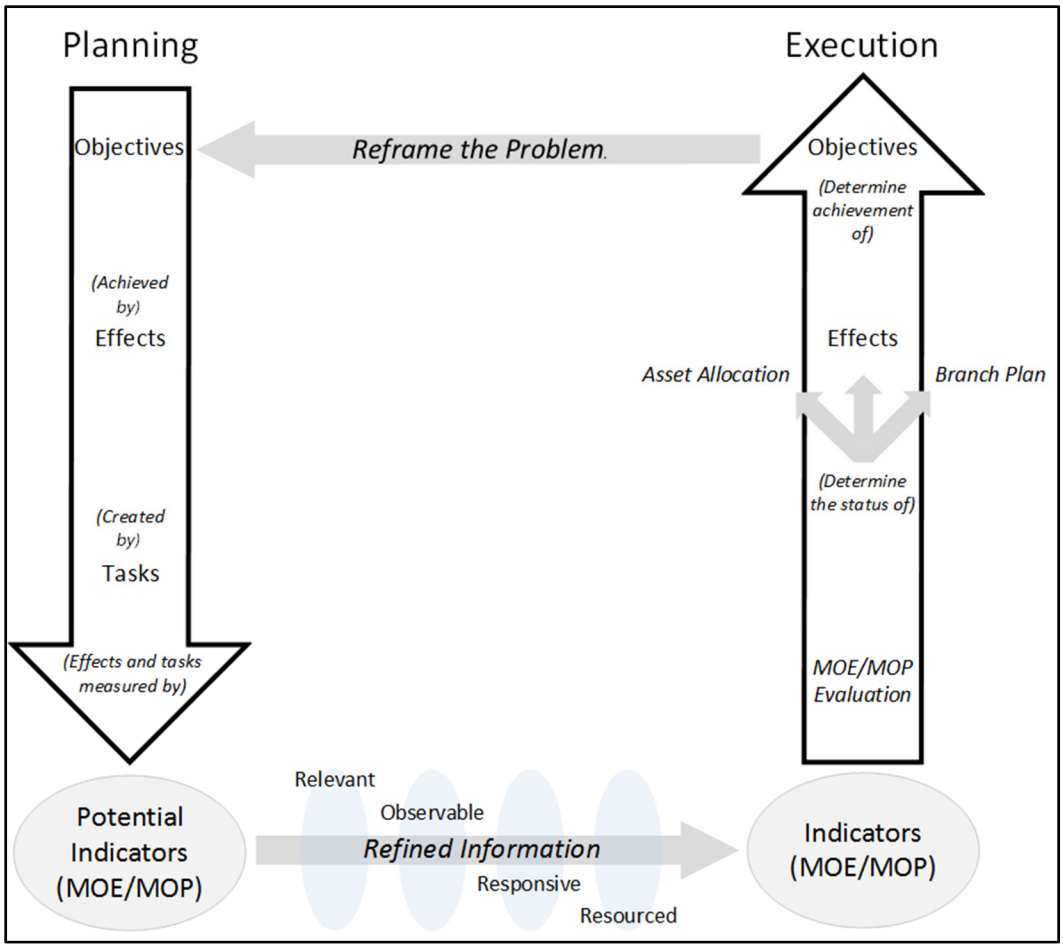


Figure H-1. Assessment in Planning and Execution

The assessment process asks three basic questions: What happened? Why does it matter? What should happen next? Operation assessment answers these questions using the following activities.

1. Collect information and intelligence to determine the current state of the operating environment. Collected data can be quantitative or qualitative but must be relevant to accomplishment of operational OBJs.

2. Analyze collected information and intelligence. Data is analyzed to determine how changes in the operating environment influence COPS and FOPS. In addition, this analysis will inform decision making and provide insight into progress toward achieving operational OBJs.

3. Make conclusions and recommendations ISO decision making. Determine the next steps of the operation and determine progress toward achieving operational OBJs. Negative trends or stalling progress must be addressed.

H.2.1 The Assessment Organization

Within this appendix the assessment organization includes both the assessment cell and assessment working group (AWG). NTTP 3-32.1 describes MOC assessment organizations.

H.2.1.1 Assessment Cell

The assessment cell is the organization within the MOC that is responsible for leading operation assessment. The assessment cell consists of five positions: assessment lead, FUPLANS (assessment) planner, FOPS (assessment) planner, task analyst, and effects analyst. Descriptions of these position can be found in OPNAV M-3500.42A.

H.2.1.2 Assessment Working Group

The AWG is a cross-functional team that assists in developing the assessment plan, validating collection requirements, and validating recommendations from the assessment cell. This WG ensures that assessment is not conducted in isolation from other staff directorates. The assessment cell lead should expect consistent staff representation in the AWG. Membership of the AWG will be designated in a seven-minute drill. Membership will vary by plan but should always include representation from intelligence, OPS, and planning directorates. Figure H-2 provides an example AWG 7-minute drill.

The AWG provides subject matter expertise in the development of the assessment plan. In addition, the AWG validates existing measures to ensure they are relevant, observable, and can be collected IAW the needs of the commander's decision cycle.

Within the AWG, assessment stakeholders strive for focused, relevant, and collectible indicators. Staff representatives develop indicators based on their functional estimates. Subordinate LNOs provide feedback to validate unit reporting that supports performance measures. Collection management representatives streamline indicator sourcing considerations. During execution, the AWG evaluates information and intelligence, forms conclusions, and develops recommendations for follow-on OPS.

H.3 CONDUCTING OPERATION ASSESSMENT

Operation assessment activities occur during planning and execution. Outlined below is the recommended process for developing an assessment plan and using that plan during execution to evaluate OPS.

1. Develop an assessment plan.[24]

   a. Organize for assessment planning.

   b. Receive IPOE/conduct systems analysis.

[24] The steps Develop an Assessment Approach and Develop and Assessment Plan, outlined in JP 5-0, have been combined under Develop an Assessment Plan for simplicity. The analytic tasks for both are contained herein.

<table>
<tr><td colspan="2">Battle rhythm event: AWG.</td></tr>
<tr><td>Meeting chair:
Assessment cell lead.</td><td>Meeting point of contact:
Assessment cell lead.</td></tr>
<tr><td colspan="2">Purpose:
Modify the assessment plan and collection requirements; recommend changes to effects, MOPs, MOEs, and indicators during execution; collect, measure, and analyze indicators and associated operational criteria; conduct operational plan review and determine whether ongoing OPS are achieving commander’s intent; assist planners in preparing recommendations to the commander to address deficiencies in achieving operational intent.</td></tr>
<tr><td>Inputs:
1. HHQ plans/orders.
2. Commander’s guidance/intent, CCIRs.
3. JIPOE/IPOE, plans in development.
4. Orders in execution.
5. ROE/RUF.
6. IO synchronization matrix.
7. Existing MOEs/MOPs and assessment/collection plan.
8. OPS update.
9. Intelligence update.
10. Situation reports.
11. Task assessments.
12. MOE indicators.</td><td>Output:
1. Identification of/revision to MOEs/MOPs.
2. Identification of operational opportunities to seize or risks to mitigate.
3. Planning recommendations (branches/sequels) based on assessment of phase OPS for maritime planning groups.
4. Input to HHQ assessment.
5. OPS assessment update.
6. Commander’s assessment brief (CAB).
7. ROE supplemental requests.
8. Collection requirements and adjustments.
9. CCIR change recommendations.</td></tr>
<tr><td>Preceding battle rhythm events:
IO WG, fires WG, commander’s update brief.</td><td>Follow-on battle rhythm events:
Collection management WG, decision boards.</td></tr>
<tr><td>Meeting frequency:
Weekly during normal OPS.
As required (daily) during crisis or exercise.</td><td>Next meeting date:
To be determined.</td></tr>
<tr><td colspan="2">Meeting venue:
Will vary, typically in classified spaces. May be done via secure VTC or in an online venue.</td></tr>
<tr><td colspan="2">Premeeting coordination:
Assessment cell will develop the assessment plan or develop an assessment and recommendations for consideration.</td></tr>
<tr><td colspan="2">Composition:
Assessment cell (lead), admin/personnel, intelligence collections manager, lead intelligence analyst, FCC, fires, staff judge advocate, LOG readiness center, medical (in humanitarian assistance (HA)/disaster relief (DR) events), political advisor, leads from COPS/FOPS/FUPLANS and OPT planning efforts, subordinate command LNOs, and coalition LNOs. Other representatives may include IO and protection.</td></tr>
<tr><td colspan="2">Note: The information provided above is notional. Specific inputs will vary by command.</td></tr>
</table>

Figure H-2. Example Assessment Working Group 7-minute Drill

c. Analyze OBJs, effects, and task.

d. Develop MOEs and MOPs.

e. Develop a collection plan.

f. Establish assessment responsibilities.

g. Communicate the assessment plan.

2. Collect information and intelligence.

   a. Collect MOPs.

   b. Collect MOEs.

3. Analyze information and intelligence.

   a. Conduct effects assessment.

   b. Conduct risk assessment.

   c. Develop assessment recommendations.

4. Communicate feedback and recommendations.

5. Adapt plans and OPS.

H.4 ASSESSMENT PLANNING

Assessment planning is a continuous effort that is aligned with commander's mission, intent, guidance, and operational plan. The conclusions and recommendations developed by the assessment organization are a result of comparing expected outcomes to actual results. These comparisons are used to inform the commander as to the efficacy of OPS and to identify opportunities and risk in FOPS. Assessment planning coincides with Navy planning (see Figure H-3).

The assessment plan provides indicators to recognize the following: progress of the operation, the efficient use of the force, operational risk, and decision support criteria. In addition, the assessment organization may develop indicators in response to HHQ assessment requirements.

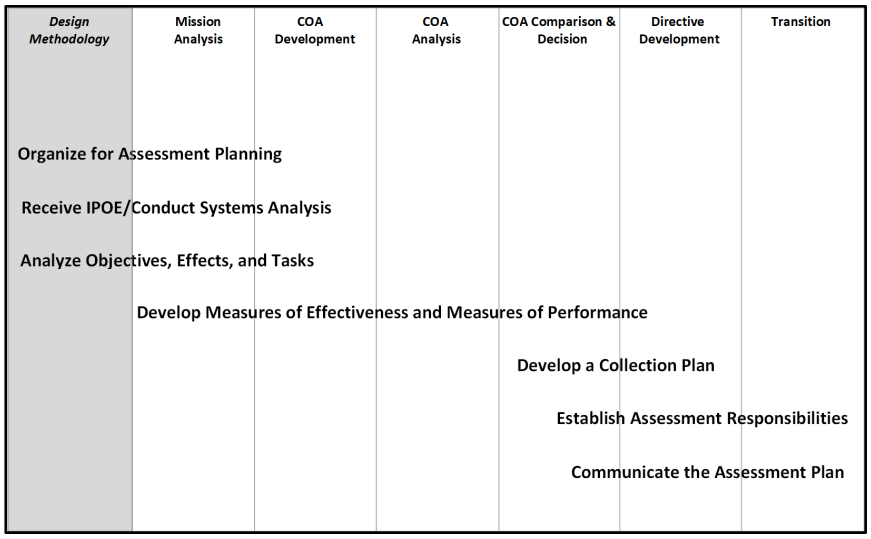


Figure H-3. Developing an Assessment Plan within Navy Planning

The assessment plan must nest with the operational plan. The focus of assessment planning may evolve based on the commander's expectations as well as the development of the operational plan. The OPT lead should expect the assessment cell to actively participate in planning.

An assessment plan is developed at the level of the operational plan that it supports. However, it may also be necessary to support the HHQ's operation assessment as well. These requirements may not be satisfied by assessment plan developed to support the maritime operational plan. The assessment cell should be prepared to augment the assessment of HHQ.

Other considerations particular to assessment planning include:

1. Determine required assessment product formats (e.g., collection plans, appendix directive, data visualization).

2. Determine applicable terms of reference, socialized throughout the staff.

3. Use the AWG as the entity to develop and vet the assessment plan.

4. Recognize that assessment plan development will lag operational planning until a COA is approved.

5. Using both observation-based (quantitative) and judgment-based (qualitative) measures provides both a mathematical rigor to offset bias and a focus on intangibles based on human judgment.

6. Do not limit measures development only by what is provided by functional areas in the AWG for information and intelligence. Be proactive in systems analysis with intelligence representatives IOT better develop MOEs.

7. Be proactive in understanding and shaping subordinate TF reporting requirements to COPS IOT focus the resources of performance indicators.

8. Coordinate and synchronize through the WG what measures "answer the mail" in terms of task completion, desired effects condition recognition, and OBJ achievement.

What follows is a description of each step of the assessment planning process.

H.4.1 Organize for Assessment Planning

Operation assessment begins with the initiation of planning. The assessment cell leader must organize assessment personnel, gather the appropriate directives and guidance, and develop a planning timeline. Inputs to assessment planning include HHQ directive, HHQ assessment plan (if available), maritime commander's intent and planning guidance, IPOE products, and maritime planning products (as they become available), to include briefs, orders, CONOPS, synchronization matrix, and DSM. Assessment planners establish a battle rhythm for the assessment organization, publish an initial 7-minute drill for the AWG, and determine the key events with the commander and staff leadership. Finally, assessment planners will need to identify potential data sources and obtain IPOE products.

H.4.2 Receive Intelligence Preparation of the Operational Environment/Conduct Systems Analysis

The assessment organization leverages the IPOE, other staff estimates, and OPT products (e.g., COG analysis) to gain insight into the OE. In addition, the assessment organization may conduct their own systems analysis in anticipation of indicator development.[25] This substep provides the context for understanding the operational plan and developing the assessment plan.

[25] There are several techniques available to conduct systems analysis. One way is to analyze the PMESII sources of power within the operating environment. Another is to analyze the relevant actors, their functions, relationships, and tensions. This technique is described in conceptual planning.

H.4.3 Analyze Objectives, Effects, and Tasks

An assessment plan is unique to the operation. Prior to developing an assessment plan, the assessment cell must understand the commander's intent, operational OBJs, effects, and tasks. This is an opportunity to evaluate, understand and seek clarification of a plan's logical linkages and organizational structure. A commander will provide initial commander's intent and planning guidance as an output of conceptual planning. Commander's guidance and intent provides direction for the assessment plan development.

An assessment plan can be based on conceptual or detailed planning. Organize the assessment plan from the structure of the operational approach determined during conceptual planning or from the operational phasing determined during detailed planning. An assessment plan that is framed by operational approach will be organized via OBJs that relate to lines of effort or LOOs. An assessment plan that is framed by detailed planning will be organized via OBJs that relate to a phasing construct. In either case, the assessment organization must appreciate how a plan organizes OBJs, effects, and tasks. Figure H-4 depicts a conceptual overview of an assessment plan.

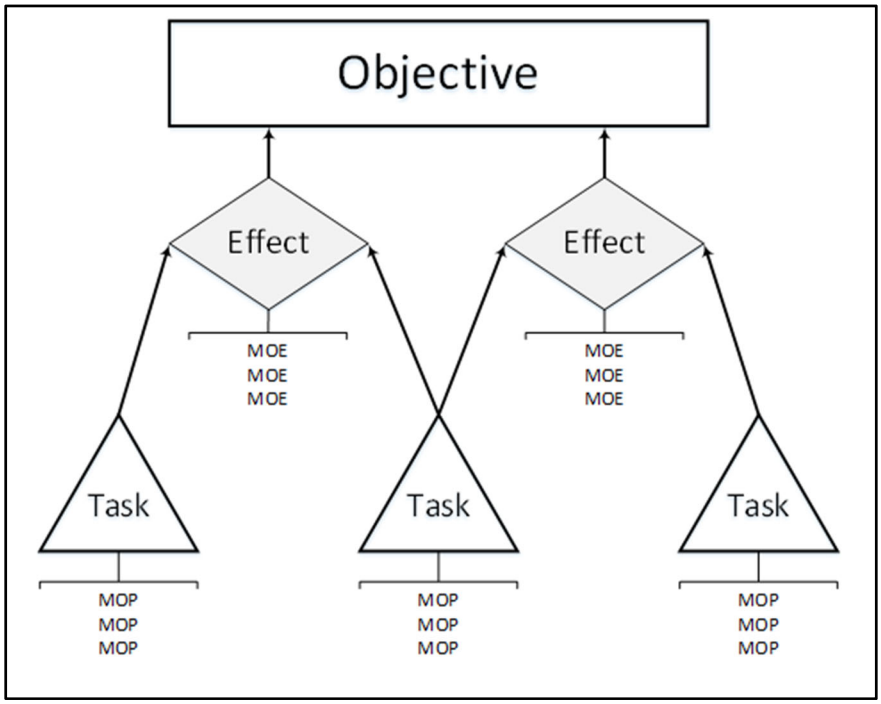


Figure H-4. Assessment Plan Concept Overview

Assessment planners must keep in mind that it is not feasible to assess every element of a plan. The assessment plan must focus data collection and analysis to determine operational progress and to support the commander's decision making. It is important to vet the assessment plan through the AWG and staff leadership to focus the assessment effort to best support the commander's decision cycle.

Decision points are another part of the operational plan that must be considered by the assessment organization. A decision point may result from achieving an OBJ or effect, completion of a task, validation of an assumption, analysis of risk, or meeting phase or LOO transition criteria. The evaluation of indicators related to decision points is an essential part of operation assessment and will provide additional focus for the assessment organization. The assessment cell should develop indicators that enhance the recognition of decision support criteria based on the decision point and its related CCIRs. Decision points may include:

1. Potential adjustment decision recommendations:

    a. Shift forces and/or resources ISO the ME or SE.

b. Shift or delay the action of decisive efforts, SEs, or SHs.

c. Adjust force actions based on anticipated risk.

d. Recognize anticipated opportunities to seize or threats to counter.

2. Potential execution decision recommendations:

a. Adjust the plan or realign the approach.

b. Shift the ME.

c. Shift geographical focus of the decisive efforts, SEs, or SHs.

d. Execute a branch or sequel.

e. Transition phases, stages, or steps.

Figure H-5 illustrates how the JFMCC assessment cell organized the assessment plan for phase II of Operation FA.

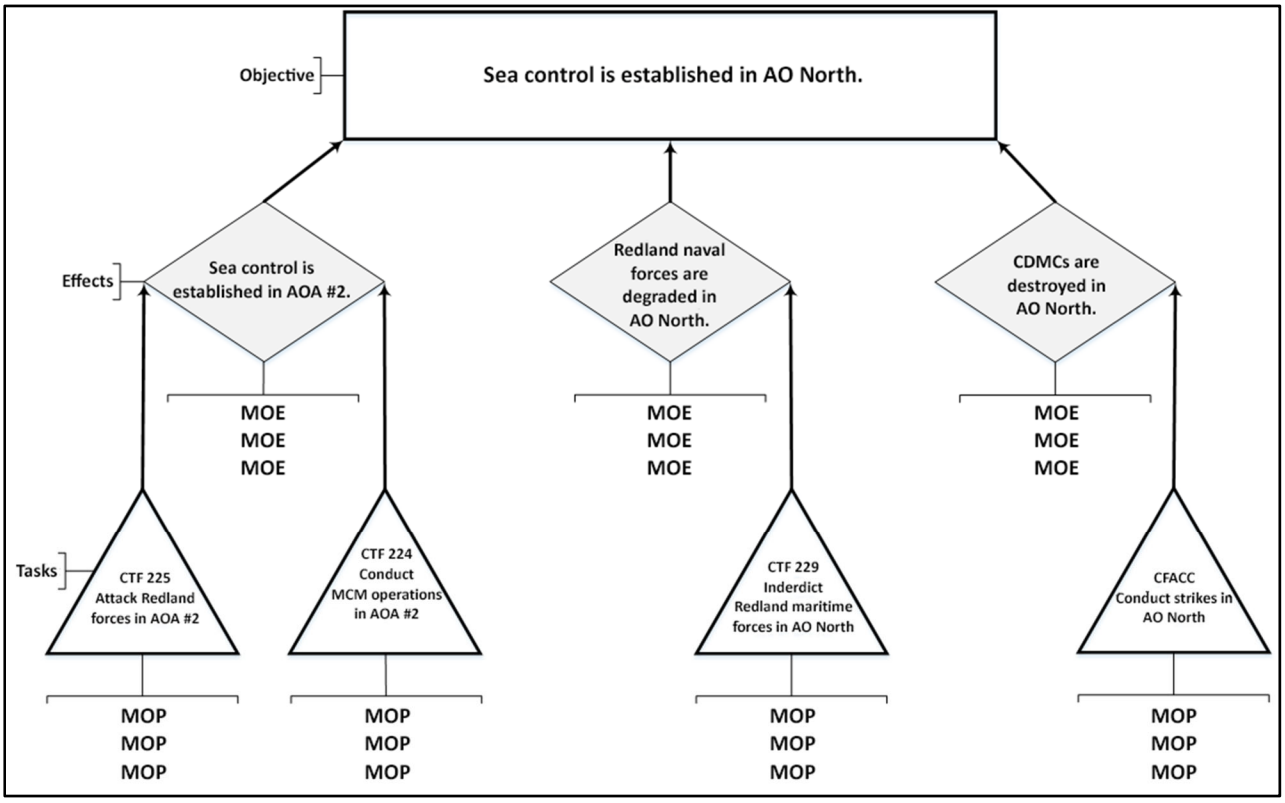


Figure H-5. Example of Organization of Assessment Plan

H.4.4 Develop Measures of Effectiveness and Measures of Performance

The development of indicators is central to assessment planning. Indicators come in two categories. MOEs are indicators used to measure the current state of external systems (enemy forces, commercial shipping, population, etc.) over time. They are measures of output or result. MOEs answer the question, "What changed in the operating environment?" MOPs are metrics used to measure task accomplishment. MOPs answer the question, "What did

we do to create the change in the operating environment?" Indicators, whether MOEs or MOPs, must be relevant to the OBJ, effect, or task, observable in a way that can be measured, responsive to the information needs of the assessment organization and commander, and resourced via a means of collection.

In developing indicators, an understanding of information characterization and type will help focus and align assessment plan development. Information can be characterized by a combination of any two of the following characteristics: quantitative, qualitative, OBJ, or subjective. Quantitative data is numerical. Qualitative data is descriptive. OBJ data is factual and observable. Subjective data is based on opinion and judgment.

Data will be presented as one of four types. An understanding of data type will provide insight into the kinds of analysis that may be performed. Figure H-6 describes the types of data and associated analysis. Types of data are:

ER-1 1. Nominal data is organized by name. Examples of nominal data include sex, eye color, or a friendly force description by nation.

Data Type	Frequency Distribution	Median and Percentiles	Add or Subtract	Mean and Standard Deviation	Ratio
Nominal	Yes	No	No	No	No
Ordinal	Yes	Yes	No	No	No
Interval	Yes	Yes	Yes	Yes	No
Ratio	Yes	Yes	Yes	Yes	Yes

Figure H-6. Data Type and Permitted Computation

2. Ordinal data is organized by some order or scale without specific value from one score to the next. Examples of ordinal data include race finish positions (1st, 2nd, 3rd) and Likert scales (rate pain from 1–10).

3. Interval data is ordinal data that has a standardized value between points, without a true zero. Temperature is an example of interval data.

4. Ratio data is interval data with a true zero. Distance in miles from one location to another is an example of ratio data.

There are different techniques used to develop indicators. One links indicators to outcomes based on a series of assessment questions (see the example below). This approach is described in detail in the JP 5-0. MOE development should be informed by the intelligence estimate, COG analysis, task analysis, risk assessment, and DSM. MOP development is informed by task analysis, TF LNOs, and subject matter experts. In addition, the UJTL and UNTL can provide insight into indicator development.

The following is an example of how this method was used to develop MOEs for the JFMCC assessment plan for phase II of Operation FA.

OBJ: Sea control is established in AO NORTH.

Effect: Sea control is established in AOA 2.

(1) Assessment Question: How do we recognize that sea control is established in AOA 2?

IRs:

(a) Are Redland surface combatants patrolling in AOA 2?

i. MOE: number of Redland surface combatants located within AOA 2.

ii. MOE: type of surface combatant located within AOA 2

iii. MOE: amount of time Redland surface combatants remain within AOA 2.

(b) Are Redland subs patrolling in AOA 2?

i. MOE: number of Redland subs within AOA 2.

ii. MOE: description of Redland sub OPS within AOA 2 (combat, ISR, surfaced, submerged).

(c) Has Redland conducted offensive mining in AOA 2?

i. MOE: Minefield location within AOA 2.

ii. MOE: Size and type of minefield located within AOA 2.

The following is an example of how this method was used to develop MOPs for the JFMCC assessment plan for phase II of Operation FA.

OBJ: Sea control is established in AO NORTH.

Effect: Sea control is established in AOA 2.

(1) Assessment question: What friendly force information is required to ensure JFMCC forces have completed tasks to standard?

(a) Task: CTF 225 attacks Redland forces in AOA 2.

i. MOP: 75 percent or more of CTF 225 combat power located within AOA 2.

ii. MOP: average on-station time for CTF 225 units within AOA 2.

iii. MOP: CTF 225 is positioned to interdict Redland naval forces attempting to enter AOA 2.

(b) Task: CTF 224 conduct mine countermeasures (MCM) OPS within AOA 2.

i. MOP: Q-route is established within AOA 2.

ii. MOP: percent probability of detection attained within AOA 2 (goal is 80 percent).

iii. MOP: time to establish Q-route to 80 percent probability of detection in AOA 2.

(c) Task: CTF 227 conducts airborne surveillance and reconnaissance in AO NORTH and AO WEST.

i. MOP: CTF 227 on-station time over AO NORTH.

ii. MOP: CTF 227 on-station time over AOA 2.

iii. MOP: number of CTF 227 sorties dedicated to AO NORTH.

iv. MOP: number of CTF 227 sorties dedicated to AOA 2.

Once an indicator (MOE or MOP) has been identified, baseline information (normal patterns of life, typical operational tempo, etc.) can be established. IPOE and systems analysis will provide the baselines associated with

the selected indicators. Other necessary information relative to indicators include the desired or acceptable condition and/or rate of change, collection source, frequency of collection, and information type or format (number, location, physical description). In execution, changes in both MOEs and MOPs are evaluated against baselines to reveal trends over time. These trends inform task and effect assessment.

H.4.5 Develop a Collection Plan

The assessment organization, in coordination with the collection management WG, develops the assessment collection plan. The collection plan is a tab to the assessment appendix within the directive. The collection plan will adjust over time as the operation progresses and the environment and organization priorities evolve. The collection plan is continually refined and vetted to keep pace with the speed of the operation, the changing environment, and the commander's decision cycle. The format for an assessment collection plan will vary by command. A note of caution: Over-metricizing the assessment collection plan burdens both the staff and subordinate TFs.

Figure H-7 is the completed JFMCC assessment framework for Operation FA. It provides a comprehensive view of all indicators (MOEs, MOPs), including specific data parameters and indicators linked to decision points.

H.4.6 Establish Assessment Responsibilities

Operation assessment requires a cross-functional approach to data collection and analysis. Clearly delineating responsibilities for data collection, analysis, and product development is essential when moving an assessment plan from planning to execution.

MOPs are typically collected within the MOC by the maritime intelligence operations center (MIOC). Similarly, MOPs are typically tracked within the MOC by COPS. Understanding where the information and intelligence will be processed and distributed highlights the importance of MIOC and COPS representation within the AWG. Once information and intelligence are received, they are normally analyzed within the assessment cell and the AWG. The assessment cell is responsible for creating products that depict the results of the analysis and any conclusions and recommendations regarding follow-on OPS.

H.4.7 Communicate the Assessment Plan

The assessment plan may consist of several products. The plan should result in a formal assessment annex (or an appendix to Annex C, Operations) to the directive being produced. It should be supported by a collection plan that is synchronized through the collection management WG. The collection plan should not only include requisite sourced indicators that support the recognition of effects and OBJ achievement, but also additional indicators organized to support anticipated decisions and operational mission risk.

Members of the AWG should socialize the collection plan with their respective sections internally to the MOC. The assessment cell lead will socialize the plan with staff principals and the command group through various event venues in the OPS process and adjust the plan as required prior to the commander authorizing the directive. The planning lead will socialize the plan with HHQ and adjust the plan as required.

A final consideration is the medium that will convey the assessment. The development of products to convey the assessment are numerous and unique to each commander perspective. Thermographs, stoplight charts, spider charts, geospatial examples, graphs, matrices, qualitative narratives, and a host of other options exist in current doctrine and operational research that may best convey the assessment to the commander. For more information on these methods, see NTTP 5-01.3.

Phase II *Operation FREEDOM ASSURANCE*

Objective: Sea control is established in OA North

			Baseline	Acceptable Condition, Desired Rate of Change	Data Requirements (Source, Periodicity, Format)	Remarks
Effect	**Sea control is established in AOA #2.**					**DECISION POINT: Sea control in phase II is transition criteria for phase III.**
	MOE:	Number of Redland surface combatants located within AOA #2.	2x Redland surface combatants patrol IVO AOA #2 daily.	No Redland surface combatants within AOA #2.	Multiple sources, MIOC; reported daily during phase II; via number, type, and geolocation.	Redland surface combatants include all types and classes.
	MOE:	Time Redland surface combatants remain within AOA #2.	Daily patrol IVO AOA #2 normally 8 hours.	Zero hours is the goal. Intial a DROC indicating an observed reduction from 8 hours daily.	Multiple sources, MIOC; reported daily during phase II; in hours.	Redland surface patrols last between 4-8 hours depending on vessel type and patrol orders.
	MOE:	Redland submarines located within AOA #2.	1X submarine patrols IVO AOA #2 as part of weekly patrol.	No Redland submarines within in AOA #2.	Multiple sources; reported daily during phase II; geolocation.	Redland submarine patrols occur sporadically and are often less than 1 week in duration.
	MOE:	Minefield location IVO AOA #2.	Minefiled located IVO AOA #2. Quantity and type unkown.	Redland mines do not impede follow on amphibious operations.	Source CTF 224, report daily, clearance time remaining and POD percent.	This MOE links to tasking for CTF 224.
	MOE:	Number of neutral commercial ships transiting IVO AOA #2.	4–6 commercial ships transit IVO AOA #2 daily.	DROC trend to zero transits IVO AOA #2 by end of phase II.	Multi-source, NCAGS, MIOC; report daily, geo-location, numbers and flag-state.	Notice to Mariners has been submitted to re-route commercial traffic.
Task	**CTF 225 attack Redland forces in AOA #2.**					
	MOP:	Percent of CTF 225 combat power located within AOA #2.	100 percent	75 percent of CTF 225 combat power is available to indirect Redland forces within AOA #2.	Position reported in GCCS.	This requirment was determined by CTF 225 and takes into account logistical requirements outside of AOA #2.
	MOP:	CTF 225 positioned to interdict Redland naval forces attempting to enter AOA #2.	NA	All Redland forces attempting to enter AOA #2 are interdicted.	CTF 225 daily situation reports.	At the discretion of CTF 225. Interdiction may include non-lethal and lethal fires.
Task	**CTF 224 conduct MCM operations in AOA #2.**					
	MOP:	Q-route is established in AOA #2.	NA	Q route complete to 80 percent POD within 72-96 hours.	Source CTF 224, report daily, clearance time remaining and POD percent.	Q-route determination is at the discretion of CTF 224.
	MOP:	Time to achieve an 80 percent POD.	72 hours to clear an area the size of AOA #2 to 80 percent POD.	Q route complete to 80 percent POD within 72-96 hours.	Source CTF 224, report daily, clearance time remaining and POD percent.	**DECISION POINT: If 80 percent POD not achieved, within 96 hours the decision to conduct the amphibious landing in phase III will be reconsidered by the Commander.**
Task	**CTF 227 conduct surveillience and reconnaissance in OA.**					
	MOP:	CTF 227 on-station time in OA North.	Planned is 8 flight hours in OA North daily.	6 hours daily.	Source CTF 224, report daily, time on-station in hours.	CTF 227 tasking includes airborne S&R in OA North and OA West. Times are for OA North only. In phase II OA North has priority of coverage.
	MOP:	CTF 227 on-station time in AOA #2.	Multiple visits for a total of 4 hours of coverage in AOA #2 daily.	3 hours daily.	Source CTF 224, report daily, time on-station in hours.	AOA #2 is within OA North. Within OA North AOA #2 has priority of coverage..

Figure H-7. Example Joint Force Maritime Component Commander Assessment Framework

H.5 OPERATION ASSESSMENT DURING EXECUTION

During the execution of an operation, the assessment organization will use the metrics established in the assessment plan to support the commander in determining progress toward achieving operational OBJs. The assessment process consists of three distinct parts—information collection, evaluation, and resulting conclusions and recommendations/decisions—and is clearly evident within the commander's decision cycle.

Information and intelligence are collected IAW the commander's priorities and established collection plans. A formalized assessment plan developed ICW operational planning will focus assessment plan collection efforts. Information and intelligence are then evaluated as to their importance to the commander. Recommendations and decisions regarding FOPS are formed and distributed to the appropriate organization for action within the MOC.

H.5.1 Collect Information and Intelligence

Collection of MOEs and MOPs is at the heart of operation assessment. The collection plan will determine the collection priorities for MOEs and MOPs.

H.5.1.1 Collect Measures of Effectiveness

MOEs are indicators that measure the current state of systems (enemy naval forces, commercial shipping, a local population, etc.). Given the external nature of MOEs (i.e., not friendly force information), many are collected and analyzed by the MIOC before being distributed to the assessment organization. That said, some MOEs may be collected via other sources, depending on the nature of the mission. For example, in freedom of navigation OPS, information on commercial shipping may be collected by naval cooperation and guidance for shipping activities, whereas, during HA/DR OPS, information may come from NGOs, DOS, or U.S. Agency for International Development.

H.5.1.2 Collect Measures of Performance

MOPs are used to assess friendly activity and task accomplishment. They answer whether the activity took place and if it was conducted to standard. Given the internal nature of MOPs, they are typically collected by COPS as part of normal TF reporting. The assessment organization will use the collected MOPs to determine whether friendly force activities were conducted to standard.

H.5.2 Analyze Information and Intelligence

In analyzing information and intelligence, the assessment organization endeavors to identify operationally significant trends and changes within the OE and develop conclusions regarding the impact to maritime OPS and recommendations regarding FOPS.

H.5.2.1 Conduct Effects Assessment

Effects assessment provides the commander with the current status of progress toward or away from attaining operational effects and OBJs. Effects assessment is the process of analyzing trends in system behavior or capabilities that result from activities conducted by the maritime commander to determine progress toward achieving operational OBJs. The trend analysis is based on MOEs and MOPs developed during planning. Effects assessment includes effects analysis, task analysis, and the identification of any deficiencies that inhibit the achievement of OBJs. Effect assessment should also include assessment of information that correlate to decision points, transition criteria, and risk.

Effects assessment asks two basic questions: Was the task completed to standard, and was the desired effect achieved? The aim of effect assessment is to understand why an effect was or was not achieved and use that analysis to understand progress toward achieving operational OBJs. Accordingly, four broad outcomes of effects assessment are defined below.

1. Effect achieved/task completed: This is the desired outcome as outlined in the plan. In this instance, it is worth determining if there is a clear association between the task and effect.

2. Effect achieved/task not completed: In this case, a favorable outcome is achieved without the associated friendly activity. This may indicate a faulty assumption regarding the relationship between the task and effect. Analysis will focus on why the effect was achieved and why the task was not completed.

3. Effect not achieved/task completed: In this case, there may be a faulty assumption regarding the relationship between task and effect similar to that in case two. In this case, the analysis will focus on whether the planned friendly activity was correctly chosen.

4. Effect not achieved/task not completed: In this case, it is important to understand why the task was not completed. This may result a change in asset allocation or other changes to COPS to complete the task to standards outlined during planning.

The key in each case is to understand why the broad outcome has occurred and then communicate any conclusions and recommendations regarding risk and opportunities for follow-on OPS.

H.5.2.2 Conduct Risk Assessment

Operational risk is identified and mitigated throughout planning. The assessment organization should develop indicators that support the recognition of operational risk. This effort helps key leaders recognize opportunities to seize or threats to counter depending on the speed of the problem and timing. The additional indicators are captured in the assessment plan and resourced through collection management. During task analysis, the assessment cell will determine whether the tasks specifically assigned to mitigate risk were accomplished to standard and whether conditions were established that further diminish risk. Additionally, in developing conclusions and recommendations, the assessment organization will identify new opportunities and associated risk.

H.5.2.3 Develop Assessment Recommendations

Upon completion of effects assessment, the assessment organization will develop conclusions and make recommendations for follow-on OPS. Generally, recommendations will influence COPS or planning. Assessment recommendations may include continuing OPS as planned, small adjustments to OPS to include realignment of resources, changes to priority of effort, adjustments to command relationships, planning and/or execution of branches and sequels in response to a decision point, or reframing the problem to begin planning anew. The nature and specificity of the recommendation is situationally dependent. The importance of vetting the assessment recommendation via the AWG and other staff decision boards prior to briefing the commander cannot be overstated.

H.5.3 Communicate Feedback and Recommendations

Assessment products include the assessment plan, written reports and summaries, and data visualization products for delivery at established battle rhythm events. OPNAV M-3500.42A depicts a Commander's Assessment Board as the venue to provide operation assessment feedback and commander's decisions on OPS and planning. In practice, the venue for communicating operation assessments to the commander varies but includes the commander's update brief, decision board, and the planners' board.

H.5.4 Adapt Plans and Operations

Upon receiving the feedback and recommendations from the staff, the commander will provide direction and guidance for follow-on OPS. This may result in changes to COPS, conducting crisis planning, and/or new contingency planning efforts. These efforts will require adaptation by the assessment organization. This may include refocusing the assessment plan to ensure that the assessment remains aligned to the operational plan. This occurs as frequently as necessary to ensure that the assessment plan remains relevant.

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APPENDIX I
Staff Estimates

I.1 PURPOSE OF ESTIMATES

Staff estimates provide the commander with relevant information to support decision making throughout planning and execution. An estimate is a detailed evaluation of how factors in a staff section's functional area or subordinate commander's warfare area affect the mission.

I.2 TYPES OF ESTIMATES

There are three types of estimates: staff estimates, estimates of supportability, and commander's estimates (often synonymous with a level 1 plan or level 1 planning detail). The annexes in this appendix provide sample formats for general and specific staff estimates (including a prefilled sample estimate), estimates of supportability, and commander's estimates. The samples are consistent with common practice, and not prescriptive. Planners should modify these formats to suit their needs or the commander's expectations.

I.2.1 Staff Estimates

During planning, staff estimates support staff contributions to planning and form the basis for annexes and appendixes of orders or plans. During execution, they provide functional insight and assessment of ongoing OPS as well as a visualization of implications for FOPS. There are several types of estimates that a maritime staff may be required to provide or support in development. Generally, any staff representative that provides input to a commander's decision process or participates in planning should produce or contribute to a staff estimate.

Estimates may be presented as text documents or graphic or oral presentations. Their form depends on the time available, command SOPs, and the level of command. However, the format should not be an impediment for providing timely information that is relevant to the commander's decisions.

Types of staff estimates generated by maritime staffs include, but are not limited to:

1. OPS estimate
2. Personnel estimate
3. Intelligence estimate
4. LOG estimate
5. Communications estimate
6. Civil-military operations (CMO) estimate
7. IO estimate
8. Special staff estimates (e.g., legal, PA, medical).

See annexes 1 through 5 of this appendix for sample formats of select staff estimates.

I.2.2 Estimates of Supportability

The JFMCC or NCC may require subordinate commanders (e.g., CTFs) to submit estimates of supportability. The JFC may also request the JFMCC or NCC to submit an estimate of supportability to the joint force. In this sense, estimates of supportability ensure alignment among senior and subordinate commanders and staffs with regard to proposed COAs, tasks or missions, and capabilities. The estimate of supportability should indicate the subordinate unit's ability to support each COA and identify risks associated in supporting them. See Annex I-6 for a sample estimate of supportability.

I.2.3 Commander's Estimate

As part of their response to an emerging crisis, geographic combatant commanders (GCCs) may be required to produce a commander's estimate or level 1 planning detail. The commander's estimate provides the SECDEF with military options to meet a potential contingency. Component commanders may be required to submit supporting plans to the commander's estimate. In some cases, especially if the maritime HQ serves as the framework for a JTF HQ, the GCC may request the JTF to submit a commander's estimate. See Annex I-7 for a sample commander's estimate.

I.3 ESTIMATES IN THE NAVY PLANNING PROCESS

Staff estimates are developed and continuously refined throughout planning and execution to ensure that COAs and COPS are supportable and sustainable. In addition, subordinate commanders provide estimates of supportability throughout the planning process. Figure I-1 demonstrates how the staff estimates process aligns with the NPP and how subordinate commanders provide estimates of supportability throughout the planning process.

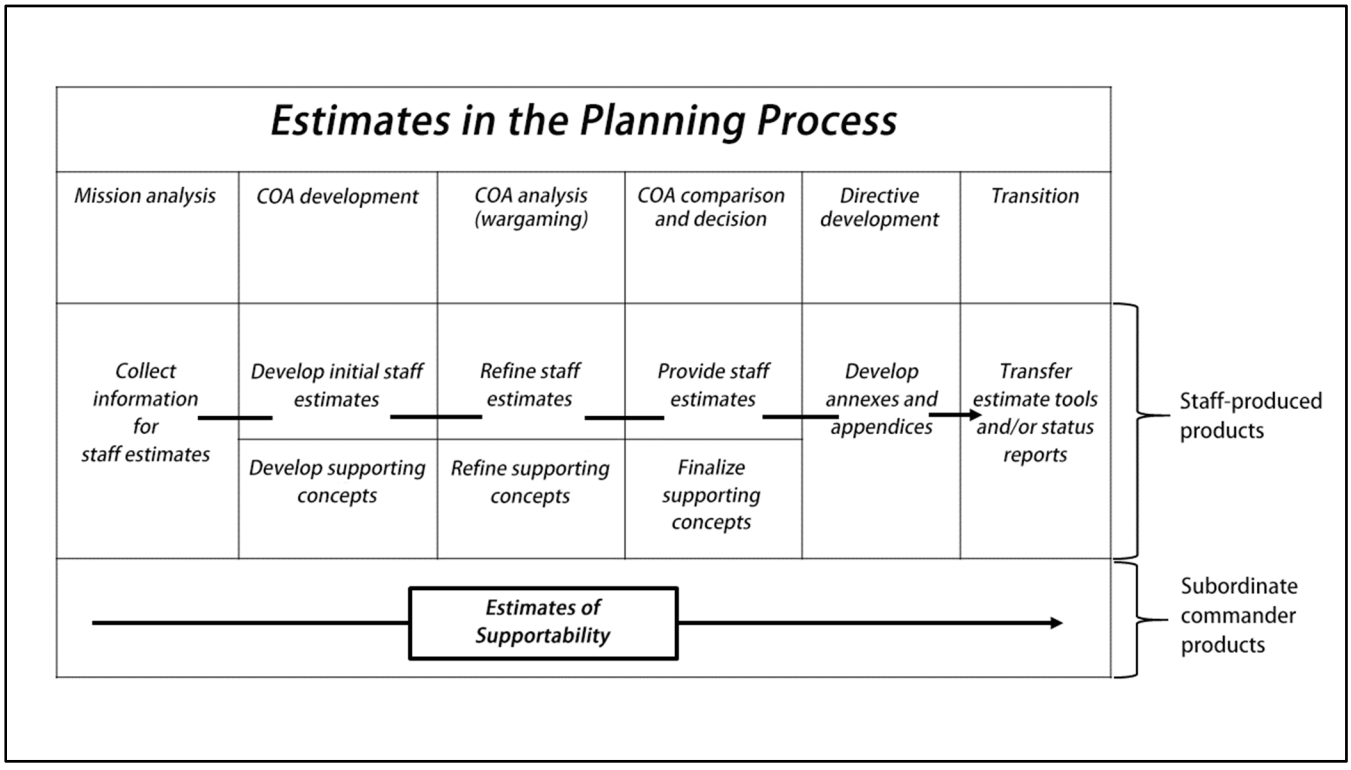


Figure I-1. Estimates in the Planning Process

I.3.1 Prior to Mission Analysis

Upon receipt of the mission or as directed, the functional and special staffs analyze all relevant information to include plans, orders, and directives to determine initial support requirements. This, along with mission analysis, is the information-gathering phase of staff estimates.

The development of facts, assumptions, and the situation analysis (of the AO, area of interest, and enemy, friendly, and support requirements) furnishes the basis for the staff estimate. The estimate consists of significant facts, events, and conclusions based on analyzed data. It recommends how to best use available resources. Rapid decision making and planning hinge on thorough and timely estimates. They are the basis for forming viable COAs.

Staff elements should stay ahead of the planning effort IOT provide key facts necessary to begin understanding the OE and friendly situation. Tardy relevant data can lead to critical errors and omissions during subsequent planning. Staff sections should begin their initial assessment of their specific functional areas and provide this information to the OPT through their representative in the responsible planning group. Staff estimates at this point of planning will most likely be informal because the situation is still unfolding.

I.3.2 Mission Analysis

During mission analysis, staff sections examine the mission from their functional point of view to assist the commander and the staff in gaining a deeper understanding of the operation. Information and analysis developed by the staff sections is forwarded through their representatives to the OPT to inform the overall mission analysis. Depending on the mission, information from specific staff estimates critical to planning should be included in the overall mission analysis briefing to the commander. Staff estimates at this point can be either formal or informal, but consideration should be made to putting the estimate on paper for follow-on plan or order development.

I.3.3 Course of Action Development

After completion of the mission analysis briefing, the staff has an approved mission statement, an initial commander's intent, initial CCIRs, and planning guidance to focus efforts during COA development. With this information, the planning team begins to develop options to accomplish the assigned mission. During the initial stage of COA development, staff sections should develop their formal staff estimates and share them with adjacent and subordinate commands to enhance information flow.

Once the planning team has identified draft COAs, the staff sections change their focus from information gathering to developing concepts of support for these COAs. The first step is to determine the functional area requirements for each individual COA from their staff perspective. For instance, determine the personnel, LOG, protection, and medical support requirements of each COA. The next step is for each staff section to conduct an initial validity test of each COA from their specific staff or functional point of view. This validity check should examine whether the COAs are suitable, feasible, acceptable, distinct, and complete. It is intended to prevent the planning team from presenting COAs to the commander that, for one reason or another, cannot be executed (see Chapter 3 for COA tests for validity). Once the validity check is complete, staff sections should develop individual concepts for how their functions can support the overall operation and any changes required to support the COA variations. This could entail recommendations for basing, LOG flow, force placement, SLOCs, employment options to comply with specific legal regimes, etc. Taken together, the staff estimates provide more detail for the planning team's COAs. The estimates should highlight from a functional perspective the risks (and potential mitigations) and opportunities afforded by each COA.

The next step for the staff sections should be to identify potential tasks for their counterparts in subordinate units to accomplish for their functional area. Those tasks that are important for the overall operation should become part of the base plan or order while the others are incorporated into applicable annexes/appendixes. Additionally, functional staff planners should be prepared to brief their respective concepts of support, by phase, with tasks to subordinates, identifying functional limitations and capability gaps. Lastly, staff sections should assist the OPT in creating the COA development briefing, including providing input for potential COA evaluation criteria.

I.3.4 Course of Action Analysis (Wargaming)

IOT provide input to COA analysis, the staff supports the war-gaming process. This includes not only sending representatives to the war game to provide functional area expertise, but also by providing those representatives for the concepts of support developed by their individual staff sections during the COA development phase. These concepts of support, while not yet finalized, should contain important planning factors and risks associated with each COA. Lastly, the concepts of support assist in development of the synchronization matrix that displays in tabular format how each subordinate TF or organization and operational function should be employed throughout the phases of the operation.

During the war game, the individual concepts of support are the basis for friendly force action based on the operational functions. This information provides fidelity for friendly moves and countermoves. While conducting the war game, functional area representatives should assist the planning team in identifying critical events and decision points from their perspective for the individual COAs. Lastly, functional area representatives should assist in developing evaluation criteria to assess the COAs.

After the war game, the functional area representatives brief their staff sections on what was learned from their perspective. In particular, if enemy or threat actions require drastic alteration to an individual concept of support, the staff section quickly makes appropriate changes and ensures the rest of the staff is informed in case the new concept of support affects others. In the event of a major change in a functional concept of support, the planning team may be required to reanalyze the affected COAs to ensure they remain viable and to consider potential unintended consequences. Individual staff estimates should be updated and included during planning for potential branch plans.

I.3.5 Course of Action Comparison and Decision

Based on the analysis of the war game, staff sections should identify the advantages and disadvantages of each COA from their functional area perspective. The staff estimates should provide a recommended COA to the commander. After a COA has been selected, the staff continues to finalize their concepts of support, ensure the synchronization matrix is complete, and inform the CONOPS.

I.3.6 Directive Development

Once the staff has developed the CONOPS (see Chapter 5), staff estimates should be used to help develop much of the content of the base plan or order, annexes, and appendixes. Figure I-2 shows the relationship between individual staff estimates and portions of a plan or order. Staff estimates in text format can be easily transferred into directives with minimal changes. Maps, overlays, and graphics can be used for clarification.

I.3.7 Transition

During the transition step of the NPP, the staff should transition all relevant information and tools to whomever is monitoring the execution or refining the transitioned plan. Tools that track readiness, such as the status of supply, networks, ISR platforms, etc., help the staff to maintain situational awareness and identify deficiencies that may affect OPS or other staff sections.

I.3.7.1 Lesson Learned

Staffs often forget that the major purposes of estimates during execution are to support the commander's decisions as well to assist in effecting timely staff actions (e.g., resupply, reorienting, etc.). This means that the functional estimates should be tightly linked to the commander's FFIRs and PIRs. Information collected that does not support either FFIRs or PIRs and is not necessary for functional staff actions is probably superfluous and an unnecessary reporting burden.

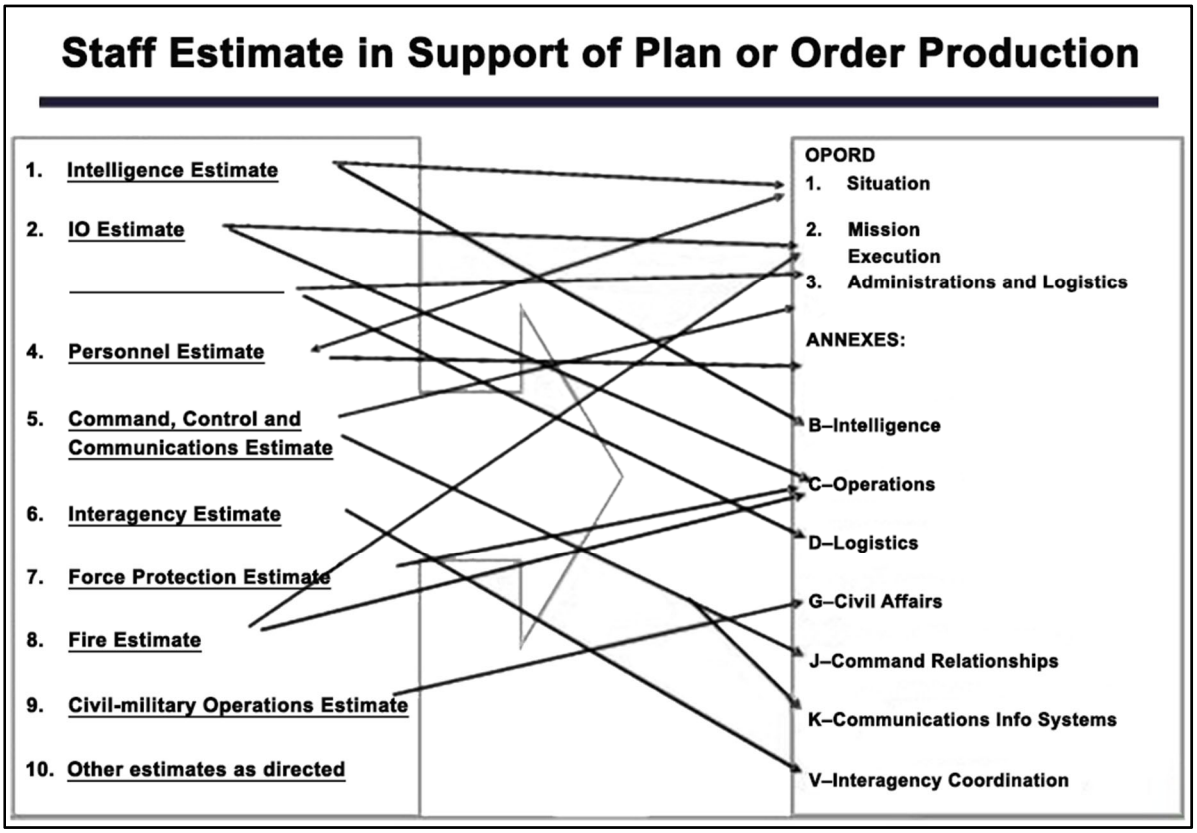


Figure I-2. Relationship between Staff Estimates and Directive Development

I.4 STAFF ESTIMATES DURING EXECUTION—THE RUNNING ESTIMATE

During execution, staff estimates track current status using the tools that either were transitioned from planning or are part of routine OPS. Staff estimates that support COPS are updated frequently to maintain relevancy and are often referred to as running estimates. Figure I-3 provides a sample of a LOG running estimate to track the level of sustainment for the force. Any conclusions and recommendations that are developed during planning should be revised based on the impact of new facts, assumption validation or invalidation, and updated commander's guidance. These modifications are especially useful in operation assessment, decision support, and in preparing for a change of phase or mission. Additionally, continually updated staff estimates enable quicker development of required branch and sequel plans.

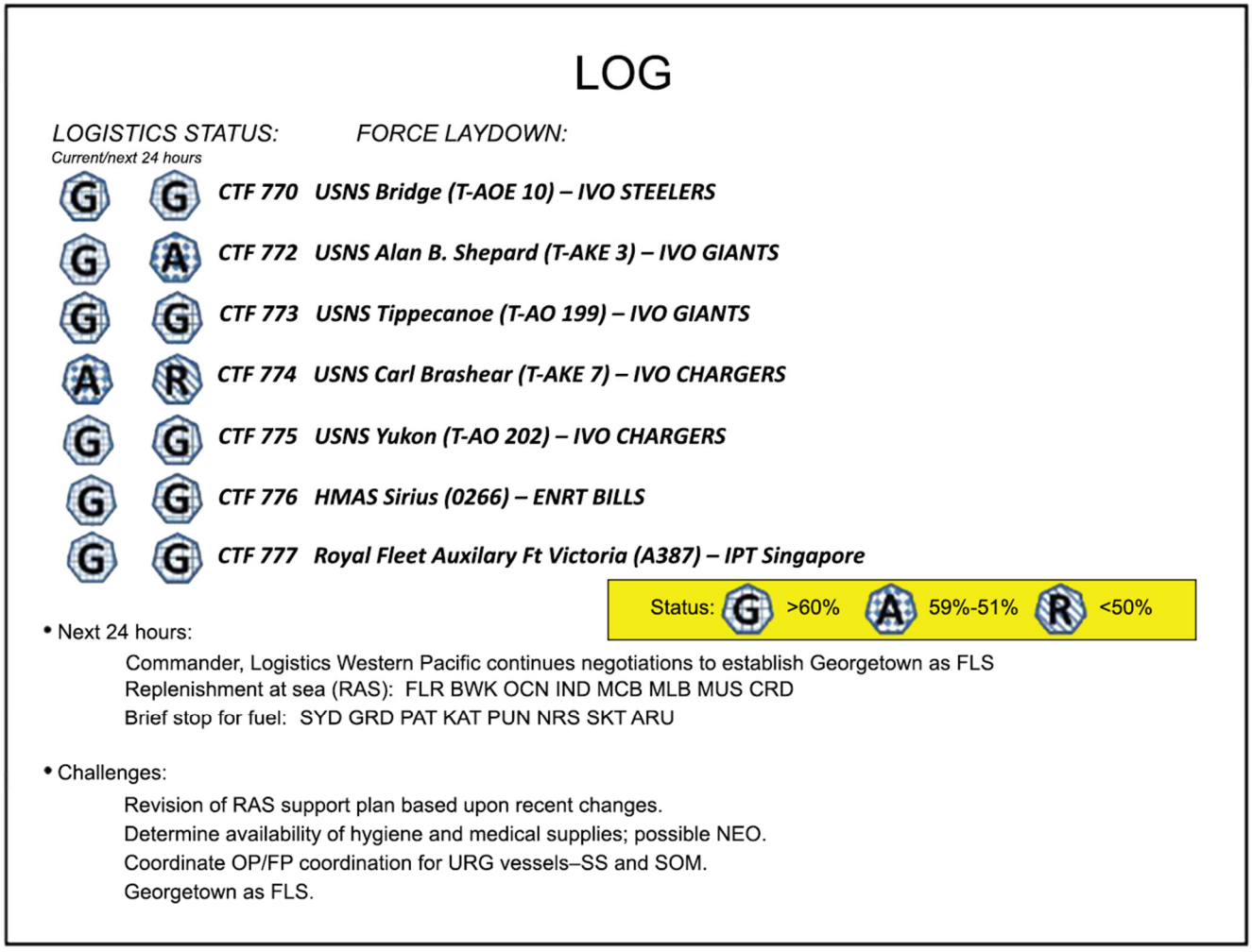


Figure I-3. Sample Logistics Running Estimate Used During Execution

ANNEX 1 TO APPENDIX I

Generic Staff Estimate Format

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

STAFF ESTIMATE NUMBER ____ of _____

( ) REFERENCES:
a. Maps and charts
b. Other relevant documents.

1. ( ) MISSION. State the mission of the command as a whole, taken from the commander's mission analysis, planning guidance, or other statements.

2. ( ) SITUATION:

a. ( ) Characteristics of the AO.

(1) ( ) Weather. How different military aspects of weather could affect specific staff areas of concern and resources.

(2) ( ) Geography. How aspects of physical space could influence specific staff areas of concern and resources (maritime, land, air space).

(3) ( ) Other pertinent facts. Pertinent political, economic, sociological, and psychological factors and infrastructure.

b. ( ) Enemy forces. Enemy disposition, composition, strength, capabilities, and COA(s) as they affect specific staff areas of concern.

c. ( ) Friendly forces.

(1) ( ) Friendly COA(s)

(2) ( ) Current status of resources

(3) ( ) Comparison of requirements versus capabilities and recommended solutions

(4) ( ) Key considerations (governing factors/evaluation criteria) for COA supportability.

d. ( ) Assumptions.

3. ( ) ANALYSIS. Analyze each COA using key considerations (evaluation criteria) to determine advantages and disadvantages.

4. ( ) COMPARISON. Compare COA(s) using key considerations (evaluation criteria). Rank COA(s) for each key consideration. Support each comparison with a decision matrix.

5. ( ) RECOMMENDATIONS AND CONCLUSIONS:

a. ( ) Recommend a COA based on the comparison (most supportable from specific staff perspective).

b. ( ) Discuss issues, deficiencies, and risks with impact mitigations.

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ANNEX 2 TO APPENDIX I

Intelligence Estimate Format

[CLASSIFICATION]

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

INTELLIGENCE ESTIMATE NUMBER ____ of _____

( ) REFERENCES:
a. Maps and charts
b. Other relevant documents.

1. ( ) MISSION. State the mission of the command as a whole, taken from the commander's mission analysis, planning guidance, or other statements.

2. ( ) ENEMY SITUATION. State conditions that exist and indicate how they affect enemy capabilities and the assigned mission. This paragraph describes the AO, the enemy military situation, and the effect of these two factors on enemy capabilities.

a. ( ) Characteristics of the AO. Discuss the physical characteristics of the AO on military activities of all combatants. If an analysis of the area has been prepared separately, this paragraph in the intelligence estimate may simply refer to it and then discuss how it influences the existing situation on the military AO.

(1) ( ) Military geography.

(a) ( ) Topography.

i. ( ) Situation. Describe relief and drainage, vegetation, surface materials, cultural features, and other characteristics in terms of their influence on key terrain, observation, fields of fire, obstacles, cover and concealment, avenues of approach, lines of communication, and landing areas and zones.

ii. ( ) Effect on enemy capabilities. Discuss the influence of topography on broad enemy capabilities, such as attack and defense, describing generally how the topography affects each type of activity. The effect on employment of chemical, biological, radiological, nuclear, and high-yield explosives (CBRNE) weapons; amphibious, airborne, or air-landed forces; surveillance devices and systems; communications equipment and systems; EW; MISO, OPSEC, and military deception; logistic support; and other appropriate considerations should be included.

iii. ( ) Effect on friendly COAs. Discuss the influence of topography on friendly force's military OPS (attack, defense, etc.) in the same fashion as for the enemy in the preceding subparagraphs.

(b) ( ) Oceanography and hydrography.

i. ( ) Situation. Describe the nature of the coastline; adjacent islands; location, extent, and capacity of landing beaches and their approaches and exits; nature of the offshore approaches, including bottom type and gradients; natural obstacles; surf, tides, and current conditions.

ii. ( ) Effect on enemy capabilities. Discuss the impact of the existing situation on broad enemy capabilities.

iii. ( ) Effect on friendly COAs. Discuss the impact of the existing situation on broad friendly COAs.

(c) ( ) Climate and weather.

i. ( ) Situation. Describe temperature, cloud cover, visibility, precipitation, day and night illumination data, and other climate and weather conditions and their general effects on roads, rivers, soil trafficability, and observation.

ii. ( ) Effect on enemy capabilities. Discuss the impact of the existing climate and weather situation on broad enemy capabilities.

iii. ( ) Effect on friendly COAs. Discuss the impact of the existing climate and weather situation on broad friendly COAs.

(2) ( ) Transportation.

(a) ( ) Situation. Describe roads, railways, inland waterways, airfields, and other physical characteristics of the transportation system and their capabilities in terms of rolling stock, barge capacities, terminal facilities, and other pertinent data.

(b) ( ) Effect on enemy capabilities. Discuss the impact of the existing transportation system and capabilities on broad enemy capabilities.

(c) ( ) Effect on friendly COAs. Discuss the impact of the existing transportation system and capabilities on broad friendly COAs.

(3) ( ) Telecommunications.

(a) ( ) Situation. Describe telecommunications facilities and capabilities in the area.

(b) ( ) Effect on enemy capabilities. Discuss the impact of the existing telecommunications situation on broad enemy capabilities.

(c) ( ) Effect on friendly COAs. Discuss the impact of the existing telecommunications situation on broad friendly COAs.

(4) ( ) Politics.

(a) ( ) Situation. Describe the organization and operation of civil government in the AO.

(b) ( ) Effect on enemy capabilities. Consider the impact of the political situation on broad enemy capabilities.

(c) ( ) Effect on friendly COAs. Consider the impact of the political situation on broad COAs for friendly forces.

(5) ( ) Economics.

(a) ( ) Situation. Describe industry, public works and utilities, finance, banking, currency, commerce, agriculture, trades and professions, labor force, and other related factors.

(b) ( ) Effect on enemy capabilities. Discuss the impact of the economic situation on broad enemy capabilities.

(c) ( ) Effect on friendly COAs. Discuss the impact of the economic situation on broad COAs for friendly forces.

(6) ( ) Sociology.

(a) ( ) Situation. Describe language, religion, social institutions and attitudes, minority groups, population distribution, health and sanitation, and other related factors.

(b) ( ) Effect on enemy capabilities. Discuss the impact of the sociological situation on broad enemy capabilities.

(c) ( ) Effect on friendly COAs. Discuss the impact of the sociological situation on broad COAs for friendly forces.

(7) ( ) Science and technology.

(a) ( ) Situation. Describe the level of science and technology in the AO.

(b) ( ) Effect on enemy capabilities. Discuss the impact of science and technology on broad adversary capabilities.

(c) ( ) Effect on friendly COAs. Discuss the impact of science and technology on broad COAs for friendly forces.

b. ( ) Enemy military situation (ground, naval, air, other services).

(1) ( ) Strength. State the number and size of enemy units committed and enemy reinforcements available in the AO. Consider ground strength; air power; naval forces; nuclear, biological, and chemical weapons; EW; unconventional warfare (UW); surveillance potential; and all other strengths (that might be significant).

(2) ( ) Composition. Outline the structure of enemy forces (order of battle) and describe significant organizational features, identity, armament, and weapon systems.

(3) ( ) Location and disposition. Describe the geographic location of enemy forces in the area, including fire support elements; C2 facilities; air, naval, and missile forces; and bases.

(4) ( ) Availability of reinforcements. Describe enemy reinforcement capabilities in terms of ground, air, naval, missile, nuclear, biological, and chemical forces and weapons; terrain and weather; road and rail networks and transportation; replacements and labor forces; prisoner of war policy; and possible aid from sympathetic or participating neighbors.

(5) ( ) Movements and activities. Describe the latest known enemy activities in the area.

(6) ( ) LOG. Describe levels of supply, resupply ability, and capacity of beaches, ports, roads, railways, airfields, and other facilities to support supply and resupply. Consider hospitalization and evacuation, military construction, labor resources, and maintenance of combat equipment.

(7) ( ) Operational capability to launch missiles. Describe the total missile capability that can be brought to bear on forces operating in the area, including characteristics of missile systems, location and capacity of launch or delivery units, initial and sustained launch rates, size and location of stockpiles, and other pertinent factors.

(8) ( ) Serviceability and operational rates of aircraft. Describe the total aircraft inventory by type, performance characteristics of operational aircraft, initial and sustained sortie rates of aircraft by type, and other pertinent factors.

(9) ( ) Operational capabilities of combatant vessels. Describe the number, type, and operational characteristics of ships, boats, and craft in the naval inventory; base location; and capacity for support.

(10) ( ) Technical characteristics of equipment. Describe the technical characteristics of major items of equipment in the enemy inventory not already considered, such as missiles, aircraft, and naval vessels.

(11) ( ) Electronic intelligence (ELINT). Describe the enemy intelligence-gathering capability using electronic devices.

(12) ( ) CBRNE. Describe the types and characteristics of nuclear, biological, and chemical weapons in the enemy inventory; stockpile data; delivery capabilities; nuclear, biological, and chemical employment policies and techniques; and other pertinent factors.

(13) ( ) Significant strengths and weaknesses. Discuss the significant enemy strengths and weaknesses derived from the facts presented in the preceding subparagraphs.

c. ( ) Enemy unconventional and psychological warfare situation.

(1) ( ) Guerrilla. Describe the enemy capability for policy with regard to, and current status in the area of, guerrilla or insurgent OPS.

(2) ( ) Psychological. Describe enemy doctrine, techniques, methods, organization for, and conduct of MISO in the AO.

(3) ( ) Subversion. Describe enemy doctrine, techniques, methods, organization for, and conduct of subversion in the AO.

(4) ( ) Sabotage. Outline the enemy organization and potential for and conduct of sabotage in the AO.

3. ( ) ENEMY CAPABILITIES. List each enemy capability that can affect the accomplishment of the assigned friendly mission. Each enemy capability should contain information on what the enemy can do; where it can do it; when it can start and finish it; and what strength it can devote to the task. The N2 should be able to tell the commander what the enemy can do using its forces in a joint effort. First, of course, the N2 should assess the enemy's ground, naval, and air forces. It is customary to enumerate the weapons of mass destruction and UW capacities separately. Hypothetical examples follow:

a. ( ) Ground capabilities.

(1) ( ) The enemy can attack at any time along our front with an estimated six infantry Divs and two tank Divs supported by 24 battalions (BNs) of artillery.

(2) ( ) The enemy can defend now in its present position with seven infantry Divs supported by two tank Divs and 16 BNs of medium and light artillery.

(3) ( ) The enemy can reinforce its attack (or defense) with all or part of the following units in the times and places indicated:

Unit	Place	Time
315th Airborne Div	Vicinity RESOGA	8 hours after starting time
41st Motorized Div	Vicinity CARDINAL	6 hours after starting time

b. ( ) Air capabilities.

(1) ( ) Starting now and based on an estimated strength of 300 fighters and 100 medium bomber aircraft, the enemy can attack in the AO with 240 fighter sorties per day for the first 2 days, followed by a sustained rate of 150 sorties per day and 60 bomber sorties per day for 1 day, followed by a sustained rate of 48 sorties per day.

(2) ( ) Using airfields in the vicinity of ____, the enemy has sufficient transport sorties to lift one division in a single lift to airfields in the vicinity of ___, ___, and ___ within 4 hours of flying time.

c. ( ) Naval capabilities. Starting now, the enemy can conduct sustained sea and air OPS in the entire area with six guided missile destroyers (DDGs), four guided missile cruisers (CGs), one aircraft carrier (nuclear) (CVN), seven subs (nuclear-powered) (SSNs), a mine force of 20 craft, and 70 gunboats and smaller craft now on station in the area.

d. ( ) Nuclear capabilities. The enemy can employ, at any time and in any part of the AO, an estimated 40–60 nuclear weapons of yields from 2–50 kilotons delivered by cannon and rocket artillery, guided missiles, and aircraft.

e. ( ) Chemical-biological capabilities. The enemy can employ the chemical-biological agents _____, _____, and _____in the AO at any time delivered by air, cannon, rocket artillery, and guided missile.

f. ( ) UW capability. The enemy can conduct UW OPS in the area within 10 days after starting the operation using dissident ethnic elements and the political enemies of the current government.

g. ( ) Joint capabilities. The enemy has limited experience in fighting as a joint force and is more accustomed to fighting independently as services. The enemy can continue to defend in its present position with six infantry Divs, supported by 16 artillery BNs and reinforced by three mechanized Divs, within 8 hours after starting movement. Enemy defense also can be supported by 150 fighter sorties daily for a sustained period and by continuous naval surface and air OPS employing six DDGs, four guided missile frigates, seven SSNs, and one CVN.

4. ( ) ANALYSIS OF ENEMY POTENTIAL COAS. Analyze each capability in light of the assigned mission, considering all applicable factors from paragraph 2, and attempt to determine and give reasons for the relative order of probability of adoption by the enemy. An examination of each enemy COA should include a discussion of the factors that favor or militate against their adoption by the adversary and, when applicable, enemy vulnerabilities attendant to that COA (i.e., conditions or circumstances of the enemy situation that render the enemy especially liable to damage, deception, or defeat). Finally, the analysis should also include a discussion of any indications that point to possible adoption of the COA. For example:

a. ( ) Attack now with forces along the forward edge of the battle area.

(1) ( ) The following factors favor the enemy's adoption of this COA:

(a) ( ) . . .

(b) ( ) . . .

(2) ( ) The following factors mitigate against the enemy's adoption of this COA:

(a) ( ) Road and rail networks will not support large-scale troop and supply movements necessary for an attack in the area.

(b) ( ) Terrain in the area does not favor an attack.

(3) ( ) Adoption of this COA will expose the enemy's west flank to counterattack.

(4) ( ) Except for minor patrol activity in the area, there are no indications of adoption of this COA.

b. ( ) Delay from present positions along the river line.

(1) ( ) The following factors favor the enemy's adoption of this COA:

(a) ( ) There are several excellent natural barriers between the ___ River and the ___ Mountains.

(b) ( ) The effectiveness of the water barriers will improve and trafficability on the upland slopes of the terrain barriers will deteriorate with advent of the monsoon.

(2) ( ) The following factors mitigate against the enemy's adoption of this COA:

(a) ( ) . . .

(b) ( ) . . .

(3) ( ) In the adoption of this COA, the enemy's lines of communication will be restricted by a limited road and rail network that can easily be interdicted.

(4) ( ) The following facts indicate adoption of this COA:

(a) ( ) Aerial photography indicates some preparation of barriers in successive positions.

(b) ( ) Considerable troop movement and pre-positioning of floating bridge equipment along the water barriers have been detected.

5. ( ) CONCLUSIONS. Conclusions resulting from discussion in paragraph 4 include, when possible, a concise statement of the impact of each capability on the accomplishment of the assigned friendly mission. Cite enemy vulnerabilities where applicable. This paragraph contains a summary of enemy COAs most likely to be adopted, listed in the order of relative probability if sufficient information is available to permit such an estimate. Exploitable vulnerabilities should also be listed where applicable.

a. ( ) Enemy COAs in relative probability of adoption

(1) ( ) Defend in present locations with . . .

(2) ( ) Delay from present positions along . . .

(3) ( ) Reinforce the defense or delay with . . .

(4) ( ) Conduct UW OPS in the area.

b. ( ) Vulnerabilities.

(1) ( ) Enemy left (west) flank is open to envelopment by amphibious assault. . . .

(2) ( ) The enemy's air search radar coverage is poor in the left (west) portion of its defensive sector.

(Signed) _____

N2

(ANNEXES: (By letter and title.) Annexes should be included where the information is in graphs or of such detail and volume that inclusion makes the body of the estimate cumbersome. They should be lettered sequentially as they occur throughout the estimate.

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[CLASSIFICATION]

ANNEX 3 TO APPENDIX I

Logistics Estimate Format

[CLASSIFICATION]

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

LOG ESTIMATE NUMBER ___

( ) REFERENCES:
a. Maps and charts
b. Other pertinent documents.

1. ( ) MISSION. State the mission of the command as a whole, taken from the commander's mission analysis, planning guidance, or other statements.

2. ( ) SITUATION AND CONSIDERATIONS.

a. ( ) Characteristics of the AO. Summarize data about the area, taken from the intelligence estimate or area study, with specific emphasis on significant factors affecting LOG activities.

b. ( ) Enemy forces. Refer to current issuing HQ staff intelligence estimate.

c. ( ) Friendly forces.

(1) ( ) Present disposition of major elements. Include an estimate of their strengths.

(2) ( ) Own COAs. State the proposed COAs under consideration obtained from OPS or plans Div.

(3) ( ) Probable tactical developments. Review major deployments and LOG preparations necessary in all phases of the operation proposed.

d. ( ) LOG situation. State known personnel problems, if any, which may affect the LOG situation.

e. ( ) Coordination and communications situation. State the coordination and communications situation, emphasizing known coordination problems that may affect the LOG situation.

f. ( ) Assumptions. State assumptions about the logistical aspects of the situation made for this estimate. Because basic assumptions for the operation already have been made and will appear in planning guidance and in the plan itself, they should not be repeated here. Certain LOG assumptions may have been made in preparing this estimate, and those should be stated.

g. ( ) Special features. Special features not covered elsewhere in the estimate that may influence the LOG situation.

h. ( ) LOG situation.

(1) ( ) Supply and service installations. Describe and give the location of key supply and service installations to be used to support the operation.

(2) ( ) Supply. State availability of pre-positioned war reserve stock, authorized levels of supply, known deficiencies of supply stocks and supply systems, and responsibilities and policies regarding supply.

(3) ( ) Transportation. List air, sea, and surface transportation availability; coordination; regulations; lift capability; and responsibilities and policies regarding supply.

(4) ( ) Medical services. Describe availability of evacuation, hospital facilities, and medical responsibilities and policies, including the anticipated evacuation policy.

(5) ( ) Civil engineering support. List responsibilities for civil engineering support, limiting factors, and other appropriate considerations.

(6) ( ) Miscellaneous. Include other LOG matters not considered elsewhere that may influence selection of a specific COA. Include identification of known deficiencies of combat service support (CSS). Include civil and indigenous materiel resources available or essential to support military OPS. Also, consider the requirement to meet minimum essential needs of civil populace for whom the commander may become responsible.

3. ( ) LOG ANALYSIS OF OWN COAS. Make an orderly examination of the LOG factors influencing the proposed friendly COAs. The OBJ of this analysis is to determine if the LOG requirements can be met and to detail the LOG implications that should be weighed by the commander. Throughout the analysis, keep LOG considerations foremost in mind. The analysis is not intended to produce a decision; it is intended to ensure that all applicable LOG factors have been properly considered and to serve as the basis for the comparisons in paragraph 4.

a. ( ) Analyze each COA from the LOG point of view. The detail in which the analysis is made is determined by considering the level of command, scope of contemplated OPS, and urgency.

b. ( ) For each COA under consideration, analyze the LOG factors described in paragraph 2. Examine these factors realistically from the standpoint of requirements versus actual or programmed capabilities, climate and weather, oceanography and hydrography, time and space, enemy capabilities, and other significant factors that may have an impact on the LOG situation as it affects the COAs.

4. ( ) COMPARISON OF OWN COAS.

a. ( ) List the advantages and disadvantages of each proposed COA from the N4's point of view.

b. ( ) Use a worksheet similar to that used for the commander's estimate, if necessary.

5. ( ) CONCLUSIONS.

a. ( ) State whether or not the mission set forth in paragraph 1 can be supported from a logistical standpoint.

b. ( ) State which COA under consideration can best be supported from a logistical standpoint.

c. ( ) Identify the major LOG deficiencies that must be brought to the commander's attention. Include recommendations concerning the methods to eliminate or reduce the impact of those deficiencies.

(Signed) _____

N4

ANNEXES: (By letter and title.) Use annexes when the information is in graphs or is of such detail and volume that inclusion in the body makes the estimates too cumbersome. Annexes should be lettered sequentially as they occur throughout the estimate.

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Note

NTTP 5-01.4, Navy Planning, Logistics, provides further information on the LOG estimate and development of the COLS.

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ANNEX 4 TO APPENDIX I

Information Operations Estimate Format

[CLASSIFICATION]

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

IO ESTIMATE NUMBER ___

( ) REFERENCES:
a. Maps and charts
b. Other pertinent documents.

1. ( ) MISSION. State the mission of the command as a whole, taken from the commander's mission analysis, planning guidance, or other statements.

2. ( ) SITUATION AND CONSIDERATIONS.

a. ( ) Characteristics of the AO. Summarize data about the area, taken from the intelligence estimate or area study, with specific emphasis on significant factors affecting IO activities. Include strategic communication and guidance from DOS and the CCDR.

b. ( ) Enemy forces.

(1) ( ) Strength and dispositions. Refer to current intelligence estimate.

(2) ( ) Enemy capabilities. Discuss enemy capabilities, taken from the current intelligence estimate, with specific emphasis on their impact on the IO situation.

c. ( ) Friendly forces.

(1) ( ) Present disposition of major elements. Include an estimate of their strengths.

(2) ( ) Own COAs. State the proposed COAs under consideration, obtained from OPS or plans Div.

(3) ( ) Probable tactical developments. Review major developments and IO preparations necessary in all phases of the proposed operation. IO capabilities against the enemy should be included.

d. ( ) Personnel situation. State known personnel problems that may affect the IO situation.

e. ( ) LOG situation. State known LOG problems that may affect the IO situation.

f. ( ) Assumptions. State assumptions about the IO aspects of the situation. Because basic assumptions for the operation already have been made and will appear in planning guidance and in the plan itself, they should not be repeated here. Certain IO assumptions may have been made in preparing this estimate, and those should be stated here.

g. ( ) Special features. State special features that are not covered elsewhere in the estimate but that may influence the IO situation.

h. ( ) Command, control, communications, and computer situation. Consider line-of-sight communications, satellite communications (SATCOM), ground mobile CPs, the DSCS ground mobile segment, Defense Courier Service interface, etc.

(1) ( ) C2 communications

(2) ( ) Administrative communications

(3) ( ) Communications intelligence

(4) ( ) Communications security (COMSEC)

(5) ( ) Communications support for combat OPS

(a) ( ) Joint tactical air OPS

(b) ( ) Air-to-ground OPS (CAS and air interdiction)

(c) ( ) Naval surface fire support OPS.

(6) ( ) Communications control and aids for supporting arms

(7) ( ) Communications requirements for other activities

(8) ( ) Computer networks.

3. ( ) SYSTEM SUPPORT AND ANALYSIS OF OWN COAS. Make an orderly examination of the IO factors influencing the proposed COAs. The OBJ of this analysis is to determine the IO implications that should be weighed by the commander.

a. ( ) Analyze each COA from an IO point of view. The detail in which the analysis is made is determined by considering the level of command, scope of contemplated OPS, and urgency.

b. ( ) The IO factors in paragraph 2 are analyzed for each COA under consideration. Examine these factors realistically and include appropriate considerations of local culture, enemy capabilities and vulnerabilities, and other significant factors that may affect the IO situation as it affects the COAs.

c. ( ) Throughout the analysis, keep IO foremost in mind. The analysis is not intended to produce a decision but to ensure that all applicable factors have been properly considered and to serve as the basis for the comparisons in paragraph 4.

4. ( ) COMPARISON OF OWN COAS.

a. ( ) List the advantages and disadvantages of each proposed COA from the Navy staff communications/information systems directorate's point of view.

b. ( ) Use a worksheet similar to the one in the commander's estimate, if necessary.

5. ( ) CONCLUSIONS.

a. ( ) State whether or not the mission set forth in paragraph 1 can be supported from an IO standpoint.

b. ( ) State which COA under consideration can best be supported from an IO standpoint.

c. ( ) Identify the major IO deficiencies that must be brought to the commander's attention. Include recommendations concerning the methods of eliminating or reducing the impact of those deficiencies.

(Signed) _____

N39

ANNEXES: (By letter and title.) Use annexes when the information is in graphs or is of such detail and volume that inclusion in the body makes the estimates too cumbersome. They should be lettered sequentially as they occur throughout the estimate. Subject areas that should be discussed are COMSEC, IO systems protection (including identification of initial nodes), and communications planning.

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ANNEX 5 TO APPENDIX I

Civil-military Operations Estimate Format

[CLASSIFICATION]

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

CMO ESTIMATE NUMBER ___

( ) REFERENCES:
a. Maps and charts
b. Other pertinent documents.

1. ( ) MISSION. State the mission of the command as a whole, taken from the commander's mission analysis, planning guidance, or other statements.

2. ( ) SITUATION AND CONSIDERATIONS.

a. ( ) Intelligence situation. Include relevant information obtained from the intelligence estimate.

(1) ( ) Characteristics of the AO. Identify physical features, climate, and basic political, economic, and psychological factors that are significant to CMO.

(a) ( ) Attitudes of the population (cooperative or uncooperative)

(b) ( ) Availability of basic necessities (food, clothing, water, shelter, and medical care), including civilian capabilities of self-support

(c) ( ) Availability of local material and personnel to support military OPS

(d) ( ) Number of dislocated civilians in the area

(e) ( ) Amount and type of war damage suffered by the economy (particularly in transportation, public utilities, and communications)

(f) ( ) Status and character of the civil government

(g) ( ) Status and location of NGOs and international organizations in the area

(h) ( ) State of health of the civilian populace.

(2) ( ) Enemy strength and dispositions.

(3) ( ) Enemy capabilities. Consider sabotage, espionage, subversion, terrorism, and movement of dislocated civilians.

(a) ( ) Affecting the mission

(b) ( ) Affecting CMO activities.

b. ( ) Friendly situation. Include information on friendly forces/organizations/capabilities/resources that can affect CMO.

(1) ( ) Present dispositions of major tactical elements

(2) ( ) Possible COAs to accomplish the mission

(3) ( ) Projected OPS and other planning factors required for coordination and integration of staff estimates.

c. ( ) Personnel situation. Include information on CMO personnel capacity.

(1) ( ) Present dispositions of personnel and administration units and installations that affect the CMO situation

(2) ( ) Projected developments within the personnel field likely to influence CMO.

d. ( ) LOG situation. Include information obtained from the LOG officer.

(1) ( ) Present dispositions of LOG units and installations that affect the CMO situation

(2) ( ) Projected developments within the LOG field likely to influence CMO.

e. ( ) CMO situation. Discuss the status of the CMO situation. In the case of detailed information at higher levels of command, a summary may appear with reference to an annex to the estimate.

(1) ( ) Disposition and status of CA elements and related significant military and nonmilitary elements.

(2) ( ) Current problems faced by the command. Estimate the impact of FUPLANS of the supported unit's operation pertinent to the CMO mission.

(3) ( ) Projected impact of civilian interference with military OPS.

(4) ( ) Government functions.

(a) ( ) Legal

(b) ( ) Public administration

(c) ( ) Public education

(d) ( ) Public health

(e) ( ) Public safety.

(5) ( ) Economic functions.

(a) ( ) Civilian supply

(b) ( ) Economic development

(c) ( ) Food and agriculture.

(6) ( ) Public facilities functions.

(a) ( ) Public communications

(b) ( ) Transportation

(c) ( ) Public works and utilities.

(7) ( ) Special functions.

(a) ( ) Civil information

(b) ( ) Cultural relations

(c) ( ) Dislocated civilians

(d) ( ) Emergency services

(e) ( ) Environmental management.

f. ( ) Assumptions. Until specific planning guidance becomes available, give CMO assumptions required to initiate planning or to prepare the estimate. Modify the assumptions as factual data become available.

3. ( ) ANALYSIS OF COAS. Analyze all CMO factors indicating problems and deficiencies of each COA.

4. ( ) COMPARISON OF COAS.

a. ( ) Evaluate CMO deficiencies and list the advantages and disadvantages of each proposed COA.

b. ( ) Discuss the advantages and disadvantages of each tactical COA under consideration from the CMO standpoint. Eliminate those that are common to all COAs or that are minor. Include methods of overcoming deficiencies or modifications required in each COA. Priority is on one major CA activity that most directly relates to the mission—for example, preventing civilian interference with tactical and logistical OPS.

5. ( ) CONCLUSIONS OR RECOMMENDATIONS.

a. ( ) Indicate whether the stated mission can be supported from the CMO standpoint.

b. ( ) Indicate the COA best supported from the CMO standpoint.

c. ( ) List the primary reasons other COAs are not favored.

d. ( ) List the major CMO problems that must be brought to the commander's attention. Include specific recommendations on the methods of eliminating or reducing the effects of these deficiencies.

(Signed) _____

Title

ANNEXES: (By letter and title.) Use annexes when the information is in graphs or is of such detail and volume that inclusion in the body makes the estimates too cumbersome. They should be lettered sequentially as they occur throughout the estimate.

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ANNEX 6 TO APPENDIX I

Estimate of Supportability Format

[CLASSIFICATION]

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

ESTIMATE OF SUPPORTABILITY

( ) REFERENCES:
a. Maps and charts
b. Other pertinent documents.

1. ( ) MISSION

a. ( ) Basic mission. State the mission of the command as a whole, taken from the commander's mission analysis, planning guidance, or other statements. If this is a running estimate early in the planning process, the mission statement may not be complete. In this case, identify what actions need to be taken as part of the HHQ COA(s).

b. ( ) Previous decisions.

(1) ( ) List in detail each essential task already assigned to the issuing HQ.

(2) ( ) These essential tasks should contribute to the current HHQ COA(s) under consideration.

2. ( ) SITUATION AND CONSIDERATIONS

a. ( ) Enemy.

(1) ( ) Present disposition of major elements (see intelligence estimate).

(2) ( ) Capabilities (see intelligence estimate).

b. ( ) Population. Disposition of the population such as attitudes toward the insurgents, HN, and allied or coalition forces.

c. ( ) Own forces.

(1) ( ) Present disposition of the issuing HQ major subelements.

(2) ( ) Probable tactical developments

(a) ( ) Summarize intended actions required to support each COA.

(b) ( ) Estimate timelines when assigned tasks (or envisioned actions) are expected to be achieved under each COA being considered.

(3) ( ) Own COAs.

(a) ( ) COA #1. Describe in general terms (enough to give an overall picture and to distinguish it from the other COA[s]). If COAs are still in development, describe the distinguishing features. For example, "This COA will have an amphibious assault."

(b) ( ) COA #2 (and subsequent COAs). Describe in general terms.

(c) ( ) Describe the characteristics of the geographic area in which the operation is to be conducted. Provide cultural considerations or other aspects of the area that may affect OPS. Readers may be referred to the intelligence estimate.

(d) ( ) Assumptions.

i. ( ) List all assumptions regarding the enemy, friendly, and population situation.

ii. ( ) List assumptions about enemy and friendly capabilities.

iii. ( ) List assumptions about enemy and population intentions.

d. ( ) Special factors. List any special aspects of the situation that could affect resource allocation.

3. ( ) COA ANALYSIS

a. ( ) Ranks the COAs under consideration by HHQ according to the ability of the commander to support them.

b. ( ) Concept of employment. Describe the concept of employment for each of the HHQ COA(s).

c. ( ) Enemy situation and capabilities. Describe how the enemy situation and capabilities might affect the ability to support each or any of the HHQ COA(s).

d. ( ) Requirement for support. Explain how and to what degree the commander can support each of the HHQ COA(s). In order for the maritime commander to determine the ability to support HHQ COAs, the commander must be aware of the requirements for all other elements of the maritime force (e.g., consider all demands on the force that could be independent of and running concurrently with the plan or order being evaluated).

e. ( ) Topography. As applicable, describe how the topography might affect the ability to support each or any of the HHQ COA(s).

f. ( ) Weather. Describe how the weather might affect the ability to support each or any of the HHQ COA(s).

g. ( ) Observation and surveillance. The issuing HQ describes how the observation and surveillance issues might affects its ability to support each or any of the HHQ COA(s).

h. ( ) Communications requirements. The issuing HQ describes how the communications issues might affect its ability to support each or any of the HHQ COA(s).

i. ( ) LOG. The issuing HQ describes how the LOG issues might affect its ability to support each or any of the HHQ COA(s).

j. ( ) Hydrographic conditions. The issuing HQ describes how the hydrographic issues might affect its ability to support each or any of the HHQ COA(s) during amphibious OPS.

k. ( ) Other considerations. The issuing HQ describes any other considerations that the commander and staff assess as important to its organization's ability to support HHQ COAs.

4. ( ) EVALUATION

   a. ( ) COA #1

      (1) ( ) Advantages

         (a) ( ) List the advantages of this COA specific to the issuing HQ, not necessarily to the force as a whole.

         (b) ( ) List of advantages continued.

      (2) ( ) Disadvantages.

         (a) ( ) List the disadvantages of this COA specific to the issuing HQ, not necessarily to the force as a whole.

         (b) List of disadvantages continued.

   b. ( ) COA #2.

      (1) ( ) Advantages

      (2) ( ) Disadvantages.

5. ( ) CONCLUSIONS.

   a. ( ) Identify the preferred COA and rationale for its choice.

   b. ( ) Rank any other COAs with the rationale for their order.

   c. ( ) Recommend any changes to one or more of the COA(s) based on the ability to furnish support.

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ANNEX 7 TO APPENDIX I

Commander's Estimate Format

[CLASSIFICATION]

Issuing HQ Place of Issue
Day, Month, Year, Hour, Zone

COMMANDER'S ESTIMATE

( ) REFERENCES:
a. Maps and charts
b. Other pertinent documents.

1. ( ) MISSION. State the assigned or deduced task and its purpose. If there are multiple missions, determine priorities. List any intermediate tasks, prescribed or implied, necessary to the accomplishment of the mission.

2. ( ) THE SITUATION AND COAS.

a. ( ) Considerations affecting the possible COAs. Determine and analyze those factors that will influence the choice of a COA as well as those that affect the capabilities of the enemy. Consider each of the following and other factors and include under each a statement of each fact (or assumption if necessary) and deduce the probable influence on enemy or friendly actions.

(1) ( ) Characteristics of the AO.

(a) ( ) Military geography.

i. ( ) Topography. Consider factors of relief, vegetation, surface materials, and similar characteristics as they affect such elements of an operation as observation, maneuver, fire support, concealment, cover, air and surface movement, lines of communication, avenues of approach, key terrain, biological and chemical weapons employment, electronic emissions of all types, and unconventional, psychological, and other significant activities.

ii. ( ) Oceanography and hydrography. Include the characteristics of offshore sea areas, approaches to the beaches, currents, tides, the beaches themselves, ports, docks, and similar maritime considerations.

iii. ( ) Climate and weather. Extremes of temperature, wind velocities, cloud cover, visibility, precipitation, and other factors that can affect military OPS should be determined and presented. Sunrise, sunset, and twilight data are normally given in this subparagraph.

(b) ( ) Transportation. Indicate characteristics of roads, railways, inland waterways, and airfields, including such factors as size, capacity, conditions, and other facts that affect enemy capabilities and friendly COAs.

(c) ( ) Telecommunications. List radio, SATCOM, landline, and other communications facilities in the AO that might aid in C2 of military forces. Facilities considered in this subparagraph are not those in the organic capability of the opposing forces but rather those already established in the area.

(d) ( ) Politics. Include such considerations as political stability, alliances, relations with other countries, aspects of international law, control over subversion and dissidence, and other factors that may influence the selection of a COA. Neutrality or belligerency of neighboring states is often listed here.

(e) ( ) Economics. Include the organization of the economy and its mobilization capacity, the industrial base of the enemy to support hostilities, finance, foreign trade, and similar influences as they affect selection of a COA.

(f) ( ) Sociology. Consider social conditions that might range from the psychological ability of the populace to withstand the rigors of war, to health and sanitation conditions in the AO. Language, social institutions and attitudes, and other factors that may affect the selection of a COA should be considered.

(g) ( ) Science and technology. Although little immediate military impact may result from the state of science and technology in a target area, consider the long-range effects of such factors as the technical skill level of the population and scientific and technical resources in manpower and facilities, and how they may affect the choice of COA.

(2) ( ) Relative combat power.

(a) ( ) Enemy.

i. ( ) Strength. Give number and size of enemy units committed and those available for reinforcement. This is not intended to be just a tabulation of numbers of aircraft, ships, missiles, or other military weaponry but a study of what the enemy commander can bring to bear in the area in terms of ground units committed and reinforcing, aircraft sortie rates, missile delivery rates, unconventional, psychological, and other strengths.

ii. ( ) Composition. Include major enemy combat formations, equivalent strengths of enemy and friendly units, and major weapon systems and armaments in the enemy arsenal and their operational characteristics.

iii. ( ) Location and disposition. Indicate the geographic location of enemy units; fire support elements; C2 facilities; air, naval, and missile forces; and other combat power in or deployable to the AO.

iv. ( ) Reinforcements. Estimate enemy reinforcement capabilities that can influence the AO. This study should include ground, air, naval, and missile forces; nuclear, biological, chemical, and other advanced weapon systems; and an estimate of the relative capacity to move these forces to—and within—the AO.

v. ( ) LOG. Summarize the enemy capabilities with which they have been credited, and include such considerations as supply, maintenance, hospitalization and evacuation, transportation, labor, construction, and other essential logistic means. Broadly speaking, this is a feasibility test for enemy capabilities.

vi. ( ) Time and space factors. Estimate where and when enemy forces and reinforcements can initially be deployed and employed. Such a study normally includes distances and travel times by land, sea, and air from major bases or mounting areas into the AO.

vii. ( ) Combat efficiency. Estimate the enemy state of training, readiness, combat experience, physical condition, morale, leadership, motivation, doctrine, discipline, and whatever significant strengths or weaknesses may be derived from the preceding paragraphs.

(b) ( ) Friendly. In general, follow the same pattern used for analysis of the enemy when appraising the commander's own force. The description of what to consider and the approach outlined in subparagraph (2)(a) apply to the analysis of friendly forces.

(3) ( ) Assumptions. Assumptions are important factors on which the conduct of the operation is based and shall be noted as such.

b. ( ) Enemy capabilities. State the enemy capabilities that can affect the accomplishment of the commander's mission (enemy capabilities are obtained from the intelligence estimate of the situation).

c. ( ) Own COAs. State all valid COAs open to the commander that could accomplish the mission.

3. ( ) ANALYSIS OF OPPOSING COAS. Determine the probable effect of each enemy capability on the success of each of the commander's own COAs.

4. ( ) COMPARISON OF OWN COAS. Weigh the advantages and disadvantages of each friendly COA with respect to the evaluation criteria/governing factors. Decide which COA promises to be the most successful in accomplishing the mission.

5. ( ) DECISION. Translate the COA selected into a concise statement of what the force as a whole is to do and as much of the when, where, how, and why as may be appropriate.

ANNEXES: (As required, by letter and title.)

(Signed) _____

Commander

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APPENDIX J
Format for Plans and Orders

J.1 GENERAL CONSIDERATIONS

This appendix focuses on preparing directives for Navy staffs whose commanders are serving in a JFMCC or NCC role. For guidance on preparing joint OPS plans and orders (e.g., JTF role), see JP 5-0 and CJCSM 3130 (series). For guidance on OPGEN and other tactical directives, see NWP 3-56 and other tactical publications.

The entire staff develops the directive. Planners develop and refine functional concepts throughout the process, such as the concept of LOG or scheme of fires. The staff principals and their respective sections are responsible for completing the annexes as directed by the command, usually through the COS or MOC director. The staff should ensure all tasks are either in the base plan or order, in the tasks paragraph or coordinating instructions, or no lower than an appendix to an annex. Tasks that appear in tabs, exhibits, or attachments will lose visibility by the subordinate units and may not be accounted for and accomplished.

J.2 DIRECTIVES CONTENT AND FORMAT

Written directives may contain both text and graphics. With today's multiple means to convey information, it is not uncommon for orders to reference graphics or overlays that are developed and reside in other mediums (e.g., collaboration at sea websites, Global Command and Control System-Maritime (GCCS-M), etc.). While this can contribute to brevity, commanders and staffs should ensure that these other mediums are readily accessible by those supporting and supported units and agencies affected by the order. When communications throughputs are limited, the small file-size of text-only directives can allow a much faster distribution than a graphics-heavy document.

The five-paragraph SMEAC format remains the standard structure for written plans and when issuing WARNORDs, EXORDs, OPORDs, and FRAGORDs. These SMEAC paragraph headings should be shown in the written directive. See Figure J-1.

Acronym	Order Type
ALERTORD	Alert Order
DEPORD	Deployment Order
EXORD	Execute Order
FRAGORD	Fragmentary Order
OPORD	Operations Order
PLANORD	Planning Order
PTDO	Prepare to deploy Order
WARNORD	Warning Order

Figure J-1. Types of Orders

Acronyms and abbreviations should be used to save space and time. Consider who the recipients of the directive are and use recognized/defined acronyms and abbreviations (e.g., for U.S. joint forces, use acronyms and abbreviations found in the DOD Dictionary; for NAVFOR, also use NTRP 1-02, the Navy Supplement to the DOD Dictionary of Military and Associated Terms; for allied or other multinational OPS, use NATO or other recognized acronyms and abbreviations, as appropriate). Spell out the entire acronym or abbreviation and place the acronym or abbreviation between parentheses at first use in the order. Thereafter, the acronym or abbreviation can be used throughout the rest of the plan or order. If the definition of a term is unclear, include its definition in the text of the order to ensure clarity for all readers.

When specifying geographic latitude/longitude or grid coordinates with respect to a specific chart or map, be sure to include the chart/map numbers, map datum, and grid reference system used.

If necessary, to ensure clarity when describing force/unit movement or points outlining force/unit OPAREAs, apply a consistent method and consider a common frame of reference, whether that be cardinal points (e.g., north, south, etc.), true/magnetic bearings, clockwise/counterclockwise direction, etc.

When referring to and depicting control measure boundaries (e.g., CSG/ESG OPAREAs), routes and force movement, be cognizant of naming and depiction conventions used by the other components/Services and deconflict as appropriate to avoid confusion.

Use specific letters to designate dates and times as appropriate (see Figure J-2). State the letter used and their meanings.

Term	Definition
C-day	The unnamed day on which a deployment operation commences or is to commence.
L-hour	1. The specific hour on C-day at which a deployment operation commences or is to commence.
D-day	The unnamed day on which a particular operation commences or is to commence.
H-hour	1. The specific hour on D-day at which a particular operation commences. 2. In amphibious OPS, the time the first landing craft or amphibious vehicle of the waterborne wave lands or is scheduled to land on the beach, and in some cases, the commencement of countermine breaching OPS.
E-day	The day landing force personnel, supplies, and equipment begin to embark aboard amphibious warfare or commercial ships.
M-day	Mobilization day; unnamed day on which mobilization of forces begins.
F-hour	The effective time of announcement by the SECDEF to the military departments of a decision to mobilize Reserve units.
N-day	The unnamed day an active duty unit is notified for deployment or redeployment.
O-Day	Offload day; an unnamed day when the maritime prepositioning force offload commences.
R-day	The day on which redeployment of major combat, combat support, and CSS forces begins in an operation.
S-day	The day the President authorizes Selective Reserve call-up (not more than 200,000).
T-day	The effective day coincident with Presidential declaration of national emergency and authorization of partial mobilization (not more than 1,000,000 personnel exclusive of the 200,000 call-up).
W-day	Declared by the President, W-day is associated with an enemy decision to prepare for war (unambiguous strategic warning).

Figure J-2. Operational Time Definitions

Refer to days preceding or following C-, D-, or M-day by using a plus or minus sign and an Arabic number after the letter. For example, D-3 is 3 days before D-day; D+7 is 7 days after D-day. When using a time element other than days, spell it out. For example, D + 3 months.

Refer to hours preceding or following (H- or L-hour) by a plus or minus sign and an Arabic number after the letter. For example, H–3 is 3 hours before H-hour; H+7 is 7 hours after H-hour. When using a time element other than hours, spell it out. For example, H+30 minutes.

Where it is necessary to identify a particular operation or exercise, place a nickname or codewords before the letter such as ENDURING FREEDOM (D-day) or IRAQI FREEDOM (M-day).

Express the date and time as a six-digit date-time group. The first two digits indicate the day of the month; the last four digits indicate the time. The letter at the end of the time indicates the time zone. Add the month or the month and year to the date-time group (in standard message format) when necessary to avoid confusion. Express all times in an order in terms of one time zone. If using local time use the term LOCAL rather than L to avoid confusion with the LIMA (L) time zone. If the units involved in the operation are in different time zones, use ZULU (Z) time.

J.3 ANNEXES, APPENDIXES, AND TABS

Annexes, appendixes, and tabs (hereafter referred to as attachments for brevity) are used to keep the base directive simple and clear. Attachments include information, administrative support details, and instructions that expand upon the base order. The staff section responsible for the functional area addressed in the attachment prepares it (e.g., Annex D, Logistics, would be prepared by the LOG readiness center/N4 staff). Attachments should contain information specific to the operation and refer to SOPs as appropriate.

Attachments provide structure for organizing information. The general rule of OPORD attachments is that only those attachments that are required are used. The number and type of attachments also depend on the commander, level of command, and complexity or needs of the particular operation. Minimizing the number of attachments keeps the order consistent with completeness and clarity.

While the list of attachments and their formats set forth in CJCSM 3130 (series) is extensive, there may be a need for the development of other annexes, appendixes, and tabs to meet requirements specific to maritime OPS. The goal should be to make the order as clear and simple as possible. Thus, staffs must weigh the need for additional annexes, appendixes, and tabs against its return on investment in terms of order clarity, simplicity, and brevity. If the information relating to an attachment's subject is brief, place that information in the base order and omit the attachment.

Formatting and content requirements for annexes and their assigned appendixes and tabs are contained in CJCSM 3130 (series). Basic annex, appendix, and tab organization are summarized below.

ANNEX A TASK ORGANIZATION

Appendix 1 Time-phased Force and Deployment List

Appendix 2 Shortfall Identification

Appendix 3 Military Deterrent Options

ANNEX B INTELLIGENCE

Appendix 1 Priority Intelligence Requirements

Appendix 2 Signals Intelligence

Tab A Communications Intelligence Collection Requirements

Tab B Operational Electronic Intelligence Collection Requirements

Appendix 3 Counterintelligence

Tab A Counterintelligence Target List

Tab B Multidiscipline Counterintelligence Threat Report

Tab C Umbrella Counterintelligence Force Protection Source Operation Proposal

Appendix 4 Targeting

Tab A Target List (Conventional)

Tab B Target List (No-Strike)

Tab C Target List (Restricted)

Tab D Sensitive Target List

Appendix 5 Human-resource Intelligence

Tab A Human Intelligence Operations Cell

Tab B Enemy Prisoners of War/Civilian Detainees

Appendix 6 Intelligence Support to Information Operations

Appendix 7 Imagery Intelligence

Appendix 8 Measurement and Signature Intelligence

Appendix 9 Captured Enemy Equipment

Tab A Specific Prioritized Intelligence Collection Requirements

Tab B Equipment Releasable for Operational Purposes

Appendix 10 National Intelligence Support Team

ANNEX C OPERATIONS

Appendix 1 Nuclear Operations

Tab A Target List

Appendix 2 Combating Weapons of Mass Destruction

Appendix 3 Information Operations

Tab A Military Information Support Operations

Tab B Military Deception

Tab C Operations Security

Tab D Electromagnetic Warfare

Appendix 4 Special Operations

Appendix 5 Personnel Recovery

Tab A Search and Rescue

Tab B Nonconventional Assisted Recovery

Tab C Survival, Evasion, Resistance, and Escape

Tab D Reintegration

Tab E Accounting

Appendix 6 Cyber

Tab A Department of Defense Information Network Operations

Tab B Offensive Cyber Operations

Tab C Defensive Cyber Operations

Appendix 7 (Removed; included in Tab E to Appendix 5)

Appendix 8 Rules of Engagement

Appendix 9 Reconnaissance

Appendix 10 Air Base Operability

Appendix 11 Friendly Force Information Requirements

Appendix 12 Noncombatant Evacuation Operations

Appendix 13 Explosive Ordnance Disposal

Appendix 14 Amphibious Operations

Appendix 15 Force Protection

Tab A Combating Terrorism

Tab B Physical Security

Tab C Base Defense

Appendix 16 Critical Infrastructure Protection

ANNEX D LOGISTICS

Appendix 1 Petroleum, Oils, and Lubricants Supply

Tab A Estimate of Petroleum, Oils, and Lubricants Support Requirements

Appendix 2 Joint Subsistence, Food Service Support, and Water Management

Appendix 3 Mortuary Affairs

Appendix 4 Sustainability Analysis

Appendix 5 Mobility and Transportation

Tab A Enroute Support Requirements

Tab B Reception, Staging, Onward Movement, and Integration

Tab C Primary and Alternate Ports

Appendix 6 Engineering Support Plan

Tab A Construction Standards

Appendix 7 Nonnuclear Ammunition

Tab A Munitions Matrix

Appendix 8 Logistics Automation

ANNEX E PERSONNEL

Appendix 1 Enemy Prisoners of War, Civilian Internees, and Other Detained Persons

Appendix 2 Processing of Formerly Captured, Missing, or Detained United States Personnel

Appendix 3 Finance and Disbursing

Appendix 4 Legal

Appendix 5 Military Postal Service

Tab A Aerial Mail Terminals

Tab B Military Post Offices

Appendix 6 Chaplain Activities

Tab A Inter-Service Chaplain Support

Tab B HN Religious Support

Appendix 7 Removed (included as ANNEX W)

ANNEX F PUBLIC AFFAIRS

Appendix 1 Personnel Requirements for Joint Information Bureaus (JIBs) and Sub-JIBs

Appendix 2 Equipment and Support Requirements for JIBs and Sub-JIBs

Appendix 3 General Ground Rules for the Media

Appendix 4 DOD National Media Pool

ANNEX G CIVIL AFFAIRS

ANNEX H METEOROLOGICAL AND OCEANOGRAPHIC OPERATIONS

ANNEX J COMMAND RELATIONSHIPS

Appendix 1 Command Relationships Diagram

ANNEX K COMMAND, CONTROL, COMMUNICATIONS, AND COMPUTER SYSTEMS

Appendix 1 Information Assurance

Tab A Information Security

Appendix 2 Satellite Communications

Tab A Ultrahigh Frequency Satellite Communications Network List

Tab B Super High Frequency Satellite Communications Network List

Tab C Extreme High Frequency Satellite Communications Network List

Tab D Commercial Satellite Communications Network List

Appendix 3 Defense Courier Service

Appendix 4 Foreign Data Exchanges

Appendix 5 Electromagnetic Spectrum Management

Tab A Electromagnetic Interference Reporting

Tab B Joint Task Force Joint Communications-electronics Operating Instructions Concept

ANNEX L ENVIRONMENTAL CONSIDERATIONS

Appendix 1 Environmental Assessments

Tab A Environmental Assessment Exemptions

ANNEX M GEOSPATIAL INFORMATION AND SERVICES

Appendix 1 Geospatial Information and Services Requirements List

Appendix 2 Geospatial Information and Services Transportation Requirements

Appendix 3 Geospatial Information and Services Reports (Optional)

ANNEX N SPACE OPERATIONS

ANNEX P HOST-NATION SUPPORT

Appendix 1 Host-nation Support Agreements List

Appendix 2 Host-nation Support Reliability

Appendix 3 Presumed Host-nation Support

ANNEX Q MEDICAL SERVICES

Appendix 1 Joint Patient Movement System

Appendix 2 Joint Blood Program

Appendix 3 Hospitalization

Appendix 4 Returns to Duty

Appendix 5 Medical Logistics (Class 8A) System

Appendix 6 Force Health Protection

Appendix 7 Host Nation Health Support

Appendix 8 Medical Planning Responsibilities and Task Identification

ANNEX R REPORTS

ANNEX S SPECIAL TECHNICAL OPERATIONS

ANNEX T CONSEQUENCE MANAGEMENT

ANNEX U NOTIONAL COUNTERPROLIFERATION DECISION GUIDE

ANNEX V INTERAGENCY COORDINATION

Appendix 1 Humanitarian

Appendix 2 Economic

Appendix 3 Political

ANNEX W CONTINGENCY CONTRACTING

ANNEX X EXECUTION CHECKLIST

ANNEX Y STRATEGIC COMMUNICATION

ANNEX Z DISTRIBUTION

Note

The letters I and O are intentionally omitted; I is not authorized; Y may be used, if required. For annexes that are not used, list the annex and annotate "not used." This helps the recipients ensure they received the entire order and are not missing certain annexes.

J.4 ORDER FORMAT GUIDE INTRODUCTION

The Appendix J annexes that follow are provided to serve as a detailed guide to assist in base directive development. While the guide's focus on orders, the guideline for paragraph content also applies to plans.

The guide is descriptive in nature and identifies the content that should be placed in the appropriate paragraph. The formats and contents presented in the guide are standard templates that are usually followed, except when in the judgment of the commander, modifications are required.

Paragraphs and subparagraph headings indicated in the guide should appear in each directive. If information or instructions are not required in a particular paragraph, then that paragraph is noted as "not applicable" to show that consideration has been given to that part of the order. Examples of a WARNORD, OPORD, and FRAGORD, are presented to provide clarity.

Note

The following examples in this appendix are generic examples only; plain language address directory (PLAD), general admin (message) (GENADMIN) and content

should reflect existing requirements, situations and missions. The examples provided are specific to the scenario presented in this publication. They do not display every option or element of amplifying information that is in the narrative description provided in Annex J-2 and it is not meant to be used as a template, but rather as an example for the reader to better understand the format and basic flow of an order. Merely taking the example order provided and using them as templates to draft orders may not be appropriate. Order content and format (especially with respect to optional supporting paragraphs) is specific to the commander's intent and the mission. For example, content in an order for an HA/DR operation, maritime interception operations (MIO), NEO, etc. mission will be significantly different from the content provided in the examples in this publication which are based upon a fictitious conventional warfighting operation.

Formats for specific types of plans and orders:

Annex J-1: Basic Warning Order

Annex J-2: Basic Operation Order

Annex J-3: Plan or Order Annexes, Appendixes, and Tabs

Annex J-4: Basic Fragmentary Order.

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ANNEX 1 TO APPENDIX J

Basic Warning Order Format

J.1.1 GENERAL

The basic WARNORD format serves as a higher command's means of instructing subordinate units to take certain measures in anticipation of some future event. It is issued at the earliest practicable time following recognition of a crisis. If the crisis warrants change in the alert status of units or pre-positioning of units, then the WARNORD can contain a deployment preparation or deployment order (DEPORD). In practice, a WARNORD is sent to subordinate commanders after mission analysis by the HHQ staff IOT enable vertical and horizontal collaborative and parallel planning. Depending on time available, the commander may send additional, more detailed WARNORDs upon the completion of the COA development or the COA comparison and decision steps as a means to keep subordinate and adjacent commanders and their staffs informed, and to enable the continuation of their planning efforts.

The WARNORD normally is issued by record communication using a precedence of immediate or flash, as appropriate. If the situation is time-sensitive, voice communications or electronic orders should be used initially to pass WARNORD information. A voice order may be acted on immediately; however, a record communication is forwarded as soon as practicable to confirm oral or electronic orders, tasks, etc., and to keep all crisis participants informed.

J.1.2 CONTENTS

The precise contents of the WARNORD may vary widely, depending on the nature of the crisis and the degree of prior planning. Where little or no prior planning exists to meet a crisis, the subordinate commanders are provided with essential guidance necessary to permit crisis planning to commence. The WARNORD should be issued as soon as possible, even if detailed guidance is not available.

Normally, the WARNORD at the operational level either allocates forces available for planning or requests the subordinate commander's assessment of forces required to accomplish the mission. Additional information should be sent as soon as possible in message form and should reference the initial WARNORD.

The WARNORD defines the OBJs, anticipated mission or tasks, pertinent constraints, command relationships and, if applicable, tentative combat forces available to the commander for planning and strategic lift allocations. Further guidance relating to the crisis, such as changes to existing ROE or any specific directions from the joint force HQ, is also provided as necessary, but maximum flexibility is left to the subordinate commander in determining how to carry out the assigned mission and tasks. Major paragraphs and items of information that should be considered for inclusion in the WARNORD are discussed below.

Purpose statement: Statement that the message is a WARNORD. Indicate specific tasking or requests to supported and supporting commanders.

1. Situation: Short summary of the situation, including as appropriate:

   a. Political situation and possible enemy forces in the expected AO and a brief description of the AO

   b. Anticipated attitude and actions of friendly nations

   c. Type, level, and source of major combat forces available for planning, or a request for the subordinate commander's assessment of forces and strategic lift required

   d. Assumptions that may significantly affect the subordinate commander's planning.

2. Mission: If the mission statement has been approved by the commander, it is included word for word. If not, a draft mission statement is included that is clearly identified as DRAFT.

3. Execution.

a. COAs: If the commander desires that specific COAs be examined, they are listed here. Otherwise, the commander solicits specific input or actions from subordinate forces. Reference is made to an existing OPLAN or CONPLAN if applicable.

b. Any additional guidance that enables planning and COA development by the subordinate commander. Examples are provided in C through G as follows:

c. OPSEC and deception guidance.

d. Intelligence guidance:

(1) Intelligence personnel and equipment available to augment the supported commander

(2) Availability of national intelligence collection and communication assets

(3) Delegation of signals intelligence (SIGINT) operational tasking authority

(4) ROE for intelligence collection OPS.

e. Counterintelligence (CI) guidance:

(1) Designate Service(s) to provide CI element(s).

(2) Establish CI liaison responsibilities.

(3) Develop CI collection requirements.

f. CA guidance.

(1) CA mission. List required actions and specific results sought such as minimizing interference and maximizing influence regarding the civilian population's impact on military OPS; satisfying legal and moral obligations of the commander to the civil population; determining the availability of host-nation support (HNS) resources; providing support for HA and disaster response OPS; enhancing friendly nation stability and infrastructure development; and facilitating post-conflict restoration or transition activities.

(2) CA OBJs. List specific results sought, such as determination assessment of civil, indigenous, and HNS resources; support for HA and population or resource control OPS; assistance to civil requirements; facilitation of post-conflict transition activities; and enhanced friendly nation self-help capabilities to provide socioeconomic services.

g. Coordinating instructions.

(1) Tentative C-day and L-hour (if other than 0001Z) for planning.

(2) Anticipated date of execution (D-day). The date may be highly tentative at this time, but it provides the commander with a relative timeframe for planning based on the NCA perception of urgency.

(3) Anticipated duration of OPS.

(4) Known operational limitations (constraints/restraints (e.g., overflight, port clearances)).

(5) ROE guidance.

(6) Supporting commander coordination or monitoring instructions.

(7) Authorization for direct liaison between commanders.

4. Administration and LOG.

a. Transportation (movement planning guidance as appropriate)

b. Known LOG constraints

c. Personnel deployment criteria

d. Codewords or nicknames of the operation

e. Reporting instructions (special instructions and suspense for the submission of reports)

f. Classification and declassification guidance

g. PA guidance

h. Restricted access special category handling.

5. C2.

a. Communications guidance.

b. Command relationships: Provide higher authority (CCDR/JFC/commander, joint task force (CJTF)) guidance of the command relationships between commanders of forces assigned specified to include relationships with other forces or agencies. Specify the command relationships of subordinate commanders (e.g., CTFs); provide coordination instructions and a listing of the assigned forces and relationships.

J.1.3 DEPLOYMENT PREPARATION AND DEPLOYMENT ORDERS

If required by prevailing circumstances, the WARNORD may include a deployment preparation order or DEPORD (i.e., changes to alert status of units and movement of selected forces to pre-position for impending OPS). If the WARNORD contains such information, the first paragraph states, “This is a WARNORD. The Secretary of Defense has authorized…”

The following is an example of a JFMCC WARNORD (generic example only; PLAD and GENADMIN formats should reflect existing requirements/situations).

UNCLAS

261200ZJULXX

FM JFMCC BLUE SWORD

TO CTF 220
CTF 223
CTF 225
CTF 226
CTF 227
CTF 229

INFO CJTF BLUE SWORD CJCS WASHINGTON DC
COMUSEASTCOM MACDILL AFB FL//00/CCJ3//
COMUSSOCOM MACDILL AFB FL
COMNAVSPECWARCOM CORONADO CA
COMUSNORTHCOM PETERSON AFB
COMUSTRANSCOM SCOTT AFB IL
COMUSSTRATCOM OFFUTT AFB NE
COMUSMARFOREASTCOM
COMFLTFORCOM NORFOLK VA
AMEMBASSY GREYLAND
AMEMBASSY WHITELAND
AMEMBASSY PINKLAND
AMEMBASSY BLUESKIES
AMEMBASSY GREENACRES
CIA LANGLEY VA
DIA WASHINGTON DC

UNCLAS//N03000//OPER/BLUE SWORD//

MSGID/ORDER/JFMCC BLUE SWORD//
AMPN/SPECIAL HANDLING INSTRUCTIONS//
ORDTYP/WARNORD/JFMCC BLUE SWORD//
TIME ZONE/ZULU//
REF/A/RMG/CJCS/192200ZJUL19/NOTAL//
REF/B/RMG/COMUSEASTCOM/242100ZJUL19/NOTAL//
REF/C/DOC/CJCSI 3121.01A/15JAN00//
REF/D/RMG/COMUSEASTCOM/201600ZJUL19//
REF/E/RMG/COMUSEASTCOM/221500ZJUL19//
REF/F/DOC/UNSCR 1655//
REF/G/DOC/UNSCR 1658//
REF/H/DOC/UNSCR 1659 EMBARGO//
REF/I/DOC/UNSCR 1660 MIO//
REF/J/RMG/CJTF BLUE SWORD/WARNORD/260900ZJUL19
NARR/REF A IS CJCS WARNORD ON SITUATION IN EASTERN REDLAND SEA. REF B IS USEASTCOM ESTIMATE FOR FORCES REQUIRED ISO PINKLAND. REF C IS CJCS ROE. REF D IS ROE SERIAL ONE. REF E IS EMIO ROE. REF F IS UNSCR AUTHORIZING MULTINATIONAL FORCE RESPONSE OR UNILATERAL U.S. RESPONSE. REF G IS UNSCR WARNING REDLAND TO STOP SUPPORTING TERRORIST ACTIVITY IN THE PINKLAND COAST REGION. REF H IS UNSCR ESTABLISHING AN ARMS EMBARGO AGAINST THE GOVERNMENTS OF REDLAND AND

CONDEMNS THOSE NATIONS WHO SUPPLY ARMS TO REDLAND OR TERRORISTS WITHIN THE BOUNDARIES OF REDLAND. REF I IS UNSCR WHICH CALLS UPON MEMBER NATIONS THAT DEPLOY MARITIME FORCES TO THE AREA TO USE SUCH MEASURES AS MAY BE NECESSARY TO HALT ALL INWARD AND OUTWARD MARITIME SHIPPING IOT INSPECT AND VERIFY THEIR CARGO AND DESTINATIONS, AND TO ENSURE STRICT IMPLEMENTATION OF THE PROVISION RELATED TO SUCH SHIPPING LAID DOWN IN UNSCR 1969. REF J IS THE CJTF BLUE SWORD WARNORD ISO ABOVE REFERENCED DIRECTIVE.//

GENERAL TEXT (GENTEXT)/AUTHORITY/

(U) This is a WARNORD. The SECDEF has authorized CJCS to expand the scope of U.S. involvement in the vicinity of Redland. CJTF BLUE SWORD has been activated. JFMCC forces will make preparations for immediate deployment to the eastern Redland Sea.//

HEADING/TASK ORGANIZATION//

UNIT	/UNITLOC	/COMMENTS
/USEASTCOM	/CONUS	/CCDR
/CJTF BLUE SWORD	/PINKLAND	/USARFOREAST
/CJTF BLUE SWORD JFMCC	/EMB UND	/COMEASTFLT
/USS UNDERWAY (UND)	/ONSTA JOA	/CJTF BLUE SWORD JFMCC EMB
/CTF 220	/EMB TR	/CCSG-2
/TF 220	/ENR JOA	/TRCSG
/CTG 220.1	/ENR JOA	/CCSG-2
/CTU 220.1.1	/ENR JOA	/USS THEODORE ROOSEVELT (TR)
/CTU 220.1.2	/ENR JOA	/USS CHANCELLORSVILLE (CHV)(CG 62)
/CTE 220.1.2.1	/ENR JOA	/HSL 42 DET B
/CTU 220.1.3	/ENR JOA	/USS SHILOH (SHI) (CG 67)
/CTE 220.1.3.1	/EMB SHI	/HSL 42 DET C
/CTU 220.1.4	/ENR JOA	/USS CAMDEN (CAM) (AOE 2)
/CTG 220.2	/EMB TR	/COMCARAIRWING 14 (CVW-14)
/TG 220.2	/EMB TR	/CARAIRWING 14 (CVW-14)
/CTU 220.2.1	/EMB TR	/COMCARAIRWING 14 (CVW-14)
/CTU 220.2.2	/EMB TR	/VFA-1
/CTU 220.2.3	/EMB TR	/VFA-2
/CTU 220.2.4	/EMB TR	/VMFA-22
/CTU 220.2.5	/EMB TR	/VAW-120
/CTU 220.2.6	/EMB TR	/VAQ-127
/CTU 220.2.7	/EMB TR	/HS-12
/CTG 220.3	/EMB TR	/COMDESRON 22 (CDS-22)
/TG 220.3	/ENR JOA	/DESRON 22
/CTU 220.3.1	/ENR JOA	/USS STETHAM (STE) (DDG 63)
/CTU 220.3.2	/ENR JOA	/USS SAMPSON (SAM) (DDG 102)
/CTE 220.3.2.1	/EMB SAM	/HSL 42 DET A
/CTU 220.3.3	/ENR JOA	/USS FREEDOM (FRE) (LCS 1)
/CTE 220.3.3.1	/EMB FRE	/HSL 44 DET A
/CTF 223	/CONUS	/COMLOGFORNAVEAST
/CTG 223.1	/EMB RIC	/COMLOGRON EAST
/TG 223.1	/ENR JOA	/UNDERWAY REPLENISHMENT GROUP
/CTU 223.1.1		/USNS RICHARD E. BYRD (RIC) (T-AKE-4)
/CTU 223.1.2		/USNS SUPPLY (SUP) (T-AOE 6)
/CTU 223.1.3		/USNS PECOS (PEC) (T-AO 197)

/CTG 223.2	/EMB DTW	/COMMANDER MPS SQUADRON THREE
/TG 223.2	/ENR JOA	/MPSRON 3
/CTU 223.2.1		/MV PFC DEWAYNE T WILLIAMS (DTW) (T-AK 3009)
/CTU 223.2.2		/MV 1ST LT BALDOMERO LOPEZ (LBL) (T-AK 3010)
/CTU 223.2.3		/MV 1ST LT JACK LUMMUS (LJL) (T-AK 3011)
/CTU 223.2.4		/MV SGT WILLIAM R BUTTON (WRB) (T-AK 3012)
/CTF 224	/EMB NAS	/NASESG
/TF 224	/ENR JOA	
/CTG 224.1	/EMB NAS	/CPR 6
/CTU 224.1.1	/ENR JOA	/USS NASSAU (NAS) (LHA-4)
/CTU 224.1.2	/ENR JOA	/USS SAN ANTONIO (SAT) (LPD 17)
/CTU 224.1.3	/ENR JOA	/USS NEW ORLEANS (ORL) (LPD 18)
/CTU 224.1.4	/ENR JOA	/USS GERMANTOWN (GER) (LSD 42)
/CTG 224.2	/EMB NAS	/31ST MEU
/CTG 224.3	/EMB COR	/COMMCMRON TWO
/CTU 224.3.1	/ENR JOA	/USS CORONADO (COR) (LCS 4)
/CTE 224.3.1.1	/EMB COR	/HSL 44 DET P
/CTU 224.3.2	/ENR JOA	/USS DEVASTATOR (DEV) (MCM 6)
/CTU 224.3.3	/ENR JOA	/USS FORT WORTH (FTW) (LCS 3)
/CTE 224.3.3.1	/EMB FTW	/MCM DET ONE
/CTU 224.3.4	/ENR JOA	/USS AVENGER (AVE) (MCM 1)
/CTF 225	/EMB ANT	/CO ANT
/CTG 225.1	/ENR JOA	/USS ANTIETAM (ANT) (CG 54)
/CTG 225.2	/ENR JOA	/USS MCCAMPBELL (MCB) (DDG 85)
/CTG 225.3	/ENR JOA	/USS WILLIAM S SIMS (WSS) (DDG 113)
/CTF 227	/CONUS	/COMPATRECONFORCE
/CTG 227.1	/U.S. NAV AIR STA	/VP-40
/CTU 227.1.1	/PINKLAND ISB	/VP-40 DET
/CTG 227.2	/U.S. NAV AIR STA	/VQ-2
/CTU 227.2.1	/PINKLAND ISB	/VQ-2 DET
/CTF 229	/U.S. NAVY BASE	/COMSUBGRUEAST
/CTG 229.1	/JOA	/USS CHICAGO (CHI) (SSN 721)
/CTG 229.2	/ENR JOA	/USS TUCSON (TUS) (SSN 770)
/PINKLAND	/JOA	/2X TYPE 22, 1X TYPE 42

GENTEXT/SITUATION//

1. (U) Situation.

1.A. (U) General. The inability of the government of Redland to eliminate safe havens for terrorist organizations continues to pose a threat to the region. Radical fundamentalist groups with known ties to international and regional terrorist organizations continue to plan, train, and resource terrorist activities from within the sovereign borders of Redland. These groups intend to disrupt international shipping throughout the Redland Sea and pose a direct threat to Redland's neighbors, including Pinkland. Terrorists continue to undermine the authority of the Redland government politically, economically, and militarily; and Redland continues to ignore international

demands that the terrorist organizations be destroyed. Additionally, the Redland navy operates in an aggressive manner to establish regional maritime dominance and protect Redland commercial exclusive economic zone (EEZ) activity that violates international law. Fishing and offshore oil are major Redland economic activities. Pinkland is Redland's principal regional competitor and a primary U.S. economic and defense partner. Redland and terrorists based in Redland have become increasingly aggressive toward Pinkland with numerous, recent acts of violence linked to paramilitary organizations based in Redland. Redland is seeking to topple Pinkland's government and replace it with a more sympathetic regime that will acquiesce to its national goals. Because of its close ties with the United States, the government of Pinkland has requested that the United States and coalition partners provide necessary military assistance to disrupt terrorist activities within Redland and to help prevent the influx of aid to these groups from outside Redland's borders.

1.B. (U) Enemy. The Redland navy's main source of strength is its sub force, which consists of four sub (conventional propulsion) (SS) with one underway nearly continuously and one in refit and unavailable. Redland coastal defense cruise missiles (CDCMs) have OTH range but lack effective third party targeting. Fast attack craft also have ASM capability with limited over-the-horizon targeting (OTHT). Redland has no MPRA and limited number of short range ISR-capable aircraft. All Redland navy vessels are capable of mining. Mines are stored in a depot in its navy base and must be moved to the piers for loading. Main source of strength for the government of Redland is the 23rd Guards Div, which is its best-equipped and trained ground unit that is manned exclusively by government loyalists. This unit also provides clandestine support to Home Guard units that are connected to terrorist attacks on Pinkland and Redland's other neighbors—as well as U.S. military and civilians in the region. See Annex B for order of battle and additional information.

1.C. (U) Friendly. ISO Operation FA, Pinkland has authorized limited basing rights to U.S. and coalition forces. Pinkland has agreed to allow the use of an airfield to support fixed-wing aircraft OPS and act as an ISB. See Annex A for forces deployed in Pinkland as of the issue of this OPORD. Hangars and ramp space with maintenance facilities have also been made available. Pinkland has authorized the use of its main port facility for U.S. and coalition combatant and LOG shipping. Port can support up to four destroyer-sized ships at the military pier along with 100,000-gross-ton ships at the commercial facility. Cranes and tugs available only during daylight hours. Radiation of military radars in port is not authorized. Port security provided by Pinkland contracted force.

1.D. (U) Assumptions.
1.D.1. (U) Greyland will permit overflight for carrier-based tactical aviation and Tomahawk land-attack missiles (TLAMs).
1.D.2. (U) Pinkland naval forces will cooperate with the JTF but remain under their national control.
1.D.3. (U) JFLCC Patriot missile sites are FMC at the ISBs before combat OPS begin.
1.D.4. (U) Pinkland will allow establishment of an SPOD and forward operating base (FOB) at Port X-Ray.
1.D.5. (U) The DOS-designated safe haven for noncombatant evacuees (NCEs) will be the Port X-Ray in Pinkland.
1.D.6. (U) Orangeland will not freely allow terrorist camps to be established within their borders (JTF assumption).
1.D.7. (U) Whiteland will allow support to terrorist organizations to pass over their border into Redland (JTF assumption).
1.D.8. (U) Coalition nations will remain committed throughout the length of the operation (JTF assumption).

GENTEXT/MISSION//
2. (U) O/O, JFMCC BLUE SWORD establishes maritime superiority in vicinity of southern Redland and the AOA IOT establish a secure and stable maritime environment free from Redland military interference or terrorist threats.

GENTEXT/EXECUTION//
3. (U) Execution.

3.A. (U) Commander's intent.
3.A.1. (U) Purpose. The JFMCC supports the JTF and government of Pinkland by controlling all aspects of the maritime environment to provide a stable and secure environment free from any conventional military or terrorist threats emanating from Redland.

3.A.2. (U) Method. Commencing combat OPS on D-day, the JFMCC conducts sea control OPS to create a condition of maritime advantage in the maritime AO. We will do this by destroying enemy forces, suppressing enemy sea commerce, and protecting vital sea lanes. If required, we will conduct OPS focusing our efforts in the southern portion of the maritime AO to create a degree of sea control that allows amphibious OPS to commence. Follow-on tasks include developing Port Yankee as an SPOD for follow-on forces and sustainment. Concurrent to these efforts, we will use our strike capabilities to support the land component commander's efforts in defeating the 23rd Guards Div and destroy the terrorists and their infrastructure in the JOA.
3.A.3. (U) End state. Redland maritime forces and maritime-based terrorists are neutralized and cannot conduct organized OPS. We retain the ability to strike in and from the maritime environment as well as place amphibious forces ashore into Pinkland or Redland. Port Yankee is a fully functioning SPOD. Maritime forces are prepared to transition OPS to Pinkland.
3.A.4. (U) Risk. I see a significant threat from Redland forces for our maritime forces at the outset. We must be fully prepared to transition to kinetic OPS quickly to attrite their force and reduce the risk, while holding attrition to civilian population and infrastructure to an absolute minimum.

3.B. (U) COAs. CJTF planning is ongoing. JTF CONOPS provided via separate correspondence (SEPCOR) once approved. Anticipated maritime OPS include, but are not limited to, the following: presence, show-of-force, strikes, IO, MIO, FP, amphibious OPS, MSO, ISR, and LOG.

3.C. (U) Tasks. Provided SEPCOR once CJTF CONOPS is approved.

3.D. (U) Coordinating instructions.
3.D.1. (U) Daily CTF brief to JFMCC (VTC) occurs as per published battle rhythm in DIM.
3.D.2. (U) C-day is 02AUG20XX.
3.D.3 (U) D-day is to be determined (TBD).
3.D.4. (U) Greyland is conducting a naval exercise on or about 04AUG20XX in Greyland territorial waters (TTW).
3.D.5. (U) Anticipated length of operation is less than 180 days.
3.D.6. (U) Codeword assigned to this operation is FA.
3.D.7. (U) ROE.
3.D.7.A. (U) Standing ROE are in effect.
3.D.7.B. (U) JFMCC TFs request supplemental measures for mission accomplishment IAW ref (d) procedures.
3.D.8. (U) Direct liaison authorized (DIRLAUTH) ALCON. Keep JFMCC informed of planning status.//

GENTEXT/ADMIN AND LOG//
4. (U) Administration and LOG.

4.A. (U) Concept of sustainment. CTFs identify and report shortfalls of critical supply items and maintenance to JFMCC IAW OPTASK LOG.

4.B. (U) Material and services.
4.B.1. (U) Maintenance. TBD.
4.B.2. (U) Supply. TBD.
4.B.3. (U) Petroleum, oils, and lubricants (POL). TBD.

4.C. (U) Medical services. CJTF is coordinating Pinkland hospitals to provide Level III medical care.

4.D. (U) Personnel. Submit personnel status report (PERSTAT) daily to JFMCC N1 via RMG IAW OPTASK COMM.

4.E. (U) CA. Be prepared to provide HA for IDPS within Redland flowing into Greyland, Whiteland, Orangeland, or out to sea.

4.F. (U) PA guidance.
4.F.1. (U) Public release of information about this operation is not authorized until approved by CJTF (PA) and HHQ.
4.F.2. (U) Proposed public affairs guidance (PPAG) provided in ref (e).

GENTEXT/COMMAND AND CONTROL//
5. C2.

5.A. (U) Command.

5.A.1. (U) Command relationships.

5.A.1.A. (U) COMEASTFLT is designated JTF BLUE SWORD JFMCC. JFMCC exercises OPCON over all U.S. Maritime forces listed in this OPORD. JFMCC has coordinating authority with the maritime forces of Pinkland, Greenland, and BLUESKIES IOT achieve assigned tasks.

5.B. (U) Primary/threat warning communications and link common operational picture (COP). CJTF BLUE SWORD JFMCC OPTASK COMM provides communication schedule via SEPCOR. CJTF BLUE SWORD JFMCC joint interface control officer (JICO) provides link architecture for maritime COP SEPCOR.

5.C. (U) JFMCC validates and forward shortfall requirements to CJTF J6. Because of limited satellite capacity, BPT recommend relative priorities and allocate channels and circuits within the scope of this operation.//

AKNLG/YES/INST: Contact JFMCC watch captain via e-mail/phone.//

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ANNEX 2 TO APPENDIX J

Basic Operation Order Format Guide

(The administrative data at the beginning of an OPORD should include all of the below information.)

J.2.1 STANDARD TRANSMISSION INFORMATION

The first item is the standard transmission information commonly referred to as the PLAD. Include precedence of message; generally, immediate precedence should be used unless otherwise specified. State whom the message is from; generally, the command originating or updating the information. Next, list the action addressees—these are the appropriate planning participants directly concerned as identified by the originators. List the information addressees—these are all other interested planning participants as identified by the originator. Distribution is by policy and procedure of the issuing HQ.

J.2.2 OPERATION ORDER: CODE NAME AND NUMBER

The next item of the heading is the name and copy number assigned by the issuing HQ. Number plans and orders consecutively by calendar year. Include code name, if any.

J.2.3 TIME ZONE

The third item is the time zone. State the time zone used in the AO during execution. When the OPORD applies to units in different time zones, use Greenwich Mean (ZULU) Time.

J.2.4 REFERENCES

The last item is references. List any previous orders (e.g., PLANORD, WARNORD, EXORD, etc.). List maps, charts, and other relevant documents (e.g., COMSEVENTHFLT No 7-94; commander's estimate; CTG 70.5; etc.). When using a map/chart, include the map/chart series number (and country or geographic areas if required), sheet number (and names if required), edition, and scale (if required).

J.2.5 CONTENT

GENTEXT/AUTHORITY//
(This free text paragraph should briefly describe the specific military operation for which the present scheme was developed. And the authority under which it is sent.)

Task organization: List and describe the organization of forces available to the issuing HQ and their command and support relationships as of the date of issuance of the OPORD (for example, OPCON, TACON, support, coordination, etc.). For units attached to another unit, list the time or times that attachment is effective if different from the time the order is effective. If the task organization changes during execution of the operation, these changes are best addressed in an Annex A or within the C2 paragraph. Group units (other than major subordinate commands and those units that are attached to or that support a major subordinate command) under a single heading that reflects that they are under the C2 of the force HQ. The task organization shall reflect the selected COA and decision. When detailing the task organization, give the numerical designations of units in Arabic numerals. When distinguishing between national forces of two or more nations (coalition OPS), insert the distinguishing letters of the country between the numerical designation and the unit name (for example, 3rd (DEU) Corps). Designate TFs by the last name of the commander of the TF (TF SMITH), a code name (TF STRIKE), or a number (TF 17 or TG 60.5). Refer to Annex A (Task Organization) if information is lengthy or complicated.

GENTEXT/SITUATION//
(The situation paragraph contains the necessary information on the current situation needed in the development of general COAs by subordinate commanders.)

1. (U) Situation.

1.A. (U) General. Generally describes the conditions of the OE that impact the operation and is further broken down in the following subparagraphs:

1.B. (U) Enemy forces. Summarize the enemy situation in the intended AO. Identify the enemy's OBJs and end state. Identify enemy forces and appraise their general capabilities. Describe the enemy's current disposition, location, and strength. This section may be written as a complete free text paragraph or broken down into subparagraphs. Refer to order Annex B (Intelligence) as required.

1.B.1. (U) Enemy COG. State the enemy's strategic and operational COGs.

1.B.2. (U) Enemy critical factors. State key enemy critical capabilities, critical requirements, and critical vulnerabilities.

1.B.3. (U) Probable enemy COAs. State the enemy's most likely and most dangerous COAs associated with achieving the enemy's end state. List enemy maneuver and other area capabilities that will impact friendly OPS. Include employment of enemy functional area assets to achieve end state.

1.B.4. (U) Terrorist threats. Identify known or potential terrorist threats and adversaries within the AO.

1.C. (U) Friendly forces. Briefly identify the missions of friendly forces and the OBJs, goals, and missions of civilian organizations that impact the issuing HQ or the action of subordinate commanders. Generally broken down in the following subparagraphs:

1.C.1. (U) Friendly COG. List friendly strategic and operational COGs and key critical strengths and critical weaknesses.

1.C.2. (U) Friendly critical factors. List key friendly critical capabilities, critical requirements, and critical vulnerabilities.

1.C.3. (U) HHQ mission and intent. In the subparagraphs, identify and state the mission and commander's intent for HHQ one level up.

1.C.3.A. (U) Mission. State the HHQ mission statement.

1.C.3.B. (U) Commander's intent. State the HHQ commander's intent.

1.C.4. (U) Missions of adjacent units. Identify and state the missions of adjacent units and other units whose actions have a significant impact on the issuing HQ. This information can usually be pulled from the CONOPS of these other units. Use additional subparagraphs as required.

1.D. (U) Interagency, international organizations, and NGOs. Identify and state the OBJs or goals and primary tasks of those non-Department of Defense (DOD) organizations that have a significant role within the AO. Refer to orders Annex V (Interagency Coordination) as required.

1.E. (U) Civil considerations. Describe the critical aspects of the civil situation that impact OPS. Refer to ANNEX G as required.

1.F. (U) Attachments and detachments. List units attached to or detached from the issuing HQ. State when each attachment or detachment is effective (for example, O/O, on commitment of the reserve) if different from the effective time of the OPORD. Do not repeat information already listed in Annex A.

1.G. (U) Assumptions. List any remaining assumptions that were used in the development of the OPORD that have not been proven as facts or disproven and affect the development of the CONOPS. Ensure only relevant, pending assumptions are included. Residual HHQ assumptions should be identified as such so the reader can distinguish.

1.H. (U) Operational limitations.

1.H.1. (U) Constraints.

1.H.2. (U) Restraints.

GENTEXT/MISSION//
2. (U) State the approved mission statement developed during mission analysis.

GENTEXT/EXECUTION//
(In the subparagraphs below, describe how the commander intends to accomplish the mission in terms of the commander's intent, an overarching CONOPS, schemes of employment for each operational function, assessment, specified tasks to subordinate units, and key coordinating instructions in the subparagraphs as follows.)

3. EXECUTION

3.A. (U) Commander's intent. In the subparagraphs below state approved commander's intent (purpose, method, and end state) that was developed during mission analysis.

3.A.1. (U) Purpose. State the purpose from the commander's approved intent.

3.A.2. (U) Method. State the method from the commander's approved intent.

3.A.3. (U) End state. State the end state from the commander's approved intent.

3.B. (U) CONOPS. State the approved CONOPS developed during COA comparison and decision. The CONOPS is a statement that directs the manner in which subordinate units cooperate to accomplish the mission and establishes the sequence of actions the force will use to achieve the end state. It states the principal tasks required, the responsible subordinate units, and how the principal tasks complement one another. The CONOPS paragraph is descriptive in nature and not directive. Direction to forces is given in the Tasks paragraph below. The CONOPS paragraph explains how the various tasks fit together to make a coherent CONOPS. The CONOPS projects the status of the force at the end of the operation. The CONOPS may be a single paragraph divided into two or more subparagraphs or, if unusually lengthy, summarized here with details located in order Annex C (Operations). If the CONOPS is phased, describe each phase in a subparagraph including the phase ending conditions ("Phase II ends when…"). Label these subparagraphs as phase followed by the appropriate Roman numeral, for example: phase I. If the operation is phased, all paragraphs and subparagraphs of the base order and all annexes shall mirror the phasing established in the CONOPS. The CONOPS paragraph should clearly identify the MEs, SEs, SUSs, and SHs.

Note

While functional concepts are addressed within the CONOPS, subparagraphs may be used to clarify further supporting concepts to ensure synchronization. Paragraphs 3.B.1-3.B.6 are examples of such subparagraphs. Only use the subparagraphs required. Omit those not used. Adjust paragraph numbering as necessary, but ensure the Tasks paragraph begins a new subparagraph level, i.e., 3.C or 3.D, etc. When using such subparagraphs, the same rule applies—provide the amount of detail required to clearly convey the concept and refer to the OPORD annexes and appendixes. Provide more detail, as required, in the annexes and appendixes.

3.B.1. (U) Scheme of maneuver. Describe the employment of maneuver units IAW the CONOPS. Explain how the primary tasks of maneuver units conducting decisive actions (MEs and SEs) achieve the overall OBJs. Next, explain how the primary tasks of maneuver units conducting shaping OPS, including security OPS, shape the OE to aid the accomplishment of the decisive actions. Refer to OPORD Annex C as required.

3.B.2. (U) Concept of intelligence. Describe how the commander envisions intelligence supporting the CONOPS. Include the priority of effort to situation development, targeting, and assessment. State the priority of intelligence support to units and areas. Refer to Annex B (Intelligence) as required. Describe how the commander intends to use ISR to support the CONOPS. Include the primary reconnaissance OBJs. Refer to Appendix 9 (Reconnaissance) of OPORD Annex C as required.

3.B.3. (U) Scheme of fires. Describe how the commander intends to use fires (lethal and nonlethal) to support the CONOPS with emphasis on supporting the scheme of maneuver. State the operational level fire support tasks and OBJs and how those tasks support the rest of the CONOPS. State the priorities for, allocation of, and restrictions on fires.

3.B.4. (U) Concept of protection. Describe how the commander envisions protection supporting the CONOPS. Include the priorities of protection by mission area (e.g., AMD, surface defense, subsurface defense, etc.). Address the scheme of OA security, including security for bases, and critical infrastructure. Refer to Appendix 15 of OPORD Annex C as required.

3.B.5. (U) Concept of IO. Describe how the commander visualizes executing IO. Describe how IO will support the command's operational mission. Summarize the concepts for supervision and termination of IO. The CONOPS may be a single paragraph or divided into two or more paragraphs depending upon the complexity of the operation. When an operation involves various phases, the concept of IO should be prepared in subparagraphs describing the role of IO in each phase. Refer to Appendix 3 (Information Operations) of OPORD Annex C as required.

3.B.6. (U) Assessment. Describe the priorities for assessment and identify the MOEs used to assess end state conditions and OBJs.

3.C. (U) Tasks to subordinate and supporting units. State the tasks and purposes assigned to each subordinate or supporting unit. The subordinate commanders must understand clearly what they have to accomplish and why, without any infringement on their authority to decide how. Each task and purpose should include when (the conditional clause, e.g. O/O, once the port is secure, at 0600, etc.), what (the task), where (any restrictions/direction on the location of the task, e.g., in OPAREA Blue, within the JOA, in vicinity of Redland ports, etc.), and why (purpose). Use a separate subparagraph for each unit. Identify each unit by phase if using phase construct. List units in task organization sequence. For those tasks that affect two or more units where clarification regarding deconfliction of operation is required, address the tasks in coordinating instructions in addition to listing the tasks under each individual unit. Ensure tasks use the tasks from the UJTL or UNTL as much as possible. Resort to plain language when tasks are not covered in appropriate lists. Tasks assigned to supporting units (e.g., adjacent components) should already be coordinated with those units.

3.D. (U) Coordinating instructions. List the instructions applicable to the entire force or two or more elements of the force that are necessary for proper coordination of the operation, but are not appropriate for inclusion in a particular annex. They should establish the conditions for execution and provide information about the timing of execution and deployments. Examples are below, adjust paragraph numbering as needed.

3.D.1. (U) Time or condition when the OPORD becomes effective. In the case of an OPORD that is not effective on receipt, this subparagraph should indicate the date and time the order will become effective. Tentative dates for D-day and H-hour are usually given in this subparagraph. Other key dates are also included (e.g., C-day).

3.D.2. (U) CCIRs. List the CCIRs here.

3.D.3. (U) AOs and fire support coordination measures. List AOs and fire support coordination or control measures here.

3.D.4. (U) Airspace coordinating measures. List critical airspace coordinating or control measures here.

3.D.5. (U) ROE. List ROE here. Refer to Appendix 8 to Annex C as required.

3.D.6. (U) Risk-reduction control measures. State measures specific to this operation not included in unit SOPs. They may include mission-oriented protective posture, operational exposure guidance, and fratricide prevention measures. Refer to Appendix 15 to Annex C as required.

3.D.7. (U) Personnel recovery coordination measures. Refer to Appendix 5 (Personnel Recovery) to Annex C as required.

3.D.8. (U) Environmental considerations. Refer to Annex L (Environmental Considerations) as required.

3.D.9. (U) Information themes and messages. List themes and messages here.

3.D.10. (U) Other coordinating instructions. List additional coordinating instructions and tasks that apply to two or more units as subparagraphs and other specifics needed to coordinate force activities (e.g., DIRLAUTH, daily briefing/meeting times and medium, etc.).

GENTEXT/ADMINISTRATION AND LOGISTICS//

This paragraph sets forth the manner of LOG support for the contemplated operation. For large OPS, it is almost always necessary to prepare a separate Annex D (Logistics) and Annex E (Personnel). Enough information should be included in the body of the order to make clear the basic COLS so subordinate commanders can accomplish their tasks and provide them with sufficient information needed to plan further. For paragraph 4 of the order, an appropriate sequence of presentation follows.

4. (U) Administration and LOG.

4.A. (U) Concept of sustainment. Describe the concept of sustainment, including priorities of sustainment by unit, area, or assigned mission. Briefly summarize, (as with CONOPS) the overall operation, this time from the LRC/N4 point of view. Use the following subparagraphs to provide the broad concept of support for LOG, personnel, and health system support. Provide detailed instructions for each sustainment sub function in the appendixes to OPORD Annex D.

4.B. (U) Material and services. List material and services for supply, maintenance, transportation, construction, and allocation of labor for LOG purposes. Refer to OPORD Annex D as required.

4.C. (U) Medical services. List plans and policies for hospitalization and evacuation of both military and civilian personnel. Refer to OPORD Annex Q (Medical Services) as required.

4.D. (U) Personnel. List unit strengths, replacements, and personnel policies and procedures, including those pertaining to civilians and prisoners of war. Refer to OPORD Annex E (Personnel) as required.

4.E. (U) CA. Describe control of civil populations, refugees, and related matters. Refer to OPORD Annex G as required.

4.F. (U) PA. Provide PA guidance.

4.G. (U) Miscellaneous. As required.

GENTEXT/COMMAND AND CONTROL//
This paragraph should include signal, recognition, and identification instructions, electronic policy, HQ locations, and movements, codewords, code names, and liaison.

5. (U) C2.

5.A. (U) Command.

5.A.1. (U) Command relationships. Joint OPS, by their nature, have complex command relationships. OPORDs shall be specific concerning these arrangements, including shifts that may take place as the operation progresses from one phase to the next. Refer to OPORD Annex J (Command Relationships) as required. All command relationships established in the OPORD must be addressed here, especially supporting relationships. Having a single source for command relationships within the OPORD clearly describes for subordinates their relationships with their forces and adjacent units.

5.A.2. (U) Location of commander. State the location(s) of the commander and the second-in-command during the operation, CPs, alternate CPs, flagships, and alternate flagships along with their times of activation and deactivation. State where the commander intends to be during the operation by phase if the operation is phased.

5.A.3. (U) Succession of command. State the succession of command if not covered in the unit's SOPs. Ensure those commanders within the succession of command are not collocated.

5.A.4. (U) Liaison requirements. State liaison requirements not covered in the unit's SOPs.

5.B. (U) Command, control, and communications. State information about pertinent command, control, and communications nets; operating procedures; recognition and identification procedures; electronic emission constraints; and so on. Refer to OPORD Annex K (Command, Control, Communications, and Computer Systems) as required.

5.B.1. (U) Connectivity.

5.B.2. (U) Reports. List reports not covered in SOPs. Refer to OPORD Annex R (Reports) as required.

5.C. (U) Any other C2 items necessary.

The following is an example of a JFMCC OPORD (generic example only; PLAD and GENADMIN formats should reflect existing requirements/situations).

UNCLAS

021200ZAUG20XX

FM JFMCC BLUE SWORD

TO CTF 220
CTF 223
CTF 224
CTF 225
CTF 227
CTF 229

INFO CJTF BLUE SWORD
JFACC BLUE SWORD
JFLCC BLUE SWORD
JFSOCC BLUE SWORD
CJCS WASHINGTON DC
COMUSEASTCOM MACDILL AFB FL//00/CCJ3//
COMUSSOCOM MACDILL AFB FL
COMNAVSPECWARCOM CORONADO CA
COMUSNORTHCOM PETERSON AFB
COMUSTRANSCOM SCOTT AFB IL
COMUSSTRATCOM OFFUTT AFB NE
COMUSMARFOREASTCOM
COMFLTFORCOM NORFOLK VA
AMEMBASSY BLUESKIES
AMEMBASSY GREENACRES
AMEMBASSY GREYLAND
AMEMBASSY WHITELAND
AMEMBASSY PINKLAND
CIA LANGLEY VA
DIA WASHINGTON DC

UNCLAS//N03000//OPER/BLUE SWORD//

MSGID/ORDER/JFMCC BLUE SWORD//
AMPN/SPECIAL HANDLING INSTRUCTIONS//ORDTYP/OPORD/JFMCC BLUE SWORD//
TIMEZONE/ZULU//REF/A/RMG/CJCS/190100ZJUL19/-/NOTAL//
REF/B/RMG/COMUSEASTCOM/240600ZJUL19/-/NOTAL//
REF/C/RMG/CJTF BLUE SWORD/301900ZJUL19/-/NOTAL//
REF/D/RMG/JFMCC BLUE SWORD/261200ZJUL19/-/NOTAL//
REF/E/DOC/CJCSI 3121.01A/15JAN00//
REF/F/RMG/CJTF BLUE SWORD/011201ZAUG20XX//
REF/G/GCCS-M/JFMCC BLUE SWORD/011200ZAUG20XX//
NARR/REF A IS CJCS WARNORD. REF B IS COMUSEASTCOM WARNORD. REF C IS CJTF BLUE SWORD OPORD. REF D IS JFMCC BLUE SWORD WARNORD. REF E IS CJCS STANDING 1 ROE. REF F IS CJTF BLUE SWORD PPAG. REF G IS OP BLUE SWORD JFMCC FORCE OPS AREAS OVERLAYS FOR EACH OPERATION PHASE.//

GENTEXT/AUTHORITY//
(U) This is an OPORD. The SECDEF has authorized execution of OPERATION FA. CJTF BLUE SWORD has been activated and JFMCC forces will make preparations for immediate deployment to the designated JOA.//

HEADING/TASK ORGANIZATION//

/UNIT	/UNITLOC	/COMMENTS
/USEASTCOM	/CONUS	/CCDR
/CJTF BLUE SWORD	/PINKLAND	/USARFOREAST
/CJTF BLUE SWORD JFMCC	/EMB UND	/COMEASTFLT
/USS UNDERWAY (UND)	/ONSTA JOA	/CJTF BLUE SWORD JFMCC EMB
/CTF 220	/EMB TR	/CCSG-2
/TF 220	/ENR JOA	/TRCSG
/CTG 220.1	/ENR JOA	/CCSG-2
/CTU 220.1.1	/ENR JOA	/USS THEODORE ROOSEVELT (TR)
/CTU 220.1.2	/ENR JOA	/USS CHANCELLORSVILLE (CHV) (CG 62)
/CTE 220.1.2.1	/ENR JOA	/HSL 42 DET B
/CTU 220.1.3	/ENR JOA	/USS SHILOH (SHI) (CG 67)
/CTE 220.1.3.1	/EMB SHI	/HSL 42 DET C
/CTU 220.1.4	/ENR JOA	/USS CAMDEN (CAM) (AOE 2)
/CTG 220.2	/EMB TR	/COMCARAIRWING 14 (CVW-14)
/TG 220.2	/EMB TR	/CARAIRWING 14 (CVW-14)
/CTU 220.2.1	/EMB TR	/COMCARAIRWING 14 (CVW-14)
/CTU 220.2.2	/EMB TR	/VFA-1
/CTU 220.2.3	/EMB TR	/VFA-2
/CTU 220.2.4	/EMB TR	/VMFA-22
/CTU 220.2.5	/EMB TR	/VAW-120
/CTU 220.2.6	/EMB TR	/VAQ-127
/CTU 220.2.7	/EMB TR	/HS-12
/CTG 220.3	/EMB TR	/COMDESRON 22 (CDS-22)
/TG 220.3	/ENR JOA	/DESRON 22
/CTU 220.3.1	/ENR JOA	/USS STETHAM (STE) (DDG 63)
/CTU 220.3.2	/ENR JOA	/USS SAMPSON (SAM) (DDG 102)
/CTE 220.3.2.1	/EMB SAM	/HSL 42 DET A
/CTU 220.3.3	/ENR JOA	/USS FREEDOM (FRE) (LCS 1)
/CTE 220.3.3.1	/EMB FRE	/HSL 44 DET A
/CTF 223	/CONUS	/COMLOGFORNAVEAST
/CTG 223.1	/EMB RIC	/COMLOGRON EAST
/TG 223.1	/ENR JOA	/UNDERWAY REPLENISHMENT GROUP
/CTU 223.1.1		/USNS RICHARD E. BYRD (RIC) (T-AKE-4)
/CTU 223.1.2		/USNS SUPPLY (SUP) (T-AOE 6)
/CTU 223.1.3		/USNS PECOS (PEC) (T-AO 197)
/CTG 223.2	/EMB DTW	/COMMANDER MPS SQUADRON THREE
/TG 223.2	/ENR JOA	/MPSRON 3
/CTU 223.2.1		/MV PFC DEWAYNE T WILLIAMS (DTW) (T-AK 3009)
/CTU 223.2.2		/MV 1ST LT BALDOMERO LOPEZ (LBL) (T-AK 3010)

/CTU 223.2.3		/MV 1ST LT JACK LUMMUS (LJL) (T-AK 3011)
/CTU 223.2.4		/MV SGT WILLIAM R BUTTON (WRB) (T-AK 3012)
/CTF 224	/EMB NAS	/NASESG
/TF 224		/ENR JOA
/CTG 224.1	/EMB NAS	/CPR 6
/CTU 224.1.1	/ENR JOA	/USS NASSAU (NAS) (LHA-4)
/CTU 224.1.2	/ENR JOA	/USS SAN ANTONIO (SAT) (LPD 17)
/CTU 224.1.3	/ENR JOA	/USS NEW ORLEANS (ORL) (LPD 18)
/CTU 224.1.4	/ENR JOA	/USS GERMANTOWN (GER) (LSD 42)
/CTG 224.2	/EMB NAS	/31ST MEU
/CTG 224.3	/EMB COR	/COMMCMRON TWO
/CTU 224.3.1	/ENR JOA	/USS CORONADO (COR) (LCS 4)
/CTE 224.3.1.1	/EMB COR	/HSL 44 DET P
/CTU 224.3.2	/ENR JOA	/USS DEVASTATOR (DEV) (MCM 6)
/CTU 224.3.3	/ENR JOA	/USS FORT WORTH (FTW) (LCS 3)
/CTE 224.3.3.1	/EMB FTW	/MCM DET ONE
/CTU 224.3.4	/ENR JOA	/USS AVENGER (AVE) (MCM 1)
/CTF 225	/EMB ANT	/CO ANT
/CTG 225.1	/ENR JOA	/USS ANTIETAM (ANT) (CG 54)
/CTG 225.2	/ENR JOA	/USS MCCAMPBELL (MCB) (DDG 85)
/CTG 225.3	/ENR JOA	/USS WILLIAM S SIMS (WSS) (DDG 113)
/CTF 227	/CONUS	/COMPATRECONFORCE
/CTG 227.1	/U.S. NAV AIR STA	/VP-40
/CTU 227.1.1	/PINKLAND ISB	/VP-40 DET
/CTG 227.2	/U.S. NAV AIR STA	/VQ-2
/CTU 227.2.1	/PINKLAND ISB	/VQ-2 DET
/CTF 229	/U.S. NAVY BASE	/COMSUBGRUEAST
/CTG 229.1	/JOA	/USS CHICAGO (CHI) (SSN 721)
/CTG 229.2	/ENR JOA	/USS TUCSON (TUS) (SSN 770)
/PINKLAND	/JOA	/2X TYPE 22, 1X TYPE 42

SEE ANNEX A FOR DETAILED TASK ORGANIZATION.
GENTEXT/SITUATION//

1. (U) Situation.

1.A. (U) General. The inability of the government of Redland to eliminate safe havens for terrorist organizations continues to pose a threat to the region. Radical fundamentalist groups with known ties to international and regional terrorist organizations continue to plan, train, and resource terrorist activities from within the sovereign borders of Redland. These groups intend to disrupt international shipping throughout the Redland Sea and pose a direct threat to Redland's neighbors, including Pinkland. Terrorists continue to undermine the authority of the Redland government politically, economically, and militarily; and Redland continues to ignore international demands that the terrorist organizations be destroyed. Additionally, the Redland navy operates in an aggressive manner to establish regional maritime dominance and protect Redland commercial EEZ activity that violates international law. Fishing and offshore oil are major Redland economic activities. Pinkland is Redland's principal regional competitor and a primary U.S. economic and defense partner. Redland and terrorists based in Redland have become increasingly aggressive toward Pinkland with numerous, recent acts of violence linked to paramilitary organizations based in Redland. Redland is seeking to topple Pinkland's government and replace it with a more sympathetic regime that will acquiesce to its national goals. Because of its close ties with the United States, the government of Pinkland has requested that the United States and coalition partners provide necessary military assistance to disrupt terrorist activities within Redland and to help prevent the influx of aid to these groups from outside Redland's borders.

1.B. (U) Enemy. The Redland navy's main source of strength is its sub force, which consists of four SSs with one underway nearly continuously and one in refit and unavailable. Redland CDCMs have OTH range but lack effective third-party targeting. Fast attack craft also have ASM capability with limited OTHT. Redland has no MPRA and limited number of short range ISR-capable aircraft. All Redland navy vessels are capable of mining. Mines are stored in a depot in its navy base and must be moved to the piers for loading. Main source of strength for the government of Redland is the 23rd Guards Div, which is its best-equipped and trained ground unit that is manned exclusively by government loyalists. This unit also provides clandestine support to Home Guard units that are connected to terrorist attacks on Pinkland and Redland's other neighbors—as well as U.S. military and civilians in the region. See Annex B for order of battle and additional information.

1.B.1. (U) Enemy COG. Operational COG for Redland: Kilo subs.

1.B.2. (U) Enemy critical factors.

1.B.2.A. (U) Critical capabilities:

1.B.2.A.1. (U) C2—ability to plan/task organize and direct Kilo subs against enemy ships.

1.B.2.A.2. (U) Intelligence—ability to detect enemy ship movements/analyze information on ship movements to determine where and when Kilo subs should be employed against which targets.

1.B.2.A.3. (U) Movement and maneuver—ability to manage waterspace/maneuver Kilo forces/provide indications and warnings (I&W)/direct/provide WSM to deconflict OPS against enemy shipping.

1.B.2.A.4. (U) Fires—ability to provide complementary sea denial fires by channeling blue forces into optimal kill areas.

1.B.2.A.5. (U) Sustainment—ability to resupply/repair/replenish Kilo subs.

1.B.2.A.6. (U) Protection—ability to mitigate JTF BLUE SWORD force ASW capabilities/ability to safeguard Kilo bases.

1.B.2.B. (U) Critical requirements:

1.B.2.B.1. (U) C2—CP locations/communications links (physical and electromagnetic) and the sub's ability to transmit/receive with these links.

1.B.2.B.2. (U) Intelligence—CPs/fusion centers/human intelligence (HUMINT)/coastal surveillance radars/signals intelligence/C2 structure and communications links for passing information on JTF BLUE SWORD ship locations.

1.B.2.B.3. (U) Movement and maneuver—suitable waterspace and WSM to support Kilo sub OPS/bases of OPS support/trained crews.

1.B.2.B.4. (U) Fires—antiship mobile platforms and missile storage locations/surface fleet/C2 structure/ISR capabilities to detect JTF BLUE SWORD forces/airfields to support aviation OPS/sea mines and minelayers/mine storage facilities.

1.B.2.B.5. (U) Sustainment—storage and repair facilities/sub tenders/ports/lines of communications between storage sites and ports/C2 infrastructure.

1.B.2.B.6. (U) Protection—use of OPSEC/adequate I&W of JTF BLUE SWORD movements/local air superiority/deception OPS.

1.B.2.C. (U) Critical vulnerabilities.

1.B.2.C.1. RF communications, intelligence fusion centers, sub piers, missile storage facilities, key bridges.

1.B.3. (U) Enemy COA.

1.B.3.A. (U) Most likely: See Annex B.

1.B.3.B. (U) Most dangerous: See Annex B.

1.B.4. (U) Terrorist threats. There are multiple terrorist groups using Redland as a safe haven. See INTSUM (010500ZAUG20XX) for detailed analysis of terrorist groups in Redland.

1.C. (U) Friendly. ISO Operation FA, Pinkland has authorized limited basing rights to U.S. and coalition forces. Pinkland has agreed to allow the use of an airfield to support fixed-wing aircraft OPS and act as an ISB. See Annex A for forces deployed in Pinkland as of the issue of this OPORD. Hangars and ramp space with maintenance facilities have also been made available. Pinkland has authorized the use of its main port facility for U.S. and coalition combatant and LOG shipping. The port can support up to four destroyer-sized ships at its military pier along with two 100,000 gross ton ships at its commercial facility. Cranes and tugs available only during daylight hours. Radiation of military radars in port is not authorized. Port security provided by a Pinkland contracted force.

1.C.1. (U) Friendly COG. Operational COG (JFMCC level): See Annex C.

1.C.2. (U) Friendly critical factors: See Annex C.

1.C.3. (U) HHQ mission and intent.

1.C.3.A. (U) HHQ mission. O/O, JTF BLUE SWORD defeats the Redlands 23rd Guards Div and destroys terrorist forces and their infrastructure in Redland, IOT eliminate the terrorist base of OPS in the area, secure the territorial integrity of Pinkland, and assist in stabilizing the region.

1.C.3.B. (U) HHQ intent.

1.C.3.B.1. (U) Purpose. Enable a stable and secure environment in which Redland terrorist and conventional forces no longer threaten the government of Pinkland.

1.C.3.B.2. (U) Method. There will be three primary focus areas: neutralizing Redland C2, destroying the Redland conventional forces, and eliminating terrorist groups and their infrastructure. We will respect the boundaries of neighboring neutral countries and minimize damage to civilian infrastructure within Redland. Operating within a joint organization construct, U.S. forces will cooperate with multinational partners and synchronize with Pinkland to ensure unity of effort. Maximize employment of all JTF capabilities, especially our precision fires and mobility, to dominate the OE. Speed and timing are essential to mission success.

1.C.3.B.3. (U) End state. The 23rd Guards Div is defeated, and terrorist forces and their camps in Redland are destroyed. Territorial integrity of Pinkland is secured, and Pinkland is prepared to assume responsibility for security OPS. Conditions for a stable environment that will not support further terrorist activities in Redland are established. The JTF can transition OPS to Pinkland and redeploy.

1.C.3.B.4. (U) Risk: I am prepared to accept risk by not completing foregoing completion of our shaping OPS to neutralize conventional Redland military forces if should an early opportunity presents itself to quickly eliminate the terrorists and their bases.

1.C.4. (U) Missions of adjacent units. See JFMCC BLUE SWORD SIPR Collaboration at Sea site for JFLCC, JFACC, AND JFSOCC CONOPS.

1.D. (U) Interagency, NGOs, and international organizations. DOD is ISO DOS during phase IV and phase V OPS. Detailed planning for phase IV and phase V OPS with DOS is ongoing. Details are promulgated via FRAGORD.

1.E. (U) Civil considerations. Facilitate or coordinate essential population control measures to minimize civilian interference with military OPS. All humanitarian relief activities are coordinated with American embassies. See Annex G.

1.F. (U) Attachments and detachments. None.

1.G. (U) Assumptions.

1.G.1. (U) Greyland will permit overflight for carrier-based tactical aviation and TLAMs.

1.G.2. (U) Pinkland naval forces will cooperate with the JTF but remain under their national control.

1.G.3. (U) JFLCC Patriot missile sites are FMC at the ISBs before combat OPS begin.

1.G.4. (U) Pinkland will allow establishment of an SPOD and FOB at Port X-Ray.

1.G.5. (U) The DOS-designated safe haven for NCEs will be Port X-Ray in Pinkland.

1.G.6. (U) Orangeland will not freely allow terrorist camps to be established within their borders (JTF assumption).

1.G.7. (U) Whiteland will allow support to terrorist organizations to pass over their border into Redland (JTF assumption).

1.G.8. (U) Coalition nations will remain committed throughout the length of the operation (JTF assumption).

1.H. (U) Operational limitations.

1.H.1. (U) Constraints.

1.H.1.A. (U) Strictly limit collateral damage during the employment of lethal fires.

1.H.1.B. (U) Preserve Redland civilian infrastructure.

1.H.1.C. (U) Incorporate the Pinkland military into the planning and execution of OPS.

1.H.2. (U) Restraints.

1.H.2.A. (U) Do not mine Redland TTW.

1.H.2.B. (U) JTF forces must not conduct offensive OPS from neutral country TTW.//

GENTEXT/MISSION//

2. (U) O/O, JFMCC BLUE SWORD establishes maritime superiority in vicinity of southern Redland and the AOA IOT establish a secure and stable maritime environment free from Redland military interference or terrorist threats.//

GENTEXT/EXECUTION//

3. (U) Execution.

3.A. (U) Commander’s intent.

3.A.1. (U) Purpose. JFMCC supports the JTF and government of Pinkland by controlling all aspects of the maritime environment to provide a stable and secure environment free from any conventional military or terrorist threats emanating from Redland.

3.A.2. (U) Method. Commencing combat OPS on D-day, JFMCC conducts sea control OPS to create a condition of maritime advantage in the maritime AO. We will do this by destroying enemy forces, suppressing enemy sea commerce, and protecting vital sea lanes. If required, we will conduct OPS focusing our efforts in the southern portion of the maritime AO to create a degree of sea control that allows amphibious OPS to commence. Follow on tasks include developing Port Yankee as an SPOD for follow-on forces and sustainment. Concurrent to these efforts, we will use our strike capabilities to support the land component commander efforts in defeating the 23rd Guards Div and destroy the terrorists and their infrastructure in the JOA.

3.A.3. (U) End state. Redland maritime forces and maritime-based terrorists are neutralized and cannot conduct organized operations. We retain the ability to strike in and from the maritime environment as well as place amphibious forces ashore into Pinkland or Redland. Port Yankee is a fully functioning SPOD. Maritime forces are prepared to transition OPS to Pinkland.

3.A.4. (U) Risk. I see a significant threat from Redland forces for our maritime forces at the outset. We must be fully prepared to transition to kinetic OPS quickly to attrite their force and reduce the risk, while holding attrition to civilian population and infrastructure to an absolute minimum.

3.B. (U) CONOPS. JFMCC BLUE SWORD conducts OPS over five phases. In phase I, JFMCC deploys into the JOA and conducts maritime stability OPS IOT deter Redland maritime aggression. In phase II, JFMCC conducts offensive OPS IOT gain sea control in key DPs. JFMCC executes an amphibious assault at OBJ RAT in phase III IOT fix Redland forces ISO JFLCC. During phase IV, JFMCC conducts stability OPS ICW Pinkland naval forces IOT prevent terrorist exploitation of the maritime domain. In phase V, JFMCC forces transition missions, functions, and tasks to designated military and agency partners, redeploys attached forces, and disestablishes.

3.B.1. (U) Phase I. This phase begins when C-day commences and hostilities have not yet been declared. JTF BLUE SWORD deploys joint forces into the JOA IOT protect Pinkland and deter Redland aggression. JFMCC (JTF ME) deploys forces into the JOA and conducts maritime stability OPS IOT deter Redland maritime aggression. JFACC (JTF SE) conducts DCA IOT protect JTF forces and provides aerial refueling support IOT extend TF 220's operational reach. JFSOCC (JTF SE) conducts SR in Redland IOT set conditions for FOPS. JFLCC (JTF SE) deploys forces to Pinkland IOT support follow-on offensive OPS in Redland. CTF 225 (JFMCC ME) conducts patrols in AO NOR EAST IOT deter Redland maritime aggression. CTF 227 (JFMCC SE) conducts airborne surveillance and reconnaissance IOT determine Redland maritime disposition and intentions. CTF 229 (JFMCC SE) provide SUW, ASW, and ISR support in SUB OPAREA TEXAS IOT protect JFMCC forces and enable FOPS. CTF 220 (JFMCC SH) conducts IAMD in AO NOR EAST IOT protect deploying JTF forces. CTF 224 (JFMCC SH) conducts amphibious rehearsal in AOA 1 IOT prepare for offensive operations in Redland. CTF 223 (JFMCC SUS) conducts replenishment at sea IOT extend operational reach of JFMCC forces. The Pinkland navy conducts patrols in AO NOR EAST IOT protect Pinkland ISBs. The greatest risk to mission in phase I is the employment of Redland anti-access capabilities in the vicinity of Pinkland, which would impede JFMCC (and JTF) deployment. This risk is mitigated by the execution of IAMD by CTF 220 and by patrolling Pinkland waters by CTF 225. Phase I ends when Redland agrees to diplomatic resolution or commences hostilities, and JFMCC forces have completed deployment to the JOA and are in position to conduct offensive OPS. Maritime commerce avoids the JOA.

3.B.2. (U) Phase II. This phase begins when Redland commences hostilities, JFMCC forces have completed deployment to the JOA and are in position to conduct offensive OPS. JTF BLUE SWORD conducts offensive OPS in the JOA IOT degrade Redland forces. JFMCC (JTF ME) conducts offensive OPS in the JOA IOT gain sea control in AO CENTRAL. JFACC (JTF SE) conducts strike OPS in the JOA IOT degrade Redland defensive capabilities. JFSOCC (JTF SE) conducts DA at OBJ RAT IOT degrade Redland C2 and conducts surface observations in AOA 2 IOT prepare for follow-on amphibious assault. JFLCC (JTF SH) stages forces in Pinkland IOT conduct phase III forcible entry operation. CTF 225 (JFMCC ME) attacks Redland naval forces IOT gain local sea control in AO CENTRAL. CTF 225 also conducts strike OPS IOT degrade Redland defensive capability. CTF 220 (JFMCC SE) conducts strike OPS in the JOA IOT degrade Redland maritime forces and protects CTF 224 forces in AO SOUTH to preserve combat power for phase III. CTF 229 (JFMCC SE) interdicts Redland maritime forces ISO CTF 225 to establish sea control in AO CENTRAL. CTF 227 (JFMCC SE) conducts airborne surveillance and reconnaissance in AO CENTRAL, NORTH, and WEST IOT determine

Redland maritime disposition and intentions. CTF 224 (JFMCC SH) O/O conducts MCM in AOA 2 to set the conditions for amphibious OPS. CTF 223 (JFMCC SUS) continues replenishment at sea IOT extend operational reach of JFMCC forces. Pinkland naval forces, when sea control is established in AO CENTRAL, escort CTF 224 to AOA 2 IOT prevent Redland interference. The greatest risk to mission in phase II is Redland subs preventing CTF 225 from establishing sea control in AO CENTRAL. This risk is mitigated by the prioritization of finding and engaging these forces early in the operation. Phase II ends when sea control is established in AO CENTRAL, CTF 224 is positioned to conduct amphibious OPS, and Redland forces are degraded in AO CENTRAL.

3.B.3. (U) Phase III. This phase begins when sea control is established in AO CENTRAL. CTF 224 is positioned to conduct amphibious OPS. Redland forces in AO CENTRAL (AOA 2/OBJ RAT) are degraded. JTF BLUE SWORD conducts simultaneous forcible-entry OPS to defeat Redland forces. JFLCC (JTF ME) conducts airborne forcible entry into Redland IOT destroy Redland military forces at OBJs DOG and CAT. JFMCC (JTF SE) conducts amphibious OPS in AOA-2 IOT fix Redland forces at OBJ RAT. JFACC (JTF SE) conducts joint fire support IOT degrade Redland forces in JOA. JFSOCC (JTF SE) conducts DA against high-value targets) IOT disrupt Redland reinforcements. CTF 224 (JFMCC ME) conducts amphibious assault at AOA 2 IOT fix Redland forces at OBJ RAT. CTF 225 (JFMCC SE) covers CTF 224 IOT enable amphibious OPS in AOA 2. CTF 220 (JFMCC SE) conducts CAS in AOA 2 IOT support CTF 224 amphibious operation. CTF 229 (JFMCC SE) attacks Redland naval forces IOT expand sea control in the maritime AO. CTF 227 (JFMCC SE) conducts armed reconnaissance in AOs CENTRAL, NORTH, and WEST IOT prevent Redland interference with amphibious OPS. CTF 223 (JFMCC SUS) continues replenishment at sea IOT extend operational reach of JFMCC forces. Pinkland naval forces protect CTF 223's forces in AO CENTRAL IOT prevent disruption of combat LOG. The greatest risk to mission in phase III is if Redland ground forces are not fixed at OBJ RAT. This risk is mitigated by use of IRCs to deceive Redland decision makers, and through use of airborne interdiction to destroy those ground force elements fleeing RAT toward JFLCC OBJs DOG and CAT. Phase III ends when Redland forces are defeated, JTF forces have consolidated in Redland, and JFMCC forces are postured for stability OPS.

3.B.4. (U) Phase IV. This phase begins when Redland forces are defeated. JTF forces have consolidated in Redland. JFMCC forces are postured for stability OPS. JTF BLUE SWORD conducts stability OPS throughout the JOA IOT set conditions for relief by Pinkland and other designated forces. JFLCC (JTF ME) conducts counterinsurgency and counter terrorism operations in Redland IOT stabilize security environment. JFMCC (JTF SE), in concert with Pinkland naval forces, conducts stability operations in the maritime AO IOT prevent terrorist exploitation. JFACC (JTF SE) protects JFLCC and JFSOCC forces in Redland IOT ensure freedom of maneuver. JFSOCC (JTF SE) conducts foreign internal defense (FID) with Pinkland security forces IOT enable transfer of security missions. CTF 225 (JFMCC ME) conducts MIO in OPAREAs CENTRAL and WEST IOT deny freedom of movement to suspected terrorists. CTF 227 (JFMCC SE) conducts airborne R&S in the JOA IOT discern terrorist disposition and intentions. CTF 220 (JFMCC SE) maintains CAP and SUCAP IOT protect CTF 225 and CTF 224 forces in AO CENTRAL and WEST. CTF 224 (JFMCC SE) conducts counterterrorist operations in vicinity of OBJ RAT IOT secure area in preparation for turnover to Redland forces. CTF 229 (JFMCC SE) conducts surveillance of Redland maritime forces to determine disposition and intentions. CTF 223 (JFMCC SUS) maintains replenishment at sea mission. The greatest risk to mission in phase IV is not being able to stabilize the maritime environment sufficiently to allow turnover to Pinkland or other nation forces. This risk is mitigated by working alongside Pinkland naval forces early in the operation to understand the range of support required by them which will allow turnover to occur. Phase IV ends when Redland maritime forces no longer pose a threat to Pinkland, Pinkland maritime forces are able to conduct stability OPS within the Pinkland EEZ, and commercial maritime activity has returned to precrisis levels.

3.B.5. (U) Phase V. This phase begins when Redland maritime forces no longer pose a threat to Pinkland, Pinkland maritime forces are able to conduct stability OPS within the Pinkland EEZ, and commercial maritime activity has returned to precrisis levels. JTF FA coordinates the transition of JTF missions with Pinkland MOD IOT ensure a seamless continuation of security activities. JFLCC (JTF ME) conducts relief in place with designated multinational forces (MNFs) IOT ensure continuation of security missions. JFMCC (JTF SE) conducts relief in place with Pinkland's navy IOT ensure the continuation of MSO. JFACC (JTF SE) conducts a relief in place with Pinkland's air force IOT ensure the continuation of security missions. JFSOCC (JTF SE), in concert with the JFLCC, conducts relief in place with designated MNF IOT ensure the continuation of security missions. O/O, all attached forces will redeploy and functional component commands disestablish. CTF 225 (JFMCC ME)

conducts relief in place with Pinkland's navy IOT ensure the continuation of MSO. All other CTFs, O/O redeploy attached forces. The greatest risk to mission in phase V is Pinkland being unable to assume security responsibilities. This risk is mitigated by ensuring Pinkland forces have necessary forces in place before redeployment of U.S. forces. Phase V ends when Pinkland or other designated organizations have assumed responsibility for JFMCC missions and JFMCC forces have redeployed.

3.B.6. Scheme of fires. See Annex C.

3.B.7. Concept of protection. See Annex C.

3.C. (U) Tasks.

3.C.1. (U) Phase I.

3.C.1.A. (U) CTF 220.

3.C.1.A.1. (U) Conduct IAMD in AO NOR EAST IOT protect deploying forces.

3.C.1.A.2. (U) BPT conduct aerial show of force to deter Redland aggression.

3.C.1.B. (U) CTF 223.

3.C.1.B.1. (U) Conduct LOG IOT extend operational reach of JFMCC forces.

3.C.1.B.2. (U) Conduct intratheater airlift in FREEDOM ASSURANCE JOA IOT support JFMCC deployment.

3.C.1.C. (U) CTF 224.

3.C.1.C.1. (U) Conduct amphibious rehearsal in AOA 1 IOT prepare for offensive OPS in Redland.

3.C.1.C.2. (U) BPT conduct defensive mine clearance IOT ensure access to ISBs and SLOCs.

3.C.1.D. (U) CTF 225 (ME).

3.C.1.D.1. (U) Conduct patrols in AO NOR EAST IOT deter Redland maritime aggression.

3.C.1.E. (U) CTF 227 (SE).

3.C.1.E.1. (U) Conduct airborne surveillance and reconnaissance in the JOA IOT determine Redland maritime disposition and intentions.

3.C.1.F. (U) CTF 229 (SE).

3.C.1.F.1. (U) provide SUW, ASW, and ISR support in SUB OPAREA TEXAS IOT protect JFMCC forces and enable FOPS.

3.C.1.G. (U) Pinkland.

3.C.1.G.1. (U) Conduct patrols in AO NOR EAST IOT protect Pinkland ISBs.

3.C.1.G.2. (U) BPT conduct defensive mine clearance of Pinkland ISBs IOT ensure FA force flow.

3.C.2. (U) Phase II.

3.C.2.A. (U) CTF 220 (SE).

3.C.2.A.1. (U) Conduct strike OPS in the JOA IOT degrade Redland maritime forces.

3.C.2.A.2. (U) Protect CTF 224 forces in AO SOUTH to preserve combat power for phase III.

3.C.2.B. (U) CTF 223.

3.C.2.B.1. (U) Conduct LOG IOT extend operational reach of JFMCC forces.

3.C.2.C. (U) CTF 224.

3.C.2.C.1. (U) O/O, conduct MCM in AOA 2 to set the conditions for amphibious OPS.

3.C.2.D. (U) CTF 225 (ME).

3.C.2.D.1. (U) Attack Redland forces IOT gain local sea control in AO CENTRAL.

3.C.2.D.2. (U) Conduct strike OPS IOT degrade Redland defensive capability.

3.C.2.E. (U) CTF 227 (SE).

3.C.2.E.1. (U) Conduct airborne surveillance and reconnaissance in AOs CENTRAL, NORTH, and WEST IOT determine Redland maritime disposition and intentions.

3.C.2.E.2. (U) Conduct armed reconnaissance in AOs CENTRAL, NORTH, and WEST IOT degrade Redland maritime capabilities.

3.C.2.F. (U) CTF 229 (SE).

3.C.2.F.1. (U) Interdict Redland maritime forces ISO CTF 225 to establish sea control in AO CENTRAL.

3.C.2.G. (U) Pinkland.

3.C.2.G.1. (U) O/O, escort CTF 224 to AOA 2 IOT prevent Redland interference.

3.C.3. (U) Phase III.

3.C.3.A. (U) CTF 220 (SE).

3.C.3.A.1. (U) Conduct CAS in AOA 2 IOT support CTF 224 amphibious operation.

3.C.3.B. (U) CTF 223.

3.C.3.B.1. (U) Conduct replenishment at sea IOT extend operational reach of JFMCC forces.

3.C.3.C. (U) CTF 224 (ME).

3.C.3.C.1. (U) Conduct amphibious assault at AOA 2 IOT fix Redland forces at OBJ RAT.

3.C.3.D. (U) CTF 225 (SE).

3.C.3.D.1. (U) Cover CTF 224 IOT enable amphibious OPS in AOA 2.

3.C.3.E. (U) CTF 227.

3.C.3.E.1. (U) Conduct airborne R&S in the JOA IOT determine remaining Redland capability.

3.C.3.E.2. (U) Conduct armed reconnaissance in AO CENTRAL, NORTH, and WEST IOT prevent Redland interference with amphibious OPS.

3.C.3.F. (U) CTF 229 (SE).

3.C.3.F.1. (U) Attack Redland naval forces IOT expand sea control in the maritime AO.

3.C.3.G. (U) Pinkland.

3.C.3.G.1. (U) Protect CTF 223 forces in AO CENTRAL IOT prevent disruption of combat LOG.

3.C.4. (U) Phase IV.

3.C.4.A. (U) CTF 220 (SE).

3.C.4.A.1. (U) Conduct CAP/SUCAP IOT protect CTF 225 and CTF 224 forces in AO CENTRAL and WEST.

3.C.4.A.2. (U) BPT conduct strike OPS IOT prevent resurgence of Redland aggression.

3.C.4.B. (U) CTF 223.

3.C.4.B.1. (U) Conduct replenishment at sea IOT extend operational reach of JFMCC forces.

3.C.4.C. (U) CTF 224 (SE).

3.C.4.C.1. (U) Conduct CT OPS in vicinity of OBJ RAT IOT secure area in preparation for turnover to RL forces.

3.C.4.C.2. (U) BPT conduct FHA in Greyland IOT relieve human suffering.

3.C.4.C.3. (U) BPT conduct mine clearance OPS in AO WEST IOT ensure access to commercial shipping.

3.C.4.C.4. (U) BPT conduct port reconstruction at port YANKEE IOT ensure rapid opening of SPOD.

3.C.4.D. (U) CTF 225 (ME).

3.C.4.D.1. (U) Conduct MIO in AO CENTRAL and WEST IOT deny freedom of movement to suspected terrorists.

3.C.4.E. (U) CTF 227 (SE).

3.C.4.E.1. (U) Conduct airborne R&S in the JOA IOT discern terrorist disposition and intentions.

3.C.4.F. (U) CTF 229 (SE).

3.C.4.F.1. (U) Conduct surveillance of RL maritime forces to determine disposition and intentions.

3.C.4.G. (U) Pinkland.

3.C.4.G.1. (U) Conduct MIO in AO NOR EAST IOT provide secure operating environment.

3.C.5. (U) Phase V.

3.C.5.A. (U) CTF 220 (SE).

3.C.5.A.1. (U) O/O, redeploy attached forces.

3.C.5.B. (U) CTF 223.

3.C.5.B.1. (U) ICW CTF 224 and CTF 225, redeploy attached forces.

3.C.5.C. (U) CTF 224 (SE).

3.C.5.C.1. (U) O/O, redeploy attached forces.

3.C.5.D. (U) CTF 225 (ME).

3.C.4.D.1. (U) Conduct relief in place with Pinkland navy IOT ensure continuation of MSO.

3.C.5.D.2. (U) O/O, redeploy attached forces.

3.C.5.E. (U) CTF 227 (SE).

3.C.5.E.1. (U) O/O, redeploy attached forces.

3.C.5.F. (U) CTF 229 (SE).

3.C.4.F.1. (U) O/O, redeploy attached forces.

3.C.4.G. (U) Pinkland.

3.C.5.G.1. (U) Assume sole responsibility for maritime stabilization OPS in the Pinkland EEZ.

3.D. (U) Coordinating instructions.

3.D.1. (U) Daily CTF brief to JFMCC (VTC) occurs as per published battle rhythm in DIM.

3.D.2. (U) C-day is 02AUG20XX.

3.D.3. (U) D-day is 19AUG20XX.

3.D.4. (U) Anticipated length of operation is less than 180 days.

3.D.5. (U) Codeword assigned to this operation is FA.

3.D.6. (U) ROE.

3.D.6.A. (U) Standing ROE are in effect.

3.D.6.B. (U) JFMCC TFs request supplemental measures for mission accomplishment IAW ref (d) procedures.

3.D.6.C. (U) POTUS approval is required for use of riot control agents.

3.D.7. (U) JOA: as published. JFMCC AOs per ref (g).

3.D.8. (U) DIRLAUTH ALCON. Keep JFMCC informed.

3.D.9. (U) CCIRS.

3.D.9.A. (U) Phase I.

3.D.9.A.1. (U) PIR 1—What is the current disposition and intent of Redland minelaying and CDCM forces?

3.D.9.A.2. (U) PIR 2—What is the threat to forces establishing ISB in Pinkland?

3.D.9.A.3. (U) FFIR 1—Has JFMCC IAMD capability been degraded?

3.D.9.B. (U) Phase II.

3.D.9.B.1. (U) PIR 1—What is the disposition of Redland maritime forces in AO WEST and CENTRAL?

3.D.9.B.2. (U) PIR 2—What is the disposition of Redland ground forces in vicinity of OBJ RAT?

3.D.9.B.3. (U) FFIR 1—Is sea control established in AO CENTRAL?

3.D.9.B.4. (U) FFIR 2—CTF 220 capable of conducting flight OPS?

3.D.9.C. (U) Phase III.

3.D.9.C.1. (U) PIR 1—What is the remaining Redland maritime capability?

3.D.9.C.2. (U) PIR 2—Are Redland ground forces reinforcing in vicinity of OBJ RAT?

3.D.9.C.3. (U) FFIR 1—Are conditions set to commence amphibious assault?

3.D.9.C.4. (U) FFIR 2—CTF 224 capable of conducting amphibious OPS?

3.D.9.D. (U) Phase IV.

3.D.9.D.1. (U) PIR 1—Are terrorists using the maritime domain to conduct attacks against JTF or Redland forces?

3.D.9.D.2. (U) FFIR 1—Are international partners prepared to assume security missions?

3.D.9.E. (U) Phase V. None.

GENTEXT/ADMINISTRATION AND LOGISTICS//

4. (U) Administration and LOG.

4.A. (U) Concept of sustainment. CTF 223 coordinates underway replenishment for underway units. See Annex D. CTFs identify and report shortfalls of critical supply items and maintenance to JFMCC IAW OPTASK LOG.

4.A.1. (U) Phase I. The JFMCC establishes JFMCC ISBs and FLS. See Annex D. CTF 223 establishes RAS procedures and priorities.

4.A.2. (U) Phase II. All CTFs coordinate reloading standard missile and TLAM at PL ISBs with CTF 223. Priority: CTFs 225, 220, 227, and 224.

4.A.3. (U) Phase III. Sustainment of forces ashore at RAT. Priority: CTFs 224, 220, 225, 227, and 229.

4.A.4. (U) Phase IV. CTF 223 coordinates transportation of HA supplies to Greyland and engineering support at port YANKEE. Priority: CTFs 224, 225, and 220.

4.A.5. (U) Phase V. The JFMCC disestablishes ISBs and FLS. The JFMCC N4 determines LOG support requirements to enable Pinkland maritime OPS.

4.B. (U) Material and services.

4.B.1. (U) Maintenance. See Annex D.

4.B.2. (U) Supply. See Annex D.

4.B.3. (U) POL. See Annex D.

4.C. (U) Medical services. U.S. surgical teams are established in Pinkland hospitals to provide level III medical care. See Annex Q.

4.C.1. (U) Phase II: CASEVAC/MEDEVAC of JFMCC personnel during offensive OPS is the responsibility of the JFMCC and is coordinated through JFMCC N3.

4.C.2. (U) Phase III: CASEVAC/MEDEVAC of JFMCC personnel during amphibious OPS is the responsibility of CTF 224.

4.C.3. (U) Phase IV: CASEVAC/MEDEVAC of JFMCC forces operating in Redland and Greyland is the responsibility of CTF 224.

4.D. (U) Personnel. Submit PERSTAT daily to JFMCC N1 via RMG IAW OPTASK COMM. See ANNEX E.

4.E. (U) CA. Be prepared to provide HA for IDPS within Redland flowing into Greyland, Whiteland, Orangeland, or out to sea. See Annex G.

4.F. (U) PA guidance.

4.F.1. (U) Public release of information about this operation is not authorized until approved by CJTF (PA) and HHQ.

4.F.2. (U) PPAG provided in ref (e).

GENTEXT/COMMAND AND CONTROL//

5. C2.

5.A. (U) Command.

5.A.1. (U) Command relationships.

5.A.1.A. (U) COMEASTFLT is designated JTF BLUE SWORD JFMCC. JFMCC exercises OPCON over all U.S. maritime forces listed in this OPORD. JFMCC has coordinating authority with the maritime forces of Pinkland, Greenland, and BLUESKIES IOT achieve assigned tasks.

5.A.1.B. (U) Commander, CSG TWO is designated commander, CTF 220. CTF 220 exercises TACON over forces designated in the task organization of this order. CTF 220 is the supported commander for maritime superiority OPS with CTF 227 and CTF 229 supporting with intelligence, surveillance, reconnaissance, and targeting (ISR-T). During phase III operations, CTG 220.2 is in direct support (DS) of CTG 224.2 during the amphibious assault.

5.A.1.C. (U) Commander, Navy LOG Forces East is designated commander, CTF 223. CTF 223 exercises TACON over forces designated in the task organization of this order.

5.A.1.D. (U) Commander, NASSAU ESG, is designated commander, CTF 224. CTF 224 exercises TACON over forces designated in the task organization of this order. CTF 224 is the supported commander for amphibious OPS with CTF 225 in DS. During phase III operations, CTG 220.2 is in DS of CTG 224.2 during the amphibious assault.

5.A.1.E. (U) Commander, USS *Antietam*, is designated commander, CTF 225. CTF 225 exercises TACON over forces designated in the task organization of this order. CTF 224 is the supported commander for amphibious OPS with CTF 225 in DS.

5.A.1.F. (U) Commander, Reconnaissance Patrol Force, is designated commander, CTF 227. CTF 227 exercises TACON over forces designated in the task organization of this order. CTF 220 is the supported commander for maritime superiority OPS with CTF 227 supporting with ISR-T.

5.A.1.G. (U) Commander, Submarine Group East, is designated commander, CTF 229. CTF 229 exercises TACON over forces designated in the task organization of this order. CTF 220 is the supported commander for maritime superiority OPS with CTF 229 supporting with ISR-T.

5.A.2. (U) Succession of JFMCC command is CTF 220, CTF 224, and CTG 220.1.

5.A.3. (U) JFMCC BLUE SWORD EMB USS *Theodore Roosevelt*.

5.B. (U) Command, control, and communications.

5.B.1. (U) Connectivity.

5.B.1.A. (U) Primary, secondary, and tertiary circuits per Annex K.

5.B.1.B. (U) Phone. Secure telephone in port or underway via automated public branch exchange using phone numbers provided SEPCOR for JFMCC watch captain and JFMCC assistant watch captain.

5.B.1.C. (U) E-mail. Secure and UNCLASSIFIED e-mail addresses; see Annex K.

5.B.1.D. (U) Chat. Secure chat IP address and chat room provided SEPCOR.

5.B.1.E. (U) JFMCC website at WWW.BLUESWORD_JFMCC.NAVY.SMIL.MIL.

5.B.2. (U) Reporting instructions. CTFs provide daily situation reports to JFMCC IAW Annex R of this OPORD.

5.C. (U) JFACC is CJTF JICO. JFMCC is alternate CJTF JICO. Maritime COP is managed by JFMCC. Additional communications guidance delineated in JFMCC OPTASK COMMS.

AKNLDG/Y/INST: Contact JFMCC watch captain via e-mail/phone.//
Break transmission (BT)

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ANNEX 3 TO APPENDIX J

Annexes, Appendixes, and Tabs

J.3.1 GENERAL

As mentioned previously, attachments (annexes, appendixes, and tabs) are used to keep the base plan or order simple and clear. They include information, administrative support details, and instructions that expand upon the base plan or order. Attachments also permit separate distribution from the plan or order. These attachments are often developed from functional staff estimates. Appendixes and tabs may be developed to provide detail on specific areas as the mission dictates.

While the list of attachments and their formats set forth in CJCSM 3130.03, APEX Planning Formats and Guidance, is extensive, there may be a need for the development of other attachments to meet requirements specific to maritime OPS (e.g., fires, joint SUW, or assessment annex). The goal should be to make the order as clear and simple as possible. Thus, staffs must weigh the need for additional attachments against its return on investment in terms of plan or order clarity, simplicity, and brevity.

J.3.2 CONTENT

CJCSM 3130 (series) provides guidance regarding format and content of attachments. A few basic annex guides are also provided in the rest of this appendix IOT better enable the reader to understand the basic content and format of annexes.

Page numbering:

1. Annexes. Begin with the letter of the appropriate annex followed by the page number (e.g., A-1 for the first page in Annex A).

2. Appendixes. Begin with letter of the annex, followed by appendix number, and then page number (e.g., A 1-1 for the first page in Appendix 1 to Annex A).

Tabs. Begin with letter of the annex, followed by appendix number, tab letter, and then page number (e.g., A-1-A-1 for the first page in Tab A to Appendix 1 to Annex A).

J.3.3 SAMPLE FORMAT OF ANNEX A

Organization for combat is a commander's plan for grouping organic and attached units to effectively employ forces to support the scheme of maneuver. C2 is the means by which an operational commander synchronizes or integrates force activities IOT achieve unity of effort. Unity of effort is strengthened through adherence to basic C2 tenets such as clearly defined authorities, roles, and relationships.

The organization for OPS is determined after consideration of the unit's mission, missions assigned in subordinate units, operating environment, enemy strength in each subordinate unit area, and the amount of combat power, including maneuver and fire support units, available to the unit commander. These groupings may be shown, if simple, just before paragraph 1 of the basic OPORD or OPLAN. If these groupings are complex or change during execution of the operation, the task organization will be shown in a separate appendix. At a minimum, the task organization lists all major commands or task groupings directly subordinate to the commander issuing the basic OPORD or OPLAN. In addition, all organizations that directly support the operation are listed and designated as support although they are not under the command of the supported commander. Organizations to be established specifically to implement the basic OPORD or OPLAN should appear in the task organization.

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ANNEX A TO OPORD OR PLAN (Number) (Operation CODE WORD) TASK ORGANIZATION

(U) REFERENCES: List any previous orders (e.g., PLANORD, WARNORD, EXORD, etc.). List maps, charts, and other relevant documents (e.g., COMSEVENTHFLT No 7-94; NWP 5-01; commander's estimate, CTG 70.5, etc.). When using a map/chart, include the map/chart series number (and country or geographic areas if required), sheet number (and names if required), edition, and scale (if required).

ORGANIZATION

Issuing HQ. The first entry is the organization of the issuing HQ.

Components and Major Subordinates	ORGANIZATION: At a minimum, the task organization lists all major elements directly subordinate to the HQ originating the plan or order, including forces assigned to those HQ. For example: CTF 220 • CVN XX • CVW XX • CG XX CTF 221 • DDG XX • CG XX CTF 222	COMMANDER: Indicate names of commanders of the parent organization and principal units. Commander, CSG TWO, RADM Smith Commander, Destroyer Squadron 22, CAPT Jones Commander, Naval Special Warfare Group East, CAPT Black
Supporting Commands	Defense Intelligence Agency (DIA) United States Transportation Command (USTRANSCOM) Strategic Command Special Operations Command	Director, DIA Commander, USTRANSCOM, Scott AFB, IL Commander, United States Strategic Command Commander, USSOCOM, MacDill AFB, FL

ACKNOWLEDGE RECEIPT

Appendixes:
1—Time-phased Force and Deployment List
2—Shortfall Identification
3—Military Deterrent Options

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J.3.4 SAMPLE FORMAT OF ANNEX B (INTELLIGENCE)

The purpose of Annex B (Intelligence) is to provide detailed information/intelligence on the enemy and the OE and to provide guidance on intelligence and CI functions and employment. This sample format applies to operational-level staffs and below that may be assigned joint (JFMCC/JTF) or naval (NCC/CTF) command responsibilities necessitating OPLAN/OPORD development. This template does not supersede standing theater guidance on intelligence processes.

The N2/intelligence directorate of a joint staff (J2) prepares the intelligence annex based on the previously completed intelligence estimate. This annex provides both encyclopedic data and current information on the enemy, including order of battle, location, biographical information on enemy commanders, capabilities, and intentions. One of the most important aspects covered in the enemy's intentions is the identification and discussion of the most likely and most dangerous COAs.

The OE also includes information regarding climate, topography, and geography; physical infrastructure (roads, power grids, information grids); cultural considerations that affect the operation; political structure; and leadership. Much of this information may have been previously provided in intelligence estimates and in intelligence reports and summaries provided by national sources or HHQ. This information may be referenced in the intelligence annex to reduce the size of the basic OPORD or OPLAN.

The intelligence annex normally provides IPOE products to help further planning and execution. They include such products as the situation template and MCOO.

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ANNEX B TO OPORD OR OPLAN (Number) (Operation CODE WORD)
INTELLIGENCE

(U) REFERENCES:
(a) Maps and charts required for an understanding of this annex. Reference Annex M (Geospatial Information and Services).
(b) Documents providing intelligence required for planning. Including related annexes such as Annex H (Meteorological and Oceanographic Operations).
(c) Appropriate publications on Navy and joint intelligence doctrine.
(d) Appropriate standing operating procedures and other documents providing guidance on intelligence operations.
(e) The originator of the annex should ensure that the units receiving or executing the plan or order have the cited references.

GENTEXT/SITUATION//
1.A. (U) Characteristics of the OE. Summarize the relevant characteristics of the OE. The OE encompasses the physical areas and factors, information environment, and systems perspective within the context of an analysis of all aspects of the mission and commander's intent that could influence the commander's decisions or affect friendly and enemy COAs. Include sufficient analysis of the OE to provide context to the concept of intelligence operations, but do not repeat basic information included in the general situation paragraph of the base plan. Refer the reader to the complete intelligence estimate (Appendix 11 to Annex B) for additional analysis required to inform the development of supporting plans.

1.A.1. (U) Physical areas and factors. Summarize the hydrographic data and amphibious considerations needed to support amphibious and logistic over-the-shore OPS. Refer to Annex H and Annex M. Address topographic aspects, including trafficability, key terrain, obstacles, cover, concealment, and avenues of approach. Reference Annex M. Include, as appropriate, climate and weather aspects of the OE. Coordinate with the staff weather officer or oceanographer and refer to Annex H.

1.A.2. (U) Information environment.

1.A.2.A. (U) Provide a brief summary of analysis of the aggregate of individuals, and key individuals and groups having influence among the indigenous population as well as the source of their influence.

1.A.2.B. (U) Foreign language capability needed to exploit or collect intelligence information. Identify specific language capability needed in terms of language and/or dialects for open source, signal, or HUMINT collection or exploitation.

1.A.3. Systems perspective. Provide a brief description of the interrelated PMESII systems (to include socio-cultural factors) without regard to geographic boundaries but addressing a focus area specified by the commander.

1.B. (U) Enemy.

1.B.1. (U) Evaluate the OE from the enemy's perspective in terms of a prioritized set of enemy military COAs, to include any related diplomatic, informational, or economic options.

1.B.2. (U) Prioritize enemy COAs based on how well each is supported by the overall impact of the OE.

1.B.3. (U) Outline the enemy's capability to collect, communicate to intelligence centers, process, and disseminate through telecommunications or other methods.

1.B.4. (U) Determine enemy COAs.

1.B.4.1. (U) Discern the enemy's ability to integrate air, sea, and land capabilities in combined arms OPS.

1.B.4.2. (U) Consolidate a list of all enemy COAs appropriate to the current situation and accomplishment of likely OBJs as indicated by recent activities or events. Each identified COA should meet the following five criteria: suitability; feasibility; acceptability; uniqueness; and consistency with enemy doctrine or operational patterns.

1.B.4.3. (U) Analyze each COA to identify strengths and weaknesses, COGs, critical capabilities, critical requirements, critical vulnerabilities, DPs, compatibility with the OE, and how well each COA satisfies the above five criteria.

1.C. (U) Friendly.

1.C.1. (U) Friendly intelligence capability. Identify available friendly intelligence capabilities to include interagency, allied, and coalition capabilities when appropriate.

1.C.2. (U) Employment limitations. Provide a brief description of factors affecting the employment of friendly intelligence capabilities. These may include but are not limited to a lack of access due to legal restrictions, technical limitations, or basing rights considered in the N2 staff estimate.

1.D. (U) Legal considerations. Identify any legal considerations relevant to intelligence OPS.

GENTEXT/MISSION//
Restate the base plan mission statement. Based on the approved N2/J2 staff estimate and intelligence estimate (Appendix 11), confirm the purpose of intelligence OPS and describe how federated intelligence OPS will be integrated and synchronized to support the commander's decision points and operation assessments.

GENTEXT/EXECUTION
3.A. (U) Concept of intelligence OPS.

3A.1. (U) Summarize the means and agencies employed in managing tasks associated with, collection, processing and exploitation, analysis and production, dissemination, and integration. Specify procedures for evaluation and feedback to assess the conduct of intelligence OPS.

3.A.2. (U) As applicable, identify DOD intelligence community (IC) defense (e.g., Defense Intelligence Agency (DIA), National Geospatial-Intelligence Agency, National Reconnaissance Office, National Security Agency, Service intelligence organizations), non-DOD IC (e.g., CIA, FBI, DOS, Treasury), allied, and coalition intelligence support requirements. Ensure all intelligence support is properly aligned with each phase of the operation. Submit these support requirements to the higher-level command J2 for endorsement IAW standing guidance.

3.A.3. (U) As applicable, identify national intelligence augmentation requirements including national intelligence support teams, quick reaction teams, and agency liaison requirements. Submit these augmentation requirements to the CCDR J2 for endorsement IAW standing guidance.

3.B. (U) Tasks.

3.B.1. (U) PIRs. List the PIRs required to accomplish the mission by phase. The list of PIRs should include requirements for intelligence during peacetime to confirm threat-related planning assumptions and to inform plan revisions. The list of PIRs should also include requirements associated with the execution phases to assess progress toward the achievement of OBJs and inform critical decisions. Describe the coordination or collaboration process for dynamically updating and satisfying PIR. List the PIRs in the coordinating instructions of the base plan if Annex B is not published. Provide amplifying information regarding PIRs in Appendix 1 to Annex B.

3.B.2. (U) Identify intelligence tasks required to support the plan, and the office of primary responsibility within the DOD IC for accomplishing each task. Include the responsibilities of allied nations and coalition partners for support to multinational OPS.

3.B.2.A. (U) List the PIRs required to accomplish the mission by phase. PIRs should include requirements during peace, crisis, and war, both prior to and during execution. Describe the coordination or collaboration process for satisfying PIRs. List the PIRs in the coordinating instructions of the base plan if Annex B is not published.

3.B.2.B. (U) Based on the completed N2/J2 staff estimate and the PIRs required to accomplish the mission by phase develop the intelligence task list (ITL) by phase and include the ITL as Tab A to Appendix 1 or as an additional appendix. The ITL serves as a baseline for federated analysis and production, and will be used, as required, to coordinate the supporting capabilities of combat support agencies, Services, and other DOD and non-DOD intelligence organizations.

3.B.3. (U) Orders to subordinate units. List detailed instructions for each subordinate unit performing intelligence functions in separate subparagraphs.

3.B.4. (U) Requirements to higher and supporting organizations. List intelligence support requirements to units not organic or attached, as well as to supporting DOD IC organizations, non-DOD IC organizations, intelligence agencies, interagency intelligence organizations, allied or coalition forces, and organizations providing federated support.

3.C. (U) Collection. Provide guidance for the collection of information and material. Provide guidance for managing collection activities not covered by regulation or SOP. Include OPSEC planning guidance and guidance for the use of tactical military deceptions during the planning and conduct of intelligence collection activities.

3.C.1. (U) SIGINT. Provide guidance for assignment and coordination of communications intelligence (COMINT) and ELINT resources. Include guidance on the interaction of SIGINT activities with imagery intelligence, HUMINT, and measurement and signature intelligence (MASINT) and EW support activities.

3.C.2. (U) Geospatial intelligence (GEOINT). Provide guidance for establishing and conducting imagery and geospatial activities. Include guidance on synchronizing GEOINT collection activities with SIGINT, HUMINT, and MASINT activities.

3.C.3. (U) HUMINT. Provide information and guidance pertaining to the organization, direction, and coordination of HUMINT collection OPS and support activities. Include guidance, if appropriate, on interaction with GEOINT, SIGINT, and MASINT activities.

3.C.4. (U) MASINT. Provide guidance on obtaining intelligence by quantitative and qualitative analysis of data derived from specific technical collection sensors other than those normally associated with SIGINT, GEOINT, and HUMINT. Include guidance on the interaction of GEOINT, HUMINT, SIGINT, and technical intelligence.

3.C.5. (U) CI. Provide guidance pertaining to the assignment and coordination of OPS using CI agents and sources ISO force protection (FP) efforts.

3.C.6. (U) Other collection activities. Provide guidance for collection by other specialized means such as visual, amphibious, reconnaissance, and medical collection activities to support plan (SUPPLAN) requirements. Include guidance on how these activities are expected to interact with collection efforts discussed elsewhere in this plan.

3.D. (U) Processing and evaluation. Provide guidance for converting information into usable form, including required provisions for document translation; imagery, signals, and technical sensor processing and interpretation; and other pertinent processing activity.

3.E. (U) Analysis and production. Provide guidance on analyzing and reporting collected intelligence information by all collection sources employed ISO the plan. Include guidance on multidiscipline reports that fuse information from multiple sources and on development and integration of independent and alternative assessments by the red team. Reference appropriate regulations, directives, and SOPs specifying U.S.-only and multinational reporting procedures. Identify the production effort, including any intelligence and CI products, required to support the plan.

3.F. (U) Dissemination and integration. Provide necessary guidance for conveying intelligence to appropriate levels, including the forces of allied nations. Include criteria to satisfy expanded requirements for vertical and lateral dissemination of finished intelligence and spot reports. Identify alternate means of disseminating intelligence to combat units and HQ during crises and combat OPS. The following items may be covered in this subparagraph:

3.F.1. (U) Intelligence reports required from units with specific guidance regarding periodicity and distribution.

3.F.2. (U) Formats for intelligence reports including additional appendixes if required.

3.F.3. (U) Distribution of intelligence studies.

3.F.4. (U) Requirements for releasability or disclosure to allied nations.

3.F.5. (U) Requirements for secondary imagery dissemination.

3.G. (U) Coordinating instructions. Detail instructions applicable to two or more supporting intelligence organizations. Identify which, if any, collaborative tools will be used. The following items may be covered in this subparagraph:

3.G.1. (U) Periodic or special conferences for intelligence officers.

3.G.2. (U) Intelligence liaison with adjacent commanders, crisis intelligence federation partners, foreign government agencies or military forces, and host countries.

3.G.3. (U) Coordination of protected frequencies to be nominated for the Joint Restricted Frequency List.

3.G.4. (U) R&S conferences.

3.G.5. (U) Guidance, apportionment, and targeting conferences.

3.G.6. (U) Release or disclosure of intelligence information to coalition partners.

3.G.7. (U) Sanitization of intelligence information.

GENTEXT/ADMINISTRATION AND LOGISTICS

4.A. (U) Shortfalls and limiting factors. Based on the N2/J2 estimate, list shortfalls and limiting factors, including foreign language and regional expertise capabilities, significantly affecting intelligence support. Include an impact assessment for each shortfall. Identify resource challenges and specify key tasks that may not be accomplished.

4.B. (U) Mitigation. Identify or describe the mitigation strategy, which may include contracted capabilities, in response to shortfalls or limiting factors. Cross-reference to Tab A to Appendix 3 to Annex W.

4.C. (U) Miscellaneous. Identify OPSEC, evasion and escape, deception, disclosure of intelligence, releasability to coalition forces, PA, use of specialized intelligence personnel, MISO, exploitation of captured foreign materiel and documents, and composition of the N2/J2 staff.

4.D. (U) LOG. Identify logistic requirements specific to intelligence support of the plan.

4.E. (U) Reporting. Identify any unique reporting requirements ISO the plan.

GENTEXT/COMMAND AND CONTROL

5.A. (U) Command Relationships. Describe any unique command relationships for intelligence OPS.

5.B.(U) Communications. Summarize the U.S. and non-U.S. communications systems and procedures to be used to carry out the intelligence function or reference the appropriate paragraphs of Annex K. Include comments on interoperability of these systems.

ACKNOWLEDGE RECEIPT

Appendixes
1—Priority Intelligence Requirements
2—Signals Intelligence
3—Counterintelligence
4—Targeting
5—Human Intelligence
6—Intelligence Support to Information Operations
7—Geospatial Intelligence
8—Measurement and Signature Intelligence
9—Captured Enemy Equipment
10—National Intelligence Support Team
11—Intelligence Estimate

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(Rank Name) (Position)

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J.3.5 SAMPLE FORMAT OF ANNEX C (OPERATIONS)

Annex C provides substantive guidance for planning the conduct of OPS. Plans for the employment of non-U.S. forces should include proposed command arrangements and, as necessary, consideration of requirements for furnishing essential combat and logistic support.

A complex OPS concept of support may require a schematic to show the OPS OBJs and task relationships. It includes a discussion of the overall OPS concept of support with specific details in element subparagraphs and attachments. It refers to the execution matrix to clarify timing relationships among various OPS tasks. This annex also contains the information needed to synchronize timing relationships of each element related to OPS. It includes OPS-related constraints, if appropriate.

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ANNEX C TO OPERATION ORDER OR PLAN (Number) (Operation CODE WORD)
OPERATIONS

(U) REFERENCES: List other plans, standing operating procedures, and doctrinal guidance to be followed in the conduct of OPS.

GENTEXT/SITUATION//

1. (U) General. Summarize the physical, economic, political, medical, social, religious, and psychological aspects and conditions of the OA as they may influence the operation. Do not repeat information included in the general situation discussed in the basic order or plan. If there is no new information from what is contained in the base order then indicate this by stating “See base order.”

1.A. (U) AO. Define the AO encompassed by the basic order or plan to include land, sea, and air space. The annex should also define any areas where R&S OPS are authorized.

1.B. (U) Enemy. First, list known and templated locations and activities of enemy units for two echelons down. Second, list enemy maneuver and other capabilities that will impact friendly OPS. Third, state the enemy most likely and most dangerous COAs and employment of enemy assets. Fourth, include an assessment of terrorist or criminal activities directed against USG interests in the AO. Refer to Annex B and other sources as required.

1.C. (U) Friendly. This subparagraph uses the same format as the base order and can be shortened by using the phrase “See base order” if there is no change.

1.D. (U) Interagency, international, and NGOs. Identify and describe other organizations in the AO that may impact the conduct of the unit’s OPS or require support not identified in the base order. Also identify NGOs in the AO that want nothing to do with the U.S. military not identified in the base order. Refer to Annex V (Interagency Coordination) as required.

1.E. (U) Civil considerations. List all critical civil considerations that impact on the unit's OPS, such as cultural or religious sensitivities or resolving injury or damage claims, not established in base order. Refer to Annex B and Annex G as required.

1.F. (U) Attachments and detachments. List units attached or detached only as necessary to clarify task organization. Do not repeat information already listed under task organization in the base order or in Annex A. Try to put all information in the task organization annex and state "See Annex A."

GENTEXT/MISSION//

2. (U) Enter the unit's restated mission only if this annex is distributed separately from the base order. Otherwise, state "See base order."

GENTEXT/EXECUTION//

3. (U) CONOPS. Normally, the CONOPS is included in the basic OPORD or OPLAN. However, when it is lengthy and detailed, place it here. The format and content are similar to the CONOPS in the basic OPORD or OPLAN.

3.A. (U) Conduct of OPS. Provide any guidance required for the conduct of specific OPS. If OPS are overly complex, they may require a separate appendix for greater detail. If an annex or appendix is not listed as part of the APEX construct, commanders may use ones that are not designated.

3.A.1. (U) Aviation OPS. Provide the concept of aviation OPS in the plan.

3.A.2. (U) Maritime pre-positioning force (MPF) OPS. Provide the concept of MPF OPS in the plan.

3.A.3. (U) IO. Refer to Appendix 3 (Information Operations).

3.A.4. (U) Counterinsurgency. May be pertinent in FHA. If applicable, refer to pertinent country internal defense plans.

3.A.5. (U) Chemical, biological, radiological, and nuclear defense OPS. If applicable, refer to Appendix 2 (Combating Weapons of Mass Destruction).

3.A.6. (U) Special OPS. If applicable, refer to Appendix 4 of Annex C.

3.A.7. (U) Tactical recovery of aircraft and personnel. Refer to Appendix 5.

3.A.8. (U) ROE. Refer to Appendix 8.

3.A.9. (U) Reconnaissance. Refer to Appendix 9.

3.A.10. (U) Air base operability. If applicable, refer to Appendix 10.

3.A.11. (U) FFIRs, refer to Appendix 11.

3.A.12. (U) NEOs. If applicable, refer to Appendix 12.

3.A.13. (U) Escape and evasion operations. Refer to Appendix 5, Tab C (Survival, Evasion, Resistance and Escape).

3.A.14. (U) Explosive ordnance disposal. Refer to Appendix 13 (Explosive Ordnance Disposal).

3.A.15. (U) Amphibious OPS. If applicable, refer to Appendix 14.

3.A.16. (U) Force protection. Refer to Appendix 15 (Force Protection).

3.A.17. (U) Communications strategy. Refer to Annex Y (Strategic Communication).

3.A.18. (U) Fire support. Provide the concept of fires OPS in the plan.

3.B. (U) Assessment. Describe the priorities for assessment and identify the MOPs and MOEs used to assess end state conditions and OBJs. Designate an annex or appendix as required.

3.C. (U) Operational constraints. List any constraints to the conduct of combat OPS not enumerated elsewhere, such as the impact of deployment or employment of forces and materiel on airfield ramp space, including possible HNS. Estimate the impact of these operational constraints and indicate how the CONOPS and tasks to subordinate commanders would be modified if these constraints were removed. State the effect of incremental removal of constraints.

3.D. (U) Tasks to subordinate units. List tasks assigned to specific subordinate units not contained in the base order. Each task should include who (the subordinate unit assigned the task), what (the task itself), when, where, and why (purpose). Use a separate subparagraph for each unit. List units in sequence of task organization. Place tasks that affect two or more units in coordinating instructions.

3.E. (U) Coordinating instructions. List only instructions applicable to two or more subordinate units not covered in the base plan or order.

GENTEXT/ADMIN AND LOG//
Summarize key administrative and LOG issues affecting the mission by phase. Detail should be contained in Annex D.

4. Sustainment. Describe priorities of sustainment by unit or area. Include instructions for deployment or redeployment. Identify priorities of sustainment for OPS key tasks and specify additional instructions as required. Refer to Annex D (Logistics) as required.

GENTEXT/COMMAND AND CONTROL//
List information in this paragraph and its subparagraphs only if the annex distributed separately from the base order. Otherwise, state “Same as base order.”

5. (U) C2

5.A. (U) Command.

5.A.1. (U) Command relationships. Refer to OPORD Annex J (Command Relationships) as required.

5.A.2. (U) Location of commander. State the location(s) of the commander and second-in-command during the operation, CPs, alternate CPs, flagships, and alternate flagships along with their times of activation and deactivation.

5.A.3 (U) Succession of command. State the succession of command if not covered in the unit’s SOPs.

5.A.4. (U) Liaison requirements. State the liaison requirements not covered in the base order.

5.B. (U) Command, control, and communications. State information about pertinent command, control, and communications nets; operating procedures; recognition and identification procedures; electronic emission constraints; and so on. Refer to OPORD Annex K (Communications Systems Support) as required.

5.C. (U) Reports. List reports not covered in SOPs. Refer to OPORD Annex R (Reports) as required.

Appendixes
1—Nuclear Operations (Not used in Navy plans or orders. Included to conform to CJCSM 3122.03C, JOINT OPERATIONS PLANNING AND EXECUTION.)
2—Combating Weapons of Mass Destruction
3—Information Operations
4—Special Operations
5—Personnel Recovery Operations
6—Cyber
7—{Removed}
8—Rules of Engagement 9—Reconnaissance
10—Air Base Operability
11—Friendly Force Information Requirements
12—Noncombatant Evacuation Operations
13—Explosive Ordnance Disposal
14—Amphibious Operations
15—Force Protection
16—Critical Infrastructure Protection

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J.3.6 SAMPLE FORMAT OF ANNEX D (LOGISTICS)

Annex D (Logistics) provides direction and guidance to the subordinate commanders and staffs on the provision of LOG and sustainment ISO OPS described in the OPORD, OPLAN, or CONPLAN. The theory and philosophy of LOG as practiced by the Navy is provided in NWP 4-0M, Naval Logistics. Enclosure U of CJCSM 3122.01A, Joint Operation Planning and Execution System, Volume I (Planning Policies and Procedures), lists pertinent LOG references.

The C2 of LOG and sustainment organizations, to include command relationships and C2 support requirements, should be addressed in Annex D. It provides a general discussion of how the operation will be supported and is fully integrated with other critical concepts such as maneuver, fires, and FP. As with all orders writing, it requires as much depth as is necessary to ensure understanding of envisioned LOG OPS by subordinate commanders and staffs two levels down. The N4 is normally responsible for the preparation of Annex D, however, there should be a collaborative planning effort, leveraging logisticians from other Service components, subordinate commands, and logisticians from Service and even defense agencies, under the guidance of the fleet N4. Phasing and significant anticipated changes in mission, tasks to subordinates, and priorities of sustainment should be reflected in the concept of support. Detailed or specialized information should be provided in other subparagraphs or in appendixes of Annex D.

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ANNEX D TO OPERATION ORDER OR PLAN (Number) (Operation CODE WORD)
LOGISTICS

(U) REFERENCES: List other plans, standing operating procedures, and doctrinal guidance to be followed in the conduct of OPS.

GENTEXT/SITUATION//

1. (U) General. Summarize the physical, economic, political, medical, social, religious, and psychological aspects and conditions of the OA as they may influence LOG. Do not repeat information included in the general situation discussed in the basic order or plan. If there is no new information from what is contained in the base order, indicate this by stating “See base order.”

1.A. (U) AO. Define the AO encompassed by the basic order or plan to include land, sea, and air space affected by LOG.

1.B. (U) Enemy. Refer to Annex B. Provide available information on enemy actions or intent to conduct actions to disrupt or degrade envisioned friendly LOG and CSS OPS. Include information on enemy capabilities or assets that can augment friendly LOG and CSS OPS.

1.C. (U) Friendly. List supporting LOG or CSS organizations not subordinate to the force and the specific missions and tasks assigned to each.

1.D. (U) Infrastructure. Refer to Annex B. Provide information on existing infrastructure, such as ports, factories, fuel and water sources, and lines of communications that can be used to support friendly LOG and CSS OPS.

1.E. (U) Attachments and detachments. Refer to Annex A. List LOG and CSS units from other Services/nations attached to the force. List all Marine Corps LOG and CSS units detached to support other friendly forces.

1.F. (U) Assumptions. State realistic assumptions that were used in the development of the OPORD that have not been proven as facts or disproven. Consider the effect of COPS on LOG capabilities.

1.G. (U) Resource availability. Identify significant competing demands for LOG resources where expected requirements may exceed resources. Include recommended solutions within resource levels available for planning, if any, and reasonably assured HNS.

1.H. (U) Planning factors. Refer to and use approved planning factors and formulas, except when experience or local conditions dictate otherwise. When deviating from planning factors, identify the factors and the reason.

GENTEXT/MISSION//
2. Enter the unit's restated mission only if this annex is distributed separately from the base order. Otherwise, state "See base order."

GENTEXT/EXECUTION//
3. (U) Execution.

3.A. (U) Concept of LOG and CSS. State the concept for LOG and CSS OPS necessary to implement the order or plan. Describe how sustainment supports the commander's intent and CONOPS. Establish the priorities of sustainment support to units for each phase of the operation. Refer to Annex C as required. Describe how the LOG and CSS assets will be organized and positioned to execute the mission. The concept may include planned employment of other Service and nation LOG and CSS forces, HNS LOG capabilities, or operation of the lines of communications.

3.B. (U) Tasks to units. List sustainment tasks assigned to specific subordinate units not contained in the base order.

3.C. (U) Additional responsibilities.

3.C.1. (U) Assign LOG and CSS responsibilities to subordinate LOG organizations.

3.C.2. (U) Identify and assign responsibility for LOG and CSS required from other commands, Services, or nations.

3.C.3. (U) Identify and assign responsibility for LOG and CSS required for forces assigned or attached from other commands, Services, or nations.

3.C.4. (U) Identify and assign responsibility for LOG and CSS required for NAVFOR assigned or attached to other commands, Services, or nations.

3.C.5. (U) Assign responsibilities to support joint boards and committees, such as transportation and procurement, and other Services or nations providing services.

GENTEXT/ADMINISTRATIONS AND LOGISTICS//
4. (U) LOG and CSS.

4.A. (U) Supply. Summarize the following, in coordination with supporting commanders and Service component commanders, if different from standard planning factors. Place detailed discussions in the appendixes and listings of supply depots, terminals, and lines of communications in tabs or the appropriate appendixes.

4.A.1. (U) Distribution and allocation.

4.A.1.A. (U) Purpose, location, and projected displacement of main and alternate supply depots or points and supporting terminals and ports to be used or considered.

4.A.1.B. (U) Pre-positioned LOG resource allocation.

4.A.1.C. (U) Existing terminals and lines of communications and the known or estimated throughput capability. Indicate the time-phased expansion necessary to support the plan.

4.A.2. (U) Level of supply.

4.A.2.A. (U) Indicate the time-phased operating and safety levels required to support the plan.

4.A.2.B. (U) Indicate the pre-positioned war reserve materiel requirements to support the time-phased deployments pending resupply.

4.A.2.C. (U) Specify significant special arrangements required for materiel support beyond normal supply procedures.

4.A.2.D. (U) Indicate anticipated shortfalls.

4.A.2.E. (U) Indicate common-user LOG supply support responsibilities and arrangements.

4.A.3. (U) Salvage. Provide instructions for and identify the LOG impact of the collection, classification, and disposition of salvage.

4.A.3.A. (U) Captured enemy materiel. Provide instructions for the collection, classification, and disposition of enemy materiel. See Annex B for further guidance.

4.A.4. (U) Local acquisition of supplies and services. See JP 4-01, Joint Doctrine for the Defense Transportation System, and the current version of DOD instruction 1100.22, Policy and Procedures for Determining Workforce Mix.

4.A.4.A. (U) Identify acquisition of goods and services in the following categories:

4.A.4.A.1. (U) The general categories of materiel and services that are available and contemplated as a supplement to regular sources.

4.A.4.A.2. (U) Those that may be used as emergency acquisition sources.

4.A.4.B. (U) Make a statement concerning the dependability of the local acquisition or labor source in each of the aforementioned categories and the joint or Service element that will obtain or manage these resources.

4.A.4.C. (U) State that all essential contractor services, to include new and existing contracts, have been reviewed to determine which services will be essential to OPLAN execution. Make a statement concerning the existence of contingency plans to ensure the continuation of these essential services.

4.A.5. (U) POL. Refer to Appendix 1.

4.B. (U) External support. Provide the required planning information, including type and quantity of support and instructions where inter-Service and cross-Service arrangements for common supply and service support are appropriate.

4.B.1. (U) Summarize major support arrangements that are presently in effect or that will be executed ISO the plan.

4.B.2. (U) Include significant inter-Service and cross-Service support arrangements. Refer to appropriate annexes or appendixes.

4.B.3. (U) Include foreign and HNS.

4.C. (U) Maintenance.

4.C.1. (U) General. Include sufficient detail to determine the requirements for maintenance facilities needed to support the plan. Indicate the level of maintenance to be performed and where it is to occur, including HN or contractor facilities, if applicable. Refer to Appendix 12.

4.C.2. (U) Specific guidance. Provide maintenance information in subsequent subparagraphs for watercraft, aircraft, and ground. Include proper procedure to request, priority of maintenance, location of facilities and collection points, repair time limits at each level of maintenance, and evacuation procedures.

4.D. (U) Transportation.

4.D.1. (U) General. Refer to Appendix 5. Provide general planning or execution guidance to subordinate and supporting organizations to facilitate transportation of the force and its sustainment. This can include movement and use priorities.

4.D.2. (U) Mobility support force and movement feasibility analysis. Provide an estimate of the mobility support and movement feasibility of the plan. Include in the analysis any appropriate remarks affecting mobility and transportation tasks. Consider the availability of adequate lift resources for movements of personnel and equipment, airfield reception capabilities, seaport and aerial port terminal capabilities, and port throughput capabilities. Also, consider any features that will adversely affect movement OPS, such as the effect of deployment or employment of forces and materiel on airfield ramp space (to include possible HNS).

4.D.2.A. (U) Identify the proper procedures to request ground and sea/water/river transportation.

4.D.2.B. (U) Container management. Describe the container management plan.

4.E. (U) General engineering SUPPLAN. Indicate the general engineering support activities applicable to the basic OPORD or OPLAN and the policies for providing these services. Refer to Appendix 6 as required.

4.F. (U) Health services. Refer to Annex Q.

4.G. (U) Mortuary affairs. Refer to Appendix 3 of Annex D (Mortuary Affairs) or, if not used, indicate the mortuary affairs activities applicable to the OPORD or OPLAN and policy for providing these affairs.

4.H. (U) Ammunition. Refer to Appendix 7 of Annex D (Nonnuclear Ammunition) or if not used, discuss any critical ammunition issues that may affect the ability of the force to accomplish the mission.

4.I. (U) Aviation LOG support. Critical aviation LOG and CSS issues may be discussed if they affect the ability of the force to accomplish the mission.

4.J. (U) Operational security planning guidance for LOG. Refer to Tab C (Operations Security) to Appendix 3 to Annex C. Provide comprehensive OPSEC planning guidance for planning, preparing, and executing LOG and CSS activities. At a minimum, address base, facility, installation, LOG stocks, physical, and line of COMSEC. Provide guidance to ensure that LOG and CSS activities promote essential secrecy for operational intentions, capabilities that will be committed to specific missions, and current preparatory operational activities.

4.K. (U) Administration. Include general administrative guidance to support LOG and CSS OPS for the basic OPORD or OPLAN. If reports are required, specify formats for preparation, time, methods, and classification of submission.

GENTEXT/COMMAND AND CONTROL//
List information in this paragraph and its subparagraphs only if the annex is distributed separately from the base order. Otherwise, state "Same as base order."

5. (U) C2.

5.A. (U) Command.

5.A.1. (U) Command relationships. Refer to OPORD Annex J (Command Relationships) as required for command relationships external to LOG units. Provide support relationships.

5.A.2. (U) Location of commander. State the location(s) State the location of LOG/sustainment area leaders.

5.A.3. (U) Succession of command. State the succession of command if not covered in the unit's SOPs.

5.A.4. (U) Liaison requirements. State the sustainment liaison requirements not covered in the base order.

5.B. (U) Command, control, and communications. Provide a general statement of the scope and type of communications required. Address any sustainment-specific communications requirements. Refer to OPORD Annex K (Communications Systems Support) as required.

5.C. (U) Reports. List sustainment-specific reports not covered in SOPs. Refer to Annex R (Reports) as required.

Appendixes
1—Petroleum, Oils, and Lubricants Supply
2—Joint Subsistence, Food Service Support and Water Management
3—Mortuary Affairs
4—Sustainability Analysis
5—Mobility and Transportation
6—Engineering Support Plan
7—Nonnuclear Ammunition
8—Logistics Automation

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J.3.7 SAMPLE FORMAT OF ANNEX J (COMMAND RELATIONSHIPS)

Command relationships are the interrelated responsibilities between commanders and the authority of commanders in the chain of command. Unity of effort is, in large part, achieved through the application of a flexible range of command relationships. The JFC exercises command during joint OPS according to the provisions of JP 1, Doctrine for the Armed Forces of the United States; JP 3-0; Naval Doctrine Publication 1, Naval Warfare; and NWP 3-32. These publications describe possible command relationships between the JFC, the NCC, the JFMCC, and subordinate commanders of assigned or attached forces. This annex discusses:

1. Requirements to coordinate support between forces in the same or adjacent areas according to JP 1 and the common HHQ OPORD or OPLAN.

2. Planning for succession of command and change of command location (alternate C2 procedures). Refer to paragraph 5 (C2) of the OPORD or OPLAN or Annex K (Combat Information Systems).

3. DOD Directive 3025.14, Protection and Evacuation of United States Citizens and Designated Aliens in Danger Areas Abroad, as amended by changes 1 and 2, delineates the responsibilities for protection of U.S. citizens abroad. ISO this directive, give special attention to cooperation and coordination between U.S. diplomatic and military activities during periods of tension and hostilities.

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ANNEX J TO OPERATION ORDER OR PLAN (Number) (Operation CODE WORD)
COMMAND RELATIONSHIPS

(U) REFERENCES: List documents that provide necessary guidance on the command relationships of forces concerned.

GENTEXT/SITUATION//
1. (U) Purpose. To establish the relationships between combatant commands; international commands and organizations; Commander, U.S. forces country, Service and functional component commanders; major subordinate commanders; coordinating authorities; other subordinate military activities; U.S. diplomatic missions; and Government departments or agencies that support the OPS, forces, and agencies of other nations.

1.A. (U) AO. Define the AO encompassed by the basic order or plan to include land, sea, and air space. Discuss the establishment of combat zones and communications zones. Indicate whether other OAs such as JOAs, AOs, AOAs, joint rear areas, and joint special OPS areas will be designated for commanders. A diagram or map may be used.

1.B. (U) Scope. Specify the scope and applicability of the command relationships established in this annex for specific military OPS or functions within an assigned geographic area; or for specific military OPS or functions not limited to a geographic area and the times or circumstances when the relationships become effective.

GENTEXT/MISSION//
2. Enter the unit's restated mission only if this annex is distributed separately from the base order. Otherwise, state "See base order."

GENTEXT/EXECUTION//

3. (U) Execution

3.A. (U) Identify command relationships required for the operation. Identify command lines for the following, as required:

3.A.1. (U) Service and functional components. Indicate the command lines to Service and functional components of the force and to subordinate elements, as appropriate.

3.A.2. (U) Other subordinate commands. Indicate the established command lines to subordinate commanders for conducting this operation and the conditions under which forces will be transferred to their OPCON.

3.A.3. (U) Multinational/allied/coalition commands and organizations. Identify command arrangements or relations with multinational commands and organizations, foreign military commands, or guerrilla organizations. Indicate the conditions under which such relations would become effective.

3.A.4. (U) Augmentation forces. Indicate the purpose, time, and approximate duration of the attachment and the degree of authority over and responsibility for the augmentation forces.

3.A.5. (U) Alternate procedures. Discuss procedures for succession of command and change of command location (alternate C2 procedures).

3.A.6. (U) Specialized assignments. Indicate the assignment of specialized coordination and control functions and responsibilities (e.g., airspace control authority, and area AD commander). Discuss the authority, duties, and responsibilities of these assignments.

3.B. (U) Support and coordination relationships.

3.B.1. (U) Supporting military forces. Indicate established relationships with military organizations operating ISO the originating command.

3.B.2. (U) Coordinating authorities. As necessary, assign a commander or another person the responsibility for coordinating specific functions or activities.

3.B.3. (U) Supporting agencies. Indicate the relationships between the elements of the force and any supporting agencies, such as DIA and Defense Logistics Agency (refer to other annexes or appendixes, as appropriate).

3.B.4. (U) Inter-Service support arrangements. Refer to Annex D (Logistics) for inter-Service logistic support.

3.B.5. (U) Relationships with international and foreign commands and organizations. Indicate established command arrangements or relations with international commands and organizations, foreign military commands, or guerrilla organizations. Also indicate the conditions under which such relations would become effective.

GENTEXT/ADMIN AND LOG//

4. Identify priorities of sustainment for command relationships tasks and specify additional instructions as required. Refer to Annex D (Logistics).

GENTEXT/COMMAND AND CONTROL//
5. (U) C2.

5.A. (U) Command. Identify any unique command and control capability required to support the command relationships identified above.

5.A.1. (U) Planning relationships. Specify established relationships between military commands for developing supporting plans. Include any requirements for coordination with other-nation commands and nonmilitary agencies.

5.A.2. (U) Location of commander. State the location(s) of leaders.

5.A.3. (U) Succession of command. State the succession of command if not covered.

5.A.4. (U) Liaison requirements. State the liaison requirements not covered to support the command relationships.

5.B. (U) Command, control, and communications. Provide a general statement of the scope and type of communications required. Address any specific communications requirements. Refer to OPORD Annex K (Communications Systems Support) as required.

Appendixes
1—Command Relationships Diagram

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J.3.8 SAMPLE FORMAT OF ANNEX V (INTERAGENCY COORDINATION)

Annex V (Interagency Coordination) provides military and interagency personnel with detailed information (mission, scheme, and tasks) to direct the necessary coordination and interaction between NAVFOR and interagency organizations. It describes how the commander intends to cooperate, provide support, and receive support from interagency organizations throughout the operation. For an OPLAN/CONPLAN, it is based on planning factors and estimates available at the time of preparation and is subject to modification based on the actual conditions or situation existing at the time of execution. This annex follows the five-paragraph order format; however, some subparagraphs are modified to accommodate communication with the interagency.

The Navy operations directorate (N3), ICW the N9, develops Annex V (Interagency Coordination). Interagency organizations of the USG include the following:

1. Central Intelligence Agency
2. Department of Commerce
3. DOD
4. Department of Energy
5. Department of Homeland Security
6. Department of Justice
7. DOS
8. Department of the Treasury
9. Department of Transportation
10. Environmental Protection Agency
11. National Security Council
12. Peace Corps
13. United States Agency for International Development/Office of Foreign Disaster Assistance
14. United States Department of Agriculture.

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ANNEX V TO OPERATION ORDER OR PLAN (Number) (Operation CODE WORD)
INTERAGENCY COORDINATION

(U) REFERENCES: List documents that provide necessary guidance to this annex.

GENTEXT/SITUATION//

1. (U) General. Include information affecting interagency coordination that paragraph 1 of the OPLAN or OPORD does not cover or that needs expansion.

1.A. (U) AO. Describe the area of interest as it relates to interagency coordination. Refer to Annex B as required.

1.B. (U) Politico-military situation. Summarize the politico-military situation that would establish the preconditions under which this plan might be executed. At a minimum, identify the U.S. national security objectives and interests served by this plan and the interagency capabilities needed to return to normalcy or to establish a new normalcy.

1.C. (U) Policy coordination. Identify what coordination and support requirements might be necessary to initiate interagency planning.

1.D. (U) Planning and execution coordination. Describe the proposed concept for interagency coordination during both planning and execution to ensure unity of effort and appropriate deconfliction. Outline how the process supports the operation.

1.E. (U) Assumptions. List key assumptions that might impact or influence interagency planning.

1.F. (U) Legal considerations. List any legal considerations that may affect interagency participation.
GENTEXT/MISSION//

2. Enter the unit's restated mission only if this annex is distributed separately from the base order. Otherwise, state "See base order."

GENTEXT/EXECUTION//

3. Execution.

3.A. (U) CONOPS. Outline the primary OBJs and desired effects of each phase. Describe the concept for interagency coordination and how it supports the concept of military OPS. Outline the commander's interagency coordination for each phase and what resources, capabilities, and liaison from other USG agencies can support each of these OBJs. Comment on the desirability and feasibility of international organization/NGO participation in the operation. Identify the resources or capabilities from the interagency that will support each of these OBJs and comment on the desirability and level of nongovernmental participation in the operation.

3.B. (U) Commander's intent. Describe the commander's intent and optimal level of involvement by other USG agencies for each phase. Be sure to identify the desired end state for each phase and list the anticipated desired actions of the major USG agencies to support these end states.

3.C. (U) Major areas of USG interagency response. Describe the areas of responsibility from USG agencies by major areas of response: humanitarian, economic, political or diplomatic, and others as required. The operational variables are another method to organize major areas of response: political, military, economic, social, information, infrastructure, physical environment, and time (abbreviated as PMESII-PT).

3.C.1. (U) Humanitarian. Define, in broad terms, the desired actions and responsibilities for USG agencies in rebuilding and shaping the humanitarian structure and health of the affected nation. Coordinate these requested actions with the commander's phase development.

3.C.2. (U) Economic. Define in broad terms the desired actions and responsibilities for USG agencies in rebuilding and shaping the economic structure and health of the affected nation. Coordinate these requested actions with the supported commander's phase.

3.C.3. (U) Political/diplomatic. Define, in broad terms, the desired actions and responsibilities for USG agencies in rebuilding and shaping the political and diplomatic structure and health of the affected nation. Coordinate these requested actions with the supported commander's phase development.

3.C.4. (U) Others. As required.

3.D. (U) Level of integration. Describe the level of integration envisioned between the military, USG agencies, and international organizations/NGOs as OPS transition between phases.

3.E. (U) Tasks and milestones. Identify the foreseen tasks and required milestones necessary before handing off responsibilities to civilian authorities.

3.F. (U) Coordinating instructions. List only instructions applicable to two or more subordinate units not covered in the base plan or order. Identify and list general instructions applicable to other USG agencies such as agreements with the host country and allied forces.

GENTEXT/ADMIN AND LOG//

4. (U) General. USG agencies, IOs, and NGOs often have limited resources to support themselves in foreign areas. Provide a concept for furnishing administrative and LOG support to USG agencies and NGOs participating in the operation. Some support requirements will be identified during interagency planning; however, as a minimum include the following:

4.A. (U) LOG. Use subparagraphs to identify availability, priorities, and specific instructions for maintenance, transportation, supply, field services, distribution, contracting, and general engineering support. Refer to Annex D (Logistics) and Annex P (Host-nation Support) as required.

4.B. (U) Personnel. Use subparagraphs to identify availability, priorities, and specific instructions for human resources support, financial management, legal support, and religious support. Refer to Annex D (Logistics) as required.

4.C. (U) Health System Support. Identify availability, priorities, and instructions for medical care. Refer to Annex Q as required.

GENTEXT/COMMAND AND CONTROL//

5. (U) C2.

5.A. (U) Command. Identify any unique command relationships established for the purpose of interagency coordination. Identify any interagency coordination forms or bodies such as an interagency coordination WG.

5.A.1. (U) Location of commander. Identify current or future locations of the interagency coordination leadership.

5.A.2. (U) Succession of command. Describe the proposed organizational relationship and chain of responsibility between the commander and other USG agencies and international organizations.

5.A.2.A. (U) USG. Identify the chain of authority for USG agencies.

5.A.2.B. (U) International organizations. Identify the expected chain of authority for international organizations should they become involved.

5.A.3. (U) Liaison requirements. State the interagency coordination liaison requirements not covered in the base order.

5.B. (U) Command, control, and communications. Describe the communication plan used among the issuing force and interagency organizations to include the primary and alternate means of communications. Consider OPSEC requirements. Refer to OPORD Annex K as required.

5.C. (U) Reports. List interagency coordination specific reports not covered in SOPs. Refer to Annex R (Reports) as required.

Appendixes
1—Humanitarian
2—Economic
3—Political

OFFICIAL
s/t/

(Rank Name) (Position)

Page number

[CLASSIFICATION]

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ANNEX 4 TO APPENDIX J

Basic Fragmentary Order Format

J.4.1 GENERAL

A FRAGORD references an existing OPORD which the staff has already released. Staffs send FRAGORDs to their own OPORDs, they do not draft and release a FRAGORD to an OPORD from HHQ. A FRAGORD addresses only those parts of the original OPORD that have changed. The sequence of the OPORD is used and all five paragraph headings shall be used.

J.4.2 CONTENTS

After each heading, the issuing commander or HQ sends either "no change" or the new information, regardless of the paragraph. This ensures that subordinate commanders know (especially if the message is sent over the radio) that they have received the entire FRAGORD.

In general, a FRAGORD should provide:

1. The mission statement, if changed

2. The commander's intent and concept of operation, if changed

3. Pertinent extracts taken from more detailed orders that relate to the changes addressed in the FRAGORD

4. Task organization if modified

5. Updated control measures that promote initiative, synchronization, and agility while minimizing exposure to fratricide

6. Timely changes to existing orders.

When possible, the FRAGORD includes a brief outline of the situation that includes those elements relevant to the changes in the FRAGORD. It also refers to previous orders and provides a brief and specific set of instructions. The issuing command designates a FRAGORD with the proper classification and requests acknowledgment from the command to which it is issued.

During the execution phase of a military action, it might be necessary to issue supplementary orders to address a new or changed situation. Because modern means of communication allow this to be done quickly, commanders are frequently tempted to intervene. However, excessive use of FRAGORDs tends to confuse the execution of even the best of plans.

The following is an example of a JFMCC FRAGORD (generic example only; PLAD and GENADMIN formats should reflect existing requirements/situations).

UNCLAS

090800ZAUG20XX

FM JFMCC BLUE SWORD
TO CTF 221
CTF 223
CTF 224
CTF 226

INFO CTF 220
CTF 225
CTF 226
CTF 227
CTF 230

CJCS WASHINGTON DC
COMUSEASTCOM MACDILL AFB FL
COMUSNORTHCOM PETERSON AFB
COMFLTFORCOM NORFOLK VA
MARFOREAST
AMEMBASSY PINKLAND
AMEMBASSY GREYLAND
AMEMBASSY WHITELAND
TF BLUE SWORD

UNCLAS//N030000//
OPER/BLUE SWORD//

MSGID/ORDER/JTF BLUE SWORD//
AMPN/FRAGORD/JTF BLUE SWORD//
TIMEZONE/ZULU//

REF/A/MSG/COMUSEASTCOM/202100ZAPR19/NOTAL//
REF/B/MSG/JTF BLUE SWORD/301900ZJUL19//
REF/C/MSG/JFMCC BLUE SWORD/021200ZAUG20XX//
REF/D/MSG/AMEMB PINKLAND/042300ZAUG20XX/NOTAL//
REF/E/DOC/CJCSI 3121.01A/15JAN00//

REF/F/DOC/OPERATION BLUE SWORD ROE SERIAL ONE/022120ZMAY19//

NARR/REF A IS JTF BLUE SWORD WARNORD FOR OPERATION BLUE SWORD. REF B IS JTF BLUE SWORD OPORD. REF C IS JFMCC BLUE SWORD OPORD. REF D IS AM EMB PINKLAND REQUEST FOR ASSISTANCE IN OIL INFRASTRUCTURE SECURITY. REF E IS CJCS STANDING ROE. REF F IS OPERATION BLUE SWORD ROE SERIAL ONE.

ORDTYPE/FRAGORD/SERIAL 001/

NARR/THIS IS A FRAGORD TO REF C. THE SECDEF HAS AUTHORIZED THE CJCS TO EXPAND THE SCOPE OF U.S. INVOLVEMENT TO SUPPORT THE GOVERNMENT OF PINKLAND.

GENTEXT/SITUATION/
1. (U) IN VIEW OF A SPECIFIC THREAT FROM TERRORISTS OPERATING OUT OF REDLAND, THE GOVERNMENT OF PINKLAND HAS FORMALLY REQUESTED U.S. AND COALITION ASSISTANCE IN PROTECTING THEIR NORTHERN OIL TERMINAL AND PUMPING FACILITY. JFMCC BLUE SWORD WILL SUPPORT THIS OPERATION UNTIL SECURITY ELEMENTS (USCG PSU AND CRITICAL INFRASTRUCTURE SECURITY FORCE) CAN ARRIVE FROM CONUS. THESE SECURITY FORCES WILL BE RESPONSIBLE FOR PROVIDING FID SUPPORT TO PINKLAND MILITARY AND PARAMILITARY FORCES.

GENTEXT/MISSION/
2. (U) NO CHANGE.

GENTEXT/EXECUTION/
3. (U) COMMANDER'S INTENT: NO CHANGE.
3.A. (U) TASKS.

3.A.1. (U) TASKS BELOW ARE IN ADDITION TO PHASE 1 (DETER) TASKS IN REF C.

3.A.2. (U) CTF 221.
3.A.2.A. (U) CONDUCT MARITIME SECURITY AND DEFENSE OPS OF THE NORTHERN PINKLAND OIL TERMINAL UNTIL RELIEVED ON STATION BY USCG PSU.
3.A.2.B. (U) CONDUCT HOURLY SURVEILLANCE OF NAI 26 AS PART OF OVERALL SECURITY EFFORT.
3.A.2.C. (U) PROVIDE ONE SHIP AS ON-SCENE COMMANDER DURING THE DESIGNATED PERIOD.
3.A.2.D. (U) COORDINATE WITH PINKLAND AUTHORITIES ASSIGNED TO SECURITY MISSION.

3.A.3. (U) CTF 223.
3.A.3.A. (U) CONDUCT THEATER ISR MISSIONS PER AIR TASKING ORDER TASKING.
3.A.3.B. (U) SUPPORT MSO AS REQUIRED.

3.A.4. (U) CTF 224.
3.A.4.A. (U) PROVIDE TRANSPORTATION FROM PINKLAND ISB TO THE NORTHERN PUMPING STATION AND OIL TERMINAL FOR SECURITY PERSONNEL AND EQUIPMENT ARRIVING FROM CONUS.
3.A.4.B. (U) SUPPORT OIL TERMINAL DEFENSE OPS AS REQUIRED.
3.A.4.C. (U) BPT PROVIDE LOG SUPPORT FOR USCG PSU.

3.A.5. (U) CTF 226.
3.A.5.A. (U) PROVIDE DETACHMENT TO CONDUCT SECURITY OPS FOR THE NORTHERN PINKLAND PUMPING STATION UNTIL RELIEVED IN PLACE BY USCG PSU ARRIVING FROM CONUS.
3.A.5.B. (U) SUPPORT OIL TERMINAL DEFENSE OPS AS REQUIRED.

3.B. (U) COORDINATING INSTRUCTIONS.
3.B.1. (U) H-HOUR/D-DAY/FOR OIL TERMINAL AND PUMPING STATION SECURITY IS 0900 LOCAL 21 AUG 20XX. ANTICIPATED TURNOVER OF ALL RESPONSIBILITIES IS 1400 LOCAL 26 AUG 20XX.
3.B.2. (U) ETA FOR CONUS SECURITY FORCES IN PINKLAND IS 0900 LOCAL 24 AUG 20XX.
3.B.3. (U) NAVY LNOS ARE ASSIGNED TO PINKLAND WTB(S) AS PART OF MISSION.
3.B.4. (U) COORDINATE WITH PINKLAND AUTHORITIES AND TACTICAL UNITS TO ENSURE MUTUAL UNDERSTANDING OF ROE FOR EACH FORCE.
3.B.5. (U) KNOWN OPERATIONAL CONSTRAINTS: MAXIMUM FP MEASURES IN PLACE. MINIMIZE COLLATERAL DAMAGE. EMPHASIZE RESPECT FOR LOCAL RELIGIONS/CULTURES.
3.B.6. (U) ANTICIPATED LENGTH OF OPERATION 5 DAYS.
3.B.7. (U) DIRLAUTH ALCON FOR DEVELOPMENT OF SUPPORTING PLANS.
4. (U) ADMINISTRATION AND LOG: NO CHANGE.

GENTEXT/COMMAND AND CONTROL/
5. (U) COMMAND RELATIONSHIPS: FOR THE CONDUCT OF OIL TERMINAL SECURITY, CTF 221 IS THE SUPPORTED COMMANDER. FOR THE CONDUCT OF PUMPING STATION SECURITY CTF 226 IS THE SUPPORTED COMMANDER.
6. (U) DIRLAUTH ALCON.
7. (U) AKNLG/NO/INST: NONE.//
BT

APPENDIX K
Planning Symbology

K.1 PLANNING SYMBOLOGY

Joint and multinational military OPS require use of a common language that is clearly understood by members of all Services. Graphical representation of objects of interest (e.g., units, installations, equipment, control measures, activities, and meteorological occurrences) are observed and readily understood faster than merely text alone. This is even more valid for OPS that include personnel with widely different backgrounds of language, knowledge, and experience. A common standard of joint military symbols enhances efficiency and contributes to success all military OPS.

This appendix describes use of planning symbology that is based on MIL-STD-2525D, Joint Military Symbology, and APP-06, NATO Joint Military Symbology. These documents provide a standardized, structured set of graphical symbols for the display of information in automated C2 systems and applications. They also provide the basis for (and originated from) symbology used in manually created planning charts and sketches. Complete explanation of planning symbology is beyond the scope of this appendix. Planners should refer to MIL-STD-2525D or APP-06 for details while keeping in mind that track symbol specifications are far more rigorous than what is needed for most planning products, e.g., COA sketches. Figure K-1 provides a partial overview of military planning symbology.

K.2 GENERAL CONSIDERATIONS

K.2.1 Unit Symbols

K.2.1.1 Units Versus Tracks

To declutter planning sketches, use a single object to denote multiple units (ships) that are operating in the same formation.

K.2.1.2 Frame Shapes and Colors

Frame shapes indicate unit affiliation—friendly/enemy/neutral—and the domain in which the units operate—surface/subsurface/air. Colors may be used in addition to the shape to indicate affiliation. Purple is only used to denote civilian (nonmilitary) objects.

K.2.1.3 Equipment and Unit Icons

An icon within the frame indicates unit type or function. For formations (e.g., TFs or task units), the icon should indicate the unit's primary function. For example, a CSG that includes a cruiser and destroyer would be represented by an aircraft carrier icon, while a SAG consisting of cruisers and/or destroyers only would be represented by a surface combatant icon. The generic task organization icon may also be used (e.g., to represent large, multifunction formations or forces).

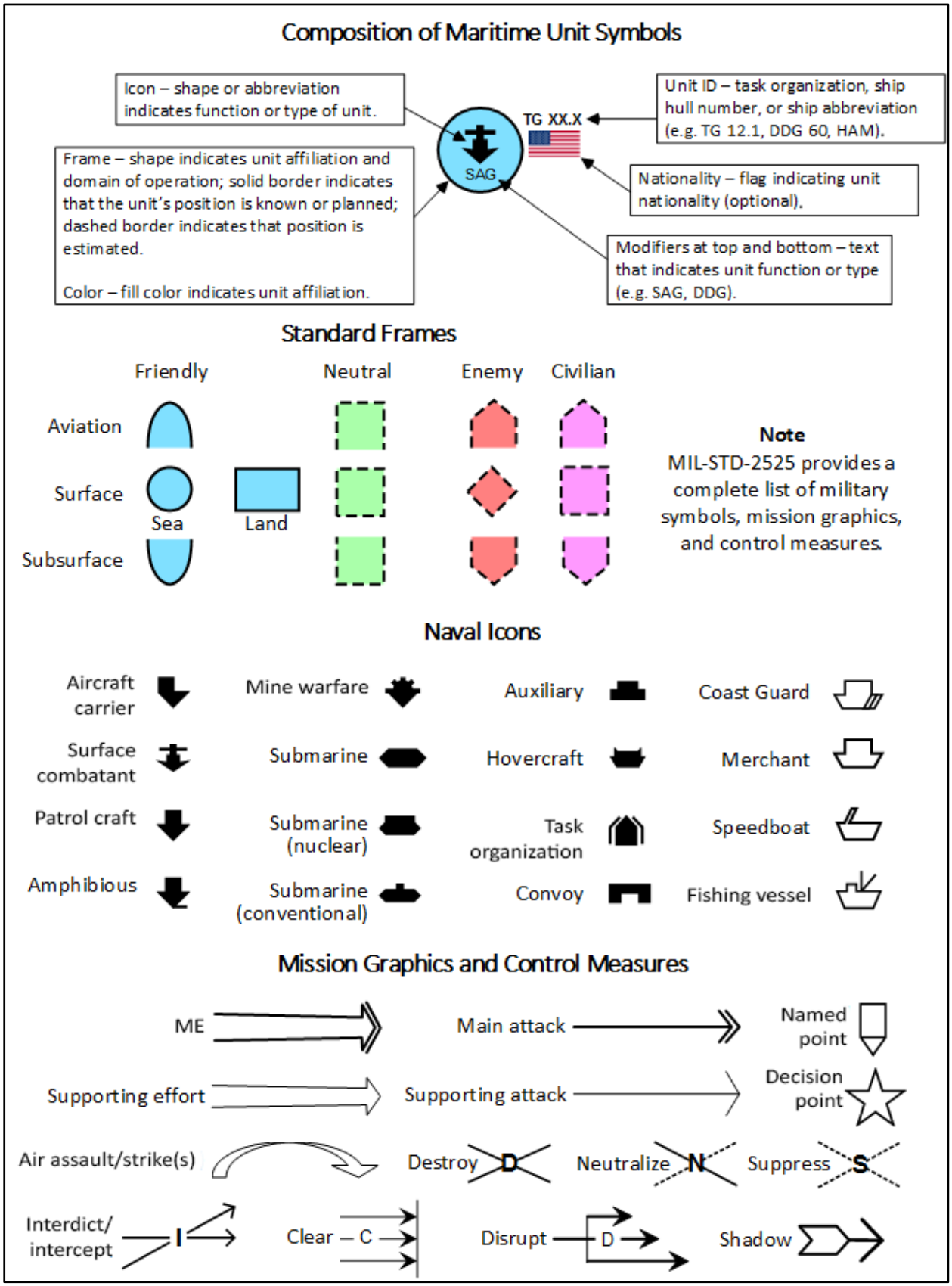


Figure K-1. Overview of Planning Symbols, Control Measures, and Mission Graphics

K.2.1.4 Unit Identity and Nationality

Each object should clearly identify the unit or task organization that it represents. Nationality and the commander's identity should be included to enhance clarity. For example:

TF 220 (United States Army)

TR CSG

(Commander, CSG TWO)

K.2.2 Control Measures

Control measure symbols graphically depict unit OPAREAs and movements. Control measure symbols generally fall into one of three categories: points, lines, or areas. Considerations for coloring and labeling of control measure symbols include:

1. Control measure symbols should be used consistently and clearly labelled and identified.

2. Decision points should be marked with the decision point (star) symbol.

3. Friendly graphic control measures are shown in black.

4. Enemy graphic control measures are shown in red. If red cannot be used, place the abbreviation ENY on the graphic in at least two places.

5. If other colors are used to show friendly or enemy factions, use a legend to ensure understanding of the use of colors on the overlay.

6. Neutral or unknown graphic control measures should be shown in green and yellow if available. If green and yellow are not available, draw neutral or unknown graphics in black and include the abbreviations of NEU for neutral or UNK for unknown.

7. Obstacles (friendly, hostile, neutral, unknown, or factional) are shown in green. If green is not available, draw obstacles in black.

8. NAIs should use the same line style and color, as should each type of unit OPAREA.

9. Clearly label each control measure to indicate its identity and, if applicable, the time period for which it is effective.

K.2.3 Tactical Mission Graphics (Tasks)

A tactical mission task is a specific activity performed by a unit while executing a form of tactical operation or form of maneuver. Tactical mission graphics are used in COA sketches to communicate these tasks as either an action by a friendly force or desired effect on an enemy force.

Note

Many, but not all, tactical mission task symbols correspond to a Navy tactical task, and their most commonly understood meanings apply to land OPS. The precise meanings of tactical mission task symbols and their usage must be clearly established when using them in maritime planning graphics.

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REFERENCES

NWP 2-0, Naval Intelligence

NWP 3-32, Maritime Operations at the Operational Level of War

NWP 3-56, Composite Warfare: Maritime Operations at the Tactical Level of War

NWP 4-0M, Naval Logistics

NTTP 3-32.1, Maritime Operations Center

NTTP 5-01.3, Multi-Service Tactics, Techniques, and Procedures for Operation Assessment

NTTP 5-01.4, Navy Planning, Logistics

NTRP 1-02, Navy Supplement to the DOD Dictionary of Military and Associated Terms

OPNAVINST 3500.38B, Universal Naval Task List

OPNAV M-3500.42A, Maritime Operations Center (MOC) Standardization Manual

CJCSM 3130 (series), Adaptive Planning and Execution (APEX) System

CJCSM 3130.03, APEX Planning Formats and Guidance

CJCSM 3105.01, Joint Risk Analysis

CJCSM 3122.01A, Joint Operation Planning and Execution System, Volume I (Planning Policies and Procedures)

CJCSM 3122.03C, Joint Operations Planning and Execution

DOD Dictionary of Military and Associated Terms

DOD Directive 3025.14, Protection and Evacuation of United States Citizens and Designated Aliens in Danger Areas Abroad

DOD Instruction 1100.22, Policy and Procedures for Determining Workforce Mix

JP 1, Doctrine for the Armed Forces of the United States

JP 2-01, Joint and National Intelligence Support to Military Operations

JP 2-01.3, Joint Intelligence Preparation of the Operational Environment

JP 3-0, Joint Operations

JP 3-32, Joint Maritime Operations

JP 5-0, Joint Planning

MIL-STD-2525D, Joint Military Symbology

APP-06, NATO Joint Military Symbology

NATO Operations Assessment Handbook

Naval Manual of Operational Planning, Navy Department, Office of the Chief of Naval Operations, Washington, DC (1948)

Universal Joint Task List

SUGGESTED READING

APP-11, NATO Message Catalogue

ATP-01 Volume I, Allied Maritime Tactical Instructions and Procedures

Sound Military Decision, Naval War College, Newport, Rhode Island (1942)

GLOSSARY

acceptability. The plan review criterion for assessing whether the contemplated course of action is proportional, worth the cost, consistent with the law of war, and is militarily and politically supportable. (DOD Dictionary)

Adaptive Planning and Execution (APEX). A Department of Defense enterprise of joint policies, processes, procedures, and reporting structures, supported by communications and information technology, that is used by the joint planning and execution community to monitor, plan, and execute mobilization, deployment, employment, sustainment, redeployment, and demobilization activities associated with joint operations.

adequacy. The plan review criterion for assessing whether the scope and concept of planned operations can accomplish the assigned mission and comply with the planning guidance provided. (DOD Dictionary)

adversary. A party acknowledged as potentially hostile to a friendly party and against which the use of force may be envisaged. (DOD Dictionary)

air defense. Defensive measures designed to destroy attacking enemy aircraft or aerodynamic missiles, or to nullify or reduce the effectiveness of such attack. (DOD Dictionary)

allocation. 1. Distribution of limited forces and resources for employment among competing requirements. 2. The temporary transfer of forces to meet the operational demand of combatant commanders, including rotational requirements and requests for capabilities or forces (unit or individual) in response to crisis or emergent contingencies. (DOD Dictionary)

amphibious assault. A type of amphibious operation that involves establishing a force on a hostile or potentially hostile shore. (DOD Dictionary)

amphibious demonstration. A type of amphibious operation conducted for the purpose of deceiving the enemy by a show of force with the expectation of deluding the enemy into following an unfavorable course of action. (DOD Dictionary)

amphibious objective area (AOA). A geographical area of sufficient size for conducting necessary sea, air, and land operations and within which is located the objective(s) to be secured by the amphibious force. (DOD Dictionary)

amphibious operation. A military operation launched from the sea by an amphibious force to conduct landing force operations within the littorals. (DOD Dictionary)

amphibious raid. A type of amphibious operation involving swift incursion into, or temporary occupation of, an objective area followed by a planned withdrawal. (DOD Dictionary)

amphibious ready group. A Navy task organization formed to conduct amphibious operations, commanded by an amphibious squadron commander. (DOD Dictionary)

amphibious support to other operations. A type of amphibious operation that contributes to conflict prevention or crisis mitigation. These may include operations such as security cooperation, foreign humanitarian assistance, civil support, noncombatant evacuations, peace operations, recovery operations, or disaster relief.

amphibious withdrawal. A type of amphibious operation involving the extraction of forces by sea in ships or craft from a hostile or potentially hostile shore. (DOD Dictionary)

antisubmarine warfare (ASW). 1. Operations conducted with the intention of denying the enemy the effective use of submarines. 2. That segment of naval warfare that involves sensors, weapons, platforms, and targets in the subsurface environment.

apportionment. The quantities of force capabilities and resources provided for planning purposes only, but not necessarily an identification of the actual forces that may be allocated for use when a plan transitions to execution. (DOD Dictionary)

area of interest. That area of concern to the commander, including the area of influence, areas adjacent thereto, and extending into enemy territory. (DOD Dictionary)

area of operations (AO). An operational area defined by a commander for land and maritime forces that should be large enough to accomplish their missions and protect their forces. (DOD Dictionary)

assessment. 1. A continuous process that measures the overall effectiveness of employing capabilities during military operations. 2. Determination of the progress toward accomplishing a task, creating a condition, or achieving an objective. (DOD Dictionary)

assign. 1. To place units or personnel in an organization where such placement is relatively permanent, and/or where such organization controls and administers the units or personnel for the primary function, or greater portion of the functions, of the unit or personnel. 2. To detail individuals to specific duties or functions where such duties or functions are primary and/or relatively permanent. (DOD Dictionary)

assigned forces. Those forces and resources that have been placed under the combatant command (command authority) of a unified commander in the Forces for Unified Commands Memorandum. Forces and resources so assigned are available for normal peacetime operations of that command. (NTRP 1-02)

assumption. A specific supposition of the operational environment that is assumed to be true, in the absence of positive proof, essential for the continuation of planning. (DOD Dictionary)

attach. 1. The placement of units or personnel in an organization where such placement is relatively temporary. 2. The detailing of individuals to specific functions where such functions are secondary or relatively temporary, e.g., attached for quarters and rations; attached for flying duty. (DOD Dictionary)

battle rhythm. A deliberate, daily schedule of command, staff, and unit activities intended to maximize use of time and synchronize staff actions. (DOD Dictionary)

battlespace. The environment, factors, and conditions that must be understood to successfully apply combat power, protect the force, or complete the mission. This includes the air, land, sea, space, and the included enemy and friendly forces; facilities; weather; terrain; the electromagnetic spectrum; and the information environment within the operational areas, areas of interest, and areas of influence. 2. All aspects of air, surface, and subsurface, land, space, and the electromagnetic spectrum that encompass the area of influence and area of interest. (NTRP 1-02)

branch. 1. A subdivision of any organization. 2. A geographically separate unit of an activity, which performs all or part of the primary functions of the parent activity on a smaller scale. 3. An arm or service of the Army. 4. The contingency options built into the base plan used for changing the mission, orientation, or direction of movement of a force to aid success of the operation based on anticipated events, opportunities, or disruptions caused by enemy actions and reactions. (DOD Dictionary)

campaign. A series of related operations aimed at achieving strategic and operational objectives within a given time and space. (DOD Dictionary)

campaign plan. A joint operation plan for a series of related major operations aimed at achieving strategic or operational objectives within a given time and space. (DOD Dictionary)

capability. The ability to execute a specified course of action. (A capability may or may not be accompanied by an intention.) (NTRP 1-02)

carrier strike group. A standing naval task group consisting of a carrier, embarked air wing, surface combatants, and submarines as assigned in direct support, operating in mutual support with the task of destroying hostile submarine, surface, and air forces within the group's assigned operational area and striking at targets along hostile shore lines or projecting power inland. (DOD Dictionary)

C-day. The unnamed day on which a deployment operation commences or is to commence. (DOD Dictionary)

center of gravity. The source of power that provides moral or physical strength, freedom of action, or will to act. (DOD Dictionary)

civil-military operations. Activities of a commander performed by designated military forces that establish, maintain, influence, or exploit relations between military forces and indigenous populations and institutions by directly supporting the achievement of objectives relating to the reestablishment or maintenance of stability within a region or host nation. (DOD Dictionary)

close support. The action of the supporting force against targets or objectives that are sufficiently near the supported force as to require detailed integration or coordination of the supporting action. (DOD Dictionary)

collection plan. A systematic scheme to optimize the employment of all available collection capabilities and associated processing, exploitation, and dissemination resources to satisfy specific information requirements. (DOD Dictionary)

collection requirement. A valid need to close a specific gap in intelligence holdings in direct response to a request for information. (DOD Dictionary)

combatant command. A unified or specified command with a broad continuing mission under a single commander established and so designated by the President, through the Secretary of Defense and with the advice and assistance of the Chairman of the Joint Chiefs of Staff. (DOD Dictionary)

combatant commander. A commander of one of the unified or specified combatant commands established by the President. (DOD Dictionary)

combat power. The total means of destructive and/or disruptive force that a military unit/formation can apply against the opponent at a given time. (DOD Dictionary)

command and control. The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of the mission. (DOD Dictionary)

commander's critical information requirement. An information requirement identified by the commander as being critical to facilitating timely decision making. (DOD Dictionary)

commander's intent. A clear and concise expression of the purpose of the operation and the desired military end state that supports mission command, provides focus to the staff, and helps subordinate and supporting commanders act to achieve the commander's desired results without further orders, even when the operation does not unfold as planned. (DOD Dictionary)

commander's planning guidance. The commander's vision of decisive and shaping actions used to assist the planning team in determining the main effort, phases of the operation, location of critical events, and other aspects of the operation the commander deems pertinent to course of action development. (NTRP 1-02)

command relationships. The interrelated responsibilities between commanders, as well as the operational authority exercised by commanders in the chain of command; defined further as combatant command (command authority), operational control, tactical control, or support. (DOD Dictionary)

common operational picture. A single identical display of relevant information shared by more than one command that facilitates collaborative planning and assists all echelons to achieve situational awareness. (DOD Dictionary)

concept of logistics support. A verbal or graphic statement, in a broad outline, of how a commander intends to support and integrate with a concept of operations in an operation or campaign. (DOD Dictionary)

concept of operations. A verbal or graphic statement that clearly and concisely expresses what the commander intends to accomplish and how it will be done using available resources. (DOD Dictionary)

concept plan. An operation plan in an abbreviated format that may require considerable expansion or alteration to convert it into a complete operation plan or operation order. (DOD Dictionary)

condition. 1. Those variables of an operational environment or situation in which a unit, system, or individual is expected to operate and may affect performance. 2. A physical or behavioral state of a system that is required for the achievement of an objective. (DOD Dictionary)

constraint. In the context of planning, a requirement placed on the command by a higher command that dictates an action, thus restricting freedom of action. (DOD Dictionary)

contingency. A situation requiring military operations in response to natural disasters, terrorists, subversives, or as otherwise directed by appropriate authority to protect United States interests. (DOD Dictionary)

course of action. 1. Any sequence of activities that an individual or unit may follow. 2. A scheme developed to accomplish a mission. (DOD Dictionary)

critical asset. A specific entity that is of such extraordinary importance that its incapacitation or destruction would have a very serious, debilitating effect on the ability of a nation to continue to function effectively. (DOD Dictionary)

critical capability. A means that is considered a crucial enabler for a center of gravity to function as such and is essential to the accomplishment of the specified or assumed objective(s). (DOD Dictionary)

critical requirement. An essential condition, resource, and means for a critical capability to be fully operational. (DOD Dictionary)

critical strength. A military or nonmilitary capability considered essential to the accomplishment of one's or the adversary's military objective(s); the most important among the critical strengths is the center of gravity. (NTRP 1-02)

critical vulnerability. An aspect of a critical requirement which is deficient or vulnerable to direct or indirect attack that will create decisive or significant effects. (DOD Dictionary)

critical weakness. A military or nonmilitary capability considered essential to the accomplishment of one's or the adversary's military objectives but, in terms of quantity, quality or both, insufficient or inadequate to perform their intended functions. (NTRP 1-02)

daily intentions message (DIM). An unformatted message with an immediate impact on operations, intended to convey direction from the latest iteration of the commander's decision cycle. (NTRP 1-02)

D-day. The unnamed day on which a particular operation commences or is to commence. (DOD Dictionary)

deception. Those measures designed to mislead the adversary by manipulation, distortion, or falsification of evidence to induce the adversary to react in a manner prejudicial to the adversary's interests. (NTRP 1-02)

decision point. A point in space and the latest time when the commander or staff anticipates making a key decision concerning a specific course of action. (DOD Dictionary)

decision support matrix (DSM). A matrix associated with a decision support template that lists decision points and their associated decision options and decision criteria.

decision support template (DST). A combined intelligence and operations graphic based on the results of wargaming that depicts decision points, timelines associated with movement of forces and the flow of the operation, and other key items of information required to execute a specific friendly course of action. (DOD Dictionary)

decisive action. Any action the commander deems fundamental to achieving mission success. (MCRP 1-10.2)

decisive point. Key terrain, key event, critical factor, or function that, when acted upon, enables commanders to gain a marked advantage over an enemy or contribute materially to achieving success. See also center of gravity. (DOD Dictionary)

direct support. A mission requiring a force to support another specific force and authorizing it to answer directly to the supported force's request for assistance. (DOD Dictionary)

economy of force. The judicious employment and distribution of forces so as to expend the minimum essential combat power on secondary efforts to allocate the maximum possible combat power on primary efforts. (DOD Dictionary)

E-day. The day landing force personnel, supplies, and equipment begin to embark aboard amphibious warfare or commercial ships. (DOD Dictionary)

effect. 1. The physical or behavioral state of a system that results from an action, a set of actions, or another effect. 2. The result, outcome, or consequence of an action. 3. A change to a condition, behavior, or degree of freedom. (DOD Dictionary)

electromagnetic warfare. Military action involving the use of electromagnetic and directed energy to control the electromagnetic spectrum or to attack the enemy. (DOD Dictionary)

end state. The set of required conditions that defines achievement of the commander's objectives. (DOD Dictionary)

essential elements of friendly information (EEFI). Specific facts about friendly intentions, capabilities, and activities needed by adversaries to plan and execute effective operations against our forces. (NTRP 1-02)

essential task. A specified or implied task that an organization must perform to accomplish the mission. (DOD Dictionary)

evaluation criteria. Standards used by the staff during course of action analysis and comparison to help identify advantages and disadvantages of the various courses of action with the intent of making a decision recommendation to the commander. (NTRP 1-02)

execute order. 1. An order issued by the Chairman of the Joint Chiefs of Staff, at the direction of the Secretary of Defense, to implement a decision by the President to initiate military operations. 2. An order to initiate military operations as directed. (DOD Dictionary)

feasibility. The plan review criterion for assessing whether the assigned mission can be accomplished using available resources within the time contemplated by the plan. (DOD Dictionary)

F-hour. The effective time of announcement by the Secretary of Defense to the Military Departments of a decision to mobilize Reserve units. (NTRP 1-02)

fires. The use of weapon systems or other actions to create specific lethal or nonlethal effects on a target. (DOD Dictionary)

force protection. Preventive measures taken to mitigate hostile actions against Department of Defense personnel (to include family members), resources, facilities, and critical information. (DOD Dictionary)

foreign humanitarian assistance. Department of Defense activities conducted outside the United States and its territories to directly relieve or reduce human suffering, disease, hunger, or privation. (DOD Dictionary)

fragmentary order. An abbreviated operation order issued as needed to change or modify an order or to execute a branch or sequel. (DOD Dictionary)

friendly force information requirement. Information the commander and staff need to understand the status of friendly force and supporting capabilities. (DOD Dictionary)

functional component command. A command normally, but not necessarily, composed of forces of two or more Military Departments which may be established across the range of military operations to perform particular operational missions that may be of short duration or may extend over a period of time. (DOD Dictionary)

general support. Support given to the supported force as a whole and not to any particular subdivision thereof. (DOD Dictionary)

governing factors. In the context of planning, those aspects of the situation (or externally imposed factors) that the commander deems critical to the accomplishment of the mission.

H-hour. 1. The specific hour on D-day at which a particular operation commences. 2. In amphibious operations, the time the first landing craft or amphibious vehicle of the waterborne wave lands or is scheduled to land on the beach, and in some cases, the commencement of countermine breaching operations. (DOD Dictionary)

implied task. In the context of planning, a task derived during mission analysis that an organization must perform or prepare to perform to accomplish a specified task or the mission, but which is not stated in the higher headquarters order. (DOD Dictionary)

information operations. The integrated employment, during military operations, of information-related capabilities in concert with other lines of operation to influence, disrupt, corrupt, or usurp the decision-making of adversaries and potential adversaries while protecting our own. (DOD Dictionary)

integrated air and missile defense. The integration of capabilities and overlapping operations to defend the homeland and United States national interests, protect the joint force, and enable freedom of action by negating an enemy's ability to create adverse effects from their air and missile capabilities. (DOD Dictionary)

intelligence. 1. The product resulting from the collection, processing, integration, evaluation, analysis, and interpretation of available information concerning foreign nations, hostile or potentially hostile forces or elements, or areas of actual or potential operations. 2. The activities that result in the product. 3. The organizations engaged in such activities. (DOD Dictionary)

intelligence estimate. The appraisal, expressed in writing or orally, of available intelligence relating to a specific situation or condition with a view to determining the courses of action open to the enemy or adversary and the order of probability of their adoption. (DOD Dictionary)

intelligence preparation of the operational environment (IPOE). An analytical methodology employed to reduce uncertainties concerning the enemy, environment, and terrain for all types of operations. Intelligence preparation of the operational environment builds an extensive database for each potential area in which a unit may be required to operate. The database is then analyzed in detail to determine the impact of the enemy, environment, and terrain on operations and presents it in graphic form. Intelligence preparation of the operational environment is a continuing process. (NTRP 1-02)

intelligence requirement. 1. Any subject, general or specific, upon which there is a need for the collection of information, or the production of intelligence. 2. A requirement for intelligence to fill a gap in the command's knowledge or understanding of the operational environment or threat forces. (DOD Dictionary)

intelligence, surveillance, and reconnaissance. 1. An integrated operations and intelligence activity that synchronizes and integrates the planning and operation of sensors, assets, and processing, exploitation, and dissemination systems in direct support of current and future operations. 2. The organizations or assets conducting such activities. (DOD Dictionary)

interdiction. 1. An action to divert, disrupt, delay, or destroy the enemy's military surface capability before it can be used effectively against friendly forces, or to achieve enemy objectives. 2. In support of law enforcement, activities conducted to divert, disrupt, delay, intercept, board, detain, or destroy, under lawful authority, vessels, vehicles, aircraft, people, cargo, and money. (DOD Dictionary)

interoperability. 1. The ability to act together coherently, effectively, and efficiently to achieve tactical, operational, and strategic objectives. 2. The condition achieved among communications-electronics systems or items of communications-electronics equipment when information or services can be exchanged directly and satisfactorily between them and/or their users. (DOD Dictionary)

joint force. A force composed of elements, assigned or attached, of two or more Military Departments operating under a single joint force commander. (DOD Dictionary)

joint force air component commander. The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and/or made available for tasking air forces; planning and coordinating air operations; or accomplishing such operational missions as may be assigned. (DOD Dictionary)

joint force commander. A general term applied to a combatant commander, subunified commander, or joint task force commander authorized to exercise combatant command (command authority) or operational control over a joint force. (DOD Dictionary)

joint force land component commander. The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and/or made available for tasking land forces; planning and coordinating land operations; or accomplishing such operational missions as may be assigned. (DOD Dictionary)

joint force maritime component commander. The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and/or made available for tasking maritime forces and assets; planning and coordinating maritime operations; or accomplishing such operational missions as may be assigned. (DOD Dictionary)

joint force special operations component commander. The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and/or made available for tasking special operations forces and assets; planning and coordinating special operations; or accomplishing such operational missions as may be assigned. (DOD Dictionary)

joint intelligence preparation of the operational environment. The analytical process used by joint intelligence organizations to produce intelligence estimates and other intelligence products in support of the joint force commander's decision-making process. (DOD Dictionary)

joint operations. Military actions conducted by joint forces and those Service forces employed in specified command relationships with each other, which of themselves, do not establish joint forces. (DOD Dictionary)

joint operations area. An area of land, sea, and airspace, defined by a geographic combatant commander or subordinate unified commander, in which a joint force commander (normally a joint task force commander) conducts military operations to accomplish a specific mission. (DOD Dictionary)

joint task force. A joint force that is constituted and so designated by the Secretary of Defense, a combatant commander, a subunified commander, or an existing joint task force commander. (DOD Dictionary)

L-hour. 1. The specific hour on C-day at which a deployment operation commences or is to commence. 2. In amphibious operations, the time at which the first helicopter or tiltrotor aircraft of the airborne ship-to-shore movement wave touches down or is scheduled to touch down in a landing zone. (DOD Dictionary)

line of effort. In the context of planning, using the purpose (cause and effect) to focus efforts toward establishing operational and strategic conditions by linking multiple tasks and missions. (DOD Dictionary)

line of operation. A line that defines the interior or exterior orientation of the force in relation to the enemy or that connects actions on nodes and/or decisive points related in time and space to an objective(s). (DOD Dictionary)

littoral. The littoral comprises two segments of operational environment: 1. Seaward: the area from the open ocean to the shore, which must be controlled to support operations ashore. 2. Landward: the area inland from the shore that can be supported and defended directly from the sea. (DOD Dictionary)

logistics. Planning and executing the movement and support of forces. (DOD Dictionary)

main effort. 1. The designated activity or subordinate organization whose mission at a given time is most critical to overall mission success. It is usually weighted with the preponderance of resources by the higher organization. 2. In naval special warfare, the main effort is defined as the primary group of assets and personnel used to commit to a target. Also sometimes referred to as the assault team. (NTRP 1-02)

maneuver. 1. A movement to place ships, aircraft, or land forces in a position of advantage over the enemy. 2. A tactical exercise carried out at sea, in the air, on the ground, or on a map in imitation of war. 3. The operation of a ship, aircraft, or vehicle, to cause it to perform desired movements. 4. Employment of forces in the operational area, through movement in combination with fires and information, to achieve a position of advantage in respect to the enemy. (DOD Dictionary)

maritime interception operations. Efforts to monitor, query, and board merchant vessels in international waters to enforce sanctions against other nations such as those in support of United Nations Security Council resolutions and/or prevent the transport of restricted goods. (DOD Dictionary)

maritime operations center (MOC). The collective name for the boards, bureaus, cells, centers, and working groups that execute the maritime headquarters maritime operations functions. (NTRP 1-02)

maritime pre-positioning force (MPF). A task organization of units under one commander formed for the purpose of introducing a Marine air-ground task force and its associated equipment and supplies into a secure area. The maritime pre-positioning force is composed of a command element, a maritime pre-positioning ships squadron, a Marine air-ground task force, and a Navy support element. (NTRP 1-02)

maritime superiority. That degree of dominance of one force over another that permits the conduct of maritime operations by the former and its related land, maritime, and air forces at a given time and place without prohibitive interference by the opposing force. (DOD Dictionary)

maritime supporting plan (MARSUPPLAN). A maritime operations plan prepared by the maritime component commander to satisfy the requests or requirements of the supported commander's plan. (NTRP 1-02)

maritime supremacy. That degree of maritime superiority wherein the opposing force is incapable of effective interference. (DOD Dictionary)

M-day. The term used to designate the unnamed day on which full mobilization commences or is due to commence. (NTRP 1-02)

measure of effectiveness. An indicator used to measure a current system state, with change indicated by comparing multiple observations over time. (DOD Dictionary)

measure of performance. An indicator used to measure a friendly action that is tied to measuring task accomplishment. (DOD Dictionary)

military information support operations (MISO). Planned operations to convey selected information and indicators to foreign audiences to influence their emotions, motives, objective reasoning, and ultimately the behavior of foreign governments, organizations, groups, and individuals in a manner favorable to the originator's objectives. (DOD Dictionary)

mine warfare. The strategic, operational, and tactical use of mines and mine countermeasures either by emplacing mines to degrade the enemy's capabilities to wage land, air, and maritime warfare or by countering of enemy-emplaced mines to permit friendly maneuver or use of selected land or sea areas. (DOD Dictionary)

mission. 1. The task, together with the purpose, that clearly indicates the action to be taken and the reason therefore. 2. In common usage, especially when applied to lower military units, a duty assigned to an individual or unit; a task. 3. The dispatching of one or more aircraft to accomplish one particular task. (DOD Dictionary)

mission command. The conduct of military operations through decentralized execution based upon mission-type orders. (DOD Dictionary)

mission statement. A short sentence or paragraph that describes the organization's essential task(s), purpose, and action containing the elements of who, what, when, where, and why. (DOD Dictionary)

mission type order. 1. An order issued to a lower unit that includes the accomplishment of the total mission assigned to the higher headquarters. 2. An order to a unit to perform a mission without specifying how it is to be accomplished. (DOD Dictionary)

modified combined obstacle overlay. A joint intelligence preparation of the operational environment product used to portray the militarily significant aspects of the operational environment, such as obstacles restricting military movement, key geography, and military objectives. (DOD Dictionary)

mutual support. That support which units render each other against an enemy, because of their assigned tasks, their position relative to each other and to the enemy, and their inherent capabilities. (DOD Dictionary)

named area of interest. The geospatial area or systems node or link against which information that will satisfy a specific information requirement can be collected, usually to capture indications of adversary courses of action. (DOD Dictionary)

national military strategy. A document approved by the Chairman of the Joint Chiefs of Staff for distributing and applying military power to attain national security strategy and Defense Strategic Guidance objectives. (DOD Dictionary)

Navy component commander (NCC). The commander of a naval component assigned or attached to a joint force (unified command) constituted and so designated by the Joint Chiefs of Staff or by a commander of an existing unified command that was established by the Joint Chiefs of Staff. (NTRP 1-02)

N-day. The unnamed day an active duty unit is notified for deployment or redeployment. (NTRP 1-02)

node. 1. A location in a mobility system where a movement requirement is originated, processed for onward movement, or terminated. 2. In communications and computer systems, the physical location that provides terminating, switching, and gateway access services to support information exchange. 3. An element of a system that represents a person, place, or physical thing. (DOD Dictionary)

objective. 1. The clearly defined, decisive, and attainable goal toward which an operation is directed. 2. The specific goal of the action taken which is essential to the commander's plan. (DOD Dictionary)

offload day (O-Day). An unnamed day when the maritime pre-positioning force offload commences. (NTRP 1-02)

operation. 1. A sequence of tactical actions with a common purpose or unifying theme. 2. A military action or the carrying out of a strategic, operational, tactical, service, training, or administrative military mission. (DOD Dictionary)

operational approach. A broad description of the mission, operational concepts, tasks, and actions required to accomplish the mission. (DOD Dictionary)

operational art. The cognitive approach by commander and staffs—supported by their skill, knowledge, experience, creativity, and judgment—to develop strategies, campaigns, and operations to organize and employ military forces by integrating ends, ways, and means. (DOD Dictionary)

operational control. 1. The authority to perform those functions of command over subordinate forces involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction necessary to accomplish the mission. (DOD Dictionary)

operational design. The conception and construction of the framework that underpins a campaign or operation plan or order. (DOD Dictionary)

operational environment. A composite of the conditions, circumstances, and influences that affect the employment of capabilities and bear on the decisions of the commander. (DOD Dictionary)

operational functions. Those actions by which the commander achieves unity of effort and builds, projects, and sustains combat power. Their effective application, in concert with one another, facilitates the planning and conduct of naval operations. Functions include fires, command and control, intelligence, movement and maneuver, protection, and sustainment.

operational general (message) (OPGEN). Maritime-unique formatted message used by both the United States Navy and NATO to promulgate general matters of policy and instructions and common aspects of operations; also may include detailed instructions for warfare responsibilities. (NTRP 1-02)

operational level of warfare. The level of warfare at which campaigns and major operations are planned, conducted, and sustained to achieve strategic objectives within theaters or other operational areas. (DOD Dictionary)

operational tasking (message) (OPTASK). Maritime-unique formatted message used by both the United States Navy and NATO to provide detailed information for specific aspects within individual areas of warfare and for tasking resources. This includes logistics, may be issued at all levels above the unit, and may be Navy-wide or focused on a particular theater or strike group. (NTRP 1-02)

operation assessment. 1) A continuous process that measures the overall effectiveness of employing capabilities during military operations in achieving stated objectives. 2) Determination of the progress toward accomplishing a task, creating a condition, or achieving an objective. (DOD Dictionary)

operation order. A directive issued by a commander to subordinate commanders for the purpose of effecting the coordinated execution of an operation. (DOD Dictionary)

operation plan. A complete and detailed joint plan containing a full description of the concept of operations, all annexes applicable to the plan, and a time-phased force and deployment list. (DOD Dictionary)

operations security. A capability that identifies and controls critical information, indicators of friendly force actions attendant to military operations, and incorporates countermeasures to reduce the risk of an adversary exploiting vulnerabilities. (DOD Dictionary)

planning order. A planning directive that provides essential planning guidance and directs the development, adaptation, or refinement of a plan/order. (DOD Dictionary)

priority intelligence requirement. An intelligence requirement that the commander and staff need to understand the threat and other aspects of the operational environment. (DOD Dictionary)

protection. Preservation of the effectiveness and survivability of mission-related military and nonmilitary personnel, equipment, facilities, information, and infrastructure deployed or located within or outside the boundaries of a given operational area. (DOD Dictionary)

public affairs. Communication activities with external and internal audiences. (DOD Dictionary)

R-day. The day on which redeployment of major combat, combat support, and combat service support forces begins in an operation. (NTRP 1-02)

request for information. 1. Any specific time-sensitive ad hoc requirement for intelligence information or products to support an ongoing crisis or operation not necessarily related to standing requirements or scheduled intelligence production. 2. A term used by the National Security Agency/Central Security Service to state ad hoc signals intelligence requirements. (DOD Dictionary)

restraint. In the context of planning, a requirement placed on the command by a higher command that prohibits an action, thus restricting freedom of action. (DOD Dictionary)

rules of engagement. Directives issued by competent military authority that delineate the circumstances and limitations under which United States forces will initiate and/or continue combat engagement with other forces encountered. (DOD Dictionary)

running estimate. A staff estimate, that is not static, but instead is continuously updated with new information as the operation proceeds. (NTRP 1-02)

scheme of fires. The detailed, logical sequence of targets and fire support events to find and engage targets to support the commander's objectives. (DOD Dictionary)

scheme of maneuver. The central expression of the commander's concept for operations that governs the development of supporting plans or annexes of how arrayed forces will accomplish the mission. (DOD Dictionary)

S-day. The day the President authorizes Selective Reserve call-up (not more than 200,000). (NTRP 1-02)

sea control. The condition in which one has freedom of action to use the sea for one's own purposes in specified areas and for specified periods of time and, where necessary, to deny or limit its use to the enemy. Sea control includes the airspace above the surface and the water volume and seabed below. (NTRP 1-02)

sea control operations. The employment of forces to destroy enemy naval forces, suppress enemy sea commerce, protect vital sea lanes, and establish local military superiority in vital sea areas. (DOD Dictionary)

sea denial. Partially or completely denying the adversary the use of the sea with a force that may be insufficient to ensure the use of the sea by one's own forces. (NTRP 1-02)

search and rescue. The use of aircraft, surface craft, submarines, and specialized rescue teams and equipment to search for and rescue distressed persons on land or at sea in a permissive environment. (DOD Dictionary)

sequel. The subsequent operation or phase based on the possible outcomes of the current operation or phase. (DOD Dictionary)

Service component command. A command consisting of the Service component commander and all those Service forces, such as individuals, units, detachments, organizations, and installations under that command, including the support forces that have been assigned to a combatant command or further assigned to a subordinate unified command or joint task force. (DOD Dictionary)

shaping actions. Those activities conducted by friendly forces designed to set conditions for the success of subsequent operations. Normally associated with supporting forces or efforts. (NTRP 1-02)

shaping effort. The designated activity or subordinate organization(s) whose mission at a given time creates desired conditions or effects for current or future activities but does not directly support the main effort. (NTRP 1-02)

situation template. A depiction of assumed adversary dispositions, based on that adversary's preferred method of operations and the impact of the operational environment if the adversary should adopt a particular course of action. (DOD Dictionary)

situational awareness (SA). Knowledge and understanding of the current situation that promotes timely, relevant, and accurate assessment of friendly, enemy, and other operations within the battlespace in order to facilitate decision-making. An informational perspective and skill that fosters an ability to determine quickly the context and relevance of events that are unfolding. (NTRP 1-02)

specified task. In the context of planning, a task that is specifically assigned to an organization by its higher headquarters. (DOD Dictionary)

staff estimate. A continual evaluation of how factors in a staff section's functional area support and impact the planning and execution of the mission. (DOD Dictionary)

strike. An attack to damage or destroy an objective or a capability. (DOD Dictionary)

support. 1. The action of a force that aids, protects, complements, or sustains another force in accordance with a directive requiring such action. 2. A unit that helps another unit in battle. 3. An element of a command that assists, protects, or supplies other forces in combat. (DOD Dictionary)

supported commander. 1. The commander having primary responsibility for all aspects of a task assigned by Chairman of the Joint Chiefs of Staff Instruction 3110.01J, (U) 2015 Joint Strategic Capabilities Plan (JSCP), or other joint planning authority. 2. In the context of operation planning, the commander who prepares operation plans or operation orders in response to requirements of the Chairman of the Joint Chiefs of Staff. 3. In the context of a support command relationship, the commander who receives assistance from another commander's force or capabilities, and who is responsible for ensuring that the supporting commander understands the assistance required. (DOD Dictionary)

supporting commander. 1. A commander who provides augmentation forces or other support to a supported commander or who develops a supporting plan. 2. In the context of a support command relationship, the commander who aids, protects, complements, or sustains another commander's force and who is responsible for providing the assistance required by the supported commander. (DOD Dictionary)

supporting effort. 1. The designated activity or subordinate organization(s) whose mission at a given time is designed to directly contribute to the success of the main effort. 2. In naval special warfare, the supporting effort is a group of assets and personnel used to support the main effort, usually from a standoff position in the target area. Also referred to as the reconnaissance team. (NTRP 1-02)

surface action group. A temporary or standing organization of combatant ships, other than carriers, tailored for a specific tactical mission. (DOD Dictionary)

surface combatant. A ship designed to engage in attacks against airborne, surface, subsurface, and shore targets. (DOD Dictionary)

surface warfare. That portion of maritime warfare in which operations are conducted to destroy or neutralize enemy naval surface forces and merchant vessels. (DOD Dictionary)

sustainability. The ability to maintain the necessary level and duration of operational activity to achieve military objectives. Sustainability is a function of providing for and maintaining those levels of ready forces, materiel, and consumables necessary to support military effort. (NTRP 1-02)

sustainment. The provision of logistics and personnel services required to maintain and prolong operations until successful mission accomplishment. (DOD Dictionary)

sustaining actions. Those activities conducted by friendly forces to provide logistics and personnel services. (NTRP 1-02)

sustaining effort. The designated activity or subordinate organization(s) whose mission is directed at sustaining friendly forces to continue or prepare for activity. (NTRP 1-02)

synchronization. 1. The arrangement of military actions in time, space, and purpose to produce maximum relative combat power at a decisive place and time. 2. In the intelligence context, application of intelligence sources and methods in concert with the operation plan to ensure intelligence requirements are answered in time to influence the decisions they support. (DOD Dictionary)

synchronization matrix. A staff planning and execution aid that depicts the timing of tasks during execution of an operation to ensure that the operation is synchronized across time, space, and purpose in relation to its objectives.

tactical command (TACOM). Command authority granted to an allied/multinational maritime commander by an allied/multinational maritime commander with either operational command or operational control. Tactical command is authority delegated to an allied/multinational commander to assign subordinate forces for the accomplishment of the mission assigned by higher authority. It involves the responsibility for the conduct of the tasks pertaining to the mission(s), (i.e., issuing detailed orders and ensuring their correct execution). It also involves responsibility for the general safety of attached units, although ultimate responsibility remains with the commanding officers. Tactical command of units temporarily attached does not include the authority to give them tasks inconsistent with the mission previously allocated to them. (NTRP 1-02)

tactical control. The authority over forces that is limited to the detailed direction and control of movements or maneuvers within the operational area necessary to accomplish missions or tasks assigned. (DOD Dictionary)

tactical level of warfare. The level of warfare at which battles and engagements are planned and executed to achieve military objectives assigned to tactical units or task forces. (DOD Dictionary)

tactical recovery of aircraft and personnel. A Marine Corps mission performed by an assigned and briefed aircrew for the specific purpose of the recovery of personnel, equipment, and/or aircraft when the tactical situation precludes search and rescue assets from responding and when survivors and their location have been confirmed. (DOD Dictionary)

target area of interest. The geographical area where high-value targets can be acquired and engaged by friendly forces. (DOD Dictionary)

targeting. The process of selecting and prioritizing targets and matching the appropriate response to them, considering operational requirements and capabilities.

task. A clearly defined action or activity specifically assigned to an individual or organization that must be done as it is imposed by an appropriate authority. (DOD Dictionary)

task organization. An organization that assigns to responsible commanders the means with which to accomplish their assigned tasks in any planned action. (DOD Dictionary)

T-day. The effective day coincident with Presidential declaration of national emergency and authorization of partial mobilization (not more than 1,000,000 personnel exclusive of the 200,000 call-up). (NTRP 1-02)

time-phased force and deployment data. The time-phased force, non-unit cargo, and personnel data combined with movement data for the operation plan or operation order, or ongoing rotation of forces. (DOD Dictionary)

times. The Chairman of the Joint Chiefs of Staff coordinates the proposed dates and times with the commanders of the appropriate unified and specified commands, as well as any recommended changes to when specified operations are to occur (C-, D-, M-days end at 2400 hours Universal time [Zulu time] and are assumed to be 24 hours long for planning). (DOD Dictionary)

time-sensitive target (TST). A joint force commander-validated target or set of targets requiring immediate response because it is a highly lucrative, fleeting target of opportunity or it poses (or will soon pose) a danger to friendly forces. (DOD Dictionary)

undersea warfare. Military operations based within the undersea domain including the strategic, operational, and tactical employment of undersea systems, vehicles, weapons, sensors, and infrastructure.

warning order. 1. A preliminary notice of an order or action that is to follow. 2. A planning directive that initiates the development and evaluation of military courses of action by a commander. (DOD Dictionary)

waterspace management. The allocation of waterspace in terms of antisubmarine warfare attack procedures to permit the rapid and effective engagement of hostile submarines while preventing inadvertent attacks on friendly submarines. (DOD Dictionary)

W-day. Declared by the President, W-day is associated with an adversary decision to prepare for war (unambiguous strategic warning). (NTRP 1-02)

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LIST OF ACRONYMS AND ABBREVIATIONS

This list is intended to be a planner tool and therefore includes acronyms and abbreviations normally encountered by naval planners. As a result, not all entries are found in the text. For complete lists of all abbreviations used in joint and Navy doctrine see the DOD Dictionary of Military and Associated Terms, and NTRP 1-02, Navy Supplement to the DOD Dictionary of Military and Associated Terms. These online lists are updated continuously and may be downloaded and printed.

DOD Dictionary of Military and Associated Terms: http://www.jcs.mil/Doctrine

Navy Supplement to the DOD Dictionary of Military and Associated Terms: doctrine.navy.mil

AD	air defense
AMD	air and missile defense
AO	area of operations
AOA	amphibious objective area
APEX	Adaptive Planning and Execution
ARG	amphibious ready group
ASCM	antiship cruise missile
ASW	antisubmarine warfare
AWG	assessment working group
BMD	ballistic missile defense
BN	battalion
BPT	be prepared to
BT	break transmission
C2	command and control
CA	civil affairs
CAB	commander's assessment brief
CAP	combat air patrol
CAS	close air support
CBRNE	chemical, biological, radiological, nuclear, and high-yield explosives
CCDR	combatant commander
CCIR	commander's critical information requirement
CDCM	coastal defense cruise missile
CG	guided missile cruiser
CI	counterintelligence
CJCS	Chairman of the Joint Chiefs of Staff
CJCSM	Chairman of the Joint Chiefs of Staff manual
CJTF	commander, joint task force
CMO	civil-military operations
COA	course of action
COG	center of gravity
COLS	concept of logistics support

COMSEC	communications security
CONOPS	concept of operations
CONPLAN	concept plan
COP	common operational picture
COPS	current operations
COS	chief of staff
CP	command post
CRUDES	cruiser-destroyer
CSG	carrier strike group
CSS	combat service support
CTF	commander, task force
CTG	commander, task group
CTU	commander, task unit
CVN	aircraft carrier (nuclear)
CVW	carrier air wing
CWC	composite warfare commander
DA	direct action
DCA	defensive counterair
DDG	guided missile destroyer
DEPORD	deployment order
DIA	Defense Intelligence Agency
DIM	daily intentions message
DIRLAUTH	direct liaison authorized
Div	division
DOD	Department of Defense
DOS	Department of State
DP	decisive point
DR	disaster relief
DS	direct support
DSM	decision support matrix
DST	decision support template
EEFI	essential elements of friendly information
EEZ	exclusive economic zone
ELINT	electronic intelligence
ESG	expeditionary strike group
EW	electromagnetic warfare
EXORD	execute order
FA	FREEDOM ASSURANCE
FFIR	friendly force information requirement
FHA	foreign humanitarian assistance
FLS	forward logistics site
FMC	fully mission capable

FOB	forward operating base
FID	foreign internal defense
FOPS	future operations
FP	force protection
FRAGORD	fragmentary order
FUPLANS	future plans
GCC	geographic combatant commander
GCCS-M	Global Command and Control System-Maritime
GENADMIN	general admin (message)
GENTEXT	general text
HA	humanitarian assistance
HHQ	higher headquarters
HN	host nation
HNS	host-nation support
HQ	headquarters
HUMINT	human intelligence
I&W	indications and warnings
IADS	integrated air defense system
IAMD	integrated air and missile defense
IAW	in accordance with
IC	intelligence community
ICW	in conjunction with
IO	information operations
IOT	in order to
IPOE	intelligence preparation of the operational environment
IR	intelligence requirement
IRC	information-related capability
ISB	intermediate staging base
ISO	in support of
ISR	intelligence, surveillance, and reconnaissance
ISR-T	intelligence, surveillance, reconnaissance, and targeting
ITL	intelligence task list
JFACC	joint force air component commander
JFC	joint force commander
JFLCC	joint force land component commander
JFMCC	joint force maritime component commander
JFSOCC	joint force special operations component commander
JIB	joint information bureau
JICO	joint interface control officer
JIPOE	joint intelligence preparation of the operational environment
JOA	joint operations area
JP	joint publication

JRAM	Joint Risk Analysis Methodology
JTF	joint task force
LNO	liaison officer
LOAC	law of armed conflict
LOE	line of effort
LOG	logistics
LOO	line of operation
LPD	amphibious transport dock
MARSUPPLAN	maritime supporting plan
MASINT	measurement and signature intelligence
MCC	maritime component commander
MCM	mine countermeasures
MCMRON	mine countermeasures squadron
MCOO	modified combined obstacle overlay
ME	main effort
MIO	maritime interception operations (USN)
MIOC	maritime intelligence operations center
MISO	military information support operations
MIW	mine warfare
MMF	multinational maritime force
MNF	multinational force
MOC	maritime operations center
MOE	measure of effectiveness
MOP	measure of performance
MPC	maritime planning center
MPF	maritime pre-positioning force
MPRA	maritime patrol and reconnaissance aircraft
msg	message
MSO	maritime security operations
NAI	named area of interest
NATO	North Atlantic Treaty Organization
NAVFOR	Navy forces
NCC	Navy component commander
NCE	noncombatant evacuee
NEO	noncombatant evacuation operation
NFC	numbered fleet commander
NGO	nongovernmental organization
NIMS	National Incident Management System
NPP	Navy planning process
NTTP	Navy tactics, techniques, and procedures
NWP	Navy warfare publication
OA	operational area

OBJ	objective
OE	operational environment
O/O	on order
OPAREA	operating area
OPCON	operational control
OPGEN	operational general (message)
OPLAN	operation plan
OPNAV	Office of the Chief of Naval Operations
OPNAVINST	Chief of Naval Operations instruction
OPORD	operation order
OPS	operations
OPSEC	operations security
OPT	operational planning team
OPTASK	operation tasking (message)
OTHT	over-the-horizon targeting
PA	public affairs
PERSTAT	personnel status report
PIR	priority intelligence requirement
PLAD	plain language address directory
PLANORD	planning order
PMESII	political, military, economic, social, information, and infrastructure
POD	port of debarkation
POL	petroleum, oils, and lubricants
PPAG	proposed public affairs guidance
R&S	reconnaissance and surveillance
RAS	replenishment at sea
RCPA	relative combat power analysis
RFC	request for capabilities
RFF	request for forces
RFI	request for information
ROC	rehearsal of concept
ROE	rules of engagement
RUF	rules for the use of force
SAG	surface action group
SATCOM	satellite communications
SE	supporting effort
SECDEF	Secretary of Defense
SEPCOR	separate correspondence
SH	shaping effort
SLOC	sea line of communications
SMEAC	situation, mission, execution, administration and logistics, and command and control

SOF	special operations forces
SOP	standard operating procedure
SPOD	seaport of debarkation
SR	special reconnaissance
SS	submarine (conventional propulsion)
SSE	sensitive site exploitation
SSN	submarine (nuclear-powered)
sub	submarine
SUCAP	surface combat air patrol
SUPPLAN	support plan
SUS	sustaining effort
SUW	surface warfare
TACON	tactical control
TACTAS	tactical towed-array sonar
TBD	to be determined
TBM	theater ballistic missile
TF	task force
TG	task group
TLAM	Tomahawk land-attack missile
TPFDD	time-phased force and deployment data
TTP	tactics, techniques, and procedures
TTW	territorial waters
UJTL	Universal Joint Task List
UNSCR	United Nations Security Council resolution
UNTL	Universal Naval Task List
U.S.	United States
USG	United States Government
UW	unconventional warfare
VTC	video teleconferencing
WARNORD	warning order
WG	working group
WSM	waterspace management

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